Manual

Production Data Manager
SCS-PDM 8.1

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1 HYDRA Production Data Manager

Purpose

The HYDRA Production Data Manager is a data interface that enables all kinds of data to be exchanged using the HYDRA system. They might include, for example, order, human resources, machine and process data provided by external BDE/ PZE systems, data concentrators, machine controls or similar systems.

The data are transferred to the HYDRA database where they are processed further in the installed HYDRA applications.

Requirement: external systems are able to operate the HYDRA Production Data Manager in HYDRA standard format.

The HYDRA Production Data Manager can be used, for example, to transfer entry data for MDE/ BDE/ PZE/ LLE/ ZKS directly into the HYDRA system or to retrieve data from HYDRA without using a terminal for entry.

Data can also be transferred or retrieved that would otherwise normally be entered into or retrieved from HYDRA clients.

The data can be transferred to HYDRA online or offline using an external application.

Implementation considerations

You use this component if:

- You want to exchange data with HYDRA yourself or via 3rd party applications
- You want to connect existing applications to HYDRA

Integration

The HYDRA Production Data Manager provides its own program library as well as related interface calls for integration.

Features

- Programming libraries:
    - For integration into the user-defined application
- Function descriptions:

    - Based on the HYDRA objects and their functions

- Special workshops:
    - for objective-oriented integration and use of the HYDRA Production Data Manager

2 HYDRA Production Data Manager - Technology

2.1 Overview

There are different mechanisms to connect to third-party systems. The sections that follow describe the individual options in more detail:

Online connection “external interface” (ddcom)

In online mode data are directly sent from the external system (on HYDRA server or another server within the network) to HYDRA and processed there by a library (ddcom) using TCP/IP. The library is available as 32 bit DLL in Windows and as library in different UNIX derivatives.

Online/offline connection "communication-DLL“ (hyextcom)

If an external system is supposed to be connected, which however does not have the above-mentioned library, data can be sent directly to HYDRA where they are buffered and processed at a later point in time by means of a special communication service using a communication DLL.

- This communication DLL is only available for Windows at the moment.

Offline connection "Batch“

In offline mode ASCII files, which are checked and taken over to the system by a HYDRA process, are transferred to HYDRA.

Online connection “external interface“ to Windows terminal (ctcom)

In online mode data are directly sent from the external system (on Windows terminal or another Windows computer within the network) to a Windows terminal and processed by means of a library (ctcom) using TCP/IP.

- Subject to the sent data, a special processing is required for the Windows terminal if this interface is used for the connection. The sent data are prepared, posted locally and sent to the HYDRA server.
- The library is only available as DLL for Windows at the moment.

Using these interfaces, order postings, personnel postings, quantity postings and status postings, among other things, or other data records can be transferred to HYDRA in online or offline mode via an external application. Moreover order information, machine information etc. can be requested in list form in online mode.

2.2 General notes on using the HYDRA Production Data Managers

The user has to consider the following points when using HYD-PDM:

☞ **To avoid errors:**

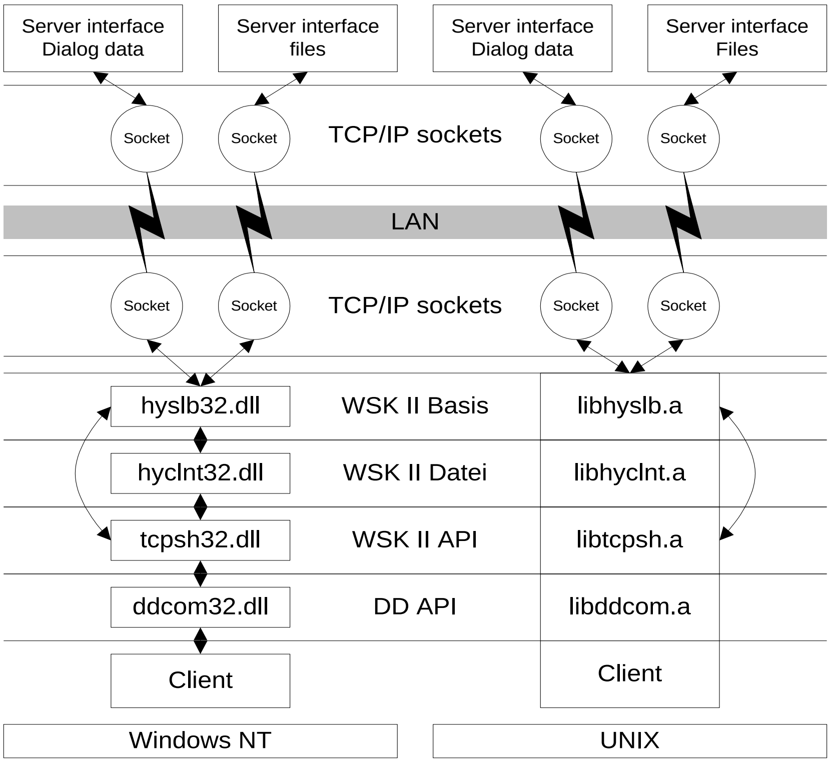
The SCS-PDM interface requires the dialog data to be in the correct chronological order.

If data are sent and their chronological order is wrong errors in determining quantities and activities will occur. In addition to this, unintentional plausibility errors might arise due to wrong plausibility checks. Consequently, data records might be rejected.

2.3 SCS-PDM Client Libraries

2.3.1 Online connection “external interface“ (ddcom)

The library ddcom exchanges data with the server via TCP/IP using several network libraries of NET-WSK (pl. see figure). The server answers requests and, if necessary, generates lists that may also be read using the library ddcom. In Windows NT the libraries are combined as DLL and in UNIX they are combined as libraries.



2.3.1.1 Initialization of the interface

The interface to HYDRA has to be initialized when starting the program. If the initialization process is not successful the external interface can continue working OFFLINE but it has to be initialized successfully before data will be sent.

int ddinit(char *host, short user);

host Host name or IP address

Using a host name can lead to delays in the communication if the resolution of the host name into the IP address via a DNS server is slow. We recommend using the IP address.

user HYDRA user number

Unique user number of the shop floor device or concentrator. The user number is determined by 3000 + “terminal number“. Terminal 1 (if the shop floor device is directly connected) or concentrator 1 (if several shop floor devices are connected) corresponds to the HYDRA user number 3001 etc.

Please note:

The numbering 3000 + "terminal number“ is intended. As separate processes are established, which do not depend on real terminal processes. However, the number “USR” entered in the dialog string (pl. see below) must be 2000 + "terminal number“.

Return values of the function (communication errors)

0	**Initialization successful**
1	**Initialization of Windows sockets incorrect (ddinit() only)**
2	**No port found for “Hydra file server“ (ddinit() only)**
3	**No port found for a HYDRA interface (e.g. “HYDRA LANT-DD server 1“) (ddinit() only)**
4	Unable to build up a connection
5	Dispatch cancelled with error
6	Reception cancelled with error
7	Connected to wrong interface
8	(Local) interface operates offline at the moment
9	Error in opening a file via the “HYDRA file server“
10	Error in reading a file via the “HYDRA file server“
11	Error in writing in a file via the “HYDRA file server“
12	Error in closing a file via the “HYDRA file server“
13	Data is not yet available
23	Interface has not yet been initialized successfully
24	Synchronization with server failed (automatic re-initialization)

2.3.1.2 Sending of dialog data

int ddsend(char *dd, short *dberr);

dd Dialog data, max. 1024 Bytes

Please note:

If the ddcom32.dll is used with Delphi , the passed pointer dd has to point at a data field of at least 1024 bytes.

dberr 0 = successful, > 0 errors occurred (data base error)

This parameter has to be indicated because of compatibility reasons but it is no longer used for dialog data. The value set in dberr is undefined.

Return values of the function (communication errors)

0	**Data sent**
1	Initialization of Windows sockets is incorrect (ddinit() only)
2	No port found for “HYDRA file server“ (ddinit() only)
3	No port found for a HYDRA interface (e.g. “HYDRA LANT-DD-Server 1“) (ddinit() only)
4	**Unable to build up a connection**
5	**Dispatch with error cancelled**
6	**Reception with error cancelled**
7	**Connected to wrong interface**

8	**(Local) interface offline at the moment**
9	Error in opening a file via the “HYDRA file server“
10	Error in reading a file via the “HYDRA file server“
11	Error in writing in a file via the “HYDRA file server“
12	Error in closing a file via the “HYDRA file server“
13	Data is not yet available
23	**Interface has not yet been initialized successfully**
24	**Synchronization with server failed (automatic re-initialization)**

2.3.1.3 Receiving the result

int ddreceive(char *dd, short *dberr);

dd Dialog data, max. 1024 bytes

Please note:

If the ddcom32.dll is used with Delphi the transferred pointer dd has to point at a data field of at least 1024 bytes.

dberr 0 = successful, > 0 errors occurred (data base error)

This parameter has to be indicated because of compatibility reasons but it is no longer used for dialog data. The value set in dberr is undefined.

Return values of the function (communication errors)

0	**Data received**
1	Initialization of Windows sockets is incorrect (ddinit() only)
2	No port found for “HYDRA file server“(ddinit() only)
3	No port found for a HYDRA interface (e.g. “HYDRA LANT-DD-Server 1“) (ddinit() only)
4	**Unable to build up a connection**
5	**Dispatch cancelled with error**
6	**Reception cancelled with error**
7	**Connected to wrong interface**
8	**(Local) interface operates OFFLINE at the moment**
9	Error in opening a file via the “HYDRA file server“
10	Error in reading a file via the “HYDRA file server“
11	Error in writing in a file via the “HYDRA file server“
12	Error in closing a file via the “HYDRA file server“
13	**Data is not yet available**
23	Interface has not yet been initialized successfully
24	Synchronization with server failed (automatic re-initialization)

The function ddreceive() has to be called as long as the function returns “13” (pl. see the example code below). When the function returns “0“ data have been received, if the function does neither return “0“ nor “13“ the connection to the HYDRA process does no longer exist. In this case, either the server did not accept the command within 10s (adjustable via the function “int ddmaxsendwait(int sec);“) or processing of the command takes more than 120s (adjustable via “int ddmaxrecvwait(int sec);“).

Having received an error code that is unequal “0“ and “13“, sending of data is impossible for 600s (adjustable via “int ddminoffline(int sec);) to prevent the connected system from always getting in a timeout when a HYDRA process is finished (e.g. for service reasons). (Please see offline phase).

Before going offline, data should be buffered in a file. The subsequent transfer to HYDRA has to be in the correct chronological order ("top down“ pl. also see chapter General notes on using the).

"Sleep“ has to be implemented between the different reception cycles to prevent the system from an overload:

```
ret = ddsend(stempelsatz,&dberr);
if (ret == 0)
{
        do
        {
                ret = ddreceive(resultat,&dberr);
                if (ret == 13)
                        sleep(1); /* Unix + Windows */

        } while (ret == 13)
        /* Fehlerverarbeitung von ret */
        ...
}
else
{
        /* Fehlerverarbeitung von ret und ggf. dberr */
        ...
}
```

If the communication is effective (return value of the function ddreceive is 0) the return values (return code RET) within dialog data have to be evaluated to see whether the command was successful or not.

2.3.1.4 Reading of files

2.3.1.5 Requesting lists

Lists are requested via special commands and filed in the HYDRADIR\spool\ directory. The procedure of requesting lists is the same as it is for *"sending dialog data"* and "receiving *the result"* with the only difference that afterwards the list is transmitted from the server to the client.

int ddsend(char *dd, short *dberr);

dd Dialog data, max. 1024 bytes

dberr 0 = successful, > 0 errors occurred (database error)

Return values of the function (communication error)

please see above

Example

Access authorizations are made available, for example, via the command DLG=LIST;27. The structure of dialog data for the shop floor device with terminal number 4 is as follows: "DLG=LIST;27|DATEI=.\spool\hyu2004.c27|DAT=03/30/2001|ZEI=36003|USR=2004|...“

2.3.1.6 Receiving the result and reading the list

int ddreceive(char *dd, short *dberr);

dd	Dialog data , max. 1024 bytes
dberr	0 = successful, > 0 errors occurred (data base error)

Return values of the function (communication error)

please see above

Provided that data have been received (>=0) and dberr = 0 the data are included in the indicated file within the directory HYDRADIR\spool\ - in our example the valid access authorizations.

Attention

When interpreting data, the field abbreviations of the 1st line are supposed to be interpreted. For future versions it might be the case that **the sequence of the columns changes** or **new columns are inserted in any place.** Such columns with unknown field abbreviation have to be ignored and mustn't lead to a program error or to the program being terminated.

The file is read with the following function:

int ddgetfile(char *remotefile, char *localfile);

Remote file	{File name} (field ID FILE) as it is indicated for the request within the dialog data.
localfile	Local file name

Return values of the function (communication error)

0	File read
< 0	Communication error
8	(Local) interface operates OFFLINE at the moment
10	Error while reading from the server file
16	Error while opening the local file
17	Error while writing into the local file

2.3.1.7 Offline phase

If the return value “8” is returned while “sending dialog data” or “receiving the result”, the interface has switched into the offline mode at the client. This is the case, when HYDRA is shut down for service reasons, for example. Now it is impossible to send data for a time of 600s (adjustable via ddminoffline()) to prevent the connected system from getting into a timeout when trying to establish the connection. Only after this time has expired, the library tries to restore the connection.

int ddminoffline(int sec)

sec	Minimum offline time in seconds after network error or offline of the interface

Return values of the function

0 Value set

-1 Invalid value. Must be stated in seconds > 0.

Attention:

Before going offline, all data should be buffered in a file. The following transfer to HYDRA has to be carried out in chronological order ("top down“).

2.3.1.8 Option “parallel requests“

By means of the function *long ddmessagetype(long user)* it is possible to send several requests at the same time to the server. The request is assigned to a unique HYDRA user number by setting ddmessagetype(**user**) before sending dialog data with ddsend(). The ID has now to be set by means of ddmessagetype(user) before receiving the result with ddreceive().

Attention:

- The HYDRA user numbers in use must be unique within the entire system.
- Since the PDM library is designed single-threaded, PDM function calls have to be secured by the application (please see example)

Example for Windows:

```
CRITICAL_SECTION csDDcom;
long user1 = 3003;

EnterCriticalSection (&csDDcom);
ddmessagetype(user1);
ret = ddsend(stempelsatz,&dberr);
LeaveCriticalSection (&csDDcom);

if (ret == 0)
{
        do
        {
                EnterCriticalSection (&csDDcom);
                ddmessagetype(user1);
                ret = ddreceive(resultat,&dberr);
                LeaveCriticalSection (&csDDcom);

                if (ret == 13)
                        Sleep(1000);

        } while (ret == 13)
        /* Error processing of ret */
        ...
}
```

```
else
{
	/* Error processing of ret and dberr */ if necessary
	...
}
```

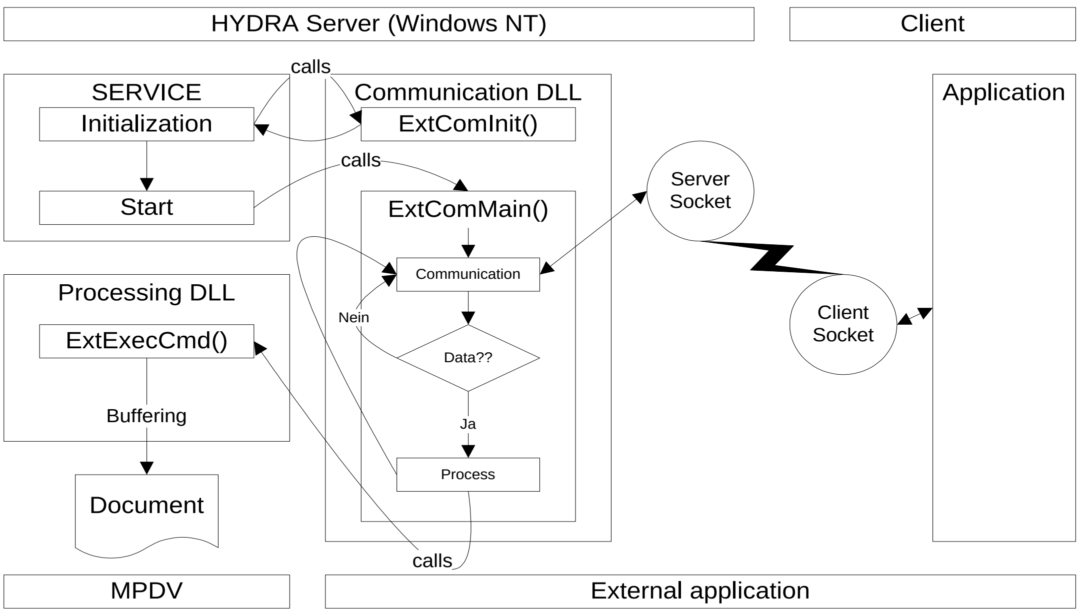
2.3.1.9 Example

```
#include <windows.h>
#include <stdio.h>
#include "ddcom.h"

char host[256];
char remotefile[256];
char localfile[256];
char dd[2048];
short user = 0;
long ms = 0;
short dberr = 0;
int rc;

void liste ( void )
{
	rc = ddsend(dd,&dberr);
	printf("ddsend: %d [%s]\n",rc,dd);

	if (rc == 0)
	{
		do
		{
			rc = ddreceive(dd,&dberr);

			if (rc == 13)
				Sleep(1000);		/* waiting intervals of 1s */
		} while (rc == 13);
		printf("ddreceive: %d [%s]\n",rc,dd);
	}

	if (rc == 0)
	{
		rc = ddgetfile(remotefile, localfile);
		printf("ddgetfile: %d\n",rc);
	}
}


int main (int argc, char *argv[])
{
	strcpy(host, "192.168.10.1");
	user = 3001;

	sprintf(remotefile,"./spool/hyu%d.c27",user);
	sprintf(dd,"DLG=LIST;27|TNR=105|DAT=04/23/1999|DATEI=%s",remotefile);

	sprintf(localfile,"list27.dat");

	rc = ddinit(host,user);
	printf("ddinit: %d\n",rc);

	if (rc != 0)
		return rc;

#ifdef TEST
	ddmaxrecvwait(10); /* wait for result 10 seconds at most */
	ddminoffline(30);  /* stay 30 seconds OFFLINE at least */
#endif

	liste();

	return 0;
}
```

2.3.2 Online/offline connection “communication DLL“ (hyextcom)

In this connection communication works e.g. on the basis of TCP/IP sockets. The passed data are accepted, buffered and asynchronously processed (no direct server response) or immediately processed synchronously (with server response, not available at the moment) by HYDRA.

2.3.2.1 Process flow of communication on the transmission level



The name of the communication DLL and the processing DLL can be adjusted in HYDRA via the registry or via environment variables. The service is started with starting HYDRA and stopped when terminating HYDRA.

2.3.2.2 Processing

This service starts the initialization of the communication DLL and thus transfers the name of the processing DLL.

Function in the communication DLL (file hyextcou.c) for the initialization:

int ExtComInit(char * ExecDLL, int Id)

ExecDLL	Name of the processing DLL incl. directory (e.g. “e:\hydra\hyprog.dll“). A debug DLL can be indicated here for debug or test reasons.
Id	ID number of the interface to be used in the communication DLL. Depending on the ID, it is possible to save, e.g., the configuration for several interfaces in the communication DLL.

Return values of the function (communication errors)

= 0	Initialization of the communication DLL successful
≠ 0	Initialization of the communication DLL was not successful → The service is finished again.

Afterwards the service starts the main function of the communication DLL, which accepts the commands of the client and returns the result “command received“ after processing by the processing DLL (e.g. ACK or NAK). The main function of the communication DLL is not terminated. Processing in the processing DLL buffers the data on the server. Then the data are cyclically read by a HYDRA module and written into the database.

Function in the communication DLL for the main function:
int ExtComMain(void);

void	- no parameters -

Return values of the function (communication errors)

= 0	Communication DLL terminates itself normally. → The service is finished again.
≠ 0	Communication DLL terminates itself successfully because of an error. → The service is finished again.

Function in the processing DLL for the data transfer:
int ExtExecCmd(char *cmd, int cmdsize, char *res, int ressize);

cmd	Data record for HYDRA, terminated by zero character („\0“), for its structure pl. see below
cmdsize	Size of data record buffer
res	Return buffer (intended for future extensions with online response)
ressize	Size of the return buffer

Return values of the function (communication errors)

= 0	Data were buffered
≠ 0	Data could not be buffered

2.3.2.3 Test possibility

In the MS-DOS input request

Instead of the processing DLL, the service is started by a debug DLL that only displays data on the screen. In this case, the service only runs as program in the MS-DOS prompt.

- The example tests the connection of two clients via the IDs 10021/10022:
  Start with *hyextsrv -d -s10021 -lhyextdbg.dll or* by *hyextsrv -d -s10022 -lhyextdbg.dll*
  The following lines are displayed (if the attached sample source code is used):
  ExtExecInit: InterfaceFile "hyextsrv.dat"
  ExtExecCmd: CMD "DLG=SYSTEM.TIME|ID=1|"
  ExtExecCmd: CMD "DLG=SYSTEM.TIME|ID=2|"
  ExtExecCmd: CMD "DLG=SYSTEM.TIME|ID=3|"
  ...
  The program is closed by pressing Ctrl and C.

As service via the service control

Two services are installed by means of which the connection can be tested as service.

- The services are installed with the command *dienste.bat –i {install}* whereas {install} stands for the installation directory.
- The services can now be started by *net start "HYDRA PDM connection 1*“ or *net start "HYDRA PDM connection 2“* (or via the service control).
- The services accept data with the IDs 10021 or 10022 and log them in float files *{install}\spool\hyextsrv.001 or {install}\spool\hyextsrv.002.*
- The services can be terminated by *net stop “HYDRA PDM connection 1“ or* by *net stop “HYDRA PDM connection 2“* (or via the service control).
- The services are removed from the service control via *dienste.bat -u {install}.*

2.3.2.4 Test possibility with MPDV

Instead of the communication DLL, the service is started with a simulation DLL, which passes on test data in intervals.

2.3.2.5 Server services in HYDRA

In HYDRA a service (server socket process) has to be started for each application that uses the PDM interface.

The IP address defines which server service the application works for. The IP address of the application is defined per service in the registry:

HYDRA server *ext. Client*

PDM-connection 1	Server socket a	⟷	Client socket	***application 1***
PDM-connection 2	Server socket b	⟷	Client socket	***application 2***
....				
PDM-connection n	Server socket z	⟷	Client socket	***application n***

The server services have to be installed as “HYDRA services“ on the HYDRA server.

2.3.3 Offline connection “batch“ or “I-Doc“

Third-party system dialog data are archived in a file to realize an offline connection within the batch. Multiple dialog data within a file are to be separated each by a word-wrap.

Within a batch or I-doc the dialogs can be treated as one or several completed database transaction(s).

There are the following alternatives regarding the communication between the external system and HYDRA:

- Transfer provided by the customer
- Use of the time-controlled host communication HYD-ZHK
- Transfer from SAP as I-Doc via the HYDRA MySap communication

2.3.3.1 Data base transactions via several dialogs

The beginning and the end of a database transaction are controlled via particular dialogs. Dialogs can only be summarized to transactions within one batch/I-Doc. The end of a batch/I-Doc corresponds to the completion of the transaction. Consequently, this is the following general procedure:

1. Dialog for starting a transaction
2. Data dialogs that are supposed to be summarized
3. Dialog for terminating a transaction
4. Dialog for starting a transaction
5. ...

The dialog WORK.BEGIN starts a transaction. The actions that are meant to be summarized are executed by the other dialogs. The dialog WORK.END completes the transaction.

Please note:

The pooling of several dialogues in a transaction is intended primarily for consistent master data maintenance. Acquisition dialogs e.g. Logging in and out of operations or persons may not be combined with WORK.BEGIN and WORK.END, otherwise dead locks can occur, which can have serious errors throughout the HYDRA system resulted. Therefore WORK.BEGIN and WORK.END should be used only on special requirements and after consultation with MPDV.

Dialog: WORK.BEGIN				
Opens a database transaction. Thus, the automatic transaction processing of the interface is deactivated for the following dialogs.				
Parameter	**Type**	**Mandatory**	**Content**	**Description**
- No -				

Dialog: WORK.END				
Completes a database transaction. There are two different cases: 1) If errors appear in dialogs within the opened transaction the transaction will be rolled back, none of the dialogs are saved in the database. 2) If no errors occur during processing of the dialogs within the transaction, the transaction will be completed and data will finally be saved in the database.				
Parameter	**Type**	**Mandatory**	**Content**	**Description**
- No -				

2.3.3.2 Transfer provided by the customer

The file has to be transmitted, secured with the transfer types copy or FTP, by the external system. This transfer can – according to the client's requirements – be carried out at a certain point in time or in defined intervals.

The dialog file is transferred from the external system to HYDRA by setting the dialog file from the external system in a defined directory on the HYDRA server: with Unix in the directory /usr/hydra; with Windows NT in \hydra. Please note that a transfer to the HYDRA server may only be realized if the file is not yet available there, to prevent a loss of data caused by "overwriting".

Integrated in the HYDRA scheduler (hysched.cfg) the file is processed by HYDRA at stipulated times or in defined intervals.

```
# Processing controlled in intervals, e.g. every 5 minutes
I 5 ./hymw.out –u9998 –L –b<File name>

# Fixed processing, e.g. daily at 0:30
F 30  0  *  *  *  ./hymw.out –u9998 –L –b<File name>
```

The dialog data in the file <file name> are processed sequentially by the HYDRA interface program. The single commands are recorded in the file ./err/hyddi_b<USR>.<PID>.pro. The file is deleted by HYDRA after posting.

In case the “-z“ parameter is indicated, the processing result is additionally recorded in the system logs and can be evaluated at the HYDRA console via File → System information → System logs → DD-BATCH application.

2.3.3.3 Using of the time-controlled host communication HYD-ZHK

HYDRA also provides the HYD-ZHK module by way of which HYDRA is able to automate the connection: Then the third-party system only has to provide the respective files locally. HYDRA takes over the actual transfer to HYDRA as well as loading the files into the HYDRA database. For further information on technical requirements please contact the MPDV project management.

More detailed information is described in the product documentation entitled “HYD-ZHK“.

2.3.3.4 HYD-ZHK with UNIX

The following lines are entered in the HYDRA scheduler subject to the respective transfer type:

- **copy**

```
# HYD-ZHK (time-controlled host connection)
# for transfer Hydra(Unix)<->PPS(Unix) via NFS
# ATTENTION:
# - mounts for directory on remote system have to be accomplished when the host computer is started
# Parameter:
# - directory:      path on mounted directory of the remote system (NFS)
# - file name: name of the PPS-file
L HYD-ZHK I 60 ./hyd_zhk.scr MOD=GET LOCAL=hydpdm.asc REMOTE="/directory/file name" CMD="./hymw.out –u9998 –L -bhydpdm.asc"
```

- **FTP**

```
# for transfer Hydra(Unix)<->PPS(WinNT/UNIX/..) via FTP
# Parameter:
# - server:    Name/IP-address of the remote system
# - ftpuser:   user name for FTP access to the remote system
# - ftppasswd: appropriate code word
# - directory:      path on remote system
# - file name: name of the PPS-file
L HYD-ZHK I 60 ./hyd_zhk.scr MOD=GET HOST=server USER=ftpuser PWD=ftppasswd LOCAL=hydpdm.asc REMOTE="/directory/file name" CMD="./hymw.out –u9998 –L -bhydpdm.asc"
```

2.3.3.5 HYD-ZHK with Windows

The following lines are entered in the HYDRA scheduler subject to the respective transfer type:

- **copy**

```
# HYD-ZHK (time controlled host connection)
# for transfer Hydra(WinNT)<->PPS(WinNT) via NT-release
# ATTENTION:
# - ggf. hyd_zhk.scr adapt!!!
# Parameter:
# - server:   name of the remote system
# - release:  release name
# - file name: name of the PPS-file
L HYD-ZHK I 60 sh.exe ./hyd_zhk.scr MOD=GET LOCAL=hydpdm.asc REMOTE="\\\\server\\release\\file name" CMD="hymw.exe –u9998 –L -bhydpdm.asc"
```

- **FTP**

```
# for transfer Hydra(WinNT)<->PPS(WinNT/UNIX/..) via FTP
# Parameter:
# - server:   name/IP-address of the remote system
# - ftpuser:  user name for FTP-access to the remote system
# - ftppasswd: appropriate code word
# - directory:    path on remote system
# - file name: name of the PPS-file
L HYD-ZHK I 60 sh.exe ./hyd_zhk.scr MOD=GET HOST=server USER=ftpuser PWD=ftppasswd LOCAL=hydpdm.asc REMOTE="/directory/file name" CMD="hymw.exe –u9998 –L -bhydpdm.asc"
```

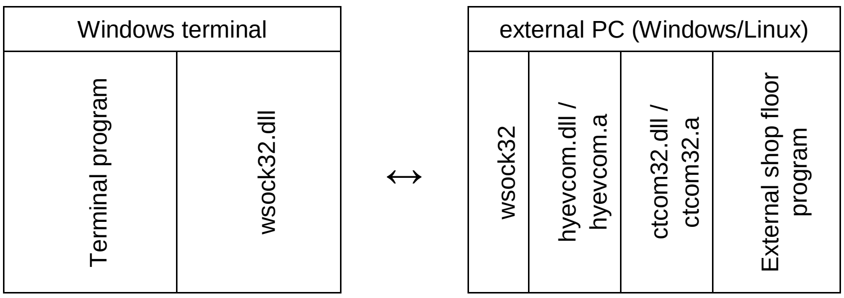
2.3.3.6 I-Docs from SAP via HYDRA-MySap

With this variant I-Docs are transferred from SAP to HYDRA. These I-Docs contain the dialogs for the production data manager. Processing of the dialogs is started in HYDRA once I-Docs have arrived.

By way of customizing services rendered by MPDV the respective settings are configured or communicated within the MySap configuration.

2.3.4 Online connection "external interface“ to Windows terminal (ctcom32)

The communication from the external computer to the HYDRA Windows terminal is realized by the *ctcom32 interface,* which needs the hyevcom library to be able to transfer data via Windows socket-32 *(wsock32.dll).* **The external computer runs with the operating systems Windows or Linux.**



Files required for Windows

ctcom.h

Definitions for the data transfer

ctcom32.lib

Library

ctcom32.dll/hyevcom.dll

Runtime libraries

ctcom32tst.exe

Test program for simulation of ext. PC

Files required for Linux:

ctcom.h

Definitions for the data transfer

ctcom32.a

Library

ctcom32tst.out

Test program for simulation of ext. PC

2.3.4.1 Setting of timeouts

Setting of timeouts (only to optimize!) for hyevcom. Values greater than 0 are accepted only.

```
int FAR PASCAL cttimeout
    (
    long t_c,        /* Timeout "Connect" in [s], default 5s */
    long t_s,        /* Timeout "Send" in [s],    default 10s */
    long t_r         /* Timeout "Receive" in [s], default 10s */
    );
```

t_c Timeout “Connect“ in seconds, default 5s

t_s Timeout “Send“ in seconds, default 10s

t_r Timeout “Receive“ in seconds, default 10s

Return values of the function (communication errors)

0 Transfer successful

2.3.4.2 Sending of dialog data

Transfer of a data buffer to the Windows terminal.

Please note:

- Each posting is to be completed by the characters EOT (End of Transmission) or the character 4 or Ctrl-D.

- **int FAR PASCAL ctcom_snd**

```
(
        char *host name,    /* Hos tname or IP-address of the CT */
        int   port,         /* Port */
        char *data,         /* Data */
        int   size,         /* Length */
        int   *err          /* Error code */
);
```

Host name	Host name or IP address of the Windows terminal. This value **should** be adjustable from the outside via the Environment, INI-file (or similar).
port	Indication of the port, on which the Windows terminal expects data. This value **should** be adjustable from the outside - just as it is the case for the host name.
data, size	Data buffer and size of the data buffer
err	Definite network error of the *hyevcom.*

- **Return values of the function (communication errors)**

0	Transfer successful
1	Initialization of Windows sockets incorrect
4	Connection cannot be established. The server module on the opposite side *host name* does possibly not run at port *port*.
5	Termination/error while sending
6	Termination/error while receiving
13	Data are not yet available
14	Size *size* of the buffer *data* is greater than MAX_CTCOM_DATA_SIZE (512 bytes)

2.3.4.3 Receiving the result

Reception of the result from the Windows terminal, once data have been sent successfully. The return value “13” indicates that data are not yet available. The program has to stop the communication independently after a certain time interval.

```
int FAR PASCAL ctcom_rcv
	(
	char *host name, /* Host name or IP-address of the CT */
	int   port,       /* Port */
	char *data,              /* Data */
	int   size,       /* Length */
	int   *err        /* Error code */
	);
```

Host name	Host name or IP address of the Windows terminal. This value **should** be able to be adjusted externally via the Environment, INI-file (or similar).
port	Indication of the port, on which the Windows terminal expects data. This value **should** be adjustable from the outside – just as it is the case for the "host name".
data, size	Data buffer and size of the data buffer
err	Definite network error of the *hyevcom.*

“Sleep“ has to be implemented between the different reception cycles in order to prevent the system from an overload:

```
int maxwarteanz = 60; /* max. 60 * 1second wait */
int anz = 0;
ret = ctcom_snd(dd,...);
if (ret == 0)
{
	do
	{
		ret = ctcom_rcv(dd, ...);
		if (ret == 13)
		{
			Sleep(1000);
			anz++;
		}
	} while ((ret == 13) && (anz < maxwarteanz))

	if (ret == 0)
	{
		/* Reception successful */
		...
	}
	else
	{
		/* Reception not successful: error processing of ret */
		...
	}
}
else
{
	/* Dispatch not successful: error processing of ret */
	...
}
```

If the communication was successful (return value of the function ctcom_rcv is 0) the return values have to be evaluated in the dialog data whether or not the command was successful.

Please note for the dialog data structure:

Dialog data must include the ID ACTION=DLG_SEND in order for the Windows terminal to forward the data string directly to the HYDRA server.

Each return string additionally includes the following data:

ID	Description

FT_ERROR	*Return code*: Return value
	FT_ERROR = 0 : Activity realized
	FT_ERROR ≠ 0 : Activity not realized, Evaluate error code
	Examples:
	`..|FT_ERROR=0|..`
	`..|FT_ERROR=4|..`

FT_ERROR_TXT	*Short text:* Error description (short) if FT_ERROR ≠ 0 or optional info if FT_ERROR = 0
	Examples:
	`..|FT_ERROR_TXT=OK [0]|..`
	`..|FT_ERROR_TXT=TNR BUSY (Process runs) [4]|..`

FT_INFO_TXT [Optional]	*From CTWIN V# 7.2.4.52 on, the following info is attached to the receiving string, when it comes to a reception timeout* `..|FT_INFO_TXT=TIMEOUT (GateWay <-> TNR) T/Wo(100/0) ServerClientSocket->No Data from Client [5/5]|..`

2.3.4.4 Setup of the test

(1) Configure the gateway communication at the terminal as described in the following section and start the Windows terminal.

(2) You can test the connection to the terminal by way of the test program *ctcom32tst.exe*. Start the test program by *ctcom32tst.exe /out{Host name/IP address} {Port} {Data}.* It will then send the character string {Data} to the port (e.g. 9002) of the specified host.

```
Example:
ctcom32tst.exe 0 192.168.1.137 9002
"ACTION=DLG_SEND|DLG=A_TR|MNR=M100|ANR=903216010100|
DAT=11/01/2008|ZEI=20367|USR=9997|"
```

(3) The command is now processed accordingly at the terminal and forwarded to the HYDRA server.

2.3.4.5 Installation

Install the **online connection “external interface” to the the Windows terminal** at the HYDRA terminal as follows:

(1) The following needs to be configured in the ctwin.ini file within the section [GateWay-Communication] of the Windows terminal:
[GateWay-Communication]
Active=true
Port=9002

(2) All files mentioned above need to be copied into the directory where the external application is started.

3 HYDRA Production Data Manager - Preface

The data transfer to HYDRA is performed by “dialog data” in the ASCII format. The string consists of header and field data. Header data are required for HYDRA to identify the data type, whereas field data contain data of the external application.

3.1 Header data of the dialog data

The following data **always** have to be transmitted in a dialog by an external acquisition system to HYDRA. The single fields are separated by the character "|“.

Identification	Content/(type)	Description
DLG	{Dialog ID}	The dialog ID indicates the required functions, e.g.: DLG=P_AN to log staff on D LG=MNR.INSERT to create a machine
USR	{N4}	HYDRA user: This HYDRA user number is the unique ID of a HYDRA client: For terminals the respective terminal number is added to the terminal number 2000. Example: USR=2004 corresponds to terminal 4 **Please note:** Terminal numbers assigned to HYDRA terminals must not be used for the HYDRA-PDM connection. We recommend using a separate number range.
DAT	{mm/dd/yyyy}	Date: current date in the format mm/dd/yyyy“ Example: DAT=03/30/2007
ZEI	{seconds}	Time: point in time in the "seconds" format, i.e. in seconds as of midnight Example: ZEI=36003 for the time: 10:00:03 am

Dialog IDs having the structure **OBJEKT.AKTION** (such as MNR.INSERT = create machine) are BAPI calls, which are described in detail in the sections that follow.

The following data **can** be transferred in a dialog to HYDRA by an external data acquisition system.

ID	Content / {Type}	Description
ID	{Identification number}	When started for the first time the ID=0 is sent. The ID is increased by 1 for each other call. The return string contains the same ID as the dialog data string.
OFF={J|N}	OFF={J|N}	Offline identification. Indicate "*OFF=J*“, if data were captured and buffered offline. In this case the data are processed separately. ("*OFF=N*" doesn't have to be indicated)
DATEI	{C256}	File name: Indication of the file name when lists are requested. The file name is created by means of the Hydra user:

		.\spool\hyu{Hydrauser}.{extension} {Hydrauser} corresponds to the user numbers of the terminal. (The terminal 1 corresponds to the Hydrauser-number 2001 etc.) {extension} can be chosen subject to the data. Standard: “txt“ Example: ...|DLG=.\spool\hyu2004.txt|...

3.2 Common notes

☞ **To avoid errors:**

The SCS-PDM interface requires the dialog data to be in the correct chronological order.

If data are sent and their chronological order is wrong errors in determining quantities and activities will occur. In addition to this, unintentional plausibility errors might arise due to wrong plausibility checks. Consequently, data records might be rejected.

- The header data DLG, USR, DAT and ZEI always have to be indicated completely within dialog data.
- OFF has only to be sent if it deals with data that were collected offline (default is OFF=N).
- It is not mandatory to keep a special order of IDs in dialog data..
- Double data records are filtered out by the process, provided that a valid ID is forwarded within header data.
- The plausibility of data is basically checked prior to posting. Incorrect data records are ignored and recorded in an error log. Processing ignores unknown field identifications.
- The field identifications ANR, AGNR, PNR, KNR and MNR have adjustable lengths and thus may vary in the HYDRA basic settings according to the respective configuration. Checks regarding data field lengths are based on the settings made within the HYDRA setup.

- All fields within a data record are separated by the "|“ character.
  Thus, the length of the data is variable but may not exceed the maximum length specified. If the characters "|“ pipe or "\“ backslash are included the data content they have to be masked by another preceding backslash :
  \ → \\
  | → \|

3.3 Field data

Further IDs (“field data”) have to or can be entered depending on the dialog ID (“DLG“).

3.4 Return values

Each return string includes the following header data:

Identification	Content/(type)	Description
RET	{N8}	Return code: Return value RET = 0: activity executed RET ≠ 0: activity has not been executed; evaluate error code Examples: ...|RET=0|... ...|RET=60|...
KT	{C20}	Short text: description of the error (short) if RET ≠ 0 or optional info if RET = 0 Examples: KT=Already logged on KT=In 2040
LT	{C40}	Long text: description of the error (long) if RET ≠ 0 or optional info if RET = 0 Example: |LT=OP cannot be logged on several times. LT=In Hans Meier

Each return string **may** optionally include the following data

Identification	Content/(type)	Description
ID	{Identification number}	The return string contains the same ID as the dialog data string, provided that it was indicated there.
	{C256}	Dialogdaten über Dialogdatei, nur in Kombination mit RETFILE (nur wenn im BAPI-Call angegeben)
RETFILE	{C256}	Rückgabewerte über Rückgabedatei, nur in Kombination mit DLGFILE (nur wenn im BAPI-Call angegeben)

Depending on the dialog ID ("DLG") additional IDs, which include, e.g., further return values (information, etc.) may also be returned.

Moreover, the following definitions apply for BAPI calls:

→ The BAPI calls ***.NEW, *.SELECT, *.LOCK** return a new or existing data record. The object name "Objekt" (value before the point) is returned as DATA=Objekt, the single values are transferred without the preceding object name.

→ The BAPI call ***.LIST** displays the selected data records in form of a list with the file name transferred in DATEI. The object name does not precede the columns. Colons are displayed as underline "_".

Error messages are described in detail in the section "HYDRA error messages" of the HYDRA-BDE manual.

3.5 BAPI call reference to the data model

The BAPI calls described in the corresponding documentation (MDM - Master Data Maintenance and DDM - Dynamic Data Management) partly refer to the HYDRA data model. For information on the data model, see the CUT-HDB course.

3.6 Lock mechanism for BAPI calls

The lock mechanism of HYDRA locks a virtual dataset on the server (e.g. "lock person 999999" or "lock machine group 145"). The virtual dataset can consist of one data record or of several data records. All BAPIs based on this virtual dataset check whether this dataset has been locked by another HYDRA client (e.g. console) prior to the BAPI calls ***.UPDATE** or ***.DELETE.**

Using the BAPI call ***.Lock** a client may lock a dataset/data record for a HYDRA client and once it has been processed by a BAPI call ***.UPDATE** it can again be released using the BAPI call ***.UNLOCK.**

In case a dataset/data record is locked at another console, the BAPI ***.LOCK** returns the error code 1666 (RET) as well as the HYDRA user (USR), the user (BEARB) and the client function (MODULE, provided that the other client has entered these details for the BAPI call ***.LOCK),** for which the dataset is locked.

To be able to delete a dataset/data record by a BAPI call ***.DELETE,** this data record should not be locked using the BAPI call ***LOCK.** The BAPI call ***.DELETE** checks in any case whether the data record/dataset is locked or not and cancels the process with an error if it is locked.

4 HYDRA Production Data Manager Basis - Data Collection

4.1 Reading time from HYDRA server

Structure of dialog data:

„DLG=SCMD;44|DAT=...|ZEI=...|USR=...|...“

The command does not expect further parameters.

The following values are returned to the terminal:

Field	Description	Example
RET=0	The return value RET is always 0.	RET=0
DAT={mm/dd/yyyy}	Current date in format "mm/dd/yyyy"	DAT=09/06/2001
ZEI={seconds}	Current time in format "seconds since midnight“ (Example: 10:00:03 is *ZEI=36003*)	ZEI=63142
GMTDAT={mm/dd/yyyy}	Current date in format "mm/dd/yyyy" in time zone GMT	GMTDAT=09/06/2001
GMTZEI={seconds}	Current time in format „seconds since midnight“ in time zone GMT	GMTZEI=55942
GMTOFF={±HHMM}	Description of the time zone local time	GMTOFF=+0200
GMTDIFF=[-]{seconds}	Deviation of local time from GMT in seconds	GMTDIFF=7200
WSDAT={mm/dd/yyyy}	Date of time shift from standard time to daylight saving time in the current year.	WSDAT=03/31/2002
WSZEI={seconds}	Time of time shift from standard time to daylight saving time in the current year.	WSZEI=7200
SWDAT={mm/dd/yyyy}	Date of time shift from daylight saving time to standard time in the current year.	WSDAT=10/27/2002
SWZEI={seconds}	Time of time shift from daylight saving time to standard time in the current year.	WSZEI=10800

4.2 Sending terminal status

Use the terminal status command to store information on the terminal status in the database. Requests asking to restart terminal, to download program or to read the authorizations are returned.

Structure of dialog data:

(The dialog ID DLG=SCMD;41 must be transferred as first entry):

„DLG=SCMD;41|DAT=...|ZEI=...|USR=...|TNR=...|...“

The following table contains the data that can be transmitted:

Field	Description	Example
IP={...}	Transfer of the IP-address. A character string of a maximum of 15 characters can be transmitted (could also be the hardware-address). With specific terminals, the hardware-address is configured in the HYDRA GUI; then it is not useful to transfer this field.	IP=192.168.20.234
PROG={...}	Program name of the terminal program or of the function list (max. 20 characters)	PROG=hyterm.exe
VERNR={...}	Version number of the terminal program (max. 10 characters)	VERNR=6.5.1.27
LOKANZ={...}	Number of local data records (after an offline phase)	LOKANZ=0
OFFDAT={...}	Date of the last offline phase	OFFDAT=05/21/1999
OFFZEI={...}	Time of the last offline phase in seconds since beginning of the day	OFFZEI=53214
ZNR={...} bzw. ZGRP={...}	Access number or access group If one of those fields is transferred with the 3 following fields the number of authorized badges, date and time are saved in the access status instead of the terminal status. This is necessary because several accesses can be administered via a terminal. You can make the request for an access or an access group.	ZNR=4 or ZGRP=5
BERANZ={...}	Number of persons authorized to access and to clock (the fields BERANZ, BERDAT and BERZEI are normally always transferred together directly after	BERANZ=124

	download of authorizations).	
BERDAT={...}	Date when the authorizations were last read	BERDAT=05/21/1999
BERZEI={...}	Time when the authorizations were last read	BERZEI=5213
SUBANZ={...}	Number of sub-systems (DS100 or LEGIC-reader)	SUBANZ=0
SUBOK={...}	Number of active sub-systems	SUBOK=0
STA={...}	Terminal status (if possible): O=Online, F=Offline: Terminal answers in the specified cycle "ZYKL" (see below) and is currently "online" or "offline". o=Online, f=Offline: Terminal does not cyclically answer, but answers only when the terminal changes to status "online" or "offline".	STA=O
NEUDAT={...}	Date of last terminal restart	NEUDAT=05/20/1999
NEUZEI={...}	Time of last terminal restart	NEUZEI=27594
PROGDAT={...}	Date of last download of terminal program	PROGDAT=05/13/1999
PROGZEI={...}	Time of last download of terminal program	PROGZEI=43834
VLISTSYNC=N	Must be transferred with terminal type 110 A-SUB (ALS sub-system) if the sequencing list has been updated. This flag is then reset.	VLISTSYNC=N
ALSSYNC=N	Must be transferred with terminal type 111 A-SYN (ALS data synchronization) if the production combinations have been updated. This flag is then reset.	ALSSYNC=N

The following values are returned to the terminal:

Field	Description	Example
RET={...}	The transfer and storage of the terminal status is only successful if RET=0. All further fields are only available if RET=0.	RET=0
ZYKL={...}	Time in seconds after which the next status message has to take place (is not processed with INCA as the time is	ZYKL=900

	read every 15 minutes).	
NEUSTART={J/N}	Do you want to restart terminal (‚J' Yes or ‚N' No)? This flag is reset if a new point in time is passed with the transfer of NEUDAT and NEUZEI. Consequently, the fields NEUDAT and NEUZEI can be passed with each terminal status.	NEUSTART=N
PROGLADEN={J/N}	Do you want to reload the terminal program? This flag is reset if a new point in time is passed with the transfer of PROGDAT and PROGZEI. Consequently, the fields PROGDAT and PROGZEI can be passed with each terminal status.	PROGLADEN=J
BERLESEN={J/N}	Do you want to read the authorizations again? This flag is reset if a new point in time is passed with the transfer of BERDAT and BERZEI. Consequently, the fields BERDAT and BERZEI can be passed with each terminal status.	BERLESEN=J
TNR={...}	Return of the terminal number for which the status was sent.	TNR=101
PROG={...}	Program name transmitted after last loading of program. If a list of the terminal programs is not available, this program name can be used to load the program.	PROG=hydra.exe
VLISTSYNC=J	Must be transferred with terminal type 110 A-SUB (ALS sub-system) if the sequencing list must be updated.	VLISTSYNC=J
ALSSYNC=J	Must be transferred with terminal type 111 A-SYN (ALS data synchronization) if the production combinations must be updated.	ALSSYNC=J
V:{module name}=…	The terminal sends version info of loaded modules (DLLs, scripts, etc.) in terminal status in format ...|V:dllname.dll=modules/function[[;version];date]|... This version information is stored in the software status as type "TERMINAL" and name "USER:{hydrauser}:{modulename}". Version: version number or time stamp in format "mm/dd/yyyy hh:mm:ss"	V:ftp.dll=FTP;7.2.1.5

	Date: date of module generation in format mm/dd/yyyy (requires hymw.out / exe 7.2.1.353)	

4.3 Reloading lists on the terminal

The terminals read their internal lists at regular intervals, e.g. the list of assigned machines, currently running orders and currently logged on persons.

If data, which is included in terminal lists and displayed on the terminal, changes after activities on other terminals or on the HYDRA server, it might be useful to immediately update the data on the terminal and not only after the next cyclic update, which might take some minutes.

Using the string command "DLG=SCMD;53|", you can reload specific lists on the terminal and immediately display the changes.

Structure of dialog data:

„DLG=SCMD;53|TYP=INFO|ACTION=LST_RELOAD|LOAD=…|TNR=…|“

The following table contains the data that can be transmitted:

Field	Description	Example
LOAD=...	Transfer of lists that must be reloaded: • MNR: Machines assigned to the terminal • ANR: Operations logged on • PNR: Staff logged on • MAT: Input materials of assigned machines • RES: Resources of machines assigned to the terminal • PPKT: CAQ inspection points. The parameters RECTYP, BER, PANNR, PAUNR, EINTTYP, EINTNR and CAUSE must additionally be specified.	LOAD=ANR,MNR|
TNR=	Terminal number	TNR=117|

Only use the string command "DLG=SCMD;53|", if necessary. A frequent reload of lists means system strain for terminal and HYDRA server.

To trigger the loading of lists, the HYDRA server sends a message to the terminal using the network. To this end, you must release the port used so it is not blocked by a firewall. Terminals normally use port 9002, the PCC port 9005.

The string command "DLG=SCMD;53|" is not suitable if you want to initialize MDE data like machine status or counter readings on the terminal from the server.

4.4 Generating MLE outbound segments

Purpose

Use this service if you want to generate MLE outbound transactions, which are then transferred to another system (ERP, WMS,...).

BAPI: HSODATA.INSERT

Identifier	Description	Example
HSODATA.SEGNAM	Segment name (CHAR30)	HSODATA.SEGNAM=E2BP_PP_ TIMTICKET
HSODATA.HYSYS	Logical target system (CHAR10)	HSODATA.HYSYS=FP
HSODATA.SDATA	Data record (CHAR1000)	HSODATA.SDATA=ABCDE
HSODATA.TYP	Type of data record: "HD" header record "CH" child record	HSODATA.TYP=HD
HSODATA.VERWEIS: HEADER	Reference to master segment when child segments are requested	HSODATA.VERWEIS:HEADER=4 711
HSODATA.LCHILD	Identifier "last child segment" "J" only if HSODATA.TYP=CH "N" any other case	HSODATA.LCHILD=J

Example: Generating a master segment

DLG=HSODATA.INSERT|HSODATA.SEGNAM=SAMPLESEGNAMHD|HSODATA.HYSYS=FP|HSODATA.SDATA=THIS IS SAMPLE DATA|HSODATA.TYP=HD|

Example: Generating master segment and child segments

DLG=HSODATA.INSERT|HSODATA.SEGNAM=SAMPLESEGNAMHD|HSODATA.HYSYS=SAP|HSODATA.SDATA=THIS IS SAMPLE DATA|HSODATA.TYP=HD|

The reference to the master segment is transferred in the BAPI result. Use this reference in subsequent requests for the child segments as *HSODATA.VERWEIS:HEADER*.

RET=0|KT=|LT=|DATA=HSODATA|VERWEIS=4711|TYP=HD|ID=|

Child segment - not the last one:

DLG=HSODATA.INSERT|HSODATA.SEGNAM=SAMPLESEGNAMCH|HSODATA.HYSYS=SAP|HSODATA.SDATA=THIS IS SAMPLE DATA CHILD|HSODATA.TYP=CH|HSODATA.VERWEIS:HEADER=**4711**|HSODATA.LCHILD=**N**|

Child segment - the last one:

DLG=HSODATA.INSERT|HSODATA.SEGNAM=ECKTESTCH|HSODATA.HYSYS=SAP|HSODATA.SDATA= THIS IS ANOTHER SAMPLE DATA CHILD |HSODATA.TYP=CH|HSODATA.VERWEIS:HEADER=**4711**|HSODATA.LCHILD=**J**|

4.5 Generating logging entry

Use the dialog described below to generate an entry in the change management (logging). This requires a relevant configuration in the *Logging - configuration*. If you additionally activate the option *Log dialog data* in this configuration, then all data included in the dialog data is transferred to the database. This data can then be displayed in the application *Logging - change management.* If you use the button *Show,* the dialog data is displayed in a separate information window.

Structure of dialog data:

„DLG=SCMD;51|USR=...|BEARB=…|ACTION=…|KEY:1=...|DAT=…|ZEI=…|“

The following table contains the data that can be transmitted:

Identifier	Description
KEY:1=	The value of this identifier must be identical to the content of field *Object* in *Logging - configuration*.
ACTION=	The value of this identifier must be identical to the content of field *Action* in *Logging - configuration*.
BEARB=	The person specified in the *Modified by* field is transferred to the logging entry.
USR=	The specified *User* is transferred to the logging entry

Activate the option *Log dialog data* in the *Logging - configuration* to save all further identifiers except the keys.

Only if the return value is 0 (`RET=0|)`, the entry was properly saved.

Example:

Content of logging configuration

Field	Content
Object	TEST
Action	INSERT
Log dialog data	Activated

Dialog data:

`DLG=SCMD;51|KEY:1=TEST|ACTION=INSERT|MNR=MyMachine|ANR=MyOperation|CNR=MyBatch|USR=2112|BEARB=12345|`

Result:

For object "TEST", an action "INSERT" is logged. Other dialog data can be viewed as detail data.

4.6 Generating entry for dialog error log

Use the dialog described below to generate an entry in the HYDRA dialog error log.

Structure of dialog data:

„DLG=SCMD;52| . . . “

The following table contains the data that can be transmitted:

Identifier	Type / max. field length	Description
STA=	C1	Status: I=Info, W=Warning, E=Error Note: Default value is E, if ERRCODE does not equal 0. It is I, if ERRCODE is 0.
ERRCODE=	N8	Error code
ERRCLASS=	C10	Error class
EREIG=	C40	Event
BEM=	C80	Comment
USR=	N4	HYDRA user
DAT=	{mm/dd/yyyy}	Date in format mm/dd/yyyy
ZEI=	{seconds}	Time: point in time in format "seconds", i.e. in seconds

		since midnight Example: ZEI=36003 for the time 10:00:03
BEARB=	C10	Modified by
MNR=	N8/C8	Machine number (numeric/alphanumeric)
MGRP=	C8	Machine group
ANR=	C	HYDRA order number (fully defined key)
CNR=	C20	Batch number
PNR=	C10	Personnel number
KNR=	C10	Badge number

4.7 Triggering escalation

Use the dialog described below to trigger a configured escalation.

Structure of dialog data:

„DLG=ESKMSG.INSERT|ESKMSG.ID=. . .| "

The following table contains the data that can be transmitted:

Identifier	Type / max. field length	Description
ESKMSG.ESKID=	C40	Escalation that must be triggered. The escalations that must be triggered are configured in the database table esk_event_cfg
ESKMSG.RCV:ART=	C1	Recipient type of message P: Person (see ESKMSG.RCV:PNR) F: Function group (see ESKMSG.RCV:FKT)
ESKMSG.RCV:PNR=	N8	Person from HR master data. If you specify the parameter, it overrides the configured recipient.
ESKMSG.RCV:FKT=	C40	Function group. If you specify the parameter, it overrides the configured recipient.
ESKMSG.ATTACH=	C250	If the mail delivery is configured, you can specify a mail attachment here. The HYDRA server must be able to reach the file.
USR=	N4	HYDRA user
DAT=	{mm/dd/yyyy}	Date in format mm/dd/yyyy
ZEI=	{seconds}	Time: point in time in format "seconds", i.e. in seconds since midnight

		Example: ZEI=36003 for the time 10:00:03
Other identifiers		For each escalation, a fixed amount of variables is defined. The defined variables can be viewed in the application *Escalation configuration.* You can attach these variables here. Note: the total length of the dialog data string is limited to 1000 characters. The available variables of each escalation are stored in the table `esk_event_reg_var`.

Example: Event ANR.REGISTER_REMARK

Available variables

```
select * from esk_event_reg_var where event_id = 'ANR.REGISTER_REMARK'
order by key_nr desc
```

event_id	ref_kenn	f_kennung	use_cond	key_nr
ANR.REGISTER_REMARK	<NULL>	ANR.ANR	J	1
ANR.REGISTER_REMARK	<NULL>	ANR.AUNR	J	0
ANR.REGISTER_REMARK	<NULL>	ANR.AGNR	J	0
ANR.REGISTER_REMARK	<NULL>	ANR.AFOLG	J	0
ANR.REGISTER_REMARK	<NULL>	ANR.UAGNR	J	0
ANR.REGISTER_REMARK	<NULL>	ANR.SPLNR	J	0
ANR.REGISTER_REMARK	<NULL>	ANR.MNR	J	0
ANR.REGISTER_REMARK	<NULL>	BEM	J	0
ANR.REGISTER_REMARK	<NULL>	ANR.PNR	J	0
ANR.REGISTER_REMARK	<NULL>	ANR.KNR	J	0
ANR.REGISTER_REMARK	<NULL>	ANR.DAT	J	0
ANR.REGISTER_REMARK	<NULL>	ANR.ZEI	J	0
ANR.REGISTER_REMARK	<NULL>	ANR.ATK	J	0
ANR.REGISTER_REMARK	<NULL>	PNR.PVORNAME	J	0
ANR.REGISTER_REMARK	<NULL>	PNR.PNAME	J	0

Be careful to always specify the key field variables (`key_nr > 0`) in the dialog string. The key fields uniquely identify an event. As long as an escalation with this key field combination is open, another escalation cannot be generated. Only if the escalation is closed, a new escalation with these key fields can be triggered.

```
DLG=ESKMSG.INSERT|ESKMSG.ESKID=ANR.REGISTER_REMARK|ANR.ANR=222222220100|ANR
.ATK=Testartikel|ESKMSG.RCV:ART=P|ESKMSG.TO:PNR=2|BEM=hello from
PDM|USR=2001|
```

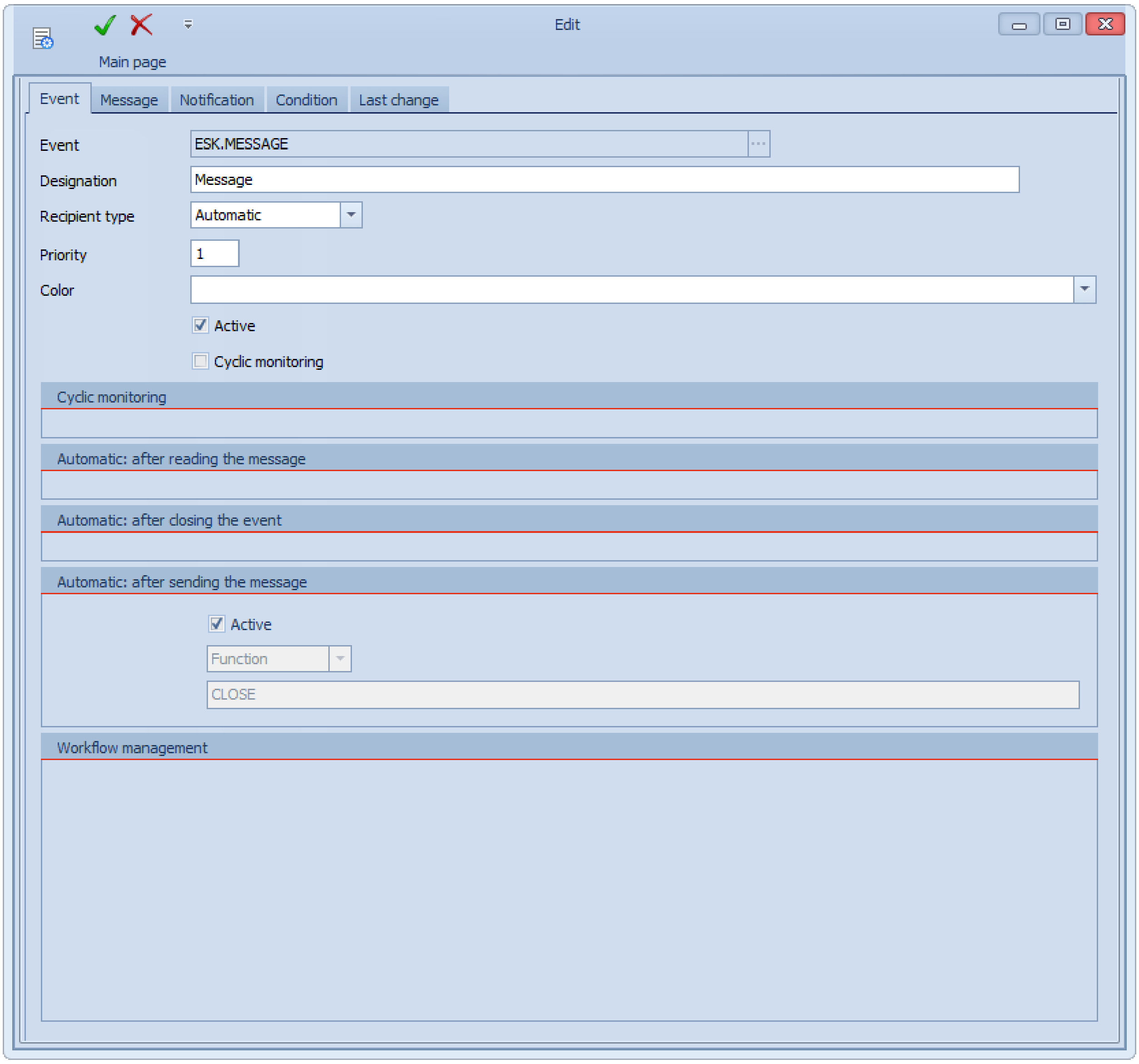
The escalation ANR.REGISTER_REMARK is triggered for operation 222222220100. The comment is "hello from PDM".

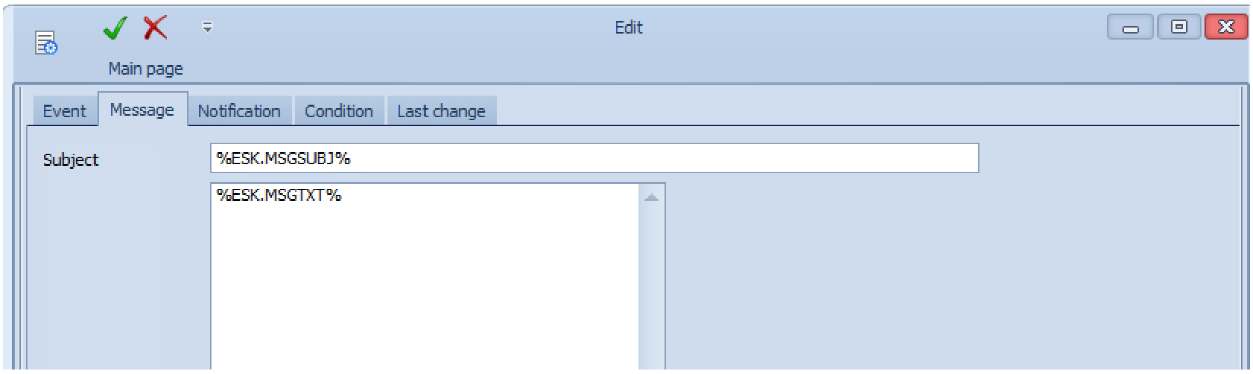
A mail attachment is added in this example.

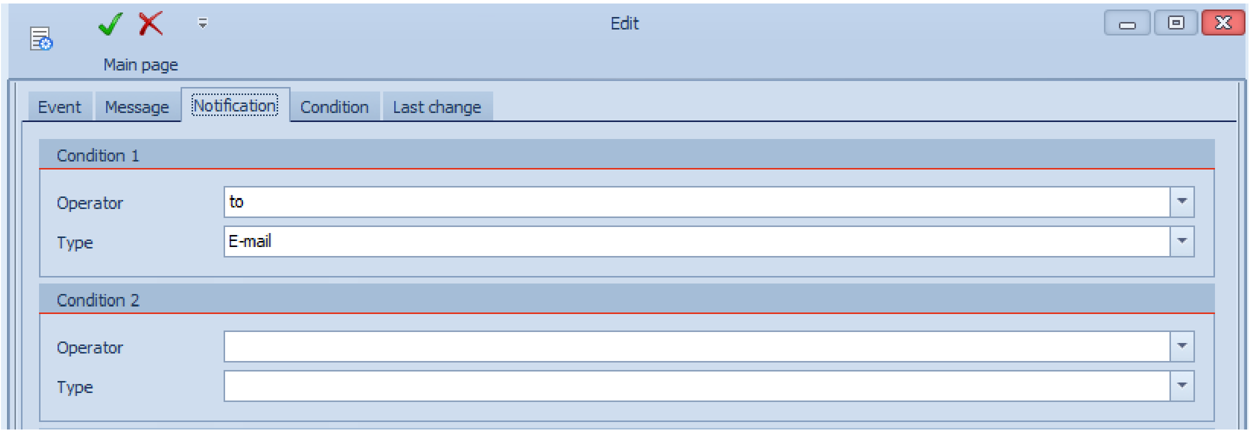
DLG=ESKMSG.INSERT|ESKMSG.ESKID=ANR.REGISTER_REMARK|ANR.ANR=222222220100|ANR.ATK=Testartikel|ESKMSG.RCV:ART=P|ESKMSG.TO:PNR=2|BEM=hello from PDM|**ESKMSG.ATTACH=.\1\grafik\bde\maschine.jpg**|USR=2001|

Example: Event ESK.MESSAGE

Use the escalation ESK.MESSAGE to send "generic" escalations. You can send mails, for example. To avoid open escalations, activate the option *Automatic: after sending the message* with CLOSE.







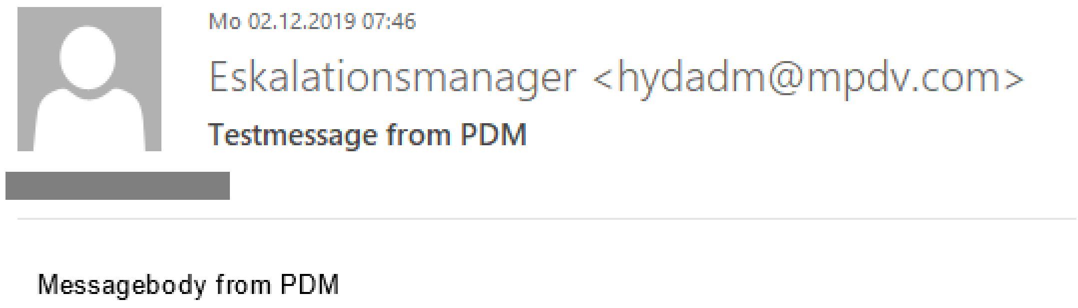
Available variables

```
select * from esk_event_reg_var where event_id = 'ESK.MESSAGE' order by
key_nr desc, f_kennung asc
```

EVENT_ID	REF_KENN	F_KENNUNG	USE_COND	KEY_NR
ESK.MESSAGE	<NULL>	ESK.MSGSUBJ	J	0
ESK.MESSAGE	<NULL>	ESKMSG.TO:ESKFKT	N	0
ESK.MESSAGE	<NULL>	ESKMSG.TO:PNR	N	0
ESK.MESSAGE	<NULL>	ESK.MSGTXT	J	0

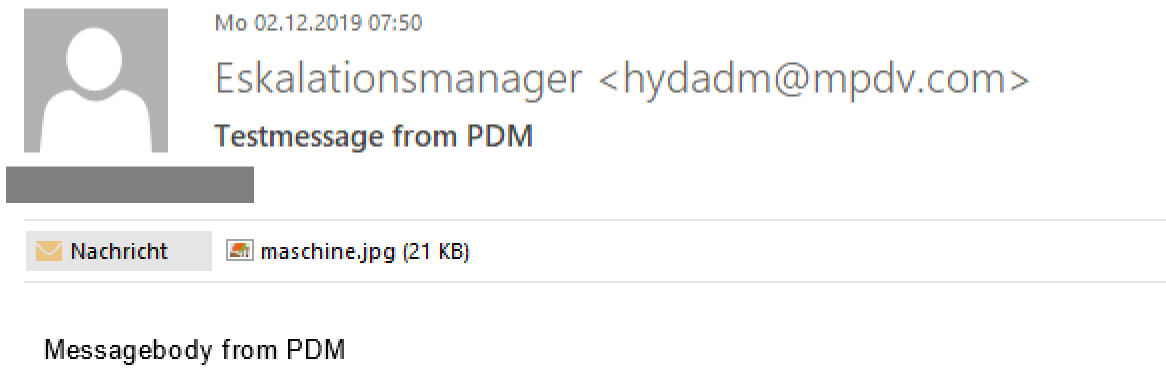
Identifier	Type / max. field length	Description
ESK.MSGSUBJ=	C50	If the mail delivery is configured, you can override the e-mail subject here.
ESK.MSGTXT=	C320	If the mail delivery is configured, you can override the e-mail body here.

```
DLG=ESKMSG.INSERT|ESKMSG.ESKID=ESK.MESSAGE|ESK.MSGSUBJ=Testmessage from
PDM|ESK.MSGTXT=Messagebody from
PDM|ESKMSG.RCV:ART=P|ESKMSG.TO:PNR=2|USR=2001|
```



You can send attachments with this escalation, too.

```
DLG=ESKMSG.INSERT|ESKMSG.ESKID=ESK.MESSAGE|ESK.MSGSUBJ=Testmessage from
PDM|ESK.MSGTXT=Messagebody from
PDM|ESKMSG.RCV:ART=P|ESKMSG.TO:PNR=2|ESKMSG.ATTACH=.\1\grafik\bde\maschine.
jpg|USR=2001|
```



5 HYDRA Production Data Manager Basis - Master Data

5.1 Note on the descriptions of the basic dialogs

All mandatory fields that must be specified have the addition PK (primary key). All other fields are optional and are processed if they are transferred.

5.2 Terminal configuration

5.2.1 Edit terminal configuration (DLG=TNR.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

You can edit the terminal configuration using the BAPI calls described in this chapter.

Tables

Table	Key field	Description
terminals	terminal no. TNR.TNR	Terminal number (PK) for the terminal label 1
pzt_kenn	user no. TNR.TNR	Terminal number (PK) for the terminal label 2
terminal_status	terminal no. TNR.TNR	Terminal number (PK) for the terminal status

BAPI call

Acronym	Content / {type}	Description
DLG	TNR.INSERT	Create terminal configuration
	TNR.UPDATE	Change terminal configuration
	TNR.DELETE	Delete terminal configuration
	TNR.COPY	Copy terminal configuration
	TNR.LOCK	Lock terminal configuration for processing
	TNR.UNLOCK	Unlock terminal configuration after processing
	TNR.NEW	Read specification for new terminal configuration
	TNR.SELECT	Select terminal configuration

TNR.TNR	{N3}	PK Terminal number
TNR.TNR:Z	{N3}	PK new (target) terminal number to copy
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Returned

Acronym	Content / {type}	Description
TNR.TNR	{N3}	TNR.INSERT, TNR.COPY: Return of the terminal number for the created configuration

Validation checks

Error codes	Description
1680	Parameter with label TNR.TNR must be specified (SELECT, LOCK, UNLOCK, INSERT, UPDATE, DELETE, COPY)
1668	The terminal must be set up in the database (UPDATE).
1683	The terminal number (TNR.TNR) must be between 1 and 999. (INSERT, UPDATE)
1683	The terminal type (TNR.ART) must be specified. (INSERT, UPDATE)
1692	The terminal type 110 and 10 are only available with license HYD-ALS and are used for the ALS sub-system and the ALS data synchronization. (INSERT, UPDATE)
1684	In certain system constellations, several terminals per machine are permitted. No machine may be assigned to this terminal which is assigned to another MDE terminal. This means that the terminal must not be a MDE terminal if an assigned machine is also assigned to another MDE terminal.
1668	The terminal with the terminal number TNR.TNR is not existent. (SELECT, LOCK, UPDATE, DELETE, COPY)
764	The terminal with the terminal number TNR.TNR exists already. (INSERT)
764	The terminal with the terminal number TNR.TNR exists already. (COPY)
1666	The data record is currently locked by another user. (UPDATE, DELETE).
1666	The data record is currently locked by another user. (LOCK).

5.2.2 List for terminal configurations (DLG=TNR.LIST)

This BAPI call list the terminals and their configurations in HYDRA.

Tables

Table	Key field	Description
terminals	terminal no TNR.TNR	Terminal number (PK) for the terminal label 1
pzt_kenn	user no TNR.TNR	Terminal number (PK) for the terminal label 2
terminal_status	terminal no TNR.TNR	Terminal number (PK) for the terminal status

BAPI call

Acronym	Contents	Description
DLG	TNR.LIST	List of terminal configuration
TNR	{N3}	PK terminal number if an action is supposed to be carried out for a terminal.
TNR:FROM	{N3}	PK terminal number from/to if the action is supposed to be carried out for several terminals in the area *from* and *to*. (In this case, no plausibility check on the existence of the entry is made).
TNR:BIS	{N3}	
MOD:AKTIV	J|N	Listing active terminals J/N
MOD:INAKTIV	J|N	Listing inactive terminals J/N
FILE	{C256}	Specification of the file name for the list

Returned

Acronym	Contents	Description
—	—	—

Validation checks

Error codes	Description
1656	The file with the name DATEI cannot be written on.

5.2.3 Leave terminal update (DLG=TNR.PROGLADEN)

You can trigger a new installation of the terminal program with this BAPI call. After a successful installation, the option with label "TNR.PROGLADEN" is automatically reset to "N". **Tables**

Table	Key field	Description
terminal_status	terminal no TNR.TNR	Terminal number (PK) for the terminal status

BAPI call

Acronym	Contents	Description
DLG	TNR.PROGLADEN	Leave or withdraw terminal update
TNR.TNR	{N3}	PK terminal number if an action is supposed to be carried out for a terminal.
TNR.TNR:VON	{N3}	PK terminal number from/to if the action is supposed to be carried out for several terminals in the area *from* and *to*. (In this case, no plausibility check regarding the existence of the entry is made).
TNR.TNR:BIS	{N3}	
TNR.PROGLADEN	J|N	J: initiate terminal update N: withdraw terminal update

Returned

Acronym	Contents	Description
—	—	—

Validation checks

Error codes	Description
1661	You must specify the parameter with label TNR.TNR or TNR.TNR:VON and TNR.TNR:BIS.
1668	The terminal (if parameter TNR.TNR is specified) must be set up in the database.

5.2.4 Restart terminal (DLG=TNR.NEUSTART)

You can trigger a restart of the terminal with this BAPI call. This setting is transferred to the terminal when the next status message is sent. After restarting the terminal, the option with label "TNR.NEUSTART" is automatically reset to "N":

Tables

Table	Key field	Description
terminal_status	terminal no	Terminal number (PK) for the terminal status

	TNR.TNR	

BAPI call

Acronym	Contents	Description
DLG	TNR.NEUSTART	Initiate or withdraw the terminal restart
TNR.TNR	{N3}	PK terminal number if an action is supposed to be carried out for a terminal.
TNR.TNR:VON	{N3}	PK terminal number from/to if the action is supposed to be carried out for several terminals in the area *from* and *to*. (In this case, no plausibility check regarding the existence of the entry is made).
TNR.TNR:BIS	{N3}	
TNR.NEUSTART	J|N	J: initiate restart terminal N: withdraw terminal restart

Returned

Acronym	Contents	Description
—	—	—

Validation checks

Error codes	Description
1661	You must specify the parameter with label TNR.TNR or TNR.TNR:VON and TNR.TNR:BIS.
1668	The terminal (if parameter TNR.TNR is specified) must be set up in the database.

5.2.5 Terminal administration (DLG=TNR.ADMIN)

You can use this BAPI call to initiate administrative tasks (terminal update, terminal restart, request a diagnosis upload, ...).

Tables

Table	Key field	Description
terminal_status	terminal no TNR.TNR	Terminal number (PK) for the terminal status

BAPI call

Acronym	Contents	Description
DLG	TNR.NEUSTART	Initiate or withdraw the terminal restart

TNR.TNR	{N3}	PK terminal number if an action is supposed to be carried out for a terminal.
TNR.TNR:VON	{N3}	PK terminal number from/to if the action is supposed to be carried out for several terminals in the area *from* and *to*. (In this case, no plausibility check regarding the existence of the entry is made).
TNR.TNR:BIS	{N3}	
TNR.PROGLADEN	J|N|U	J: initiate terminal update N: withdraw terminal update U: do not execute any changes
TNR.NEUSTART	J|N|U	J: initiate restart terminal N: withdraw terminal restart U: do not execute any changes
TNR.TLVL	{C1}	Set trace level (terminal_status.protokoll)
TNR.BERLESEN	J|N|U	J: initiate the reading of the authorizations J: initiate the reading of the authorizations U: do not execute any changes
TNR.DIAGUPLOAD	J|N|U	J: initiate the creation of a diagnosis upload N: withdraw the creation of a diagnostic upload U: do not execute any changes
TNR.AKTIV	J|N|U	J: set terminal to active N: set terminal to inactive U: do not execute any changes
TNR.ERR:TNRIP	N	N: Reset error "double HYDRA user number"

Returned

Acronym	Contents	Description
—	—	—

Validation checks

Error codes	Description
1661	You must specify the parameter with label TNR.TNR or TNR.TNR:VON and TNR.TNR:BIS.
1668	The terminal (if parameter TNR.TNR is specified) must be set up in the database.

5.3 Function authorizations

5.3.1 Edit function authorizations (DLG=BEARBFKT.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

The function authorizations are edited with these BAPI calls.

Tables

Table	Key field	Description
fkt_tab	usr prg berechtigung art BEARBFKT.BEARB BEARBFKT.FKT BEARBFKT.BERECHT BEARBFKT.ART	The combination must be unique.

BAPI call

Acronym	Content / {type}	Description
DLG	BEARBFKT.INSERT	Create function authorization
	BEARBFKT.UPDATE	Change function authorization
	BEARBFKT.DELETE	Delete function authorization
	BEARBFKT.COPY	Copy function authorization
	BEARBFKT.LOCK	Lock function authorization for processing
	BEARBFKT.UNLOCK	Unlock function authorization after processing
	BEARBFKT.NEW	Read specification for new function authorization
	BEARBFKT.SELECT	Select function authorization
BEARBFKT.BEARB	{C10}	User
BEARBFKT.FKT	{C15}	Function
BEARBFKT.BERECHT	{N4}	Authorization

BEARBFKT.ART	{C15}	Type F = function authorization P = function profile
MOD	C 1	Copy mode E - copy current selected authorization G - copy all authorizations F - copy missing authorizations
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Returned

Acronym	Content / {type}	Description
BEARB:FKT	{C15}	Current data record
FKT	{C15}	Current data record
BERECHT	{N4}	Current data record
ART	{C15}	Current data record

Validation checks

Error codes	Description
1660	The specified agent is invalid.
1661	A value is missing that is required for processing.
1662	A value relevant for processing is invalid.
1666	The function authorization is currently being maintained by another user.
1669	Data with the same key fields already exist.

5.3.2 List of function authorizations (DLG=BEARBFKT.LIST)

The BAPI call returns all specified function authorizations.

Tables

Table	Key field	Description
fkt_tab	usr prg berechtigung art BEARBFKT.BEARB BEARBFKT.FKT BEARBFKT.BERECHT BEARBFKT.ART	The combination must be unique.

BAPI call

Acronym	Contents	Description
DLG	BEARBFKT.LIST	List of function authorizations
FILE	{C256}	Specification of the file name for the list

5.4 Function profiles

5.4.1 Edit function profile (DLG=FKTPROF.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

The function authorizations are edited with these BAPI calls.

Tables

Table	Key field	Description
fkt_profil	profil prg berechtigung FKTPROF.FKTPROF FKTPROF.FKT FKTPROF.BERECHT	The combination must be unique.

BAPI call

Acronym	Content / {type}	Description
DLG	FKTPROF.INSERT	Create function profile
	FKTPROF.UPDATE	Change function profile
	FKTPROF.DELETE	Delete function profile
	FKTPROF.COPY	Copy function profile
	FKTPROF.LOCK	Lock function profile for processing
	FKTPROF.UNLOCK	Release lock for function profile after processing
	FKTPROF.NEW	Read requirement for a new function profile
	FKTPROF.SELECT	Select function profile
FKTPROF.FKTPROF	{C10}	Profiles
FKTPROF.FKT	{C15}	Function
FKTPROF.BERECHT	{C15}	Authorization
MOD	C 1	Copy mode

		G - copy all profiles F - copy missing profiles Delete mode E - delete single functions only G - delete total function
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Returned

Acronym	Content / {type}	Description
FKTPROF	{C15}	Current data record
FKT	{C15}	Current data record
BERECHT	{N4}	Current data record

Validation checks

Error codes	Description
1661	A value is missing that is required for processing.
1662	A value relevant for processing is invalid.
1666	The function profile is just edited by another function profile
1669	Data with the same key fields already exist.

5.4.2 List function profile (DLG=FKTPROF.LIST)

The BAPI call returns all defined function profiles.

Tables

Table	Key field	Description
fkt_profil	profil prg berechtigung FKTPROF.FKTPROF FKTPROF.FKT FKTPROF.BERECHT	The combination must be unique.

BAPI call

Acronym	Contents	Description
DLG	FKTPROF.LIST	Function profile list
FILE	{C256}	Specification of the file name for the list

5.5 Responsibility profiles

5.5.1 Edit responsibility profile (DLG=VABPROF.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

This BAPI call is used to edit responsibility profiles.

Tables

Table	Key field	Description
vab_profil	verantw_profil verantw_bereich VABPROF.VABPROF VABPROF.VAB	The combination must be unique.

BAPI call

Acronym	Content / {type}	Description
DLG	VABPROF.INSERT	Create responsibility profiles
	VABPROF.UPDATE	Change responsibility profile
	VABPROF.DELETE	Delete responsibility profile
	VABPROF.COPY	Copy responsibility profile
	VABPROF.LOCK	Lock processing for a responsibility profile
	VABPROF.UNLOCK	Release lock for responsibility profile after processing
	VABPROF.NEW	Read requirement for new responsibility profile
	VABPROF.SELECT	Select responsibility profile
VABPROF.VAB PROF	{C15}	Profile
VABPROF.VAB	{C15}	Area
MOD	{C1}	Copy mode G - copy all profiles F - copy missing profiles Delete mode E - Delete only responsibility area

		G - delete total profile
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Returned

Acronym	Content / {type}	Description
VABPROF	{C15}	Current data record
VAB	{C15}	Current data record
SELECT	{C1}	Current data record
USE	{C1}	Current data record
INSERT	{C1}	Current data record
UPDATE	{C1}	Current data record
DELETE	{C1}	Current data record

Validation checks

Error codes	Description
1661	A value is missing that is required for processing.
1662	A value relevant for processing is invalid.
1666	The responsibility profile is currently edited by another user
1669	Data with the same key fields already exist.

5.5.2 Responsibility Profiles (DLG=VABPROF.LIST)

The BAPI call returns all defined responsibility areas. The list can be restricted via parameter VABPROF.VABPROF (with wildcard support).

Tables

Table	Key field	Description
vab_profil	verantw_profil verantw_bereich	The combination must be unique.

	VABPROF.VABPROF VABPROF.VAB	

BAPI call

Acronym	Contents	Description
DLG	VABPROF.LIST	List responsibility
FILE	{C256}	Specification of the file name for the list

5.6 Assignment responsibility areas

5.6.1 Edit assignment responsibility areas (DLG=BEARBVABPROF.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

You can edit the assignment of the responsibility area with these BAP calls.

Tables

Table	Key field	Description
vab_berechtigung	usr berechtigung_art berechtigung_wert BEARBVABPROF.BEARB:VAB BEARBVABPROF.ART BEARBVABPROF.WERT	The combination must be unique.

BAPI call

Acronym	Content / {type}	Description
DLG	BEARBVABPROF.INSERT	Create assignment responsibility areas
	BEARBVABPROF.UPDATE	Change assignment responsibility area
	BEARBVABPROF.DELETE	Delete assignment responsibility area
	BEARBVABPROF.COPY	Copy assignment responsibility area
	BEARBVABPROF.LOCK	Lock assignment for responsibliy area for processing
	BEARBVABPROF.UNLOCK	Release lock for assignment for responsibility are after processing
	BEARBVABPROF.NEW	Read the requirement for new assigment of responsibility area
	BEARBVABPROF.SELECT	Select assignment for responsibility area
BEARBVABPROF.BEARB:VAB	{C15}	User

BEARBVABPROF.ART	{C1}	Type V – responsibility area P - profile
BEARBVABPROF.WERT	{C15}	Responsibility area or profile
MOD	{C1}	Copy mode G - copy all assignments of a user F - copy all missing assignments Delete mode E - delete only separate assignments G - delete all entries of a user
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
1661	A value is missing that is required for processing.
1662	A value relevant for processing is invalid.
1666	The assignment is currently edited by another user
1669	Data with the same key fields already exist.
1660	The specified agent is invalid.
425	The responsibility profile is not existent

5.6.2 Assignment list (DLG=BEARBVABPROF.LIST)

The BAPI call returns all defined assignments. The list can be restricted with the following parameter: BEARBVABPROF.BEARB:VAB und BEARBVABPROF.WERT (with wildcard support).

Tables

Table	Key field	Description
vab_berechtigung	usr berechtigung_art berechtigung_wert BEARBVABPROF.BEARB:VAB	The combination must be unique.

	BEARBVABPROF.ART BEARBVABPROF.WERT	

BAPI call

Acronym	Contents	Description
DLG	BEARBVABPROF. LIST	Assignment list
FILE	{C256}	Specification of the file name for the list

5.7 User administration

5.7.1 Edit user (DLG=BEARB.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

You use these BAPI call to edit users.

Tables

Table	Key field	Description
user_tab	usr BEARB.BEARB	PK user

BAPI call

Acronym	Content / {type}	Description
DLG	BEARB.INSERT	Create users
	BEARB.UPDATE	Change user
	BEARB.DELETE	Delete user
	BEARB.COPY	Copy user
	BEARB.LOCK	Lock user for processing
	BEARB.UNLOCK	Release lock for user after processing
	BEARB.NEW	Read requirement for new user
	BEARB.SELECT	Select user
BEARB.BEARB	{C10}	User
BEARB.BEARB:Z	{C10}	Target user during copying
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Returned

Acronym	Content / {type}	Description
BEARB	{C10}	Current data record

Validation checks

Error codes	Description
1661	A value is missing that is required for processing.
1662	A value relevant for processing is invalid.
1666	The user data is currently being edited by another user.
1669	Data with the same key fields already exist.
3020	The user account is locked!

Plausibility checks for password guidelines

Error codes	Description
3274	According to the password guideline, the password must not contain the user name.
3704	The entered password is not valid because the password is in the exclusion list.
3275	According to the password guideline, the password is not made-up of enough letters.
3276	According to the password guideline, the password is not made-up of enough numbers.
3277	According to the password guideline, the password is not made-up of enough special characters.
3278	According to the password guideline, the pasword is too short.
2379	According to the password guideline, the password has invalid characters.
3283	The entered password is equivalent to the current one.
3280	The entered password is already in use.
3281	Internal error during the access to the password history

5.7.2 User list (DLG=BEARB.LIST)

The BAPI call returns all defined users.

Tables

Table	Key field	Description
user_tab	usr BEARB.BEARB	PK user

BAPI call

Acronym	Contents	Description
DLG	BEARB.LIST	User list
FILE	{C256}	Specification of the file name for the list

5.7.3 User login (DLG=BEARB.LOGIN)

The BAPI call perfoms the login process of a user

Tables

Table	Key field	Description
user_tab	usr BEARB.BEARB	PK user

BAPI call

Acronym	Content / {type}	Description
DLG	BEARB.LOGIN	Logon user
BEARB.BEARB	{C10}	User
TYP	{C10	T = Hydra Mobile Login

Returned

Acronym	Content / {type}	Description
SYSPRO	{C20}	System logs If full: There is an alarm file alarm.txt available!
LANG	{N4}	Language key
MSGSYNC	{C1}	Escalation management: J = There are new messages including the automatic display of messages when you log on to the console.
PWDW	{C1}	J = password has run out

Validation checks

Error codes	Description

1661	A value is missing that is required for processing.
1667	Number of licenses were exceeded.
1665	Editor/ and or the password is invalid

5.7.4 User logout (DLG=BEARB.LOGOUT)

The BAPI call performs a logout process of a user.

Tables

Table	Key field	Description
user_tab	usr BEARB.BEARB	PK user

BAPI call

Acronym	Content / {type}	Description
DLG	BEARB.LOGOUT	Log off user
BEARB.BEARB	{C10}	User
TYP	{C10	T = Hydra Mobile Login

5.8 Locked data records

5.8.1 Delete locked data records (DLG=BEARBFKT.DELETE)

You can use these BAPI calls to delete locked data records.

Tables

Table	Key field	Description
hyd_lock	key1 key2 key3 key4 key5 param1 param2 hydrauser bearb LOCK.KEY:1 LOCK.KEY:2 LOCK.KEY:3 LOCK.KEY:4 LOCK.KEY:5 LOCK.PARAM:1 LOCK.PARAM:2 LOCK.USR LOCK.BEARB	The combination must be unique. Note: The fields key1..key5 und param1 and param2 can also be zero.

BAPI call

Acronym	Content / {type}	Description
DLG	LOCK.DELETE	Delete the locked data record
LOCK.BEARB	{C10}	User
LOCK.USR	{N4}	User number
LOCK.KEY:1	{C40}	BAPI – Object name e.g. MNR
LOCK.KEY:2	{C40}	Value of the BAPI object e.g. machin number
LOCK.KEY:3	{C40}	
LOCK.KEY:4	{C40}	
LOCK.KEY:5	{C40}	

LOCK.PARAM:1	{C40}	
LOCK.PARAM:2	{C40}	
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
1661	A value is missing that is required for processing.
101	General error message that appears if the selected data (tables or files) is not available.

5.9 Paths

5.9.1 Edit paths (DLG=PATH.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

You use these BAPI call to edit HYDRA paths.

Tables

Table	Key field	Description
hy_path	path PATH.PATH	PK path

BAPI call

Acronym	Content / {type}	Description
DLG	PATH.INSERT	Create path
	PATH.UPDATE	Change path
	PATH.DELETE	Delete path
	PATH.COPY	Copy path
	PATH.LOCK	Lock the path for processing
	PATH.UNLOCK	Release lock for path after processing
	PATH.NEW	Read requirement for new path
	PATH.SELECT	Select path
PATH.PATH	{C8}	Path
PATH.SCHEMA	{C10}	Schema
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
1661	A value is missing that is required for processing.
1669	Data with the same key fields already exist.

1666	The path is currently edited by another user.

5.9.2 Path list (DLG=PATH.LIST)

The BAPI call returns all defined paths.

Tables

Table	Key field	Description
hy_path	path PATH.PATH	PK path

BAPI call

Acronym	Contents	Description
DLG	PATH.LIST	List paths
FILE	{C256}	Specification of the file name for the list

5.10 Licensing

5.10.1 Edit licenses (DLG=LIC.INSERT, DELETE)

You use these BAPI calls to edit licenses.

Tables

Table	Key field	Description
hyd_license	productkey category value valid_till licensedate majorrelease minorrelease LIC.PROKEY LIC.CAT LIC.VAL LIC.VALIDTILL LIC.LICDAT LIC.VER:MAJ LIC.VER:MIN	PK combination from all fields

BAPI call

Acronym	Content / {type}	Description
DLG	LIC.INSERT	Create license
	LIC.DELETE	Delete license
	LIC.USE	Assign license
LIC.PROKEY	{C10}	Product name
LIC.CAT	{C10}	Category, fix "HYDRA"
LIC.VAL	{C10}	
LIC.VALIDTILL	{DATE}	
LIC.LICDAT	{DATE}	
LIC.VER:MAJ	{N4}	
LIC.VER:MIN	{N4}	
As an alternative, a license file can be transferred.		

MOD	{C1}	D – file
FILE	{C255}	File name
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
1661	A value is missing that is required for processing.
1669	Data with the same key fields already exist.
1671	The license data are incorrect or are not suitable to your system.
2027	The processing mode is for this posting not allowed.

5.10.2 License list (DLG=LIC.LIST)

The BAPI call returns licenses.

BAPI call

Acronym	Contents	Description
DLG	LIC.LIST	Liste Lizenzen
FILE	{C256}	Specification of the file name for the list
MOD	{C1}	List mode V - available licenses U- list of used licenses L - list of all licenses

5.11 Client

5.11.1 Client login and logout (DLG=CLIENT.LOGIN, LOGOUT)

These BAPI calls are used to assign a SessionID to a HYDRA USR number.

Tables

Table	Key field	Description
client_status	sessionid usr CLIENT.SID CLIENT.USR	PK SessionID PK HYDRA user number

BAPI call

Acronym	Content / {type}	Description
DLG	CLIENT.LOGIN	Create assignment between a Session ID and a HYDRA user number.
	CLIENT.LOGOUT	Delete assignment between a SessionID and a HYDRA number.
CLIENT.SID	{C100}	SessionID of the client
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Returned

Acronym	Content / {type}	Description
USR	{N5}	HYDRA user number

5.12 INI configuration

5.12.1 INI - edit configuration (DLG=INI.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT, IMPORT, EXPORT)

You use these BAPI calls to edit the INI configuration

Tables

Table	Key field	Description
hyd_ini	hydrauser ininame INI.USR INI.INI	The combination must be unique.

BAPI call

Acronym	Content / {type}	Description
DLG	INI.INSERT	INI - create configuration
	INI.UPDATE	INI - change configuration
	INI.DELETE	INI - delete configuration
	INI.LOCK	INI – configuration for processing
	INI.UNLOCK	Release lock for INI - configuration after processing
	INI.NEW	Read requirement for new INI configuration
	INI.SELECT	INI - select configuration
	INI.IMPORT	Import a complete INI configuration
	INI.EXPORT	Export of a complete INI configuration in a file.
INI.USR	{N4}	HYDRA user number
INI.INI	{C80}	INI - name
FILE	{C255}	Filename for import/export
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
1661	A value is missing that is required for processing.
1666	The configuration is currently edited by another user
1669	Data with the same key fields already exist.
101	General error message that appears if the selected data (tables or files) is not available.
1611	General database fields

410	

5.12.2 INI list - configurations (DLG=INI.LIST)

The BAPI call returns all defined INI configurations. The list can be restricted via parameter INI.INI (with wildcard support).

Tables

Table	Key field	Description
hyd_ini	hydrauser ininame INI.USR INI.INI	The combination must be unique.

BAPI call

Acronym	Contents	Description
DLG	INI.LIST	List INI configurations
FILE	{C256}	Specification of the file name for the list

5.12.3 Edit INI sections (DLG=INIDATA.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

You use these BAPI calls to edit INI sections.

Tables

Table	Key field	Description
hyd_ini_data	ini_verweis section ident INIDATA.INIVERWEIS INIDATA.SECTION INIDATA.KEY	The combination must be unique.

BAPI call

Acronym	Content / {type}	Description
DLG	INIDATA.INSERT	Create INI section

	INIDATA.UPDATE	Change INI section
	INIDATA.DELETE	Delete INI section
	INIDATA.LOCK	Lock INI section for processing
	INIDATA.UNLOCK	Release lock for INI section after processing
	INIDATA.NEW	Read requirement for new INI section
	INIDATA.SELECT	Select INI section
INIDATA.INVERWEIS	{N8}	Reference to the corresponding INI configuration in the hyd_ini table.
INIDATA.SECTION	{C80}	Section name
INIDATA.KEY	{C80}	Key name
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
1661	A value is missing that is required for processing.
1666	The configuration is currently edited by another user
1669	Data with the same key fields already exist.
101	General error message that appears if the selected data (tables or files) is not available.
1611	General database fields

5.12.4 List of the INI sections (DLG=INIDATA.LIST)

The BAPI call returns all defined sections for the INI entry. The list can be restricted via parameter INIDATA.SECTION (with wildcard support). The INI configuration must be specified via parameters INI.INI and INI.USR in the list request.

Tables

Table	Key field	Description
hyd_ini_data	ini_verweis	The combination must be unique.

	section ident INIDATA.INIVERWEIS INIDATA.SECTION INIDATA.KEY	

BAPI call

Acronym	Contents	Description
DLG	INIDATA.LIST	INI sections list
FILE	{C256}	Specification of the file name for the list

5.13 Number ranges

5.13.1 Edit number ranges (DLG=NRKREIS.INSERT, UPDATE, DELETE, LOCK, UNLOCK)

Number ranges are the basis to assign numbers automatically, e.g. order numbers, posting numbers, etc.

Example: When creating an order and no order number was specified, then the order number can be automatically calculated according to the specified number range.

You can use two different keys to process existing data records.

1. The internal number of the data record (NRKREIS.VERWEIS)
2. The combination of the logical key fields Object (NRKREIS.OBJTYP), Type (NRKREIS.ART), Key (NRKREIS.KEY) and Key Value (NRKREIS.VAL).

BAPI call

Acronym	Content / {type}	Description
DLG	NRKREIS.INSERT	Create number range
	NRKREIS.UPDATE	Change number range
	NRKREIS.DELETE	Delete number range

	NRKREIS.LOCK	Release lock number ranges
	NRKREIS.UNLOCK	Release lock for number ranges after processing
NRKREIS.OBJTYP	C 20	Generation object
NRKREIS.ART	C 6	Generation type (V=automatic number assignment)
[NRKREIS.ART:1]	C 1	Optional access to the first sign of the generation type
[NRKREIS.ART:2]	C 1	Optional access to the second sign of the generation type
NRKREIS.KEY	C 20	Optional key The value of the key is managed with identification NRKREIS.VAL NRKREIS.KEY and NRKREIS.VAL can in principle be assigned as required. The following keys are specified for number ranges processed by default in the system: „AART“: order type „SAG“: merged operation (only for object = "AUNR") „MELDART“: posting type (only for object IHNR for WRM-IH)
NRKREIS.VAL	C 20	Wert zu NRKREIS.KEY e.g. PP01 = order type for NRKREIS.KEY=AART. If "*" , then no differentiation is made. All order types without a specific number range get this entry.
NRKREIS.VERWEIS	N	Internal data record number (not for NRKREIS.INSERT)
NRKREIS.PRAEFIX	C 5	Prefix placed before the number, e.g. "GK (overheads)" could be the prefix for GK orders.
NRKREIS.BER:VON	C 20	Start of the value range
NRKREIS.BER:BIS	C 20	End of the value range
NRKREIS.BER:VAL	C 20	Last assigned value
NRKREIS.VERGCODE	C 6	Assignment code "NUM" = numeric assignment

Example

```
DLG=NRKREIS.INSERT|NRKREIS.OBJTYP=MyObj|NRKREIS.ART=V|NRKREIS.KEY=MyKey|NRKREIS.VAL=MyVal|NRKREIS.PRAEFIX=MYPRF|NRKREIS.BER:V
ON=0000000|NRKREIS.BER:BIS=9999999|NRKREIS.BER:VAL=100|NRKREIS.VERGCODE=NUM|
```

5.13.2 Number range list (DLG=NRKREIS.LIST)

The list gets all data records of the number ranges. The identification can be deduced from the editing functions.

5.13.3 Create new numbers (DLG=NRKREIS.CREATENR)

The function NRKREIS.CREATENR creates a new number in the configured number range.

The new number for allocation code "NUM" results from the value +1 last allocated.

Create a type definition for the user field in order to format values.

Field in the type definition	Contents
General / type	Number range object
General / name	Any
General / description	Any
General / length	Required length for formatting the new number. It might be the case that the length requires a prefix included in the number range.
Terminal / fill character	0: the new value is filled with leading zeros at the beginning -: the new value is filled with zeros starting at the end

BAPI call

Acronym	Contents	Description
DLG	NRKREIS.CREATENR	Generate a new number from the number range
NRKREIS.OBJTYP	C 20	Generation object
NRKREIS.ART	C 6	Generation type (V=automatic number assignment)
NRKREIS.KEY	C 20	Key Qualification Qualification within a generation object (for order number, order type, merged operation)in a generation object (for order number, e.g. order type, merged operation) AART = order type SAG = merged operation (only for object = "AUNR") „MELDART“ = posting type (only for object IHNR for

		WRM-IH)
NRKREIS.VAL	C 20	Value e.g. PP01 = order type; If "*" , then no distinction is made or all order types for which no specific number range is defined are assigned this entry.
[NRKREIS.VERWEIS]	N	Internal data record number Alternative key for the combination of NRKREIS.OBJTYP, NRKREIS.ART, NRKREIS.KEY and NRKREIS.VAL.

Returned

Acronym	Contents	Description
BER:VAL	Cx	New number not formatted
NR	Cx	Prefix + new number not formatted
NR:FILLED	Cx	Only if a type definition for the user field exists for the object of the number range:

Validation checks

Error codes	Description
3300	Incorrect assignment code Currently only the assignment code NUM is supported.
3301	If the value range of the number range is exceeded, an error message pops up.

Example

```
DLG=NRKREIS.CREATENR|NRKREIS.OBJTYP=MyObj|NRKREIS.ART=V|NRKREIS.KEY=MyKey|NRKREIS.VAL=MyVal|
```

Rückgabe:Return:

```
RET=0|KT=|LT=|DATA=NRKREIS|OBJTYP=MyObj|ART=V|KEY=MyKey|VAL=MyVal|VERWEIS=19|NR=MYPR106|PRAEFIX=MYPR|BER:VON=0000000|BER:BIS=
9999999|BER:VAL=106|VERGCODE=NUM|BEARB=12345|BEARBDAT=05/29/2019|BEARBZEI=35096|NR:FILLED=MYPR0000000000000106|
```

6 HYDRA Production Data Manager BDE/MDE - Data Collection

6.1 Note on the descriptions of the input dialogs

All fields that are mandatory and must be specified are highlighted using a gray background color. All other fields are optional and are processed when passed.

The MES order number ANR is the *fully defined key* of the object. This number always includes all specifications (in the order displayed)

- Order number
- Sequence (only relevant in combination with the additional function BDE-APF)
- Operation number
- Split number (only relevant in combination with the additional functions BDE-SGG)

The MES order number is of type C (char). The field lengths specified during the HYDRA setup must be respected. The length of the MES order number ANR is the sum total of the lengths of the different fields.

If you use the split functionality (BDE-SSG), the length of the MES order number ANR depends on whether it is a split operation or not. If it is not a split OP, the MES order number is reduced by the length of the split number.

Examples:

- Order number length: 8, operation number length: 4, no sequence, no splits:
  → ANR length: 12
- Order number length: 8, operation number length: 4, sequence number length: 1, no splits:
  → ANR length: 13

- Order number length: 8, operation number length: 4, sequence number length: 0, split number length: 1:

  → ANR length if no split operation: 12

  → ANR length if split operation: 13

- Order number length: 8, operation number length: 4, sequence number length: 1, split number length: 1:

  → ANR length if no split operation: 13
  → ANR length if split operation: 14

6.2 Order, staff and machine postings

6.2.1 Note on automatically recorded quantities

All input dialogs (e.g. A_UN, P_AB), which include manually recorded values (EGR:GUT, EGR:AUS), support the automatically recorded values AGR.C:[1..n], AGE.C:[1..n], AGG.C:[1..n], AGB.C:[1..n], AGR:HUB. For a better overview, these values are *not* displayed in the different input dialogs (exception: M_AST, A_AUN and A_ASW).

The following table generally defines the valid IDs for automatic quantities from counters in the dialog data:

Identifier	Type / max. field length	Description
AGR.C:[1..n]=	Double	Automatically recorded counter delta of counters 1...n
AGE.C:[1..n]=	C4	Unit of counter (e.g. PCS) for future use
AGG.C:[1..n]=	N4	Reason for counter (e.g. scrap reason)
AGB.C:[1..n]=	C10	Evaluation for counter (yield, scrap, rework, sample: GUT, AUS, NCH, PRB) - see note below.
AGR:HUB=	Double	Automatically recorded cycles

Do not use the following acronyms as of MW 4.0pe:		
AGR:GUT=	*Double*	*Automatic yield*
AGR:AUS=	*Double*	*Automatic scrap*
AGG:AUS=	*N4*	*Scrap reason*
AGE:GUT=	*C4*	*Unit of yield quantity*
AGE:AUS=	*C4*	*Unit of scrap quantity*

Example

AGR.C:1=47|AGB.C:1=GUT|AGR.C:2=1|AGB.C:2=AUS|AGG.C:2=12|....|AGR:HUB=..|

Notes

- Automatic counters are not yet used for calculation. A calculation using the partitioning and the pulse factor and other quantity accounts (e.g. when the total quantity is recorded) is only performed in the HYDRA server.
- The counter evaluation – identifier AGB.C:{..} – is an optional identifier and need not be transferred.
  - If the identifier AGB.C:{..} is not included in the dialog data, the counter is posted by default according to the evaluation specified in the counter configuration.
  - If the identifier AGB.C:{..} is included in the dialog data, then the counter is posted according to this evaluation. Here, the evaluation specified in the counter configuration is overridden.

6.2.2 Notes on manually recorded quantities

6.2.2.1 Recording of manual quantities in alternative quantity units

All input dialogs (e.g. A_UN, P_AB), which include manually recorded values in primary quantity unit (EGR:GUT, EGR:AUS,etc.), also support the recording of quantities in the following alternative quantity units:

- Base quantity unit
- Secondary quantity unit
- Tertiary quantity unit

The table below provides an overview of the available and valid IDs of the manual recording of quantities in the dialog data:

Identifier	Type / max. field length	Description
EGR:GUT= or EGR:GUTP=	Double	Yield recorded in primary quantity unit
EGG:GUT= or	N4	Optional reason for the yield recorded in primary quantity unit

Identifier	Type / max. field length	Description
EGG:GUTP=		
EGR:GUTB=	Double	Yield recorded in base quantity unit
EGG:GUTB=	N4	Optional reason for the yield recorded in base quantity unit
EGR:GUTS=	Double	Yield recorded in secondary quantity unit
EGG:GUTS=	N4	Optional reason for the yield recorded in secondary quantity unit
EGR:GUTT=	Double	Yield recorded in tertiary quantity unit
EGG:GUTT=	N4	Optional reason for the yield recorded in tertiary quantity unit
EGR:AUS= or EGR:AUSP=	Double	Scrap quantity recorded in primary quantity unit
EGG:AUS= or EGG:AUSP=	N4	Optional reason for the scrap recorded in primary quantity unit
EGR:AUSB=	Double	Scrap recorded in base quantity unit
EGG:AUSB=	N4	Optional reason for the scrap recorded in base quantity unit
EGR:AUSS=	Double	Scrap recorded in secondary quantity unit
EGG:AUSS=	N4	Optional reason for the scrap recorded in secondary quantity unit
EGR:AUST=	Double	Scrap recorded in tertiary quantity unit
EGG:AUST=	N4	Optional reason for the scrap recorded in tertiary quantity unit
EGR:NCH= or EGR:NCHP=	Double	Rework quantity recorded in primary quantity unit
EGG:NCH= or EGG:NCHP=	N4	Optional reason for the rework recorded in primary quantity unit
EGR:NCHB=	Double	Recorded rework quantity in base quantity unit
EGG:NCHB=	N4	Optional reason for the rework recorded in base quantity unit
EGR:NCHS=	Double	Rework quantity recorded in secondary quantity unit
EGG:NCHS=	N4	Optional reason for the rework recorded in secondary quantity unit
EGR:NCHT=	Double	Rework recorded in tertiary quantity unit
EGG:NCHT=	N4	Optional reason for the rework recorded in tertiary quantity unit
EGR:PRB= or EGR:PRBP=	Double	Open quantity / problem quantity recorded in primary quantity unit
EGG:PRB= or EGG:PRBP=	N4	Optional reason for the open quantity / problem quantity recorded in primary quantity unit
EGR:PRBB=	Double	Open quantity / problem quantity recorded in base quantity unit

Identifier	Type / max. field length	Description
EGG:PRBB=	N4	Optional reason for the open quantity / problem quantity recorded in base quantity unit
EGR:PRBS=	Double	Open quantity / problem quantity recorded in secondary quantity unit
EGG:PRBS=	N4	Optional reason for the open quantity / problem quantity recorded in secondary quantity unit
EGR:PRBT=	Double	Open quantity / problem quantity recorded in tertiary quantity unit
EGG:PRBT=	N4	Optional reason for the open quantity / problem quantity recorded in tertiary quantity unit

Examples:

- Upload of a part quantity with yield in primary quantity unit and scrap and scrap reason in secondary quantity unit
  `DLG=A_TR|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100473100200|`
  `EGR:GUTP=1|EGR:AUSS=2|EGG:AUSS=1|`
- Upload of a part quantity with yield in primary quantity unit and secondary quantity unit
  `DLG=A_TR|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100473100200|`

  `EGR:GUTP=1| EGR:GUTS=11|`

If quantities are recorded in alternative quantity units, these quantities take priority over the conversion using the conversion factors stored for the operation or over a formula stored in the system. In this case, no conversion into the quantity unit recorded is performed.

But the conversion into configured, but not recorded quantity units is performed.

Example: The yield can be recorded in primary quantity unit (EGR:GUTP) and secondary quantity unit (EGR:GUTS) in a dialog and the quantity is then converted into the base quantity and tertiary quantity if the conversion factors are stored.

The conversion of quantities into quantity units that are not manually recorded is performed according to the following priority:

1. Primary quantity: conversion into secondary quantity and tertiary quantity
2. Secondary quantity: conversion into primary quantity and tertiary quantity
3. Tertiary quantity: conversion into primary quantity and secondary quantity

The base quantity unit is generally used for conversions between primary, secondary and tertiary quantities.

For further information on the conversion of quantities and examples, refer to the document MBL_Quantity_Conversion.pdf.

6.2.2.2 Recording of quantities with different reasons

The recording of scrap, rework and open quantities with different reasons in one dialog is supported. To do so, you must complement the IDs for the scrap quantity and the reason with a numeric and consecutive index – starting with 1.

The below table for scrap also applies for rework (identifier *NCH* instead of *AUS*) and open quantities (identifier *PRB* instead of *AUS*).

Identifier	Type / max. field length	Description
EGR:AUS#[1..n]= or EGR:AUSP#[1..n]=	Double	Scrap quantity recorded in primary quantity unit with consecutive index
EGG:AUS#[1..n]= or EGG:AUSP#[1..n]=	N4	Optional reason for the scrap recorded in primary quantity unit and the relevant index
EGR:AUSB#[1..n]=	Double	Scrap recorded in base quantity unit with consecutive index
EGG:AUSB#[1..n]=	N4	Optional reason for the scrap recorded in base quantity unit and the relevant index
EGR:AUSS#[1..n]=	Double	Scrap recorded in secondary quantity unit with consecutive index
EGG:AUSS#[1..n]=	N4	Optional reason for the scrap recorded in secondary quantity unit and the relevant index
EGR:AUST#[1..n]=	Double	Scrap recorded in tertiary quantity unit with consecutive index
EGG:AUST#[1..n]=	N4	Optional reason for the scrap recorded in tertiary quantity unit and the relevant index

Example:

- Upload of a part quantity with yield quantity and two scrap quantities with reasons 20 and 30

```
DLG=A_TR|ANR=ANR=AAA2100473100200|EGR:GUTP=123|EGR:AUSP#1=2|EGG:AUSP#1=20|
EGR:AUSP#2=1|EGG:AUSP#2=20|
```

6.2.3 Collection of user fields

User fields can be collected in dialogs with the following events:

- Log on operations

- Posting of part quantities for operations
- Interrupt operations
- Log off operations
- Finish operations
- Quantity uploads for operations
- Log off staff

The collected user fields are integrated into the data of the operation and the order-related postings resulting from the posting event.

Sometimes it is unwanted that the collected user fields overwrite the user fields of the operation. If the field identification ANR:SETUSRFLD is transmitted with the value "N", the system does not integrate the collected user fields in the *Operation*, but only in the *Order-related postings*.

Identifier	Data type	Description
FU:1 to FU:6	Date	User fields for a date
FU:7 to FU:22	N (=night)	Integer with value range from -2147483647 to 2147483647
FU:23 to FU:28	DECIMAL(18,6)	Decimal number with a value range of 12 digits before comma and a precision of 6 digits after comma.
FU:29 to FU:44	C1	User field for 1 character
FU:45 to FU:50	C10	User field for a text with a maximum of 10 characters
FU:51 to FU:64	C20	User field for a text with a maximum of 20 characters
FU:65 to FU:66	C40	User field for a text with a maximum of 40 characters
ANR:SETUSRFLD	C1 = {J/N}	With ANR:SETUSRFLD=N, the user fields of the dialog (FU:1 to FU:66) are not integrated into the operation data. The user fields are only transferred to the BDE log record (if a log record is created).

6.2.4 Log operation on (DLG=A_AN)

Identifier	Type / max. field length	Description	
ANR=	C	MES order number (fully defined key)	
MNR=	C8	Workplace/machine number	
PNR=	C10	or	Personnel number
KNR=	C10		Staff badge number

Identifier	Type / max. field length	Description
CNR=	C20	Batch number (output batch)
MST=	N4	Set new machine status or automatically change into *Production* status (customer-specific)

Example 1: Log OP on

DLG=A_AN|USR=2106|DAT=02/17/2000|ZEI=47972|ANR=0004990701|MNR=4560|KNR=9999 99|

Example 2: Log OP and batch on

DLG=A_AN|USR=2106|DAT=02/17/2000|ZEI=47972|ANR=0004990701|MNR=4560|KNR=9999 99|CNR=CHV0001112|

6.2.5 Log operation and person on (DLG=A_P_AN)

Identifier	Type / max. field length	Description	
ANR=	C	MES order number (fully defined key)	
MNR=	C8	Workplace/machine number	
PNR=	C10	or	Personnel number
KNR=	C10		Staff badge number
CNR=	C20	Batch number (output batch)	
MST=	N4	Set new machine status or automatically change into *Production* status (customer-specific)	

Example 1: Log operation and person on

DLG=A_P_AN|USR=2106|DAT=02/17/2000|ZEI=47972|ANR=0004990701|MNR=4560|KNR=99 9999|

Example 2: Log batch and person on

DLG=A_P_AN|USR=2106|DAT=02/17/2000|ZEI=47972|ANR=0004990701|MNR=4560|KNR=12 3456|CNR=CHV0001112|

6.2.6 Posting of part quantity (Partial confirmation) (DLG=A_TR)

Identifier	Type / max. field length	Description	
ANR=	C	MES order number (fully defined key)	
MNR=	C8	Workplace/machine number	
PNR=	C10	or	Personnel number
KNR=	C10		Staff badge number
EGR:GUT=	N8	Yield recorded	
EGR:AUS=	N8	Scrap recorded	
EGG:GUT=	N4	Reason	
EGG:AUS=	N4	Scrap reason (required if EGR:AUS is passed)	
EGE:GUT=	C4	Unit	
EGE:AUS=	C4	Unit of scrap quantity	

Example 1: Posting of part quantity with scrap and scrap reason

DLG=A_TR|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100473100200|
MNR=60610|EGR:GUT=1|EGR:AUS=2|EGG:AUS=1|

6.2.7 Interrupt operation (DLG=A_UN)

Identifier	Type / max. field length	Description	
ANR=	C	MES order number (fully defined key)	
MNR=	C8	Workplace/machine number	
PNR=	C10	or	Personnel number
KNR=	C10		Staff badge number
EGR:GUT=	N8	Yield recorded	
EGR:AUS=	N8	Scrap recorded	
EGG:GUT=	N4	Reason	
EGG:AUS=	N4	Scrap reason	

Identifier	Type / max. field length	Description
EGE:GUT=	C4	Unit of yield quantity
EGE:AUS=	C4	Unit of scrap quantity
MST=	N4	New machine status

Example 1: Interrupt OP with input of quantity

DLG=A_UN|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100451210200|
MNR=60610|EGR:GUT=1|

Example 2: Interrupt OP with input of quantity and unit

DLG=A_UN|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100451210200|
MNR=60610|EGR:GUT=1|EGE:GUT=ST|

Example 3: Interrupt OP with input of quantity, scrap and scrap reason

DLG=A_UN|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100451210200|
MNR=60610|EGR:GUT=1|EGR:AUS=2|EGG:AUS=1|

6.2.8 Log operation off (DLG=A_AB)

Identifier	Type / max. field length	Description	
ANR=	C	MES order number (fully defined key)	
MNR=	C8	Workplace/machine number	
PNR=	C10	or	Personnel number
KNR=	C10		Staff badge number
EGR:GUT=	N8	Yield recorded	
EGR:AUS=	N8	Scrap recorded	
EGG:GUT=	N4	Reason	
EGG:AUS=	N4	Scrap reason	
EGE:GUT=	C4	Unit of yield quantity	
EGE:AUS=	C4	Unit of scrap quantity	
MST=	N4	New machine status	

Example 1: Log OP off with input of quantity

DLG=A_AB|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100451210200|
MNR=60610|EGR:GUT=1|

Example 2: Log OP off with input of quantity and unit

DLG=A_AB|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100451210200|
MNR=60610|EGR:GUT=1|EGE:GUT=ST|

Example 3: Log OP off with input of quantity, scrap and scrap reason

DLG=A_AB|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100451210200|
MNR=60610|EGR:GUT=1|EGR:AUS=2|EGG:AUS=1|

6.2.9 Finish operation (DLG=A_BE)

Finish prepared or interrupted operation.

<table>
<tr><th>Identifier</th><th>Type / max. field length</th><th colspan="2">Description</th></tr>
<tr><td>ANR=</td><td>C</td><td colspan="2">MES order number (fully defined key)</td></tr>
<tr><td>MNR=</td><td>C8</td><td colspan="2">Workplace/machine number</td></tr>
<tr><td>PNR=</td><td>C10</td><td rowspan="2">or</td><td>Personnel number</td></tr>
<tr><td>KNR=</td><td>C10</td><td>Staff badge number</td></tr>
<tr><td>EGR:GUT=</td><td>N8</td><td colspan="2">Yield recorded</td></tr>
<tr><td>EGR:AUS=</td><td>N8</td><td colspan="2">Scrap recorded</td></tr>
<tr><td>EGG:GUT=</td><td>N4</td><td colspan="2">Reason</td></tr>
<tr><td>EGG:AUS=</td><td>N4</td><td colspan="2">Scrap reason</td></tr>
<tr><td>EGE:GUT=</td><td>C4</td><td colspan="2">Unit of yield quantity</td></tr>
<tr><td>EGE:AUS=</td><td>C4</td><td colspan="2">Unit of scrap quantity</td></tr>
</table>

The time of the posting (acronym ZEI) must not be at shift change. Background: CORU-Call #118905.

Example 1: Finish OP with input of quantity

DLG=A_BE|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100451210200|
MNR=60610|EGR:GUT=1|

Example 2: Finish OP with input of quantity and unit

DLG=A_BE|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100451210200|
MNR=60610|EGR:GUT=1|EGE:GUT=ST|

Example 3: Finish OP with input of quantity, scrap and scrap reason

DLG=A_BE|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100451210200|
MNR=60610|EGR:GUT=1|EGR:AUS=2|EGG:AUS=1|

6.2.10 Quantity upload (DLG=A_MR)

You can use the quantity upload to upload quantities for orders, which are not logged on at the moment. This way, you can correct a quantity of an order without having to log the order on and off.

Note:
This dialog does not change the postings performed for the machine performance.

<table>
<tr><th>Identifier</th><th>Type / max. field length</th><th colspan="2">Description</th></tr>
<tr><td>ANR=</td><td>C</td><td colspan="2">MES order number (fully defined key)</td></tr>
<tr><td>MNR=</td><td>C8</td><td colspan="2">Workplace/machine number</td></tr>
<tr><td>PNR=</td><td>C10</td><td rowspan="2">or</td><td>Personnel number</td></tr>
<tr><td>KNR=</td><td>C10</td><td>Staff badge number</td></tr>
<tr><td>EGR:GUT=</td><td>N8</td><td colspan="2">Yield recorded</td></tr>
<tr><td>EGR:AUS=</td><td>N8</td><td colspan="2">Scrap recorded</td></tr>
<tr><td>EGG:GUT=</td><td>N4</td><td colspan="2">Reason</td></tr>
<tr><td>EGG:AUS=</td><td>N4</td><td colspan="2">Scrap reason</td></tr>
<tr><td>EGE:GUT=</td><td>C4</td><td colspan="2">Unit</td></tr>
<tr><td>EGE:AUS=</td><td>C4</td><td colspan="2">Unit of scrap quantity</td></tr>
</table>

The time of the posting (acronym ZEI) must not be at shift change. Background: CORU-Call #118905. We think that the problem that occurred with A_BE can also occur here with A_MR.

Example 1: Quantity upload with scrap and scrap reason

DLG=A_MR|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100473100200|
MNR=60610|EGR:GUT=1|EGR:AUS=2|EGG:AUS=1|

6.2.11 Log person on (DLG=P_AN)

Identifier	Type / max. field length	Description	
MNR=	C8	Workplace/machine number	
PNR=	C10	or	Personnel number
KNR=	C10		Staff badge number

Example 1: Log person on

DLG=P_AN|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|MNR=871555|

6.2.12 Log person off (DLG=P_AB)

Identifier	Type / max. field length	Description	
MNR=	C8	Workplace/machine number	
PNR=	C10	or	Personnel number
KNR=	C10		Staff badge number
EGR:GUT=	N8	Yield recorded	
EGR:AUS=	N8	Scrap recorded	
EGG:GUT=	N4	Reason	
EGG:AUS=	N4	Scrap reason	
EGE:GUT=	C4	Unit	
EGE:AUS=	C4	Unit of scrap quantity	

Example 1: Log person off

DLG=P_AB|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|MNR=871555|

Example 2: Log person off with input of quantity, scrap and scrap reason

DLG=P_AB|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|MNR=123456|EGR:GUT=1|
EGR:AUS=2|EGG:AUS=1|

6.2.13 Log off all persons from machine (DLG=P_AAB)

Identifier	Type / max. field length	Description
MNR=	C8	Workplace/machine number

Identifier	Type / max. field length	Description
EGR:GUT=	N8	Yield recorded
EGR:AUS=	N8	Scrap recorded
EGG:GUT=	N4	Reason
EGG:AUS=	N4	Scrap reason
EGE:GUT=	C4	Unit
EGE:AUS=	C4	Unit of scrap quantity

Example 1: Log off all persons from machine

DLG=P_AAB|USR=2106|DAT=02/17/2000|ZEI=47972|MNR=871555|

Example 2: Log all persons off with input of quantity, scrap and scrap reason

DLG=P_AAB|USR=2106|DAT=02/17/2000|ZEI=47972|MNR=123456|EGR:GUT=1|EGR:AUS=2|
EGG:AUS=1|

6.2.14 Change machine status (DLG=M_MST)

Identifier	Type / max. field length	Description	
MNR=	C8	Workplace/machine number	
MST=	N4	New machine status: Malfunction reason and/or machine status. Example: 1 = Production, 99 = general disturbance/malfunction The definition depends on the machine configuration.	
PNR=	C10	or	Personnel number
KNR=	C10		Staff badge number
IZY=	N8	Actual cycle in seconds per 1000 cycles	
BEM=	C40	You can enter a free comment	
PROGNDAUER=	C4	Expected time that the new status will be available. For documentation purposes, this information is stored with the status event. If the field has the value 0 or is empty, the new status will be available "for an indefinite period".	
Additional data for the escalation management			
MSGPRIO=	N1	Priority: 1 = highest, 2 = high, 3 = normal, 4 = low, 5 =lowest	

MSGCLASS=	C1	Information class/importance I = Information, W = Warning, E = Error
MSGRCV=	C40	Recipient/addressee, e.g. group of plant managers

Example 1: Person changes machine status

DLG=M_MST|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|MNR=871555|MST=3|

Note:

If the machine status is changed via PDM dialog and if the workplaces/machines are simultaneously managed by terminals (machines with MDE), the machine status change is not automatically displayed on the terminal that manages the machine.

6.2.15 Automatic status update (DLG=M_AST)

Purpose: Continuation of the machine status and posting of automatically recorded quantities and cycles.

Identifier	Type / max. field length	Description
MNR=	C8	Workplace/machine number
AGR:HUB=	N8	Automatically recorded cycles
TRGEN=	C1 = {J/N}	With TRGEN=J, the system triggers operation-related upload(s) of part quantities for the automatic quantities recorded up to now, which result in order-related log records of record type "T". If several operations are active at the machine, one upload of a part quantity is triggered per operation. If scrap counters with different reasons are recorded for the machine, several uploads of part quantities are triggered per operation.
IZY=	N8	Current actual cycle of the machine in seconds per 1000 cycles
AGR.C:[1..n]=	N8	Automatically recorded counter quantity of counter 1...n
AGE.C:[1..n]=	C4	Unit of counter (e.g. PCS) for future use
AGG.C:[1..n]=	N4	Reason for counter (e.g. scrap reason)
AGB.C:[1..n]=	C10	Evaluation of counter: - GUT = yield - AUS = scrap - NCH = rework

Identifier	Type / max. field length	Description
		- PRB = open quantity (problem quantity)

Example:

DLG=M_AST|USR=2106|DAT=02/17/2000|ZEI=47972|MNR=871555|AGR:HUB=11|AGR.C:1=47|AGB.C:1=GUT| AGR.C:2=1|AGB.C:2=AUS|AGG.C:2=12|

Note:

All input dialogs (e.g. A_UN, P_AB), which include manually recorded values (EGR:GUT, EGR:AUS), support the automatically recorded values AGR.C:[1..n], AGE.C:[1..n], AGG.C:[1..n], AGB.C:[1..n], AGR:HUB.

Important note on counters:
Automatic counters are not yet used for calculation.
A calculation using the partitioning and the pulse factor and other quantity accounts (e.g. when the total quantity is recorded) is only performed in the server.
Example:
AGR.C:1=47|AGB.C:1=GUT|AGR.C:2=1|AGB.C:2=AUS|AGG.C:2=12|....|AGR:HUB=..|

6.2.16 Change of the target quantity (DLG=A_SMG)

Identifier	Type / max. field length	Description	
ANR=	C	MES order number (fully defined key)	
MNR=	C8	Workplace/machine number	
PNR=	C10	or	Personnel number
KNR=	C10		Staff badge number
SGR:GUT=	N8	New target quantity	
SGE:GUT=	C4	Unit of the target quantity	

Example 1: Person changes target quantity

DLG=A_SMG|USR=2106|DAT=02/17/2000|ZEI=47972|MNR=100|ANR=12Y19D0001|SGR:GUT=123|SGE:GUT=ST|

6.2.17 Change of target cycle (DLG=M_SZY)

Identifier	Type / max. field length	Description
MNR=	C8	Workplace/machine number
ANR=	C	MES order number (fully defined key)

Identifier	Type / max. field length	Description	
SZY=	N8	Target cycle in seconds per 1000 cycles	
PNR=	C10	or	Personnel number
KNR=	C10		Staff badge number

Example: Person changes target quantity

DLG=M_SZY|USR=2106|DAT=02/17/2000|ZEI=58371|MNR=100|SZY=3600|

The new target cycle is set for all OPs currently logged on to the machine.

6.2.18 Change of partitioning (DLG=M_TLG)

Identifier	Type / max. field length	Description	
MNR=	C8	Workplace/machine number	
ANR=	C	MES order number (fully defined key)	
TLG=	N8	Partitioning	
PNR=	C10	or	Personnel number
KNR=	C10		Staff badge number

Example: Person changes partitioning

DLG=M_TLG|USR=2106|DAT=02/17/2000|ZEI=34392|MNR=100|TLG=2|

The new partitioning is set for all OPs currently logged on to the machine.

6.2.19 Logging of the production lock (DLG= M_PSPERRE)

Identifier	Type / max. field length	Description	
PNR=	C10	or optional	Personnel number
KNR=	C10		Staff badge number
ACTIVE=	C1	Setting or resetting the production lock: J … set N … reset	

Example: Person activates the production lock

DLG=M_PSPERRE|USR=2106|DAT=02/17/2000|ZEI=34392|ACTIVE=J|

Note:

If a person (KNR or PNR) is entered, the system checks if the person is authorized to change the production lock. The configuration is made on the MOC in the application HR master data.

6.2.20 BDE comment (DLG=HY_BEM)

Identifier	Type / max. field length	Description	
ANR	C	MES order number (fully defined key)	
BEM	C60	The BDE comment must not exceed 60 characters.	
MNR	C8	If no workplace is specified, the BDE comment is saved for the workplace where the operation is planned.	
PNR=	C10	or	Personnel number
KNR=	C10		Staff badge number

Example: Person records a BDE comment for the operation.

DLG=HY_BEM|USR=2106|ANR=210047310200|BEM=problems with material supply|KNR=1111|DAT=02/17/2000|ZEI=34392|

Notes:

- The input and display length on the AIP is limited to 60 characters.
- BDE comments can trigger an escalation. If you have configured in the escalation management that the comment text is displayed as subject of the message, then only the first 50 characters are displayed.

6.3 Postings made with shift change

Shift change records must be made in the correct order along with the other postings. All postings for the respective shift must be transferred before the shift end record/shift change record.

6.3.1 Shift end (A_AUN)

Identifier	Type / max. field length	Description
MNR=	C8	Workplace/machine number
DAT=	mm/dd/yyyy	Date of shift end in format "mm/dd/yyyy"
ZEI=	Seconds	Point in time of the shift end in seconds after midnight (example: 10:00:03 is *ZEI=36003*)

AGR:HUB=	N8	Automatically recorded cycles
AGR.C:[1..n]=	N8	Automatically recorded counter quantity of counter 1...n
AGE.C:[1..n]=	C4	Unit of counter (e.g. PCS) for future use
AGG.C:[1..n]=	N4	Reason for counter (e.g. scrap reason)
AGB.C:[1..n]=	C10	Evaluation of counter: - GUT = yield - AUS = scrap - NCH = rework - PRB = open quantity (problem quantity)

Example: **Shift end on 28-APR-2002 at 6:00**

DLG=A_AUN|DAT=04/28/2002|ZEI=21600|MNR=4560

6.3.2 Beginning of shift (A_AAN)

Identifier	Type / max. field length	Description
MNR=	C8	Workplace/machine number
DAT=	mm/dd/yyyy	Date of shift end in format "mm/dd/yyyy"
ZEI=	Seconds	Point in time of the shift end in seconds after midnight (example: 10:00:03 is *ZEI=36003*)

Example: **Beginning of shift on 28-APR-2002 at 6:00**

DLG=A_AAN|DAT=04/28/2002|ZEI=21600|MNR=4560

6.3.3 Shift change (A_ASW)

The shift change triggers a shift end and a beginning of shift.

Identifier	Type / max. field length	Description
MNR=	C8	Workplace/machine number
DAT=	mm/dd/yyyy	Date of shift end in format "mm/dd/yyyy"
ZEI=	Seconds	Point in time of the shift end in seconds after midnight (example: 10:00:03 is *ZEI=36003*)
AGR:HUB=	N8	Automatically recorded cycles
AGR.C:[1..n]=	N8	Automatically recorded counter quantity of counter 1...n

AGE.C:[1..n]=	C4	Unit of counter (e.g. PCS) for future use
AGG.C:[1..n]=	N4	Reason for counter (e.g. scrap reason)
AGB.C:[1..n]=	C10	Evaluation of counter: - GUT = yield - AUS = scrap - NCH = rework - PRB = open quantity (problem quantity)

Example: **Shift change on 28-APR-2002 at 6:00**

DLG=A_ASW|DAT=04/28/2002|ZEI=21600|MNR=4560

6.4 Reading BDE/MDE data

6.4.1 Machine info

The list of machine info is provided by the command DLG=LIST;10 and filed in the directory HYDRADIR\spool\.

The definition of the file name and the respective path is case-sensitive.

Structure of dialog data:

"DLG=LIST;10|DATEI={file name}|DAT=...|ZEI=...|USR=...|..."

The list includes the current quantity of the shift. If the unit changes when the order changes, the shift-related counters are reset to 0. The list includes the following data:

Identifier	Field designation	Description
MNR	Machines	Machine no.
MGRP	Group	Machine group
MBEZK	Mach. des.	Machine name
MBEZL	Det. machine des.	Detailed machine name
KST	Cost center	Cost center of the machine
MST	Status	No. of current machine status or 20000 = no shift 30000 = status not assigned
MSTTXT	Status text	Text of the current machine status

Identifier	Field designation	Description
PKENN	Pchar	Production control characteristic of the current status P: "Production" (defined once per machine) S: Malfunctions (defined x times per machine) A: "General malfunction" (defined once per machine) N: No shift or Status "Not assigned"
MSDATB	Date	Start date of machine status
MSZEIB	Time	Start time of machine status
MSDAUER	Duration	Time that the machine status lasts up to now
BMKNR	RPA no.	Current RPA text number
BMKTXT	RPA	Current RPA abbreviation
BMKTXTL	RPA text	Current RPA text
AGR:BMK01 to AGR:BMK12	RPA1 to RPA 12	Time posted to RPA01 to Time posted to RPA12
SKNR	S	current shift number
SKDATB	Date beg.	Date of beginning of shift
SKZEIB	Time beg.	Time of beginning of shift
SKDATE	Date end	Date of end of shift
SKZEIE	Time end	Time of end of shift
AGR:GUT	Yield	Yield quantity of current shift
AGR:AUS	Scrap	Scrap quantity of current shift
AGR:HUB	Strokes	Cycles of current shift
TLG	Partitioning	Current partitioning
SZY	Target cycle	Current target cycle
KFGKZ:1 to KFGKZ:5	Cfg_ch1 to Cfg_ch5	Configuration indicator where machine configuration data is filed
ZLO	Target	Default for: - Prepare order - Start order
ABFALL	Waste	Customer-specific
MEHRFKZ	MKz.	Customer-specific
MPUFF	Input buffer	Input buffer
MDE_MASCH	MDE machine	MDE machine

Identifier	Field designation	Description
MPL_MOD	Batch mode	Batch mode
AUTOMENGE	AutoQuantity	Autom. quantity
AGE:GUT	Unit	Unit of yield quantity
AGE:AUS	Unit	Unit of scrap quantity
IZYSM	Piece / minute	Actual cycle piece per minute
MNR.PARAM:1 MNR.PARAM:15	Mach. parameter 1 to Mach. parameter 15	General machine parameters without parameters 5 and 11
IMPFAKT	Mach. parameter 5	is machine parameter 5
STLG	Machine partitioning	is machine parameter 11
LINIE	Line	Line of the aggregate
LINIE.REF	Reference aggregate	Reference aggregate Y/N
MART	Machine type	Machine type E: Single workplace G: Group workplace
EXTTYP	External device	K: No external device J: DS100 N: MT3 E: Engel interfacing A: Arburg interfacing
EXTSNR	External device: serial number	depending on EXTTYP: A: Serial number E: Serial number
EXTID	External device: Device address / class	depending on EXTTYP Y/N: ID of master terminal is number of MT3/DS100 A: Machine class - is ARBURG control system
ADEMSDATB	Date	
ADEMSZEIB	Time	
MULTIAG	Multiple ch.	
MATPUF:OUT	Target	

Identifier	Field designation	Description
MATPUF:IN	Input buffer	
OPT:CNRAUTOGEN	Batch no. autom.	
OPT:CNRAGAN	Log on OP CEAN/UMB	
TYP	Machine type	
OPT:CHV	Batch management	
OPT:CNRPRN	Ticket print	From product version **MPL 7.2.5** no longer supported.
DIGIN:CAWL	Digital Input batch change	From product version **MPL 7.2.5** no longer supported.
MNR.VISVERBRBLZ	Consumption balance	New as of product version **MPL 7.2.5**
DIGOUT:SMENGE	Output target qty. reached	
TNR	Terminal number	
IZYABW	Cycle extension	
OPT:AUS	Scrap processing	
OPT:AUSAUTO	Scrap automatic	
OPT:AUSMANU	Scrap manual	
OPT:AUSPROT		
OPT:AUSRS232		
DIGIN:GUT		
DIGIN:AUS	Scrap counter	
DIGOUT:MSPERRE		
ANZSTAKT		
OPT:GUTMANU	Yield manual	
PARAM:12	Customer-specific	
OPT:GUTRS232		
DIGIO	Free input/output	

Identifier	Field designation	Description
CAT		
OPT:PDV	Activation of process data processing	
BDEJMOD	BDE year model	
UEBART		
DLGSTRG	Dialog control	
ICON	Symbol	
OPT:GUTAUTO		
UEBDAUER		
ANZPALMNR	No. of processing stations/palett machines	
MNR.LISTSUFFIX	Dialog control	
RESVERWEIS	Resource ID	
RESWART.BEZ	Designation	
RESWART.ART	Maintenance type	
RESWART.WARTSTA	Status	
RESWART.WARTKL	Class	
RESWART.AKTWERT	Current value	
RESWART.NAEWERT	Value next maintenance	
TNRMSPERRE	TNRMSPERRE	Internal use for CTAIP
TNRPSPERRE	TNRPSPERRE	Internal use for CTAIP

Example:

DLG=LIST;10|DAT=02/22/2005|ZEI=40000|USR=2101|DATEI= ./spool/masch_list.dat|

6.4.1.1 Dynamic extension of machine list

You can dynamically extend the machine list described and add further acronyms, if required. You can then add further fields to the standard list.

Structure of dialog data:

"DLG=LIST;10|DATEI={file name}|MOD={mode}|AKRO=OPT:VLISTMOD;ANZPALMNR;DLGSTRG;.....|....“

In the dialog data, the list of additional fields is started with the identifier AKRO=. The listed acronyms are then separated by semicolon.

MPDV Mikrolab provides a list a acronyms that can be used.

6.4.1.2 Extended machine list

Structure of dialog data:

"DLG=LIST;10|DATEI={file name}|MOD={mode}|....“

Identifier	Field designation	Description
AGR:GUTP AGR:GUT	Machine-specific yield	Primary quantity
AGR:GUTS		Secondary quantity
AGR:GUTT		Tertiary quantity
AGR:GUTB		Basic quantity
AGR:AUSP AGR:AUS	Machine-specific scrap	Primary quantity
AGR:AUSS		Secondary quantity
AGR:AUST		Tertiary quantity
AGR:AUSB		Basic quantity
AGR:NCHP	Machine-specific rework quantity	Primary quantity
AGR:NCHS		Secondary quantity
AGR:NCHT		Tertiary quantity
AGR:NCHB		Basic quantity
AGR:PRBP	Machine-specific open quantity (problem quantity)	Primary quantity
AGR:PRBS		Secondary quantity
AGR:PRBT		Tertiary quantity

Identifier	Field designation	Description
AGR:PRBB		Basic quantity
CTR:1 CTR:2 CTR:3 CTR:4 CTR:5 CTR:6	Counter 1...6	
DIV	Pulse factor	
OPT:MDETLG	Option "order-specific partitioning"	
EGE:GUTP	Primary quantity unit	
EGE:GUTS	Secondary quantity unit	
EGE:GUTT	Tertiary quantity unit	
EGE:GUTB	Base quantity unit	
UMRFAKTP:Z	Conversion of primary quantity	See note **Conversion factors for base quantity**
UMRFAKTP:N	Conversion of primary quantity	
UMRFAKTS:Z	Conversion of secondary quantity	
UMRFAKTS:N	Conversion of secondary quantity	
UMRFAKTT:Z	Conversion of tertiary quantity	
UMRFAKTT:N	Conversion of tertiary quantity	

Identifier	Field designation	Description
VERB:GUT	Offset of manual yield quantity against - AUS - NCH - PRB - empty = no offset	See note **Offset of manual quantities**
VERB:AUS	Offset of manual scrap against - GUT - NCH - PRB - empty = no offset	
VERB:NCH	Offset of manual rework quantity against - GUT - AUS - PRB - empty = no offset	
VERB:PRB	Offset of manual problem quantity = open quantity against - GUT - AUS - NCH - empty = no offset	
OPT:GUTMANUTAKT	Calculation of the relevant manual quantity additionally as cycles (i.e. additionally as AGR:HUB)	

Identifier	Field designation	Description
OPT:AUSMANUTAKT	Calculation of the relevant manual quantity additionally as cycles (i.e. additionally as AGR:HUB)	
OPT:NCHMANUTAKT	Calculation of the relevant manual quantity additionally as cycles (i.e. additionally as AGR:HUB)	
OPT:PRBMANUTAKT	Calculation of the relevant manual quantity additionally as cycles (i.e. additionally as AGR:HUB)	
VISLIST3	Display 3rd list	The third list on the terminals is dynamically controlled using this indicator.
VISFHMTNRAAN	Show material/PRT list when OP is logged on	

Conversion factors for base quantity

You use the conversion factors to convert the primary, secondary and tertiary quantities against the base quantity. You use these conversion factors, for example, when updating target quantities.

You use a numerator and denominator for the conversion factors. This way, you can also use a decimal value (meaning a figure with decimal places) as conversion factor.

Example:
- Base quantity unit: square meter M2
- Primary quantity unit: piece ST
- 1 piece = 2 square meters.

In this case to be stored as
- numerator (= UMRFAKTP:Z) 2 and
- denominator (= UMRFAKTP:N) 1.

Note: Offset of manual quantities

When a quantity is offset against another quantity account, the quantity recorded is subtracted from the other quantity account.

Example:
Total quantity and scrap is recorded → configuration VERB:AUS=GUT

6.4.2 Reading the shift calendar

The shift calendar of all machines of a terminal is provided via the command DLG=LIST;38. The calender is filed in the directory HYDRADIR\spool\. The shift calendar contains the data starting from the day before for 5 days.

Structure of dialog data:

"DLG=LIST;38|DATEI={file name}|TNR={terminal number}|DAT=…|ZEI=...|USR=...|..“

The definition of the file name and the respective path is case-sensitive.

The list includes the following data:

Identifier	Field designation	Description	
MNR	Machine	alter-native	If the mode MOD={..} is not specified, the column MNR is output with the machine number that is valid for the model.
BDEJMOD	Year model		With mode T or H MOD={..} the column BDEJMOD is output with the year model. The machine is not specified with this mode.

Identifier	Field designation	Description
SKDAT	Shift date	Date of beginning of shift
SKNR	Shift number	Shift number
SKZEIB	Beginning of shift	Time in seconds
SKZEIE	End of shift	Time in seconds
SKPAUB:1	Start of shift break 1	Time in seconds
SKPAUE:1	End of shift break 1	Time in seconds
…		
SKPAUB:6	Start of shift break 6	Time in seconds
SKPAUE:6	End of shift break 6	Time in seconds
SKAMST	Automatic machine status	Internal use in the Hydra terminal
MST999SKB	Automatic machine status	Activate/deactivate weekend status 999
SKART	Shift ID	Shift ID
BEGINNDAT	Start date	Date of beginning of shift

6.4.3 Order list

The list of order info is provided by the command DLG=LIST;11 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

„DLG=LIST;11|DATEI={file name}|DAT=...|ZEI=...|USR=...|MOD=...“

The definition of the file name and the respective path is case-sensitive.

a) Reading the sequencing lists of the terminal:

Parameter: MOD=T(terminal from user)

The list includes the data of all machines assigned to this terminal.

The list contains planned (customer-specific status X), prepared (status V), interrupted (status U) and running (status L) production orders.

Overhead orders are only included in the list when they are running (status L).

For the specific machine, the results of MOD=V and MOD=L are therefore output together.

Note: Because of the different configuration options, the list might only output a limited solution set.

b) Reading the info of an order:

Parameter: MOD=A|ANR=order+OP

c) Reading the sequencing list of a machine

Parameter: MOD=V|MNR=machine

The list includes all OPs currently planned for the machine (the configuration of the sequencing list specifies if the OPs of the machine or the OPs of the machine group are selected).

d) List of running orders on the terminal

Parameter: MOD=L

The list includes the running orders (status L) of all machines assigned to this terminal.

The list includes the following relevant data:

Identifier	Field designation	Description
ROW.IDX	ROWINDEX	
ANR	Complete order number	Complete order number contains: AUNR, AGNR, AFOLG, SPLNR
AUNR	Order number	Order number
AGNR	OP	Operation number
AFOLG	Sequence	
SPLNR	Split number	
AUART	Order type	Order type
MNR	Planned single machine	Machine number
POS	Display position	
MGRP	Group	Machine group
MDE_MASCH	MDE machine	MDE machine
ATK	Article	Material number (for output batch)
ATKBEZ	Final article des.	Article name (from order header)
AST	Status	current order status "X", "V", "L", "U"
ASTTXT	Status text	current order status - text dependent of status: planned, prepared, running, interrupted
EGR:BMK01 to EGR:BMK12	Performance posted to RPA 01 to RPA 12	
ANR:MENGEPROZ_UEBLI	Overdelivery	
ANR:OPT_UEBLI	Reaction to overdelivery	

Identifier	Field designation	Description
ANR:MENGEPROZ_UNTLI	Underdelivery	
ANR:OPT_UNTLI	Reaction to underdelivery	
EGS:GUT	since login	
SGE:B	Base quantity unit	
SGE:P	Primary input quantity unit	
SGE:S	Secondary input quantity unit	
SGE:T	Tert. Input quantity unit	
SGR:AUSB	Target scrap	
SGR:AUSP	Target scrap (prim.)	
SGR:AUSS	Target scrap (sec.)	
SGR:AUST	Target scrap (tert.)	
SGR:GUTB	Target quantity (base quantity unit)	
SGR:GUTP	Target quantity (prim.)	
SGR:GUTS	Target quantity (sec.)	
SGR:GUTT	Target quantity (ter.)	
EGR:GUTB	Yield (base quantity unit)	
EGR:GUTP	Yield (prim.)	
EGR:GUTS	Yield (sec.)	
EGR:GUTT	Yield (tert.)	
EGR:AUSB	Scrap (base quantity unit)	
EGR:AUSP	Scrap quantity (prim.)	
EGR:AUSS	Scrap quantity (sec.)	
EGR:AUST	Scrap quantity (ter.)	
EGR:NCHB	Rework quantity	
EGR:NCHP	Rework quantity (prim.)	
EGR:NCHS	Rework quantity (sec.)	
EGR:NCHT	Rework quantity (ter.)	
EGR:PRBB	Problem quantity	
EGR:PRBP	Problem quantity (prim.)	
EGR:PRBS	Problem quantity (sec.)	
EGR:PRBT	Problem quantity (ter.)	
AGE:B	Quantity unit	
AGE:P	Quantity unit (prim.)	
AGE:S	Quantity unit (sec.)	

Identifier	Field designation	Description
AGE:T	Quantity unit (ter.)	
TLG	Partitioning	Partitioning
SZY	Target cycle	Cycle time in seconds per 1000 cycles
AGBEZ	Name/designation of the OP	Operation designation/name.
ASTV	VStatus	Status of the previous OP
AGMFKZ	Parallel production	
CHPFL	Batch management required	
APRIO	External priority	
FIX	fix	Operation fixed Yes/No
DSBEZ	Data ID for ALS	
OPT:SNR	Serial numbers required	
ANR_BEARBZ	Processing time	
ANR_ANDATB	Date of logon	
ANR_ANZEIB	Time of logon	
AST_OPT_PKENN	Control characteristic "production"	
AKTIV	active	
VERARBCODE_PLAUS_MENGE	100% quantity inspection	
EGR:DAUER	Duration	
EGR:PDAUER	Workforce planning	
EGI:GUT	Yield	
EGI:AUS	Scrap	
BEM1	Comment 1	From user field 53 of operation (ANR.FU:53)
BEM2	Comment 2	From user field 54 of operation (ANR.FU:54)
PANZ	Number of staff	
MST	Machine status	
HZTYP	Semi-finished type	
HZBEZ	Name of semi-finished type	
TECHINFO	Technical information	From user field 56 of operation (ANR.FU:56)
TPE	TPUnit [OP]	
CNR	Batch number	

Identifier	Field designation	Description
DLL	Customer batch number	
DLLKZ	Customer batch identification	
LOSGR	Batch size	
HZTYP:EINH	Unit	
PLAUS:MATOK	Plaus. MatOK	
MANR	CBM: Mother operation	
RF:AGTYP	CBM: Operation type	

Example: List of all running orders

DLG=LIST;11|DAT=10/11/2000|ZEI=40000|USR=2101|MOD=L|DATEI= ./spool/auft_list.101|

6.4.3.1 Dynamic extension of the order list

As of HYDRA MW 2.0, you can dynamically complement the order list and add further acronyms, if required. You can then add further fields to the standard list.

Structure of dialog data:

"DLG=LIST;11|DATEI={file name}|MOD={mode}|AKRO=MNR_MSTDSATZ;AUART_PLAUS_MNR;AUART_VISCODE;MNR_EXTSNR; ...;.....|....“

Same procedure as for the machine list.

MPDV Mikrolab provides a list a acronyms that can be used.

6.4.3.2 Controlling the sequencing list

The sequencing list on the BDE terminal is flexible and can be adjusted to meet the customer's requirements.

The settings described below control how data is provided in the sequencing list.

Dialog	Option	Description	Possible settings
Machine configuration	Sequencing list	Via the configuration of the *sequencing list*, you can define the order pool used to generate the sequencing list. The planning in the HYDRA shop floor scheduling or in the PPS system specifies the pool of orders. The pool of orders is different if the OP is planned directly for a machine or for a machine group.	S Basic setting The value of the option with the same name in the HYDRA basic settings is used. M Pool of workplaces The terminal sequencing list only shows the operations planned for the workplace. G Pool of workplaces and groups The terminal sequencing list shows operations that are: - planned for the current workplace or - for another workplace of the group or - that are still located in the pool of groups. K Pool of workplaces and categories The sequencing list on the terminal only shows the operations that are planned for workplaces of the selected machine category. H Group control The terminal sequencing list shows the operations that are - planned for the current workplace or - for another workplace of the group.

Machine configuration	Number of OPs	You can configure the maximum number of OPs included in the sequencing list. The OPs are selected in ascending order based on how the sequencing list is sorted. By default, HYDRA sorts by the following columns: 1. sort_dat 2. sort_zeit 3. auftrag_nr of the OP	0 = no restriction (display of all existing OPs) 1-999 = maximum number of OPs
Order types	Sequencing list	On the level of orders, you use the option *Sequencing list* in the application *Order types* to configure the list. If you define an "N", all OPs of an order with this order type are not displayed in the sequencing list. Example: You do not want to show waiting period OPs in the sequencing list.	J The OP should be displayed in the sequencing list. F The OP should only be displayed in the sequencing list if it is fixed. wurde. N The OP should not be displayed in the sequencing list.
Processing codes	Sequencing list	Also in the configuration of the processing codes, you can define if an OP with this processing code is displayed in the sequencing list or not.	J The OP should be displayed in the sequencing list. N The OP should not be displayed in the sequencing list.
Status assignment	Sequencing list	Also on the level of statuses, you can make configurations. You can define if OPs with a specific status are displayed in the sequencing.	J The OP should be displayed in the sequencing list. N The OP should not be displayed in the sequencing list.

		If an "N" is defined here, an OP in this status is not displayed in the sequencing list. This is used as a standard feature to ensure that only prepared or interrupted OPs appear in the sequencing list. You can also configure that running OPs are displayed in the sequencing list.	

The following operations are not displayed in the sequencing list:

- Locked operations
- The different operations included in merged operations generated on the MOC.
- The original operation of split operations

6.4.4 Personnel list

The list of personnel is provided by the command DLG=LIST;12 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

„DLG=LIST;12|DATEI={file name}|DAT=...|ZEI=...|USR=...|MOD=...|...“

The definition of the file name and the respective path is case-sensitive.

If MOD ist not set, the list includes all persons logged on to the machines of the terminal.

If MOD=V, the list additionally includes all persons logged on in advance to the machines of the terminal.The list includes the following data:

Identifier	Field designation	Description
MNR	Machine	Machine number
ANR	Order	Order number (with all details)
AGNR	OP	Only operation number
AUNR	Order number	
AFOLG	Sequence	
SPLNR	Split number	
PNR	Personnel number	Personnel number
PNAME	Name	Name of the person
PVORNAME	First name	First name of the person
BPOS	Usr.	Operator position/function
BPBEZL	Operator position/function	Name of the operator's function
LPKZ	Wages/premium characteristics	Wages/premium indicator
KNR	Badge number	Staff badge number
VORAN	Logged on in advance	Persons logged on in advance (only with mode V)
ASTUFE	BDE authorization	

Example:

DLG=LIST;12|DAT=10/11/2000|ZEI=40000|USR=2101|DATEI= ./spool/pers_list.101|

6.4.5 Operator positions/functions of machines

The list of operator positions/functions is provided by the command DLG=LIST;14 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

"DLG=LIST;14|DATEI={File name}|DAT=...|ZEI=...|USR=...|...“

The definition of the file name and the respective path is case-sensitive.

The list includes all operator positions of the machines assigned to this terminal.

The list includes the following data:

Identifier	Field designation	Description
MNR	Machine	Machine number
BPSCHL	Operator pos.	Key
BPBEZL	Operator position/function	Description
BPTXTK	BPOS text	Short text
BPFKT	BPOS fct.	Function
BPOS	Usr.	Operator position/function

Example: Operator position/function

DLG=LIST;14|DAT=10/11/2000|ZEI=40000|USR=2101|DATEI= ./spool/bp_list.101|

6.4.7 Machine status list

The list of machine statuses is provided by the command DLG=LIST;16 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

"DLG=LIST;16|DATEI={File name}|DAT=...|ZEI=...|USR=...|...“

The definition of the file name and the respective path is case-sensitive.

a) List of all statuses of the machines assigned to the terminal.

Parameter: MOD=T|USR=HYDRA-User number|

The list is sorted by machine and list of the machine is sorted by malfunction number.

b) List of all statuses of a machine.

Parameter: MOD=M|MNR=machine number|

The list includes the following data:

Identifier	Field designation	Description
MNR	Machine	Machine number
KSTART	Cost center type	Type of cost center
MST	Status	Number of malfunction ID
ZUNR	ToNo.	Assignment number of input
MSTTNR	StNo.	Number of the malfunction text.
MSTTXT	Status text	Malfunction text
BMKTXT	RPA	RPA abbrev.
STKL	Dist. class	Class of malfunction
PKENN	Pchar	Production ID P: flags "Production"; (per machine defined once) S: flags malfunction (per machine defined x times) A: flags "general malfunction" (per machine defined once)
ZUMAN	Man.	Manual assignment J/N
ZUAUTO	Auto.	Automatic assignment J/N
MSPERR	Lock	Machine lock
PABKZ	Log off persons	Log off persons J/N
MSTUFE	Auth.lev.	Authorization level
KST01	Ccr01	Customer-specific Cost center 01
KST02	Ccr02	Customer-specific Cost center 02
KST03	Ccr03	Customer-specific Cost center 03
KST04	Ccr04	Customer-specific Cost center 04

Identifier	Field designation	Description
KST05	Ccr05	kst05[1]J=Prod. lock active kst05[2-8] time of flashing symbol of the machine status in graphic machinery
HARC:ID	Hierarchy level	Configuration **Hierarchical malfunction reasons**
HARC:TYP	Type of hierarchy level	
HARC:TXTKENN	ID: hierarchical level text	
HARC:FLDKENN	ID status field	
AUSNR:AUTO	Scrap reason	Configuration **Status-related collection of scrap reasons**
AUSNR:PSperre	Scrap reason during production lock	
PROGNDAUER	Estimated downtime	See section **Change of machine status (M_MST)**
COLOR	Color code	Configuration of the status text color RGB color - format: RRGGBB:

Example

DLG=LIST;16|DAT=10/11/2000|ZEI=40000|USR=2101|MOD=M|DATEI= ./spool/mstat_list.101|

6.4.8 Deviation reasons

The list of deviation reasons is provided by the command DLG=LIST;23 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

"DLG=LIST;23|DATEI={File name}|DAT=...|ZEI=...|USR=...|..."

The definition of the file name and the respective path is case-sensitive.

The list includes all deviation reasons created.

The list includes the following data:

Identifier	Field designation	Description
WERK	Plant	
MNR	Machine	Machine number
EGG:GUT	Deviation reason	Number of deviation reason
ABWGRTXT	Text of deviation reason	Name of deviation reason

Example:

DLG=LIST;23|DAT=10/11/2000|ZEI=40000|USR=2101|DATEI= ./spool/abgr_list.101|

6.4.9 Premium indicator

The list of premium indicators is provided by the command DLG=LIST;24 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

"DLG=LIST;24|DATEI={File name}|DAT=...|ZEI=...|USR=...|...“

Parameter: USR=HYDRA user number

The definition of the file name and the respective path is case-sensitive.

The list includes all premium indicators created for all machines of the terminal.

The list includes the following data:

Identifier	Field designation	Description
MNR	Machine number	Machine number
LGRP	Wages/premium characteristics	Wage group
LGRPTXT	Designation	Name of the wage group

Example:

DLG=LIST;24|DAT=10/11/2000|ZEI=40000|USR=2101|DATEI= ./spool/lgrp_list.101|

6.4.10 Terminal list

The list of terminals is provided by the command DLG=LIST;45 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

"DLG=LIST;45|DATEI={File name}|DAT=...|ZEI=...|USR=...|..."

Parameter: none

The definition of the file name and the respective path is case-sensitive.

The list includes all terminals created - active and inactive terminals.

The list includes the following data:

Identifier	Field designation	Description
TNR	Terminal number	
TYP	Terminal type	
CFG:1	Configuration 1	
HWADR	Hardware address	
TZ	Time Zone	
BEZK	Location	
BEZL	Designation	
BART:MDE	MDE active	
BART:ADE	ADE active	
BART:PZE	PZE active	
BART:CAQ	CAQ active	
BART:PDV	PDV active	
LANG	Language	
AKTIV	Active	
NEUSTART	Restart	
LEN_KNR	Length of badge number	
PZEBART	Operation mode PZE	
PZESTA:VORG	Default status	
SYSNR	Company number/system number	
TTXT:KOM	IN key	

Identifier	Field designation	Description
TTXT:GEH	OUT key	
TTXT:PAU	Break key	
TTXT:INFO	Info key	
FGR:1	Absence reason 1	
TTXT:FGR1	Key Absence reason 1	
...to	...to	
FGR:4	Absence reason 4	
TTXT:FGR4	Key Absence reason 4	
FGRCFG	Entry absence reason	
OPT:FGRAUTO	Absence reason collection	
PLAUS:PZEONL	Online check	
PLAUS:PZESTA	Status sequence	
ZYKLLOAD:PZE	Cycl. loading PZE	
ZEI:BERLESEN1	Time 1 load PZE	
ZEI:BERLESEN2	Time 2 load PZE	
RELZUG:DAUER	Duration opener	
PZESTA:DAUER	Duration status	
PZEINFO:DAUER	Duration info	
PZEINFO:CFG	Info display	
PZEINFO:NR	Info number	
INFOZEIB:1	Info from 1	
INFOZEIE:1	Info to 1	
...to	...to	
INFOZEIB:5	Info from 5	
INFOZEIE:5	Info to 5	
OFSLST	Delay with shift change	
NEUZEIN	Time for next restart	
NEUDATN	Date for next restart	
OPT:CNRPRN	Ticket print	
CNR:TNR	Batch number	
TGRP	Terminal group	
SAGART	MOP generat. type	
LEN:AUNR	Order length	
OPT:MDEGEN	Generation MDE postings	
OPT:RMIRUEZ	Upload Setup time	

Identifier	Field designation	Description
OPT:VGZ	Process. Standard time	
PRJ	Customer project	
OPT:MNRFTYP	Machine number format	
OPT:SAG	Proces. merged operations	
OPT:PZEPLAUS	Plaus. PZE-IN	
SYSNR:SETUP	System number	
MAXDAUERSTEMP	Time clocking supplement	
OPT:PZEAUTO	PZE controls ADE	
ADEKAR:AKTIV	Waiting time treatment ADE	
ADEKAR:DAUER	Waiting time	
ADEKAR:BMKNR	Waiting time RPA	
OPT:VLIST	Sequencing list	
OPT:AUSNR	Coll. interruption reason	
OPT:ABBRNR	Coll. scrap reason	
OPT:ERFPMENGE	Coll. person-related quantity	
OPT:ERFAMENGE	Coll. order-related quantity	
OPT:GUTMANU	Coll. yield	
OPT:AUSMANU	Coll. scrap	
OPT:CHV	Batch management	
MDEKAR:AKTIV	Waiting time treatm. MDE	
MDEKAR:UG	Lower waiting time limit MDE	
MDEKAR:BMKNPLAN	RPA for times not yet scheduled	
MDEKAR:OG	Upper waiting time limit MDE	
MDEKAR:BMKPLAN	RPA for scheduled times	
OPT:ADEKNRPLAUS	PNO entry (BDE postings)	
OPT:MDEKNRPLAUS	PNO entry (MDE postings)	
LEN:KNR	Badge number length	
VER	Version	
OPT:KDVLIST	Customer sequencing list	
OPT:WILLESYS	Wille system	
OPT:UCHGEN	Creating unknown batches	
OPT:TRMGEN	Generation partial confirmation (upload part quantity)	
OPT:KORRMNGEN	Hide correction messages	

Identifier	Field designation	Description
OPT:EDITBRMSPERR	Lock edit RM-data	
OPT:AUSPAUDAUER	Hide breaks in pers. duration	
BDEOPT:9 to BDEOPT:12	BDE-CH 9 to BDE-CH 12	
OPT:VLISTMOD	Sequencing list	
LEN:PNR	Personnel No. length	
OPT:PNRFUELL	PersonnelNofillsign	
OPT:ZST	Set access statuses	
OPT:WSTEMPFGR	Change clocking absence reason	
OPT:ADATF	FLS earliest start date	
OPT:AART	FLS display order type	
KERNELVER	Kernel version	
KDNR	Customer number	
OPT:ANTVERBPM	Proportionate posting labor times	
OPT:PZPLAUS	Process check digit	
OPT:WSTEMP	Alternate clocking	
OPT:TGLKTOBEGR	Daily account limit	
OPT:EVENTEDIT	Event update active	
VER	HYD:HYDRA version	
KK1:AKTIV	PZE:PZE as SAP subsystem	
KK2:AKTIV	BDE:BDE as SAP subsystem	
KK_KD1:AKTIV	KK_KD1:AKTIV	
KK_KD2:AKTIV	KK_KD2:AKTIV	
LEVEL	Level	
HYDDI:LEVEL	Hyddi level	
LEN:AFOLG	Length of sequence	
LEN:AGNR	Length of OP number	
LEN:SPLNR	Number of splits	
LEN:BCNR	Length of barcode	
LEN:CNR	CNR length	
OPT:CNRAUTOGEN	Generate CNR autom.	
CNRFIX:WE	Fixed incoming goods batch	

Identifier	Field designation	Description
CNRFIX:PR	Fixed production batch	
OPT:CNRAUTOAB	Log input batch autom. off	
OPT:MDETLG	Multiple partit.	
ZYKLLOAD:BDE	(empty)	
BDEJMOD	Shift model with HUPE terminal	
FIR	Comment	
ABT	Department where terminal is located	
CFG:2	Configuration flag 2	
PLAUS:OFF	Plaus. checks	
PARAM1 to PARAM5	Parameter 1 to Parameter 5	
MELDART	Control of postings	
PLAUS:ADEKNR	Configuration ADE card number	
PLAUS:MDEKNR	Configuration MDE card number	
PLAUS:MNR	Machine configuration	
PLAUS:MST	Configuration of machine status	
PLAUS:AART	Configuration of opt. entry of order type	
OPT:BERFKT	Area function active	
ART	Terminal type	
DLGSTRG	Dialog control	

Example:

DLG=LIST;45|DAT=02/11/2005|ZEI=40000|USR=2101|DATEI= ./spool/trm_list.101|

6.4.11 Comments on operations

The list of comments on operations is provided by the command DLG=LIST;61 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

"DLG=LIST;61|DATEI={File name}|DAT=...|ZEI=...|USR=...|..."

Parameter: none

The definition of the file name and the respective path is case-sensitive.

The list includes the following data:

Identifier	Field designation	Description
MNR	Machine	
ANR	Order	
AUNR	Inspection order number	
AGNR	OP	
AFOLG	Sequence	
SPLNR	Split number	
PNR	Personnel number	
PNAME	Name	
PVORNAME	First name	
DAT	Date	
ZEI	Time	
BEM	Comment	

Example:

DLG=LIST;61|DAT=02/11/2005|ZEI=40000|USR=2101|DATEI= ./spool/agkom_list.101|

6.4.12 Order components (BOM)

The list of order components ("Bill of materials") is provided by the command DLG=LIST;74 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

"DLG=LIST;74|DATEI={File name}|DAT=...|ZEI=...|USR=...|...“

The definition of the file name and the respective path is case-sensitive.

Parameter: ANR order number (mandatory)

Parameter: ART Filter indicator for components (optional parameters)

B = production resource

M = material

V = equipment

W = tools

Q = measuring equipment

D = documents

C = NC programs

U = unknown resource

The list includes the following data:

Identifier	Field designation	Description
ART	Type	Component type B = production resource M = material V = equipment W = tools Q = measuring equipment D = documents C = NC programs U = unknown resource
ATK	Article	With materials: material number
BEZ	Designation	Material designation/name
BEZ:2	Designation	Material name 2
SGR:GUT	Input quantity	Input quantity to produce 1 article in primary quantity unit in the operation. With production resources and tools, this is the quantity required of a resource for this operation.
MENGE:BED	Demand	Required quantity
SGE:GUT	Unit	Unit of the input quantity
ATKBEZ	Article	Optional component name (of the material, document, etc.)
SLP	BOM item	Item number of component (BOM item)
LAGORT	Storage location	Reserved
LAGPZ	Storage compartm.	Reserved
PATH	Path	With documents: path
FILE	File name	With documents: file name
RESTYP	Type	WNR, DOC, etc. = resource type MAT = material
MOD	Extended type	Extended type: Document :DOC Material: MAT Other: NOMAT

Example:

DLG=LIST;74|DAT=02/11/2005|ZEI=40000|USR=2101|ANR=AAA2100473100200|DATEI= ./spool/mat_list.101|

6.4.13 Scrap reason list

The list of scrap reasons is provided by the command DLG=LIST;84 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

"DLG=LIST;84|DATEI={File name}|DAT=...|ZEI=...|USR=...|..."

Parameter: MOD T terminal
M machine

ART in mode M (machine) restricted to the type
(A = scrap, N = rework, P = problem quantity, G = yield)

The definition of the file name and the respective path is case-sensitive.

The list includes the following reasons and provides the following data:

Identifier	Field designation	Description
MNR	Machine	
ART	Type	
GR	Reason	
GRTXTNR	Reason text	
BEZK	Designation	
MGR	Mother reason	
GRTXT	Scrap reason	
ATK	Article	
OPT:ATKVORG	Default article	
TNR	Terminal number	with MOD=T

Example:

DLG=LIST;84|DAT=02/11/2005|ZEI=40000|USR=2101|DATEI= ./spool/gr_list.101|MOD=T|TNR=6

6.4.14 BDE order types

The list of BDE order types is provided by the command DLG=LIST;87 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

"DLG=LIST;87|DATEI={File name}|DAT=...|ZEI=...|USR=...|..."

Parameter: none

The definition of the file name and the respective path is case-sensitive.

The list includes the following order types and provides the following data:

Identifier	Field designation	Description
AUART	Order type	
CAT	Category	The category is used to classify and to combine similar order types. The category is a logical umbrella term. Possible values: "PO" production order "PJ" Project order "PM" maintenance order "KP" capacity order "GK" overhead cost order
BEZL	Designation	
OPT:RM	Option "Upload"	With this order type, the data recorded is uploaded to the PPS system or not.
OPT:PLAN	Option "Plan"	
OPT:PLANTERM	Option "Planned dates"	
ICON	Symbol	
OPT:AUBUCH		
OPT:VLIST		
OPT:ERF		
OPT:ABE		
OPT:APAN		
OPT:AANSKBAUTO		
OPT:PANSKBAUTO		
OPT:SNR		
OPT:SNRVERG		
OPT:EDITRM		

PLAUS:STAV		
PLAUS:MINMENGEV		
PLAUS:WEIGMENGE		
FKT		
PLAUS:MNR		
PLAUS.PNR		
OPT:PRIOSTRG		
OPT:APPRN		
OPT:FERTVAR		
DLGSTRG		
VISCODE		
OPT:SYS		
AKTIV		

Example:

DLG=LIST;87|DAT=02/11/2005|ZEI=40000|USR=2101|DATEI= ./spool/auart_list.101|

6.4.15 List of counters

The counter configuration is provided in a list. You use the command DLG=LIST;131 to enable the list request for the machines according to the mode (MOD). The list is filed in the directory HYDRADIR\spool\.

Structure of dialog data:

"DLG=LIST;131|DATEI={File name}|DAT=...|ZEI=...|USR=...|...“

Parameters:

- MOD=T … List of all counters of all machines assigned to the terminal. In this mode, the acronym USR=HYDRA user number must be specified.
- MOD=M … List of all counters of a machine. In this mode, the acronym MNR=machine number must be specified.

The definition of the file name and the respective path is case-sensitive.

The list includes the following data:

Identification	Field designation	Description
MNR	Machine	
CTR	counters	Unique counter identification
BEZ	Designation	
EINH	Unit	
GR	Reason	Configuration of a reason, e.g. scrap reason
TYP	Assessment	The counter quantity is booked as yield, scrap, rework or open quantity (problem quantity).
VERB	Offset against quantity account	e.g. scrap is subtracted from the total quantity.
VERB:TLG	Allocation with partitioning	If VERB.TLG=J, you use the partitioning of machine and order to book the counter pulses.
VERB:DIV	Allocation with pulse factor	If VERB.DIV=J, you use the pulse factor of machine and order to book the counter pulses.
OPT:TAKT	Posting as cycles	If OPT:TAKT=J, the pulse factors recorded are also booked as cycles. The quantity recorded is transferred to the HYDRA server via the acronym AGR:HUB.
OPT:UEB	Monitoring	Indicator that specifies if the counter is relevant for cycle monitoring.

Example:

DLG=LIST;131|MOD=T|DAT=02/11/2005|ZEI=40000|USR=2101|DATEI= ./spool/trm_list.101|

Note:

The list is sorted by machine and counter number in ascending order.

6.4.16 List providing the assignment of workplaces to MDE shop floor clients

The list provides all assignments of workplaces of this shop floor client to another shop floor client (MDE terminal or PCC) with MDE operation mode.

The list only provides assignments that are fixed in the Workplace terminal assignment.

Workplaces, which are assigned to the executing shop floor client (parameter USR) as so-called "MDE machines", are not included in the list.

The list is provided by the command DLG=LIST;140 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

DLG=LIST;**140**|DATEI={file name}|DAT=...|ZEI=...|**USR**=...|...

Parameters:

- USR=: Terminal number of the shop floor client requesting the list.
  If the list is requested without USR=, then the system provides an empty list.

The definition of the file name and the respective path is case-sensitive.

The list includes the following data:

Identification	Field designation	Description
ID	Object ID	TYP=M: Work center numbers/machine numbers
TYP	Type	Object type: • "M" for workplace/machine
TNR	MDE terminal ID	TYP=M: Terminal number of the shop floor client where the workplace is assigned as "MDE machine".

The list is available from the following program version onwards: hymwmde72.dll / so 8.1.1.143

6.5 Reading HLS data

6.5.1 Production variants

The list of production variants is provided by the command DLG=LIST;50 and filed in the directory HYDRADIR\spool\.

Structure of dialog data:

„DLG=LIST;50|DATEI={file name}|MOD={mode}|DAT=...|ZEI=...|USR=...|...“

The definition of the file name and the respective path is case-sensitive.

The list includes all production variants (mode MOD=A) or only the changed production variants (mode MOD=G). If the mode is written in lower case letters (e.g. MOD=a), then the system only selects released production variants (STA=F).

The list includes the following data:

Identifier	Field designation	Description
VERWEIS	Reference	Unique key
VER	Version	Version in a production variant
STA	Status	F=released, S=locked, L=(logically) deleted
FIR:ATK	Company of article	Company of article/material type
ATK	Article	Article
MATTYP	Material type	Material type
FIR:MNR	Company of machine	Company of machine
MNR	Machine	Machine number of the machine
MGRP	Group	Machine groupe of the machine
MANZ	Number of machines	in format N8.2
FIR:WNR	Company of the tool	Company of the tool
WNR	Tool	Tool
WFAM	Tool family	Tool family
WANZ	Number of tools	Number of tools in format N8.2
SZY	Target cycle	Target cycle
TLG	Partitioning	Partitioning in format N8.2
RUEZ	Setup time	Setup time
ABRZ	Teardown time	Teardown/retooling time
PRIO	Prio.	Priority
BEM	Comment	Comment
DATB	Start date	Valid from
DATE	End date	Valid until
DSBEZ	Data ID	Data ID
BEARBDAT	Changed on	Date of the most recent modification
BEARBZEI	Processing time	Time of the most recent modification
BEARB	Modified by	Modified by

Example:

DLG=LIST;50|DAT=10/11/2000|ZEI=40000|USR=2101|DATEI= ./spool/fertvar.101|

6.7 Annex

6.7.1 Overview of field data BDE/MDE

The following table provides an overview of the field data with structure and short example.

Identifier	Description	Structure		Example
MNR	Machine number	...|MNR={machine number}|...		...|MNR=100|...
MGRP	Machine group	...|MGRP={machine group}|...		...|MGRP=100|...
ANR	Fully defined key	...|ANR={key}|...		...|ANR=4711001001|...
AUNR	Order number	...|AUNR={order number}		...|AUNR=4711|...
AGNR	Operation number	...|AGNR={OP}|...		...|AGNR=0010|...
AFOLG	Sequence	...|AFOLG={sequence}|...		...|AFOLG=01|...
SPLNR	Split number	...|SPLNR={split number}|...		...|SPLNR=01|...
AUART	Order type	...|AUART={type}|...		...|AUART=0|...
ATK	Article number	...|ATK={article number}|...		
EGR:*	Recorded value Manually recorded value	...|EGR:{type}={value}|... Types of recorded values: RPA01, RPA02,..., RPA12 DAUER, PDAUER HUB, GUT, AUS, LEN, GEW The value recorded is added to the relevant account in the DB.		..|EGR:GUT=10. |. ...|EGR:AUS=1 |...
EGR:*	New quantity types as of MW 2.0	GUTB	Yield (base quantity unit)	
		GUTP	Yield (primary quantity unit)	
		GUTS	Yield (secondary quantity unit)	
		GUTT	Yield (tertiary quantity unit)	
		AUSB	Scrap (base quantity unit)	
		AUSP	Scrap (primary quantity unit)	
		AUSS	Scrap (secondary quantity unit)	
		AUST	Scrap (tertiary quantity unit)	

Identifier	Description	Structure		Example
		NCHB	Rework quantity (base quantity unit)	
		NCHP	Rework quantity (primary quantity unit)	
		NCHS	Rework quantity (secondary quantity unit)	
		NCHT	Rework quantity (tertiary quantity unit)	
		PRBB	Problem quantity (base quantity unit)	
		PRBP	Problem quantity (primary quantity unit)	
		PRBS	Problem quantity (secondary quantity unit)	
		PRBT	Problem quantity (tertiary quantity unit)	
AGE:*	Units of automatically recorded values	B	Base quantity unit	
		P	Primary quantity unit	
		S	Secondary quantity unit	
		T	Tertiary quantity unit	
SGE:*	Units of target values	B	Base quantity unit	
		P	Primary quantity unit	
		S	Secondary quantity unit	
		T	Tertiary quantity unit	
EGE:*	Unit of recorded values	...|EGE:{type}={unit}|... Types see EGR		EGE:GUT=ST
EGG:*	Reasons of recorded values	...|EGG:{type}={reason}|... Types see EGR		EGG:AUS = 1
SGR:*	Target value	...|SGR:{type}={value}|... SGR:GUT =target quantity order/OP Types see EGR **New types 7.2: AUS*, GUT* (see**		...|SGR:GUT=1126|

Identifier	Description	Structure	Example
		EGR)	
TLG	Partitioning	...|TLG={partitioning}|...	...|TLG=1|... ...|TLG=0.25|...
SZY	Target cycle	...|SZY={target cycle};{unit}|... First, only unit "s"! Later conversions of the values into seconds.	...|SZY=36000;s|... ...|SZY=10;h|...
MST	Machine status	...|MST={machine status }|...	..|MST=1 |..
PNR	Personnel number	...|PNR={personnel number}|...	...|PNR=999999|...
KNR	Staff badge no.:	...|KNR={badge number}|...	...|KNR=9999|...
BPOS	Operator position/function	...|BPOS={operator function}|...	...|BPOS=A1|...
PMK	Memorize persons	...|PMK={Y/N}|...	...|PMK=J|...
CNR	Batch number	...|CNR={batch number}	...|CNR=998877|...
KST	Cost center	Cost center	
BZW	Posting required	Optional parameter. If the option is set, several validation checks are disabled. If option is not sent: = N [...|BZW={posting required}|...]	...|BZW=J|...
		Field data for BATCH	
LHW	Info on batch	Info on batch (C20)	|LHW={batch info}|...
ZLO	Target	Target	...|ZLO=121234|...
TPE	Transport unit	Transport unit (C10)	...|TPE=1234|...
SLP	BOM item	BOM item (C6)	...|SLP=AA34|...
ATTR:*	Additional attributes of a batch	...|ATTR:{no}]={attribute recorded}|.. no = 1 ..11 field types; 1 - 4 : Integer 4 – 6 : Double 7 : Char 4 8, 9 : Char 10 10, 11: Char 20	
STN	Station number	...|STN ={ station number }|...	...|STN=1|..

Identifier	Description	Structure	Example
LOSANZ	Number of parallel output batches	...|LOSANZ ={ number of batches }|...	...|LOSANZ=5|...

6.7.2 Optional field data for the premium and incentive wages LLE

The following data fields can optionally be used for the premium and incentive wages LLE in the BDE input dialogs.

You require the licenses to calculate single or group incentives.

The listed fields are optional. In most cases, these fields are not explicitly filled during data collection, but the fields are usually transferred automatically from the operation or the LLE assignment of premium groups to the BDE postings or the fields are required in special cases only.

Identifier	Type / max. field length	Description
LPGRP=	C8	Optional only with order logon: In case of Incentive Wage LLE with group bonus: The premium group can optionally only be recorded with machines having the incentive wage indicator "G"=group piecework. Here, the premium group, which is assigned to the machine in the Incentive Wage LLE, is overwritten in the U/E and B records.
LART=	C4	In case of staff logon with LLE: Optional recording of the wage type for the BDE staff postings (B record)
EGR:TE=	N8	Optional entry of the single piece specification for the person t_e in seconds per 1000 pieces.
EGR:TR=	N8	Optional entry of the setup specification for the person t_r in seconds.
EGR:TEB=	N8	Optional entry of the single piece specification for the production resource t_{eb} in seconds per 1000 pieces.
EGR:TRB=	N8	Optional entry of the setup specification for the production resource t_{rb} in seconds.
BPOS=	C10	Optional entry of an operator position/function with staff logon
LPKZ=	C10	Optional entry of a premium indicator with staff logon

6.7.3 Tips and tricks

6.7.3.1 Transfer of machine data and shift change information

If you use the PDM dialogs to integrate the machine data collection including shift change function, then you must assign the workplace/machine, for which you want to send PDM dialogs, to a terminal.

Without this assignment, the system cannot identify an active shift change via PDM dialog for this machine. Result: The system performs proportionate allocations in the order shift log that might have unexpected results.

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7.1 Note on the Descriptions of the Basic Dialogs

All mandatory fields that must be specified have the addition PK (primary key). All other fields are optional and are processed if they are transferred.

7.2 BDE Log Records

7.2.1 Create log record (DLG=ADEPRO.INSERT, COPY)

You can use the BAPI calls described in this section to create or copy log records.

Tables

Table	Key field	Description
ade_protokoll	Reference ADEPRO.VERWEIS	Reference of the posting record

BAPI call

ID	Content / {type}	Description
DLG	ADEPRO.INSERT	Create posting
	ADEPRO.COPY	Copy posting
ADEPRO.VERWEIS:Q	{N8}	PK only with ADEPRO.COPY Refers to the posting that you want to copy.
ADEPRO.SART	{C1}	PK Log record of the posting U = Order interruption E = End of order B = Staff posting H = Batch record
ADEPRO.ANR	{C40}	PK Combined order/OP number of the posting
Other option		
ADEPRO.AUNR	{C40}	PK Order number of the posting
ADEPRO.AGNR	{C40}	PK Operation number of the posting
ADEPRO.AFOLG	{C40}	PK Order sequence number of the posting (only if configured in HYDRA)

ID	Content / {type}	Description
ADEPRO.UAGNR	{C40}	PK Sub operation number of the posting (only if configured in HYDRA)
ADEPRO.SPLNR	{C40}	PK Split number of the posting (only if configured in HYDRA)
ADEPRO.DATB	{MM/DD/YYYY}	PK Start date of the posting
ADEPRO.DATE	{MM/DD/YYYY}	PK End date of the posting
…	…	For further fields, that are not MANDATORY, refer to section 7.2.4 *List of fields (acronyms)*.

Return

ID	Content / {type}	Description
VERWEIS	{N8}	Returned reference of the created posting. Example: **RET_DATA:** RET=0|KT=|LT=|DATA=ADEPRO|**VERWEIS=15673259**|RET=0|KT=|LT=|

Plausibility checks

Error codes	Description
10	The operation (ADEPRO.ANR) must be available in the backlog of orders (online dataset).
90	The workplace (ADEPRO.MNR) must be available in the workplace configuration.
1803	Check is performed if only "BEARB" is passed and BEARB does not equal "HYDRA". No responsibility area authorization for the workplace is available (ADEPRO.MNR).
1030	If ADEPRO.SART =B The person (ADEPRO.PNR) must be part of the staff.
1641	If ADEPRO.SART =H The operation (ADEPRO.ANR) must be subject to batch management.
1900	If the license LLE-GRB is active and the premium group is specified

Error codes	Description
	(ADEPRO.LEISTGRP). The premium group (ADEPRO.LEISTGRP) must be defined and valid.
1958	ADEPRO.SART =U or ADEPRO.SART =E Within the selected period of time an overlapping log record is already available for the operation and workplace, i.e. a U or E record, which reaches into or lies completely within the period of time of the new record to be created, is already available when trying to insert a log record of record type U or E.
1952	Start date (ADEPRO.DATB / ADEPRO.ZEIB) must not be greater than the end date (ADEPRO.DATE / ADEPRO.ZEIE).
806	If ADEPRO.SART =T or H The new record to be created must be within an already existing U/E record. You can deactivate this check using the parameter ADEPRO.PLAUS:AGUN=N.
806	If ADEPRO.SART = B and ADEPRO.PLAUS:AGUNB=J The center of time of the record to be created must be within an already existing log record of record type U or E.
814	If ADEPRO.SART = H The log record of record type H may only include values for one of the two quantity types: • either yield (ADEPRO.GUT/GUTP/GUTS/GUTT/GUTB) • or scrap (ADEPRO.AUS/AUSP/AUSS/AUST/AUSB)
815	If ADEPRO.SART = H A log record of record type H exists for the specified period of time. You can deactivate this check using the parameter ADEPRO.PLAUS:MULTICNR=J.
1661	You must specify the parameter with the ID ADEPRO.SART.
1661	You must specify the parameter with the ID ADEPRO.ANR (or the IDs AUNR,AGNR,AFOLG,UAGNR,SPLNR).
1661	You must specify the parameter with the IDs ADEPRO.DATB and ADEPRO.DATE.

7.2.2 Edit log record (DLG=ADEPRO.UPDATE, DELETE, LOCK, UNLOCK, SELECT)

You can use the BAPI calls described in this section to edit log records.

Tables

Table	Key field	Description
Ade_protokoll	Reference ADEPRO.VERWEIS	Reference of the log record

BAPI call

ID	Content / {type}	Description
	ADEPRO.UPDATE	Change log record
	ADEPRO.DELETE	Delete log record
	ADEPRO.LOCK	Lock log record for editing MUST be performed before ADEPRO.UPDATE
	ADEPRO.UNLOCK	Unlock log record after editing MUST be performed after ADEPRO.UPDATE
	ADEPRO.SELECT	Select log record
ADEPRO.VERWEIS	{N8}	PK Reference of the log record
ADEPRO.SART	{C1}	PK Record type U = Order interruption E = End of order B = Staff posting H = Batch record
…	…	ADEPRO.UPDATE: For further fields, that are not MANDATORY, refer to section 7.2.4 *List of fields (acronyms)*

Return

ID	Content / {type}	Description
ADEPRO.VERWEIS	{N8}	Return of reference of the changed log record. Note: This reference might not be equal to the reference of ADEPRO.UPDATE if the log record has already been uploaded and a cancel posting must be created.

Plausibility checks

Error codes	Description

Error codes	Description
10	The operation (ADEPRO.ANR) must be available in the backlog of orders.
90	The workplace (ADEPRO.MNR) must be available in the workplace configuration.
1803	Check is performed if only "BEARB" is passed and BEARB does not equal "HYDRA". No responsibility area authorization for the workplace is available (ADEPRO.MNR).
1030	Only if ADEPRO.SART=B: The person (ADEPRO.PNR) must be part of the staff.
1641	Only if ADEPRO.SART=H: The operation (ADEPRO.ANR) must be subject to batch management.
1958	Within the selected period of time, an overlapping log record is already available for the operation and workplace, i.e. a U or E record, which reaches into or lies completely within the period of time of the new record to be created, is already available when trying to insert a log record of record type U or E.
1952	Start date (ADEPRO.DATB / ADEPRO.ZEIB) must not be greater than the end date (ADEPRO.DATE / ADEPRO.ZEIE).
1900	If the license LLE-GRB is active and the premium group is specified (ADEPRO.LEISTGRP): The premium group (ADEPRO.LEISTGRP) must be defined and valid.
806	If ADEPRO.SART = T The time of the log record must be within an already existing log record of record type U or E.
806	If ADEPRO.SART=H The time of the log record must be within an already existing log record of record type U or E.
1956	UPDATE,DELETE If ADEPRO.SART=T The upload of a part quantity (partial confirmation) cannot be changed because the respective log record of record type U or E has not been generated.
814	If ADEPRO.SART =H The log record of record type H may only include values for one of the two quantity types: • either yield (ADEPRO.GUT/GUTP/GUTS/GUTT/GUTB) • or scrap (ADEPRO.AUS/AUSP/AUSS/AUST/AUSB).
815	A log record of record type H exists for the specified period of time.
1954	UPDATE.LOCK

Error codes	Description
	The cancellation log record (resp. ADEPRO.VERWEIS) cannot be changed.
1955	UPDATE.LOCK The original log record (resp. ADEPRO.VERWEIS) cannot be changed.
1957	UPDATE The record type cannot be changed (ADEPRO.SART): • the record type of a log record of record type B cannot be changed into the record type U or E. • the record type of a log record of record type U or E cannot be changed into the record type B. Note: It is possible to change the record type of a log record of record type U into the record type E and vice versa.
101	UPDATE,LOCK,UNLOCK The log record (if the parameter ADEPRO.VERWEIS is specified) must be created in the database.
1666	The log record is currently locked by another user. (UPDATE, DELETE,LOCK).
1661	UPDATE.LOCK,UNLOCK You must specify the parameter with the ID ADEPRO.VERWEIS.
1661	UPDATE.LOCK,UNLOCK You must specify the parameter with the ID ADEPRO.SART.
1661	UPDATE.LOCK,UNLOCK You must specify the parameter with the ID ADEPRO.ANR (or the IDs AUNR,AGNR,AFOLG,UAGNR,SPLNR).

Plausibility checks in the MOC

Error codes	Description
	Order type configuration: If upload cannot be changed. → Event is locked because the posting has already been uploaded.
	Order type configuration: If editing authorization is enabled and the user does not have the specified function authorization. → no authorization to change something with this order type
	Setup setting: If "Maintenance of events" is active: → BDE maintenance based on events is active. Automatically generated postings cannot be edited!
	Setup setting: If "Maintenance of events" is active:

Error codes	Description
	→ BDE maintenance based on events is active. Automatically generated postings cannot be deleted!

7.2.3 Sign log record (DLG=ADEPRO.SIGN)

You can use this BAPI call to sign a posting.

Table	Key field	Description
Ade_protokoll	Reference ADEPRO.VERWEIS	Reference of the posting

BAPI call

ID	Content / {type}	Description
DLG	ADEPRO. SIGN	PK Sign posting
ADEPRO.VERWEIS	{N8}	PK Reference
BEARB	{C10}	PK Signed by

Note: The database fields sign_dat, sign_zeit are populated using the timestamp passed (DAT/ZEI).

Return

ID	Contents	Description
—	—	—

Plausibility checks

Error codes	Description
1661	You must specify the parameter with the ID ADEPRO.VERWEIS.
101	The posting record (if the parameter ADEPRO.VERWEIS is specified) must be available in the database.

Notes on the processing

- You can insert U records even if the OP has already been finished.

- If you insert an E record, the operation status is set to finished.
- If you create or change a B record, the person-related RPAs of the respective U/E record and the order status are revised.
- If you change an H record (batch posting), the quantities of the respective U/E record and the order status are revised.
- If you change a T record (upload of part quantity), the quantities of the respective U/E record and the order status are revised.

7.2.4 List of fields (acronyms) for the ADEPRO dialog

ID	Description	DB type	Length
ADEPRO.ANR	MES order number	char	40
ADEPRO.SART	Record type: U: Order interruption; Upload of part quantity E: Order end B: Staff logoff T: Uploads of part quantities for order H: Batch logoff	char	3
ADEPRO.MST	Status text number Machine status that was active or was set when the OP was interrupted.	integer	7
ADEPRO.PANZ	reserved	smallint	7
ADEPRO.PNR	Person who triggered the log record	char	10
ADEPRO.PGRP	reserved	char	20
ADEPRO.ZEIB	Point in time of logon (time)	integer	7
ADEPRO.DATB	Point in time of logon (date)	sqldate	7
ADEPRO.ZEIE	Point in time of logoff (time)	integer	7
ADEPRO.DATE	Point in time of logoff (date)	sqldate	7
ADEPRO.EGR:DAUER	Total time of the posting, compared to the shift calendar	integer	7
ADEPRO.TNR	Terminal that made the posting	smallint	7
ADEPRO.MNR	HYDRA workplace making the posting	char	20
ADEPRO.TLG	Valid partitioning of the posting	decimal	7

ID	Description	DB type	Length
ADEPRO.SKNR	Shift number; assignment of the shift that logs off	smallint	7
ADEPRO.SKZEIB	Beginning of shift of the assigned shift	integer	7
ADEPRO.SKZEIE	End of shift of the assigned shift	integer	7
ADEPRO.SKPAUSE	Shift break in seconds	smallint	7
ADEPRO.EGR:BMK01	Time recorded, splitted onto RP accounts Activity posted in RPA 01	integer	7
ADEPRO.EGR:BMK02	Activity posted in RPA 02	integer	7
ADEPRO.EGR:BMK03	Activity posted in RPA 03	integer	7
ADEPRO.EGR:BMK04	Activity posted in RPA 04	integer	7
ADEPRO.EGR:BMK05	Activity posted in RPA 05	integer	7
ADEPRO.EGR:BMK06	Activity posted in RPA 06	integer	7
ADEPRO.EGR:BMK07	Activity posted in RPA 07	integer	7
ADEPRO.EGR:BMK08	Activity posted in RPA 08	integer	7
ADEPRO.EGR:BMK09	Activity posted in RPA 09	integer	7
ADEPRO.EGR:BMK10	Activity posted in RPA 10	integer	7
ADEPRO.EGR:BMK11	Activity posted in RPA 11	integer	7
ADEPRO.EGR:BMK12	Activity posted in RPA 12	integer	7
ADEPRO.EGR:PDAUER	Posted staff time	integer	7
ADEPRO.KST	Cost center of the workplace	char	10
ADEPRO.VERWEIS	Unique ID of the log record, important for the editing of log records	sqlserial	7
ADEPRO.RCK	J/N specifies if the log record has already been uploaded to PPS system	char	1
ADEPRO.BEARB	Initials of the user who manually makes changes	char	10
ADEPRO.BEARBDAT	Date of the manual change	sqldate	7
ADEPRO.SKDAT	Date of beginning of shift. In case of ADE	sqldate	7

ID	Description	DB type	Length
	records that extend over several shifts, the date is identified using the end posting.		
ADEPRO.RCK:01	customer-specific	char	1
ADEPRO.RCK:02	customer-specific	char	1
ADEPRO.ERFART	Is assigned if the log record is manually created using the function "Maintenance of postings" on the console: M : created manually on the console P : created automatically by the PZE. You must not change the data record in the maintenance of postings.	char	1
ADEPRO.ABREDAT	Evaluation date PZE	sqldate	7
ADEPRO.EGR:GUTB	Yield (base quantity unit)	decimal	7
ADEPRO.EGR:GUTP	Yield (primary quantity unit)	decimal	7
ADEPRO.EGR:GUTS	Yield (secondary quantity unit)	decimal	7
ADEPRO.EGR:GUTT	Yield (tertiary quantity unit)	decimal	7
ADEPRO.EGR:AUSB	Scrap (base quantity unit)	decimal	7
ADEPRO.EGR:AUSP	Scrap (primary quantity unit)	decimal	7
ADEPRO.EGR:AUSS	Scrap (secondary quantity unit)	decimal	7
ADEPRO.EGR:AUST	Scrap (tertiary quantity unit)	decimal	7
ADEPRO.EGR:NCHB	Rework quantity (base quantity unit)	decimal	7
ADEPRO.EGR:NCHP	Rework quantity (primary quantity unit)	decimal	7
ADEPRO.EGR:NCHS	Rework quantity (secondary quantity unit)	decimal	7
ADEPRO.EGR:NCHT	Rework quantity (tertiary quantity unit)	decimal	7
ADEPRO.EGR:PRBB	Problem quantity (base quantity unit)	decimal	7
ADEPRO.EGR:PRBP	Problem quantity (primary quantity unit)	decimal	7
ADEPRO.EGR:PRBS	Problem quantity (secondary quantity unit)	decimal	7
ADEPRO.EGR:PRBT	Problem quantity (tertiary quantity unit)	decimal	7
ADEPRO.EGE:GUTB	Base quantity unit	char	3
ADEPRO.EGE:GUTP	Primary quantity unit	char	3

ID	Description	DB type	Lengt h
ADEPRO.EGE:GUTS	Secondary quantity unit	char	3
ADEPRO.EGE:GUTT	Tertiary quantity unit	char	3
ADEPRO.GGR:HUB	Total of clocks recorded	decimal	7
ADEPRO.EGR:HUBG	Recorded clocks - yield	decimal	7
ADEPRO.EGG:GUTB	Deviation reason Reference: ADE_GRUND_ZUORD	integer	7
ADEPRO.EGT:GUTB	Reason text number for yield Reference: ADE_GRUND_TEXTE	integer	7
ADEPRO.EGG:AUSB	Scrap reason Reference: ADE_GRUND_ZUORD	integer	7
ADEPRO.EGT:AUSB	Reason text number for scrap Reference: ADE_GRUND_TEXTE	integer	7
ADEPRO.EGG:NCHB	Rework reason Reference: ADE_GRUND_ZUORD	integer	7
ADEPRO.EGT:NCHB	Reason text number for rework quantity Reference: ADE_GRUND_TEXTE	integer	7
ADEPRO.EGG:PRBB	Problem reason Reference: ADE_GRUND_ZUORD	integer	7
ADEPRO.EGT:PRBB	Reason text number for problem quantity Reference: ADE_GRUND_TEXTE	integer	7
ADEPRO.EGR:LST01	Posted free activity 1	decimal	7
ADEPRO.EGR:LST02	Posted free activity 2	decimal	7
ADEPRO.EGR:LST03	Posted free activity 3	decimal	7
ADEPRO.EGR:LST04	Posted free activity 4	decimal	7
ADEPRO.EGR:LST05	Posted free activity 5	decimal	7
ADEPRO.EGR:LST06	Posted free activity 6	decimal	7
ADEPRO.EGR:LST07	Posted free activity 7	decimal	7
ADEPRO.EGR:LST08	Posted free activity 8	decimal	7
ADEPRO.EGR:LST09	Posted free activity 9	decimal	7
ADEPRO.EGR:LST10	Posted free activity 10	decimal	7

ID	Description	DB type	Lengt h
ADEPRO.RGR:LST01	Current remaining activity 1	decimal	7
ADEPRO.RGR:LST02	Current remaining activity 2	decimal	7
ADEPRO.RGR:LST03	Current remaining activity 3	decimal	7
ADEPRO.RGR:LST04	Current remaining activity 4	decimal	7
ADEPRO.RGR:LST05	Current remaining activity 5	decimal	7
ADEPRO.RGR:LST06	Current remaining activity 6	decimal	7
ADEPRO.RGR:LST07	Current remaining activity 7	decimal	7
ADEPRO.RGR:LST08	Current remaining activity 8	decimal	7
ADEPRO.RGR:LST09	Current remaining activity 9	decimal	7
ADEPRO.RGR:LST10	Current remaining activity 10	decimal	7
ADEPRO:EGE:LST01	Unit of recorded activity 1	char	3
ADEPRO:EGE:LST02	Unit of recorded activity 2	char	3
ADEPRO:EGE:LST03	Unit of recorded activity 3	char	3
ADEPRO:EGE:LST04	Unit of recorded activity 4	char	3
ADEPRO:EGE:LST05	Unit of recorded activity 5	char	3
ADEPRO:EGE:LST06	Unit of recorded activity 6	char	3
ADEPRO:EGE:LST07	Unit of recorded activity 7	char	3
ADEPRO:EGE:LST08	Unit of recorded activity 8	char	3
ADEPRO:EGE:LST09	Unit of recorded activity 9	char	3
ADEPRO:EGE:LST10	Unit of recorded activity 10	char	3
ADEPRO.FU:1	User-specific field	sqldate	7
ADEPRO.FU:2	User-specific field	sqldate	7
ADEPRO.FU:3	User-specific field	sqldate	7
ADEPRO.FU:4	User-specific field	sqldate	7
ADEPRO.FU:5	User-specific field	sqldate	7

ID	Description	DB type	Length
ADEPRO.FU:6	User-specific field	sqldate	7
ADEPRO.FU:7	User-specific field	integer	7
ADEPRO.FU:8	User-specific field	integer	7
ADEPRO.FU:9	User-specific field	integer	7
ADEPRO.FU:10	User-specific field	integer	7
ADEPRO.FU:11	User-specific field	integer	7
ADEPRO.FU:12	User-specific field	integer	7
ADEPRO.FU:13	User-specific field	integer	7
ADEPRO.FU:14	User-specific field	integer	7
ADEPRO.FU:15	User-specific field	integer	7
ADEPRO.FU:16	User-specific field	integer	7
ADEPRO.FU:17	User-specific field	integer	7
ADEPRO.FU:18	User-specific field	integer	7
ADEPRO.FU:19	User-specific field	integer	7
ADEPRO.FU:20	User-specific field	integer	7
ADEPRO.FU:21	User-specific field	integer	7
ADEPRO.FU:22	User-specific field	integer	7
ADEPRO.FU:23	User-specific field	decimal	7
ADEPRO.FU:24	User-specific field	decimal	7
ADEPRO.FU:25	User-specific field	decimal	7
ADEPRO.FU:26	User-specific field	decimal	7
ADEPRO.FU:27	User-specific field	decimal	7
ADEPRO.FU:28	User-specific field	decimal	7
ADEPRO.FU:29	User-specific field	char	1
ADEPRO.FU:30	User-specific field	char	1
ADEPRO.FU:31	User-specific field	char	1

ID	Description	DB type	Length
ADEPRO.FU:32	User-specific field	char	1
ADEPRO.FU:33	User-specific field	char	1
ADEPRO.FU:34	User-specific field	char	1
ADEPRO.FU:35	User-specific field	char	1
ADEPRO.FU:36	User-specific field	char	1
ADEPRO.FU:37	User-specific field	char	1
ADEPRO.FU:38	User-specific field	char	1
ADEPRO.FU:39	User-specific field	char	1
ADEPRO.FU:40	User-specific field	char	1
ADEPRO.FU:41	User-specific field	char	1
ADEPRO.FU:42	User-specific field	char	1
ADEPRO.FU:43	User-specific field	char	1
ADEPRO.FU:44	User-specific field	char	1
ADEPRO.FU:45	User-specific field	char	10
ADEPRO.FU:46	User-specific field	char	10
ADEPRO.FU:47	User-specific field	char	10
ADEPRO.FU:48	User-specific field	char	10
ADEPRO.FU:49	User-specific field	char	10
ADEPRO.FU:50	User-specific field	char	10
ADEPRO.FU:51	User-specific field	char	20
ADEPRO.FU:52	User-specific field	char	20
ADEPRO.FU:53	User-specific field	char	20
ADEPRO.FU:54	User-specific field	char	20
ADEPRO.FU:55	User-specific field	char	20
ADEPRO.FU:56	User-specific field	char	20

ID	Description	DB type	Length
ADEPRO.FU:57	User-specific field	char	20
ADEPRO.FU:58	User-specific field	char	20
ADEPRO.FU:59	User-specific field	char	20
ADEPRO.FU:60	User-specific field	char	20
ADEPRO.FU:61	User-specific field	char	20
ADEPRO.FU:62	User-specific field	char	20
ADEPRO.FU:63	User-specific field	char	20
ADEPRO.FU:64	User-specific field	char	20
ADEPRO.FU:65	User-specific field	char	40
ADEPRO.FU:66	User-specific field	char	40
ADEPRO.EGR:PBMK01	Personnel activity posted in RPA 1	integer	7
ADEPRO.EGR:PBMK02	Personnel activity posted in RPA 2	integer	7
ADEPRO.EGR:PBMK03	Personnel activity posted in RPA 3	integer	7
ADEPRO.EGR:PBMK04	Personnel activity posted in RPA 4	integer	7
ADEPRO.EGR:PBMK05	Personnel activity posted in RPA 5	integer	7
ADEPRO.EGR:PBMK06	Personnel activity posted in RPA 6	integer	7
ADEPRO.EGR:PBMK07	Personnel activity posted in RPA 7	integer	7
ADEPRO.EGR:PBMK08	Personnel activity posted in RPA 8	integer	7
ADEPRO.EGR:PBMK09	Personnel activity posted in RPA 9	integer	7
ADEPRO.EGR:PBMK10	Personnel activity posted in RPA 10	integer	7
ADEPRO.EGR:PBMK11	Personnel activity posted in RPA 11	integer	7
ADEPRO.EGR:PBMK12	Personnel activity posted in RPA 12	integer	7
ADEPRO.RGR:PBMK01	Current remaining personnel activity RPA 1	integer	7
ADEPRO.RGR:PBMK02	Current remaining personnel activity RPA 2	integer	7
ADEPRO.RGR:PBMK03	Current remaining personnel activity RPA 3	integer	7
ADEPRO.RGR:PBMK04	Current remaining personnel activity RPA 4	integer	7

ID	Description	DB type	Length
ADEPRO.RGR:PBMK05	Current remaining personnel activity RPA 5	integer	7
ADEPRO.RGR:PBMK06	Current remaining personnel activity RPA 6	integer	7
ADEPRO.RGR:PBMK07	Current remaining personnel activity RPA 7	integer	7
ADEPRO.RGR:PBMK08	Current remaining personnel activity RPA 8	integer	7
ADEPRO.RGR:PBMK09	Current remaining personnel activity RPA 9	integer	7
ADEPRO.RGR:PBMK10	Current remaining personnel activity RPA 10	integer	7
ADEPRO.RGR:PBMK11	Current remaining personnel activity RPA 11	integer	7
ADEPRO.RGR:PBMK12	Current remaining personnel activity RPA 12	integer	7
ADEPRO.RGR:PDAUER	Current remaining labor time	integer	7
ADEPRO.RGR:BMK01	Current remaining activity RPA 1	integer	7
ADEPRO.RGR:BMK02	Current remaining activity RPA 2	integer	7
ADEPRO.RGR:BMK03	Current remaining activity RPA 3	integer	7
ADEPRO.RGR:BMK04	Current remaining activity RPA 4	integer	7
ADEPRO.RGR:BMK05	Current remaining activity RPA 5	integer	7
ADEPRO.RGR:BMK06	Current remaining activity RPA 6	integer	7
ADEPRO.RGR:BMK07	Current remaining activity RPA 7	integer	7
ADEPRO.RGR:BMK08	Current remaining activity RPA 8	integer	7
ADEPRO.RGR:BMK09	Current remaining activity RPA 9	integer	7
ADEPRO.RGR:BMK10	Current remaining activity RPA 10	integer	7
ADEPRO.RGR:BMK11	Current remaining activity RPA 11	integer	7
ADEPRO.RGR:BMK12	Current remaining activity RPA 12	integer	7
ADEPRO.RGR:DAUER	Remaining run time	integer	7
ADEPRO.ERFART	"-": original data record of data collection O: original data record if it is edited E: manually created data record (edited) S: cancellation for PPS (posting is deleted because of upload) X: posting is locked; no booking is made	char	3

ID	Description	DB type	Length
	because of plausibility error.		
ADEPRO.SKART	Shift ID of the assigned shift	char	1
ADEPRO.SKMOD	Shift model used (day type)	smallint	7
ADEPRO.LART	reserved	char	4
ADEPRO.EGR:TE	Target time per unit *te* from backlog of orders	decimal	7
ADEPRO.EGR:TR	Target setup time *tr* from backlog of orders	decimal	7
ADEPRO.EGR:TEB	Target machine time per unit *teb* from backlog of orders	decimal	7
ADEPRO.EGR:TRB	Target specified setup time *trb* from backlog of orders	decimal	7
ADEPRO.BPOS	Operator position	char	10
ADEPRO.LPKZ	Wage/premium indicator	char	10
ADEPRO.LEISTGRP	Premium group •	char	10
ADEPRO.CNR	Batch number	char	20

7.3 Configuration of Data Collection Reason Texts

7.3.1.1 Edit reason texts (DLG=GRTXT.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

You use these BAPI calls to edit reason texts.

Tables

Table	Key field	Description
ade_grund_texte	grundtext_nr GRTXT.GRTXTNR	Reason text number (PK)

BAPI call

ID	Content / {type}	Description
DLG	GRTXT.INSERT	Create reason text

ID	Content / {type}	Description
	GRTXT.UPDATE	Change reason text
	GRTXT.DELETE	Delete reason text
	GRTXT.COPY	Copy reason text
	GRTXT.LOCK	Lock reason text for editing
	GRTXT.UNLOCK	Unlock reason text after editing
	GRTXT.NEW	Read specification for new reason text
	GRTXT.SELECT	Select reason text
GRTXT.GRTXTNR	{N4}	PK reason text number
GRTXT.GRTXTNR:Z	{N4}	Target - reason text number
…	…	For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Return

ID	Content / {type}	Description
GRTXTNR	{N4}	Current data record

Plausibility checks

Error codes	Description
1661	A value is missing that is required for processing.
1666	The reason text is currently edited by another user.

7.3.1.2 List of reason texts (DLG=GRTXT.LIST)

The BAPI call returns all defined reason texts.

Tables

Table	Key field	Description
ade_grund_texte	grundtext_nr GRTXT.GRTXTNR	Reason text number (PK)

BAPI call

ID	Contents	Description
DLG	GRTXT.LIST	List of reason texts
DATEI	{C256}	Specification of the file name for the list

7.3.2 Reasons

7.3.2.1 Edit reasons (DLG=GR.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

You use these BAPI calls to edit reasons.

Tables

Table	Key field	Description
ade_grund_zuord	masch_nr art grund GR.MNR GR.ART GR.GR	The combination must be unique.

BAPI call

ID	Content / {type}	Description
DLG	GR.INSERT	Create reason
	GR.UPDATE	Change reason
	GR.DELETE	Delete reason
	GR.COPY	Copy reason
	GR.LOCK	Lock reason for editing
	GR.UNLOCK	Unlock reason after editing
	GR.NEW	Read specification for new reason
	GR.SELECT	Select reason
GR.MNR	C20	PK machine number

ID	Content / {type}	Description
GR.ART	C1	PK type A = scrap N = rework P = problem quantity G = yield L = batch reason
GR.GR	N8	PK reason number
GR.MNR:Z	C20	Target machine number
GR.ART:Z	C1	Target type
GR.GR:Z	N8	Target reason number
MOD	C1	Copy mode E- Copy currently selected reason G- Copy all reasons F- Copy missing reasons M- Copy a reason for all machines
…	…	For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Return

ID	Content / {type}	Description
MNRMNR	C20	Current data record
ART	C1	Current data record
GR	N8	Current data record

Plausibility checks

Error codes	Description
1661	A value is missing that is required for processing.
1666	The reason text is currently edited by another user.
712	Processing mode is invalid.
1401	Invalid deviation reason

Error codes	Description
1803	No responsibility area authorization. If you create workplace/machine-related reasons, the system checks the responsibility area of the workplace. To override this check, you can set the parameter VABCHECK=N.
3273	Not possible - A reason can either be assigned to SYSTEM or to a random number of machines.
709	Copy: The target machine has not been specified.
710	Copy: The target status has not been specified.

7.3.2.2 List of reasons (DLG=GR.LIST)

The BAPI call returns all defined reasons.

Tables

Table	Key field	Description
ade_grund_zuord	masch_nr art grund GR.MNR GR.ART GR.GR	The combination must be unique.

BAPI call

ID	Contents	Description
DLG	GR.LIST	List of reason texts
DATEI	{C256}	Specification of the file name for the list

7.4 Order Management and Order Sequencing

7.4.1 Lock order/operation for editing (ANR.LOCK)

You lock the operation to prevent the operation from being edited by another user.

Tables

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number if ANR.ATYP = OP

Table	Key field	Description
auftrags_bestand	ANR.AUNR	PK Order number if ANR.ATYP = AU (order)

BAPI call

ID	Content / {type}	Description
DLG	ANR.LOCK	Plan operation
ANR.ATYP	{C4}	Order type {AU} Order type {AG}
ANR.ANR	{C40}	Combined order/OP number if ANR.ATYP=OP
ANR.AUNR	{C40}	Order number if ANR.ATYP=AU (order)
BEARB	{C10}	Modified by
ANR.TABLE	{C1}	A – Backlog of orders (default value)

Return

ID	Content / {type}	Description
*		All attributes of the order or the operation

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked BEARB

7.4.2 Unlock order/operation for editing (ANR.UNLOCK)

Unlock an operation locked by a user.

Tables

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number if ANR.ATYP = OP

Table	Key field	Description
auftrags_bestand	ANR.AUNR	PK Order number if ANR.ATYP = AU (order)

BAPI call

ID	Content / {type}	Description
DLG	ANR.LOCK	Plan operation
ANR.ATYP	{C4}	Order type {AU} Order type {AG}
ANR.ANR	{C40}	Combined order/OP number if ANR.ATYP=OP
ANR.AUNR	{C40}	Order number if ANR.ATYP=AU (order)
BEARB	{C10}	Modified by
ANR.TABLE	{C1}	A – Backlog of orders (default value)

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified

7.4.3 Plan operation (ANR.EINPLANEN)

Plan an operation for a workplace.

Tables

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number

BAPI call

ID	Content / {type}	Description
DLG	ANR.EINPLANEN	Plan operation
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.ANR	{C40}	Combined order/OP number
ANR.MNR	{C8}	Workplace
BEARB	{C10}	Modified by
ANR.TABLE	{C1}	A – Backlog of orders (default value)

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
90	Machine is not available ANR.MNR is not specified
1658	No cost center authorization for machine
1666	Object is locked BEARB

7.4.4 Deallocate operation (ANR.AUSPLANEN)

Deallocate an operation from a workplace

Tables

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
…		

BAPI call

ID	Content / {type}	Description
DLG	ANR.AUSPLANEN	Deallocate operation
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.ANR	{C40}	Combined order/OP number
ANR.MGRP	{C8}	Group
BEARB	{C10}	Modified by
ANR.TABLE	{C1}	A – Backlog of orders (default value)

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1658	No cost center authorization for group
1666	Object is locked BEARB

7.4.5 Block operation (ANR.SPERREN)

Block the data collection for an operation

Tables

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
…		

BAPI call

ID	Content / {type}	Description
DLG	ANR.SPERREN	Block operation
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.ANR	{C40}	Combined order/OP number

ANR.TABLE	{C1}	A – Backlog of orders (default value)

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked
101	No data available! ANR.ANR is not available

7.4.6 Unlock operation (ANR.ENTSPERREN)

Unlock an operation that is blocked for data collection.

Tables

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
…		

BAPI call

ID	Content / {type}	Description
DLG	ANR.ENTSPERREN	Unlock operation
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.ANR	{C40}	Combined order/OP number
ANR.TABLE	{C1}	A – Backlog of orders (default value)

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked
101	No data available!

	ANR.ANR is not available

7.4.7 Update operation (ANR.AKTUALISIEREN)

Tables

Table	Key field	Description
auftrags_bestand	ANR.AUNR	PK Order number
…		

BAPI call

ID	Content / {type}	Description
DLG	ANR.AKTUALISIEREN	Update order
ANR.ATYP	{C4}	Order type {AU}
ANR.AUNR	{C40}	Order number
ANR.TABLE	{C1}	A – Backlog of orders (default value)

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
101	No data available! ANR.AUNR is not available

7.4.8 Release operation (ANR.FREIGEBEN)

Set the operation status to "Prepared".

Tables

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
…		

BAPI call

ID	Content / {type}	Description
DLG	ANR.ENTSPERREN	Unlock operation
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.ANR	{C40}	Combined order/OP number
ANR.TABLE	{C1}	A – Backlog of orders (default value)

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked
101	No data available! ANR.ANR is not available

7.4.9 Change order status (ANR.SETSTATUS)

Change order status

Tables

Table	Key field	Description
auftrags_bestand	ANR.AUNR	PK Order number
…		

BAPI call

ID	Content / {type}	Description
DLG	ANR.SETSTATUS	Change order status
ANR.ATYP	{C4}	AU – Type
ANR.AUNR	{C40}	Order number
ANR.TABLE	{C1}	A – Backlog of orders (default value)
MOD	{C1}	S → Sest status

		E → Finish order R → Reactivate order	
ANR.OPT:PKENN	{C2}	alternat ive	Production ID of the new status with MOD=S only "E" is permitted
ANR.AST	{C2}		New status with MOD=S only status with production identifier = "E" is permitted

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked
101	No data available! ANR.AUNR is not available
2026	Order is still running
712	Processing mode is invalid if MOD <> 'S'

Processing notes

<u>Reactivate (MOD=R):</u>

If you reactivate the order header, ALL operations of the order are reactivated.

The status of the order header is set as follows:

- Status V if no OP of the order has been started
- otherwise status L

<u>"Finish order" (MOD=E):</u>

If you finish the order header, ALL operations of the order are finished.

7.4.10 Change operation status (ANR.SETSTATUS)

Change operation status

Tables

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
…		

BAPI call

<table>
<tr><th>ID</th><th>Content / {type}</th><th colspan="2">Description</th></tr>
<tr><td>DLG</td><td>ANR.SETSTATUS</td><td colspan="2">Change order status</td></tr>
<tr><td>ANR.ATYP</td><td>{C4}</td><td colspan="2">OP – Type</td></tr>
<tr><td>ANR.ANR</td><td>{C40}</td><td colspan="2">MES order number</td></tr>
<tr><td>ANR.TABLE</td><td>{C1}</td><td colspan="2">A – Backlog of orders (default value)</td></tr>
<tr><td>MOD</td><td>{C1}</td><td colspan="2">S → Set status/secondary status/resource status for the operation
E → Finish operation
R → Reactivate operation</td></tr>
<tr><td>MOD:2</td><td>{C1}</td><td colspan="2">R → Set resource status for the operation
With MOD=S; ANR.STATUSTYP must not be set</td></tr>
<tr><td>ANR.STATUSTYP</td><td>{C1}</td><td colspan="2">S → Set secondary status
With MOD=S; MOD2 must not be set</td></tr>
<tr><td>ANR.OPT:PKENN</td><td>{C2}</td><td rowspan="2">Other option</td><td>Control indicator of the new status
with MOD=S</td></tr>
<tr><td>ANR.AST</td><td>{C2}</td><td>New status with MOD=S
➔ You may only set statuses with control indicator=S.
New secodary status with MOD=S and ANR.STATUSTYP=S
The statuses must be configured in the system.</td></tr>
<tr><td>ANR.RESSTA:1</td><td>{C2}</td><td colspan="2">New resource status "Person OK"
With MOD=S and MOD2:R</td></tr>
<tr><td>ANR.RESSTA:2</td><td>{C2}</td><td colspan="2">New resource status "Tool OK"
With MOD=S and MOD2:R</td></tr>
<tr><td>ANR.RESSTA:3</td><td>{C2}</td><td colspan="2">New resource status "Material OK"
With MOD=S and MOD2:R</td></tr>
</table>

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked
101	No data available! ANR.ANR is not available
2026	Order is still running
2807	Status cannot be changed, as sequence is inactive

Examples

Set operation to the status with control indicator V

DLG=ANR.SETSTATUS|ANR.ATYP=AG|ANR.ANR=S30002000700|**MOD=S|ANR.OPT:PKENN=V**|BEARB=12345|DAT=08/31/2018|ZEI=80806|

Finish operation

DLG=ANR.SETSTATUS|ANR.ATYP=AG|ANR.ANR=S30002000700|**MOD=E**|BEARB=12345|DAT=08/31/2018|ZEI=81180|

Set secondary status

DLG=ANR.SETSTATUS|ANR.ATYP=AG|ANR.ANR=S30002000700|**MOD=S|ANR.STATUSTYP=S**|ANR.AST=801|BEARB=12345|DAT=08/31/2018|ZEI=80632|

Set resource status

DLG=ANR.SETSTATUS|ANR.ATYP=AG|ANR.ANR=S30002000700|**MOD=S|MOD:2=R**|ANR.RESSTA:1=OK|ANR.RESSTA:2=OK|ANR.RESSTA:3=OK|BEARB=12345|DAT=08/31/2018|ZEI=80740|

Reactivate operation

DLG=ANR.SETSTATUS|ANR.ATYP=AG|ANR.ANR=S30002000700|**MOD=R**|BEARB=12345|DAT=08/31/2018|ZEI=81180|

7.4.11 Create order (ANR.INSERT)

Order (ANR.TABLE=A)

Work plan (ANR.TABLE=P)

Template (ANR.TABLE=P)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.AUNR	PK Order number

auftrags_leistung	ANR.AUNR	
auftrag_status	ANR.AUNR	
auftrags_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

Table	Key field	Description
arbplan_bestand	ANR.AUNR	PK Order number
arbplan_leistung	ANR.AUNR	
arbplan_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

If the entry in arbplan_bestand is a template, a respective entry in the table arbplan_verwalt must exist.

Table	Key field	Description
arbplan_verwalt	APVRW.APNR	PK Work plan number

BAPI call

ID	Content / {type}	Description
DLG	ANR.INSERT	Create order
ANR.ATYP	{C4}	Order type {AU}
ANR.AUNR	{C40}	Order number
ANR.TABLE	{C1}	A – Backlog of orders (default value)
ANR.AUART	{C5}	Order type
…		For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Return

ID	Content / {type}	Description
AUNR	{C40}	Order number
AST	{C5}	Status
OPT:PKENN	{C2}	Control characteristic "production"

ID	Content / {type}	Description
TABLE	{C1}	
ATYP	{C4}	

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1669	Data is already available → ANR.AUNR is already available
2812	Order is already available in the archive
50	Order status is not available → ANR.AUART is not specified

7.4.12 Edit order (ANR.UPDATE)

Order (ANR.TABLE=A)

Work plan (ANR.TABLE=P)

Template (ANR.TABLE=P)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.AUNR	PK Order number
auftrags_leistung	ANR.AUNR	
auftrag_status	ANR.AUNR	
auftrags_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

Table	Key field	Description
arbplan_bestand	ANR.AUNR	PK Order number
arbplan_leistung	ANR.AUNR	

arbplan_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

If the entry in arbplan_bestand is a template, a respective entry in the table arbplan_verwalt must exist.

Table	Key field	Description
arbplan_verwalt	APVRW.APNR	PK Work plan number

BAPI call

ID	Content / {type}	Description
DLG	ANR.UPDATE	Edit order
ANR.ATYP	{C4}	Order type {AU}
ANR.AUNR	{C40}	Order number
ANR.TABLE	{C1}	A – Backlog of orders (default value)
…		For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Return

ID	Content / {type}	Description
*		All attributes of the order

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked
101	No data available. ANR.AUNR is not available.
1803	No responsibility area authorization
50	Order status is not available
900	Unit is not available
2809	Order type cannot be changed because order has started

Error codes	Description
3241	User field key is not defined
1989	Maximum number of orders is exceeded for priority The maximum number of orders is exceeded for priority when using the priority management

7.4.13 Copy order (ANR.COPY)

Order (ANR.TABLE=A)

Work plan (ANR.TABLE=P)

Template (ANR.TABLE=P)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.AUNR	PK Order number
auftrags_leistung	ANR.AUNR	
auftrag_status	ANR.AUNR	
auftrags_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

Table	Key field	Description
arbplan_bestand	ANR.AUNR	PK Order number
arbplan_leistung	ANR.AUNR	
arbplan_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

If the entry in arbplan_bestand is a template, a respective entry in the table arbplan_verwalt must exist.

Table	Key field	Description
arbplan_verwalt	APVRW.APNR	PK Work plan number

BAPI call

ID	Content / {type}	Description
DLG	ANR.COPY	Copy order
ANR.ATYP	{C4}	Order type {AU}
ANR.AUNR	{C40}	Order number
ANR.TABLE	{C1}	A – Backlog of orders (default value)
ANR.TABLE:Z	{C1}	Target type: A – Order P – Work plan
ANR.AUNR:Z	{C40}	New order number
…		For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Return

ID	Content / {type}	Description
ANR		Order number of the new order

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked
1669	Data is already available → ANR.AUNR is already available
2812	Order is already available in the archive
1855	Target data record is already available
900	Unit is not available
1997	Order type is not equal If you copy an order, the order types of the source and the target order must be identical.
1998	Order template is not available

Notes on the processing

Copying a complete order:

If you copy a complete order, all values specified in the dialog data string are passed for ALL operations!

7.4.14 Delete order (ANR.DELETE)

Order (ANR.TABLE=A)

Work plan (ANR.TABLE=P)

Template (ANR.TABLE=P)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.AUNR	PK Order number
auftrags_leistung	ANR.AUNR	
auftrag_status	ANR.AUNR	
auftrags_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

Table	Key field	Description
arbplan_bestand	ANR.AUNR	PK Order number
arbplan_leistung	ANR.AUNR	
arbplan_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

If the entry in arbplan_bestand is a template, a respective entry in the table arbplan_verwalt must exist.

Table	Key field	Description
arbplan_verwalt	APVRW.APNR	PK Work plan number

BAPI call

ID	Content / {type}	Description
DLG	ANR.DELETE	Delete order

ANR.ATYP	{C4}	Order type {AU}
ANR.AUNR	{C40}	Order number
ANR.TABLE	{C1}	A – Backlog of orders (default value)

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked
101	No data available. ANR.AUNR is not available.
1803	No responsibility area authorization
1990	Order header must not be deleted

7.4.15 Select order (ANR.SELECT)

Order (ANR.TABLE=A)

Work plan (ANR.TABLE=P)

Template (ANR.TABLE=P)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.AUNR	PK Order number
auftrags_leistung	ANR.AUNR	
auftrag_status	ANR.AUNR	
auftrags_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

Table	Key field	Description
arbplan_bestand	ANR.AUNR	PK Order number
arbplan_leistung	ANR.AUNR	

arbplan_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

If the entry in arbplan_bestand is a template, a respective entry in the table arbplan_verwalt must exist.

Table	Key field	Description
arbplan_verwalt	APVRW.APNR	PK Work plan number

BAPI call

ID	Content / {type}	Description
DLG	ANR.SELECT	Select order
ANR.ATYP	{C4}	Order type {AU}
ANR.AUNR	{C40}	Order number
ANR.TABLE	{C1}	A – Backlog of orders (default value)
ANR.ARCHIV	{C1}	The archive is used

Return

ID	Content / {type}	Description
*		All attributes of the order

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
101	No data available. ANR.AUNR is not available.

7.4.16 Order – Select all operations (ANR.LIST)

Order (ANR.TABLE=A)

Work plan (ANR.TABLE=P)

Template (ANR.TABLE=P)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.AUNR	PK Order number
auftrags_leistung	ANR.AUNR	
auftrag_status	ANR.AUNR	
auftrags_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

Table	Key field	Description
arbplan_bestand	ANR.AUNR	PK Order number
arbplan_leistung	ANR.AUNR	
arbplan_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

If the entry in arbplan_bestand is a template, a respective entry in the table arbplan_verwalt must exist.

Table	Key field	Description
arbplan_verwalt	APVRW.APNR	PK Work plan number

BAPI call

ID	Content / {type}	Description
DLG	ANR.LIST	Select all operations of the order
ANR.ATYP	{C4}	Order type {AU}
ANR.AUNR	{C40}	Order number
ANR.TABLE	{C1}	A – Backlog of orders (default value)
ANR.ARCHIV	{C1}	The archive is used

Return

ID	Content / {type}	Description
*		List including all operations and all attributes of the operations

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
101	No data available. ANR.AUNR is not available.

7.4.17 Create operation (ANR.INSERT)

Operation (ANR.TABLE=A)

Work plan-operation (ANR.TABLE=P)

Template-operation (ANR.TABLE=P)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
auftrags_leistung	ANR.ANR	
auftrag_status	ANR.ANR	
auftrags_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

Table	Key field	Description
arbplan_bestand	ANR.ANR	PK Combined order/OP number
arbplan_leistung	ANR.ANR	
arbplan_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

If the entry in arbplan_bestand is a template, a respective entry in the table arbplan_verwalt must exist.

Table	Key field	Description
arbplan_verwalt	APVRW.APNR	PK Work plan number

BAPI call

ID	Content / {type}	Description
DLG	ANR.INSERT	Create operation
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.ANR	{C40}	Combined order/OP number
ANR.TABLE	{C1}	A – Backlog of orders (default value)
ANR.MNR ANR.MGRP	{C8} {C8}	Workplace Group One of the two values must be specified
…		For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Return

ID	Content / {type}	Description
ANR	{C40}	Combined order/OP number
AUNR	{C40}	Order number
AGNR	{C40}	OP number
AFOLG	{C40}	Sequence number
UAGNR	{C40}	Suboperation number
SPLNR	{C40}	Split No.
TABLE	{C1}	
ATYP	{C4}	Order type {AG} (OP)
AUART	{C5}	Order type
AST	{C2}	Status
OPT:PKENN	{C2}	Control characteristic "production"

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified

1669	Data is already available → ANR.ANR is already available
2812	Order is already available in the archive
94	OP ➔ Machine group is not available

7.4.18 Edit operation (ANR.UPDATE)

Operation (ANR.TABLE=A)

Work plan-operation (ANR.TABLE=P)

Template-operation (ANR.TABLE=P)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
auftrags_leistung	ANR.ANR	
auftrag_status	ANR.ANR	
auftrags_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

Table	Key field	Description
arbplan_bestand	ANR.ANR	PK Combined order/OP number
arbplan_leistung	ANR.ANR	
arbplan_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

If the entry in arbplan_bestand is a template, a respective entry in the table arbplan_verwalt must exist.

Table	Key field	Description
arbplan_verwalt	APVRW.APNR	PK Work plan number

BAPI call

ID	Content / {type}	Description
DLG	ANR.UPDATE	Edit operation

ID	Content / {type}	Description
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.ANR	{C40}	Combined order/OP number
ANR.TABLE	{C1}	A – Backlog of orders (default value)
…		For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Return

ID	Content / {type}	Description
*		All attributes of the order

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked
101	No data available. ANR.ANR is not available.
1803	No responsibility area authorization
50	Order status is not available
900	Unit is not available
2809	Order type cannot be changed because order has started
3241	User field key is not defined
1989	Maximum number of orders is exceeded for priority The maximum number of orders is exceeded for priority when using the priority management
911	Processing code is not available
94	Machine group ######## is not available
2808	Start date is after end date
2813	Activity code is not defined

7.4.19 Copy operation (ANR.COPY)

Operation (ANR.TABLE=A)

Work plan-operation (ANR.TABLE=P)

Template-operation (ANR.TABLE=P)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
auftrags_leistung	ANR.ANR	
auftrag_status	ANR.ANR	
auftrags_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

Table	Key field	Description
arbplan_bestand	ANR.ANR	PK Combined order/OP number
arbplan_leistung	ANR.ANR	
arbplan_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

If the entry in arbplan_bestand is a template, a respective entry in the table arbplan_verwalt must exist.

Table	Key field	Description
arbplan_verwalt	APVRW.APNR	PK Work plan number

BAPI call

ID	Content / {type}	Description
DLG	ANR.COPY	Copy order
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.ANR	{C40}	Combined order/OP number
ANR.TABLE	{C1}	A – Backlog of orders (default value)
ANR.TABLE:Z	{C1}	Target type: A – Order

ID	Content / {type}	Description
		P – Work plan
ANR.ANR:Z	{C40}	New combined order/OP number
…		For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Return

ID	Content / {type}	Description
ANR		New combined order/OP number

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked
1669	Data is already available → ANR.ANR is already available
2812	Order is already available in the archive
1855	Target data record is already available
900	Unit is not available
1997	Order type is not equal If you copy an order, the order types of the source and the target order must be identical.
1998	Order template is not available

7.4.20 Delete operation (ANR.DELETE)

Operation (ANR.TABLE=A)

Work plan-operation (ANR.TABLE=P)

Template-operation (ANR.TABLE=P)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
auftrags_leistung	ANR.ANR	
auftrag_status	ANR.ANR	
auftrags_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

Table	Key field	Description
arbplan_bestand	ANR.ANR	PK Combined order/OP number
arbplan_leistung	ANR.ANR	
arbplan_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

If the entry in arbplan_bestand is a template, a respective entry in the table arbplan_verwalt must exist.

Table	Key field	Description
arbplan_verwalt	APVRW.APNR	PK Work plan number

BAPI call

ID	Content / {type}	Description
DLG	ANR.DELETE	Delete operation
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.ANR	{C40}	Combined order/OP number
ANR.TABLE	{C1}	A – Backlog of orders (default value)

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1666	Object is locked

101	No data available. ANR.ANR is not available.
1803	No responsibility area authorization
1986	Operation cannot be deleted

7.4.21 Select operation (ANR.SELECT)

Operation (ANR.TABLE=A)

Work plan-operation (ANR.TABLE=P)

Template-operation (ANR.TABLE=P)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
auftrags_leistung	ANR.ANR	
auftrag_status	ANR.ANR	
auftrags_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

Table	Key field	Description
arbplan_bestand	ANR.ANR	PK Combined order/OP number
arbplan_leistung	ANR.ANR	
arbplan_zusatz	ANR.VERWEIS	

Tables (ANR.TABLE=P)

If the entry in arbplan_bestand is a template, a respective entry in the table arbplan_verwalt must exist.

Table	Key field	Description
arbplan_verwalt	APVRW.APNR	PK Work plan number

BAPI call

ID	Content / {type}	Description
DLG	ANR.SELECT	Select operation

ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.ANR	{C40}	Combined order/OP number
ANR.TABLE	{C1}	A – Backlog of orders (default value)
ANR.ARCHIV	{C1}	The archive is used

Return

ID	Content / {type}	Description
*		All attributes of the order

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
101	No data available. ANR.ANR is not available.

7.4.22 Split operation (ANR.SPLITCREATE)

Operation (ANR.TABLE=A)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
auftrags_leistung	ANR.ANR	
auftrag_status	ANR.ANR	
auftrags_zusatz	ANR.VERWEIS	

BAPI call

ID	Content / {type}	Description
DLG	ANR.SPLITCREATE	Split operation

ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.TABLE	{C1}	A – Backlog of orders (default value)
ANR.ANR	{C40}	Combined order/OP number
ANR.ANZSPLIT	{N8}	Number of splits

Return

ID	Content / {type}	Description
ANR.MAXANZSPLIT	{N8}	Maximum split number of operation

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
101	No data available!
1859	OP must not be a split master This operation is a so-called "split master", i.e. it is an operation that has been split up into several operations.
1860	OP must not be OP of a split OP
1862	OP must not be OP of a merged OP
1867	The OP may not be split.
1868	Maximum number of splits of the OP has been exceeded The OP that you want to split exceeds the maximum number of splits specified for the operation.
1869	The maximum number of slits has been exceeded.
30	Order is finished
20	Order is running
85	OP is locked
80	invalid OP status
31	Order is interrupted
1666	Object is locked

Error codes	Description
	BEARB

7.4.23 Cancel operation split (ANR.SPLITDELETE)

Operation (ANR.TABLE=A)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
auftrags_leistung	ANR.ANR	
auftrag_status	ANR.ANR	
auftrags_zusatz	ANR.VERWEIS	

BAPI call

ID	Content / {type}	Description
DLG	ANR.SPLITDELETE	Cancel operation split
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.TABLE	{C1}	A – Backlog of orders (default value)
ANR.ANR	{C40}	Combined order/OP number + split number

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
101	No data available!
20	Order is running

7.4.24 Enhanced operation split (ANR.ADVSPLITCREATE)

Requirements

- If you want to use the split function via PDM interface, you must configure the enhanced split function in HYDRA.

Table

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
auftrag_status	ANR.ANR	

BAPI call

ID	Content / {type}	Description
DLG	ANR.ADVSPLITCREATE	Enhanced split function
ANR.SPL	Operation (ANR)	The operation to be split (complete operation number). • An OP that is configured as OP that may be split • Or an already split OP (complete operation number incl. split)
SPL:MOD	PDM	Mode for the PDM interface
SPL:DIFFQUANTITY	J/N	Default = J: **Relevant only if you create splits using the master OP (OP has not yet been split).** Automatically creates a further split using the difference quantity. I.e. one additional split is created compared to the number specified in the IDs SPL.EGRGUTP:X,… To create a split using the difference quantity, the sum total of the target quantities of all splits to be created must be smaller than the target quantity of the master OP.
ANR.OPT:SPL_EGR_UPD	J/N	Default = J: Relevant if a split is split up a further time (MOC client function "Offset target quantity of the split OP"). The target quantity of an OP (here it is already a split), which is to be split, is reduced respectively.
ANR.OPT:PLAN	M/G	Splits planned for machine or group. If not specified, the

ID	Content / {type}	Description
		value of the OP to be split (master OP or split) is taken over.
SPL.EGRGUTP:X	Primary target quantity included in the split that is to be created	Target quantity (primary quantity) of the individual split. (X) stands for continuous numbers from 1...99
SPL.RUEZ:X	Duration in seconds	Setup time of the individual split. (X) stands for continuous numbers from 1...99, if not specified (ID is not included) the setup time of the OP is used that has been split.
BEARB	Modified by	
DAT	Date	(Example: 09/24/2014)
ZEIT	Time in seconds since midnight.	

Note

X in the IDs of the above table must be replaced successively with 1...99 for each split to be created. The ANR-BAPI automatically assigns the split numbers. For example, if an OP that has already been split up is split a new time, the split numbers are already assigned. The BAPI then automatically searches for the next free split number. If an ascending number is not assigned, the processing ends (e.g. in the string that follows, only the first two splits are created). ...|SPL.EGRGUTP:1=10|SPL.EGRGUTP:2=20| SPL.EGRGUTP:5=50|…)

Plausibility checks

Error codes	Description
10	Order not available
1859	No split possibility OP must not be a split master
1971	Split number < Minimum below min. number of splits
424	Split function is not active

Examples:

```
Split operation (use setup time of the master, no split with the difference quantity
is created):
DLG=ANR.ADVSPLITCREATE|ANR.SPL=160100400020|SPL:MOD=PDM|ANR.OPT:PLAN=G|SPL.EGRGUTP:1=
15|SPL.EGRGUTP:2=10|SPL:DIFFQUANTITY=N|BEARB=12345|DAT=09/29/2014|ZEIT=44421|
```

```
Existing split is again split up (specified setup time, quantities are not reduced by
the OP to be split. I.e. the total target quantity of the operation is increased by
the two new splits.):
DLG=ANR.ADVSPLITCREATE|ANR.SPL=16010040002002|SPL:MOD=PDM|ANR.OPT:PLAN=G|SPL.EGRGUTP:
1=5|SPL.RUEZ:1=1200|SPL.EGRGUTP:2=100|SPL.RUEZ:2=1200|ANR.OPT:SPL_EGR_UPD=N|BEARB=123
45|DAT=09/29/2014|ZEIT=44621|"
```

7.4.25 Create merged operation (ANR.SAGINSERT)

Operation (ANR.TABLE=A)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
auftrags_leistung	ANR.ANR	
auftrag_status	ANR.ANR	
auftrags_zusatz	ANR.VERWEIS	

BAPI call

ID	Content / {type}	Description
DLG	ANR.SAGINSERT	Create merged operation
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.TABLE	{C1}	A – Backlog of orders (default value)
ANR.SANR	{C40}	Merged operation Combined order/OP number
ANR.ANR	{C40}	Reference OP Combined order/OP number
ANR.ANR:1	{C40}	Operations that you want to combine into one merged operation.

ID	Content / {type}	Description
ANR.ANR:2 .. ANR.ANR:n		
ANR.SGR:GUTB	{DEC13,3}	Target quantity yield
ANR.SGR:AUSB	{DEC13,3}	Target quantity scrap
ANR.SGE:B	{C1}	Base quantity unit of the merged OP
ANR.RUEZ	{N8}	Setup time
ANR.BEARBZ	{N8}	Processing time

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
1669	Data is already available → ANR.ANR is already available
2812	Order is already available in the archive
101	No data available! → This message also pops up if no reference OP is specified.

7.4.26 Delete merged operation (ANR.SAGDELETE)

Operation (ANR.TABLE=A)

Tables (ANR.TABLE=A)

Table	Key field	Description
auftrags_bestand	ANR.ANR	PK Combined order/OP number
auftrags_leistung	ANR.ANR	
auftrag_status	ANR.ANR	
auftrags_zusatz	ANR.VERWEIS	

BAPI call

ID	Content / {type}	Description
DLG	ANR.SAGDELETE	Delete merged operation
ANR.ATYP	{C4}	Order type {AG} (OP)
ANR.TABLE	{C1}	A – Backlog of orders (default value)
ANR.SANR	{C40}	Merged operation Combined order/OP number
MOD	{C1}	E – Remove single OP from MOP G – Remove all OPs from MOP

Plausibility checks

Error codes	Description
1661	Missing parameter ANR.ATYP is not specified
101	No data available! → This message also pops up if no reference OP is specified.
1866	Specified OP is no OP of the merged OP
1865	Specified MOP is no merged OP

7.4.27 Lock order network for editing (ANETZ.LOCK)

Table	Key field	Description
ade_auftragsnetz	ANETZ.ANRV ANETZ.ANRN	PK Predecessor + successor

BAPI call

ID	Content / {type}	Description
DLG	ANETZ.LOCK	Lock order network
ANETZ.ANRV	{C40}	Predecessor Combined order/OP number
ANETZ.ANRN	{C40}	Successor

		Combined order/OP number

Return

ID	Content / {type}	Description
*		All attributes of the order network

Plausibility checks

Error codes	Description
1661	Missing parameter
1666	Object is locked BEARB

7.4.28 Unlock order network for editing (ANETZ.UNLOCK)

Table	Key field	Description
ade_auftragsnetz	ANETZ.ANRV ANETZ.ANRN	PK Predecessor + successor

BAPI call

ID	Content / {type}	Description
DLG	ANETZ.UNLOCK	Unlock order network
ANETZ.ANRV	{C40}	Predecessor Combined order/OP number
ANETZ.ANRN	{C40}	Successor Combined order/OP number

Plausibility checks

Error codes	Description
1661	Missing parameter

7.4.29 Create order network (ANETZ.INSERT)

Table	Key field	Description
ade_auftragsnetz	ANETZ.ANRV ANETZ.ANRN	PK Predecessor + successor

BAPI call

ID	Content / {type}	Description
DLG	ANETZ.INSERT	Create order network
ANETZ.ANRV	{C40}	Predecessor Combined order/OP number
ANETZ.ANRN	{C40}	Successor Combined order/OP number
ANETZ.AOB	{C2}	Relationship ES = End-Start
ANETZ.AKTIV	{C1}	J - Active
ANETZ.HERK	{C1}	Origin E = Explicit (via interface)
…		For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Return

ID	Content / {type}	Description
ANRV	{C40}	Predecessor
ANRN	{C40}	Successor

Plausibility checks

Error codes	Description
1661	Missing parameter
1669	Data is already available

7.4.30 Edit order network (ANETZ.UPDATE)

Table	Key field	Description
ade_auftragsnetz	ANETZ.ANRV ANETZ.ANRN	PK Predecessor + successor

BAPI call

ID	Content / {type}	Description
DLG	ANETZ.UPDATE	Edit order network
ANETZ.ANRV	{C40}	Predecessor Combined order/OP number
ANETZ.ANRN	{C40}	Successor Combined order/OP number
MOD	{C1}	V – predecessor – update relationships
…		For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Plausibility checks

Error codes	Description
1661	Missing parameter
1666	Object is locked

7.4.31 Delete order network (ANETZ.DELETE)

Table	Key field	Description
ade_auftragsnetz	ANETZ.ANRV ANETZ.ANRN	PK Predecessor + successor

BAPI call

ID	Content / {type}	Description
DLG	ANETZ.UPDATE	Delete order network
ANETZ.ANRV	{C40}	Predecessor Combined order/OP number

ANETZ.ANRN	{C40}	Successor Combined order/OP number
ANETZ.AUNR	{C40}	Order number with MOD=A
MOD	{C1}	A – Delete complete network

Plausibility checks

Error codes	Description
1661	Missing parameter
1666	Object is locked

7.4.32 Update order network (ANETZ.AKTUALISIEREN)

Table	Key field	Description
ade_auftragsnetz	ANETZ.ANRV ANETZ.ANRN	PK Predecessor + successor

BAPI call

ID	Content / {type}	Description
DLG	ANETZ.AKTUALISIEREN	Update order network
ANETZ.ANR	{C40}	Combined order/OP number
ANETZ.AUNR	{C40}	Order number
ANETZ.AGNR	{C40}	OP number
ANETZ.AFOLG	{C40}	Sequence number
MOD	{C1}	**MOD=I :** Recalculation of the ANETZ relationship of the added OP to the preceding and succeeding operation. **MOD=D :** Recalculation of the ANETZ relationship of the deleted OP to the preceding and succeeding operation. **MOD=A :** Recalculation of the activated ANETZ relationship of a sequence **MOD=N :** Complete order network recalculation of an order

Plausibility checks

Error codes	Description
712	Processing mode is invalid
1661	Missing parameter
1669	Data is already available with MOD=I

7.4.33 Lock material list for editing (MATLIST.LOCK)

Operation - material list (MATLIST.TABLE=A)

Workplan-operation - material list (MATLIST.TABLE=P)

Table (MATLIST.TABLE=A)

Table	Key field	Description
mlst_hy	MATLIST.ANR MATLIST.ATK MATLIST.SLP MATLIST.RESTYP	

Table (MATLIST.TABLE=P)

Table	Key field	Description
arbplan_mlst_hy	MATLIST.ANR MATLIST.ATK MATLIST.SLP MATLIST.RESTYP	

BAPI call

ID	Content / {type}	Description
DLG	MATLIST.LOCK	Lock material list
MATLIST.ANR	{C40}	Combined order/OP number
MATLIST.ATK	{C40}	Material
MATLIST.RESTYP	{C4}	Resource type
MATLIST.SLP	{C10}	BOM item (default=1)

Return

ID	Content / {type}	Description
MATLIST.*		All attributes of the newly created entry in the material list

Plausibility checks

Error codes	Description
1661	Missing parameter
101	No data available!
1666	Object is locked

7.4.34 Unlock material list for editing (MATLIST.UNLOCK)

Operation - material list (MATLIST.TABLE=A)

Workplan-operation - material list (MATLIST.TABLE=P)

Table (MATLIST.TABLE=A)

Table	Key field	Description
mlst_hy	MATLIST.ANR MATLIST.ATK MATLIST.SLP MATLIST.RESTYP	

Table (MATLIST.TABLE=P)

Table	Key field	Description
arbplan_mlst_hy	MATLIST.ANR MATLIST.ATK MATLIST.SLP MATLIST.RESTYP	

BAPI call

ID	Content / {type}	Description
DLG	MATLIST.UNLOCK	Unlock material list
MATLIST.ANR	{C40}	Combined order/OP number

MATLIST.ATK	{C40}	Material
MATLIST.RESTYP	{C4}	Resource type
MATLIST.SLP	{C10}	BOM item (default=1)

Plausibility checks

Error codes	Description
1661	Missing parameter

7.4.35 Create material list (MATLIST.INSERT)

Operation - material list (MATLIST.TABLE=A)

Workplan-operation - material list (MATLIST.TABLE=P)

Table (MATLIST.TABLE=A)

Table	Key field	Description
mlst_hy	MATLIST.ANR MATLIST.ATK MATLIST.SLP MATLIST.RESTYP	

Table (MATLIST.TABLE=P)

Table	Key field	Description
arbplan_mlst_hy	MATLIST.ANR MATLIST.ATK MATLIST.SLP MATLIST.RESTYP	

BAPI call

ID	Content / {type}	Description
DLG	MATLIST.INSERT	Create material list
MATLIST.ANR	{C40}	Combined order/OP number
MATLIST.ATK	{C40}	Material
MATLIST.RESTYP	{C4}	Resource type
MATLIST.SLP	{C10}	BOM item (default=1)

…		For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Return

ID	Content / {type}	Description
MATLIST.*		All attributes of the newly created entry in the material list

Plausibility checks

Error codes	Description
1661	Missing parameter
1669	Data is already available
3284	BOM level is invalid if MATLIST.SLS:M has been specified
1987	Operation cannot be changed
10	Order not available

7.4.36 Edit material list (MATLIST.UPDATE)

Operation - material list (MATLIST.TABLE=A)

Workplan-operation - material list (MATLIST.TABLE=P)

Table (MATLIST.TABLE=A)

Table	Key field	Description
mlst_hy	MATLIST.ANR MATLIST.ATK MATLIST.SLP MATLIST.RESTYP	

Table (MATLIST.TABLE=P)

Table	Key field	Description
arbplan_mlst_hy	MATLIST.ANR MATLIST.ATK MATLIST.SLP MATLIST.RESTYP	

BAPI call

ID	Content / {type}	Description
DLG	MATLIST.UPDATE	Edit material list
MATLIST.ANR	{C40}	Combined order/OP number
MATLIST.ATK	{C40}	Material
MATLIST.RESTYP	{C4}	Resource type
MATLIST.SLP	{C10}	BOM item (default=1)
…		For further fields, refer to the documentation HYD-HDB that includes the above listed tables

Return

ID	Content / {type}	Description
MATLIST.*		All attributes of the entry in the material list

Plausibility checks

Error codes	Description
1661	Missing parameter
101	No data available!
1666	Object is locked
3284	BOM level is invalid if MATLIST.SLS:M has been specified
1987	Operation cannot be changed
10	Order not available

7.4.37 Delete material list (MATLIST.DELETE)

Operation - material list (MATLIST.TABLE=A)

Workplan-operation - material list (MATLIST.TABLE=P)

Table (MATLIST.TABLE=A)

Table	Key field	Description
mlst_hy	MATLIST.ANR MATLIST.ATK MATLIST.SLP MATLIST.RESTYP	

Table (MATLIST.TABLE=P)

Table	Key field	Description
arbplan_mlst_hy	MATLIST.ANR MATLIST.ATK MATLIST.SLP MATLIST.RESTYP	

BAPI call

ID	Content / {type}	Description
DLG	MATLIST.DELETE	Create order network
MATLIST.ANR	{C40}	Combined order/OP number
MATLIST.ATK	{C40}	Material
MATLIST.RESTYP	{C4}	Resource type
MOD		**A** - Delete all components of an OP **E** - Delete individual components **R** - Delete all resource components of an OP **M** - Delete all material components of an OP

Plausibility checks

Error codes	Description
1661	Missing parameter
2027	Processing mode is invalid
1986	Operation cannot be deleted
1666	Object is locked

8 HYDRA Production Data Manager MDE - Master Data

8.1 Note on the descriptions of the basic dialogs

All mandatory fields that must be specified have the addition PK (primary key). All other fields are optional and are processed if they are transferred.

8.2 Machine configuration

8.2.1 Edit machine configuration (DLG=MNR.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

Use this BAPI call to create a machine or a workplace. The machine statuses 20000 "NO SHIFT" and 30000 "NOT ASSIGNED" are automatically assigned.

Tables

Table	Key field	Description
maschinen	masch_nr MNR.MNR	Machine number (PK) in machine configuration
maschinen_status	masch_nr MNR.MNR	Machine number (PK) in machine status

BAPI call

Identification	Content / {type}	Description
DLG	MNR.INSERT	Create machine/workplace
	MNR.UPDATE	Change machine/workplace
	MNR.DELETE	Delete machine/workplace
	MNR.COPY	Copy machine/workplace
	MNR.LOCK	Lock machine/workplace for editing
	MNR.UNLOCK	Unlock machine/workplace after editing
	MNR.NEW	Read specification for new machine/new workplace
	MNR.SELECT	Select machine/workplace
MNR.MNR	{C8}	PK: workplace/machine number

		The valid characters are listed in the document MOC_ResourceConfiguration.pdf. The number must not exceed 8 characters.
MNR. MNR:Z	{C8}	PK: new (target) workplace/machine number for COPY The valid characters are listed in the document MOC_ResourceConfiguration.pdf. The number must not exceed 8 characters.
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
923	The cost center of the group must match the cost center of the machine or workplace.
1611	When you delete (MNR.DELETE), the machine cannot be removed from the group assignment.
1611	When you create a machine (MNR.INSERT) and the specified machine group (MNR.MGRP) has not yet been created as capacity group, this capacity group is then automatically created in the group assignment. Creating the group MNR.MGRP as capacity group failed.
1611	When you create a machine (MNR.INSERT), this machine is automatically assigned to the group assignment of the capacity group (MNR.MGRP). The automatic assignment failed.
1661	• The value MNR.MNR has not been specified. • The value MNR.MNR:Z has not been specified with MNR.COPY • The value DAT has not been specified with MNR.SKINFO
1803	The authorization for the responsibility area is not available and the specified machine cannot be edited.
1666	The machine is currently edited by another user.

8.2.2 List of machines/workplaces (DLG=MNR.LIST)

This list shows all machines or workplaces defined in the system.

Tables

Table	Key field	Description

maschinen	masch_nr MNR.MNR	Machine number (PK) in machine configuration

BAPI call

Identification	Contents	Description
DLG	MNR.LIST	List of machines
DATEI	{C256}	Specification of the file name for the list

8.2.3 Read shift information of the machine (DLG=MNR.SKINFO)

This BAPI call returns the current shift information of the machine. Depending on the configuration of the shift calender, up to 10 shifts (index 1-10) can be returned.

BAPI call

Identification	Contents	Description
DLG	MNR.SKINFO	Read shift information
MNR.MNR	{C8}	Machine number
MNR.MOD	{C1}	1: Identify shifts following the current machine date. 2: Identify shift times of a specific shift

Return mode 1

Identification	Contents	Description
MNR	{C8}	Machine number
DAT	{mm/dd/yyyy}	Current log time for the machine
ZEI	{seconds}	Current log time for the machine
SKDAT	{mm/dd/yyyy}	Current shift date for the machine
SKNR	{N1}	Current shift number for the machine (1-4)
SKZEIB	{seconds}	Current start of shift
SZEITE	{seconds}	Current end of shift
SKDAUER:RES T	{seconds}	Remaining time of the current shift

SKDAT:1 … SKDAT:10	{mm/dd/yyyy}	Date of the shift selected
SKNR:1 … SKNR:10	{N1}	Number of the shift selected (1-4)
SKZEIB:1 … SKZEIB:10	{seconds}	Start of the shift selected
SKZEIE:1 … SKZEIE:10	{seconds}	End of the shift selected
SKDAUER:1 … SKZEIB:10	{seconds}	Duration of the shift selected

Return mode 2

Identification	Contents	Description
MNR	{C8}	Machine number
DAT	{mm/dd/yyyy}	Current log time for the machine
ZEI	{seconds}	Current log time for the machine
SKDAT	{mm/dd/yyyy}	Current shift date for the machine
SKNR	{N1}	Current shift number for the machine (1-4)
SKZEIB	{seconds}	Current start of shift
SZEITE	{seconds}	Current end of shift

8.3 Status texts

8.3.1 Edit status texts (DLG=MSTTXT.INSERT, UPDATE, DELETE, LOCK, UNLOCK, SELECT)

Use these BAPI calls to edit status texts that are assigned to the machine statuses.

Tables

Table	Key field	Description
stoertexte	stoertxt_nr	Status text number (PK) in status texts

	MSTTXT.STNR	

BAPI call

Identification	Content / {type}	Description
DLG	MSTTXT.INSERT	Create status text
	MSTTXT.UPDATE	Change status text
	MSTTXT.DELETE	Delete status text
	MSTTXT.LOCK	Lock status text for editing
	MSTTXT.UNLOCK	Unlock status text after editing
	MSTTXT.SELECT	Select status text
MSTTXT.STNR	{N4}	PK: status text number
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
734	The parameter MSTTXT.STNR has not been specified.
735	The machine status text does not exist.
733	The machine status text is referenced in the machine status configuration and cannot be deleted.
933	The status texts 20000 and 30000 must not be deleted.
736	A machine status text with this number already exists.
1666	The machine status text is currently edited by another user.

8.3.2 List of status texts (DLG=MSTTXT.LIST)

The BAPI call returns all defined status texts (incl. the special texts 20000 "NO SHIFT" and 30000 "NOT ASSIGNED").

Tables

Table	Key field	Description
stoertexte	stoertxt_nr MSTTXT.STNR	Status text number (PK) in status texts

BAPI call

Identification	Contents	Description
DLG	MSTTXT.LIST	List of status texts
DATEI	{C256}	Specification of the file name for the list

8.4 Status classes

8.4.1 Edit status classes (DLG=STKL.INSERT, UPDATE, DELETE, LOCK, UNLOCK, SELECT)

Use these BAPI calls to edit status classes that are assigned to the machine statuses.

Tables

Table	Key field	Description
mz_stklasse	nummer STKL.STKLNR	Status class number (PK) in status class configuration
mz_stklasse	kuerzel STKL.STKL	Status class (PK) in status class configuration

BAPI call

Identification	Content / {type}	Description
DLG	STKL.INSERT	Create status class
	STKL.UPDATE	Change status class
	STKL.DELETE	Delete status class
	STKL.LOCK	Lock status class for editing
	STKL.UNLOCK	Unlock status class after editing
	STKL.SELECT	Select status class
STKL.STKLNR	{N4}	PK: status class number (unique)
STKL.STKL	{C3}	PK: status class (unique)
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
1661	• The status class number (STKL.STKLNR) has not been specified • The status class (STKL.STKL) has not been specified
719	The status class is not available
750	The status class cannot be deleted because it is assigned to at least one machine status
749	The status class/status class number already exists
1666	The machine status text is currently edited by another user.

8.4.2 List of status classes (DLG=STKL.LIST)

The BAPI call returns all status classes defined.

Tables

Table	Key field	Description
mz_stklasse	nummer STKL.STKLNR	Status class number (PK) in status class configuration
mz_stklasse	kuerzel STKL.STKL	Status class (PK) in status class configuration

BAPI call

Identification	Contents	Description
DLG	STKL.LIST	List of status classes
DATEI	{C256}	Specification of the file name for the list

8.5 Machine status configuration

8.5.1 Edit valid machine statuses (DLG=MST.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

Use this BAPI call to configure the valid machine statuses for a machine or workplace. The status texts are assigned here.

Tables

Table	Key field	Description
stoer_tabelle	masch_nr MST.MNR	Machine number (PK) in machine status configuration
stoer_tabelle	stoernr MST.MST	Machine status (PK) in machine status configuration
stoer_tabelle	stoertxt_nr MST.STNR	Status text number (FK) in status texts

BAPI call

Identification	Content / {type}	Description
DLG	MST.INSERT	Create machine status configuration
	MST.UPDATE	Change machine status configuration
	MST.DELETE	Delete machine status configuration
	MST.COPY	Copy machine status configuration
	MST.LOCK	Lock machine status configuration for editing
	MST.UNLOCK	Unlock machine status configuration after editing
	MST.NEW	Read specification for new machine status configuration
	MST.SELECT	Select machine status configuration
MST.MNR	{C8}	PK: machine number
MST.MST	{N4}	PK: machine status
MST.MNR:Z	{C8}	PK: new (target) machine number for COPY with MOD=E
MST.MST:Z	{N4}	PK: new (target) machine status for COPY with MOD=E
MOD	{C1}	Copy/delete mode: G = all statuses of machine with DELETE/COPY F = missing statuses of machine with COPY E = single statuses of machine with DELETE/COPY M = single status of all machines with DELETE/COPY **Note:** The statuses 20000 and 30000 are not deleted with MST.DELETE and MOD=G because these statuses are automatically created when the machine is created (MNR.INSERT) and automatically deleted when the machine is deleted (MNR.DELETE).

…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
707	The machine number (MNR.MNR) has not been specified.
708	The machine status (MST.MST) has not been specified.
712	The parameter MOD is invalid or has not been specified.
709	The (target) machine number has not been specified.
710	The (target) machine status has not been specified.
1803	The authorization for the responsibility area is not available and the specified machine cannot be edited.
713	The machine status specified is not available.
706	The machine status specified cannot be deleted because this status is currently assigned to the machine.
714	The machine status already exists for this machine.
715	The production flag (MST.OPT:PKENN) has not been specified.
716	A status with the specified production flag (MST.OPT:PKENN) already exists for this machine.
717	The specified resource performance account (MST.BMKNR) is not available.
718	The specified status text (MST.STNR) is not available. You must create this status text in the configuration of status texts (see MSTTXT.INSERT).
719	The disturbance class (MST.STKL) is not available. You must create this class in the configuration of disturbance classes.
720	You can only set the option "Status transfer of aggregates" (MST.OPT.LINIE=J) with machines of type "Line".
809	Invalid superior machine status (MST.MST:UEB). The flag "Hierarchy level" is not set in the superior status.
810	Invalid superior machine status (MST.MST:UEB). The status is not available for the machine selected!
811	Invalid superior machine status (MST.MST:UEB). The superior and the inferior statuses are identical.
812	Invalid superior machine status (MST.MST:UEB). The higher-level/superior status refers to the lower-level/inferior status at the end of the hierarchy.

802	Status already assigned for "no shift".
1666	The machine status is currently edited by another user.

8.5.2 List machine status (DLG=MST.LIST)

This list shows all machine statuses defined in the system.

Tables

Table	Key field	Description
stoer_tabelle	masch_nr MST.MNR	Machine number (PK) in machine status configuration
stoer_tabelle	stoernr MST.MST	Machine status (PK) in machine status configuration
stoer_tabelle	stoertxt_nr MST.STNR	Status text number (FK) in status texts

BAPI call

Identification	Contents	Description
DLG	MST.LIST	List of machine statuses
DATEI	{C256}	Specification of the file name for the list

8.6 Counter configuration

8.6.1 Edit counter configuration (DLG=MNRCTR.INSERT, UPDATE, MODIFY, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

Use this BAPI call to create and edit the configuration for 1...n counters per machine or workplace.

Tables

Table	Key field	Description
maschinen_zaehler	masch_nr MNRCTR.MNR	Machine number (PK)
maschinen_zaehler	zaehler	Counter number (PK)

	MNRCTR.CTR	

BAPI call

Identification	Content / {type}	Description
DLG	MNRCTR.INSERT	Create counter configuration
	MNRCTR.UPDATE	Change counter configuration
	MNRCTR.DELETE	Delete counter configuration
	MNRCTR.COPY	Copy counter configuration
	MNRCTR.LOCK	Lock counter configuration for editing
	MNRCTR.UNLOCK	Unlock counter configuration after editing
	MNRCTR.NEW	Read specification for new counter configuration.
	MNRCTR.SELECT	Select counter configuration
MNRCTR.MNR	{N3}	PK: machine number
MNRCTR. MNR:Z	{N3}	PK: new (target) machine number for COPY
MNRCTR.CTR	{N3}	PK: counter number
MNRCTR. CTR:Z	{N3}	PK: new (target) counter number for COPY
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
1662	The parameter **Allocation with** (MNRCTR:VERB) must not have the same value as the parameter **Evaluation** (MNRCTR.TYP).
1662	The parameter **Evaluation** (MNRCTR.TYP) is not specified and the value is invalid. Possible values: GUT, AUS, NCH, PRB and empty
1662	The parameter **Allocation with** (MNRCTR:VERB) is specified and the value is invalid. Possible values: GUT, AUS, NCH, PRB and empty
1662	The parameter **Allocation with partitioning** (MNRCTR.VERB:TLG) is specified and the value is not valid. Possible values: J, N and empty
1662	The parameter **Posting as cycles** (MNRCTR.OPT:TAKT) is specified and the value is invalid. Possible values: T, N and empty

1662	The parameter **For monitoring** (MNRCTR.OPT:UEB) is specified and the value is invalid. Possible values: Z, N and empty
931	Auto-allocation is not possible.
935	The current counter configuration is only supported with CT-8xx or C-76x terminals.
1669	Counter configuration is already available (MNRCTR.UPDATE)
90	Machine does not exist (MNRCTR.MNR / MNRCTR.:Z)
930	The maximum number of counters at the machine has been exceeded.
1662	The key fields are not correctly specified: - MNRCTR.MNR - MNRCTR.MNR:Z (with MNRCTR.COPY) - MNRCTR.CTR - MNRCTR.CTR:Z (with MNRCTR.COPY)
101	No counter configuration available (with MNRCTR.UPDATE and MNRCTR.DELETE)
1669	Counter configuration is already available (with MNRCTR.INSERT)
1666	The data record is currently locked by user %s (with MNRCTR.UPDATE and MNRCTR.DELETE)
712	The parameter MOD is not valid. Possible values: E, G and F
101	Source counter configuration is not available (MNRCTR.COPY)
1855	Target counter configuration is already available (MNRCTR.COPY)

8.6.2 List of counter configuration (DLG=MNRCTR.LIST)

This list shows all defined counters of the counter configuration.

Tables

Table	Key field	Description
maschinen_zaehler	masch_nr MNRCTR.MNR	Machine number (PK)
maschinen_zaehler	zaehler MNRCTR.CTR	Counter number (PK)

BAPI call

Identification	Contents	Description
DLG	MNRCTR.LIST	List of counter configuration
DATEI	{C256}	Specification of the file name for the list

8.7 Assignment of machine/workplace to terminal

8.7.1 Edit assignment (DLG=MNRTNR.INSERT, DELETE,SELECT)

Use these BAPI calls to edit the assignments of machines to terminals.

Tables

Table	Key field	Description
masch_term_zuord	masch_nr terminal_nr MNRTNR.MNR MNRTNR.TNR	Machine and terminal must be unique.
masch_term_zuord	terminal_nrterminal_nr anzeige_pos MNRTNR.TNR MNRTNR.POS	Terminal and display position must be unique.

BAPI call

Identification	Content / {type}	Description
DLG	MNRTNR.INSERT	Create assignment
	MNRTNR.DELETE	Delete assignment
	MNRTNR.SELECT	Select assignment
MNRTNR.MNR	C 20	Machine
MNRTNR.TNR	N8	Terminal number
MNRTNR.POS	N8	Display position
…	…	For further fields, refer to the documentation HYD-HDB that describes the above listed tables

Validation checks

Error codes	Description
707	Machine has not been specified.
746	Terminal number has not been specified.
747	Position has not been specified.
510	Person is not authorized.
1672	Assignment does already exist.
1682	Assignment is not available.

8.7.2 List of assignments (DLG=MNRTNR.LIST)

The BAPI call returns all assignments for all terminals.

BAPI call

Identification	Contents	Description
DLG	MNRTNR.LIST	List of assignments
DATEI	{C256}	Specification of the file name for the list

8.8 Group assignment

8.8.1 Edit group assignment (DLG=GRPRES.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

Use these BAPI calls to edit the group assignments.

Tables

Table	Key field	Description
hy_gruppen_zuord	group res_nr res_typ GRPRES.GRP GRPRES.RESNR GRPRES.RESTYP	The combination must be unique.

BAPI call

Identification	Content / {type}	Description
DLG	GRPRES.INSERT	Create assignment
	GRPRES.UPDATE	Edit assignment
	GRPRES.DELETE	Delete assignment
	GRPRES.COPY	Copy assignment
	GRPRES.SELECT	Select assignment
	GRPRES.COPY	Copy assignment
	GRPRES.LOCK	Lock assignment for editing
	GRPRES.UNLOCK	Unlock assignment after editing
	GRPRES.NEW	Read specification for new assignment
GRPRES.GRP	C 20	Group
GRPRES.RESNR	C 40	Resource
GRPRES.RESTYP	C 4	Resource type
MOD	C 1	Copy mode E- Copy currently selected resource G- Copy all resources of the group M- Move all resources of the group

Return

Identification	Content / {type}	Description
GRP	C 20	current data record
RESNR	C 40	current data record
RESTYP	C 4	current data record
POS	N8	current data record

Validation checks

Error codes	Description
1661	• The value GRPRES.GRP has not been specified. • The value GRPRES.RESTYP has not been specified. • The value GRPRES.RESNR has not been specified.

	• The value GRPRES.GRP:Z has not been specified with GRPRES.COPY.
414	The group specified does not exist.
1669	Assignment is already available (GRPRES.UPDATE)
94	With RESTYP = MGRP Machine group does not exist.
918	With RESTYP = MGRP A resource of type MGRP must be a capacity group.
415	The RESTYP specified does not match the existing group containing only one type.
712	Processing mode is invalid.

8.8.2 List of group assignments (DLG=GRPRES.LIST)

The BAPI call lists the group assignments.

BAPI call

Identification	Contents	Description
DLG	GRPRES.LIST	List of assignments
DATEI	{C256}	Specification of the file name for the list
GRPRES.GRP	C 20	Group {wildcard permitted}
GRPRES.RESNR	C 40	Resource {wildcard permitted}
GRPRES.RESTYP	C 4	Resource type {wildcard permitted}

8.9 Groups

8.9.1 Edit groups (DLG=GRP.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

Use these BAPI calls to edit groups.

Tables

Table	Key field	Description
hy_gruppen	group user GRP.GRP GRP.BEARB	The combination must be unique.

BAPI call

Identification	Content / {type}	Description
DLG	GRP.INSERT	Create group
	GRP.UPDATE	Edit group
	GRP.DELETE	Delete group
	GRP.COPY	Copy group
	GRP.SELECT	Select group
	GRP.COPY	Copy group
	GRP.LOCK	Lock group for editing
	GRP.UNLOCK	Unlock group after editing
	GRP.NEW	Read specification for new group
GRP.GRP	C 20	Group
GRP.BEARB	C10	User
MOD	C 1	Copy mode E- copy currently selected group Z- copy currently selected group with all resources assigned

Return

Identification	Content / {type}	Description
GRP	C 20	current data record
BEARB	C10	current data record

Validation checks

Error codes	Description
1661	• The value GRPRES.GRP has not been specified. • The value GRPRES.RESTYP has not been specified. • The value GRPRES.RESNR has not been specified. • The value GRPRES.GRP:Z has not been specified with GRPRES.COPY.
414	The group specified does not exist.
1669	Assignment is already available (GRPRES.UPDATE)

94	With RESTYP = MGRP Machine group does not exist.
918	With RESTYP = MGRP A resource of type MGRP must be a capacity group.
415	The RESTYP specified does not match the existing group containing only one type.
712	Processing mode is invalid.

8.9.2 List of group assignments (DLG=GRPRES.LIST)

The BAPI call lists the group assignments.

BAPI call

Identification	Contents	Description
DLG	GRPRES.LIST	List of assignments
DATEI	{C256}	Specification of the file name for the list
GRPRES.GRP	C 20	Group {wildcard permitted}
GRPRES.RESNR	C 40	Resource {wildcard permitted}
GRPRES.RESTYP	C 4	Resource type {wildcard permitted}

8.10 MDE postings

8.10.1 Create posting (DLG=MDEPRO.INSERT, COPY)

Use the BAPI calls described in this section to create, copy and delete MDE records for postings.

Tables

Table	Key field	Description
event	Internal ID MDEPRO.VERWEIS	Internal ID of the MDE record of the posting (PK)

BAPI call

Identification	Content / {type}	Description
DLG	MDEPRO.INSERT	Create MDE posting

Identification	Content / {type}	Description
	MDEPRO.COPY	Copy MDE posting
MDEPRO.MNR	{C20}	HYDRA workplace that makes the posting (PK)
MDEPRO.DATB	{MM/DD/YYYY}	Start date of the posting (PK)
MDEPRO.DATE	{MM/DD/YYYY}	End date of the posting (PK)
MDEPRO.SART	{C1}	Record type of the posting (PK) P: log record N: end of shift record
MDEPRO.MST	{N6}	only with MDEPRO.INSERT Machine status of posting (PK)
MDEPRO.VERWEIS	{N8}	only with MDEPRO.COPY Internal ID of the posting (PK) that you want to copy.
…	…	For further fields, that are not MANDATORY, refer to section 8.10.3 List of fields (acronyms) for the MDEPRO dialog

Return

Identification	Content / {type}	Description
MDEPRO.VERWEIS	{N8}	Returns the internal ID of the posting created.

Validation checks

Error codes	Description
90	Workplace/machine (MDEPRO.MNR) must be available in the workplace/machine configuration.
1662	The parameter with the ID MDEPRO.SART must have the value "N" or "P".
1803	Check is only performed if "BEARB" is also passed and BEARB does not equal "HYDRA". Responsibility area authorization for the workplace is not available (MDEPRO.MNR).
1952	Start date (MDEPRO.DATB / MDEPRO.ZEIB) must not be greater than end date (MDEPRO.DATE / MDEPRO.ZEIE).
1661	You must specify the parameter with the ID MDEPRO.SART.

Error codes	Description
1661	You must specify the parameter with the ID MDEPRO.MNR.
1661	You must specify the parameter with the IDs MDEPRO.DATB and MDEPRO.DATE.

8.10.2 Edit posting (DLG=MDEPRO.UPDATE, DELETE, LOCK, UNLOCK)

Use the BAPI calls described in this section to edit MDE records of postings.

Tables

Table	Key field	Description
event	Internal ID MDEPRO.VERWEIS	Internal ID of the posting

BAPI call

Identification	Content / {type}	Description
	MDEPRO.UPDATE	Change posting
	MDEPRO.DELETE	Delete posting
	MDEPRO.LOCK	Lock posting for editing MUST be performed before MDEPRO.UPDATE
	MDEPRO.UNLOCK	Unlock posting after editing MUST be performed after MDEPRO.UPDATE
MDEPRO.VERWEIS	{N8}	Internal ID of the posting (PK)
MDEPRO.SART	{C1}	Record type of the posting P: log record N: end of shift record
…	…	MDEPRO.UPDATE: For further fields, that are not MANDATORY, refer to section 8.10.3 List of fields (acronyms) for the MDEPRO dialog

Return

Identification	Content / {type}	Description
event	Internal ID	Internal ID of the MDE record of the posting

Identification	Content / {type}	Description
	MDEPRO.VERWEIS	

Validation checks

Error codes	Description
90	Workplace (MDEPRO.MNR) must be available in the workplace/machine configuration.
101	UPDATE,LOCK,UNLOCK The record of the posting (if the parameter MDEPRO.VERWEIS is specified) must be available in the database.
1803	Check is only performed if "BEARB" is also passed and BEARB does not equal "HYDRA". Responsibility area authorization for the workplace is not available (MDEPRO.MNR).
1952	Start date (MDEPRO.DATB / MDEPRO.ZEIB) must not be greater than end date (MDEPRO.DATE / MDEPRO.ZEIE).
1666	The data record is currently locked by another user. (UPDATE, DELETE,LOCK).
1661	UPDATE.LOCK,UNLOCK You must specify the parameter with the ID MDEPRO.VERWEIS.
1661	UPDATE You must specify the parameter with the ID MDEPRO.SART.
1661	UPDATE You must specify the parameter with the ID MDEPRO.MNR.
1661	You must specify the parameter with the IDs MDEPRO.DATB and MDEPRO.DATE.
1662	The parameter with the ID MDEPRO.SART must have the value "N" or "P".

		dialog

8.10.3 List of fields (acronyms) for the MDEPRO dialog

Identification	Description	DB type	Length

Identification	Description		DB type	Length
MDEPRO.MNR	workplace		char	20
MDEPRO.KST	Cost center of the workplace		char	10
MDEPRO.VERWEIS	Unique ID of the log record, important for the editing of log records		sqlserial	7
MDEPRO.SART	Record type P: log record N: end of shift record Original records of corrections: 1: end of shift record 2: log record		char	1
MDEPRO.ZEIB MDEPRO.MSZEIB	or	start date (time) of status	integer	7
MDEPRO.DATB MDEPRO.MSDATB	or	start date (date) of status	sqldate	7
MDEPRO.ZEIE MDEPRO.MSZEIE	or	end time (time) of status	integer	7
MDEPRO.DATE MDEPRO.MSDATE	or	end time (date) of status	integer	7
MDEPRO.MSDAUER	Duration of the status, synchronized with the shift calendar		integer	7
MDEPRO.MST	Machine status (table stoer_tabelle)		smallint	7
MDEPRO.BMKNR	Resource performance account of machine status		smallint	7
MDEPRO.STNR	Refers to status text (table stoer_texte)		smallint	7
MDEPRO.SZY	Target cycle at time of logging		decimal	7
MDEPRO.TLG	Partitioning at time of logging		decimal	7
MDEPRO.SKNR	Shift number (1...4)		smallint	7
MDEPRO.SKZEIB	Beginning of shift		integer	7
MDEPRO.SKZEIE	End of shift		integer	7
MDEPRO.SKPDAUER	Total of breaks in this shift		smallint	7
MDEPRO.SKDATB	Beginning of shift – date		sqldate	7
MDEPRO.EGR:GUTB	Yield in base quantity unit Note: delta quantity (no absolute value)		decimal	7
MDEPRO.EGR:GUTP	Yield in primary quantity unit Note: delta quantity (no absolute value)		decimal	7

Identification	Description	DB type	Length
MDEPRO.EGR:GUTS	Yield in secondary quantity unit Note: delta quantity (no absolute value)	decimal	7
MDEPRO.EGR:GUTT	Yield in tertiary quantity unit Note: delta quantity (no absolute value)	decimal	7
MDEPRO.EGR:AUSB	Scrap in base quantity unit Note: delta quantity (no absolute value)	decimal	7
MDEPRO.EGR:AUSP	Scrap in primary quantity unit Note: delta quantity (no absolute value)	decimal	7
MDEPRO.EGR:AUSS	Scrap in secondary quantity unit Note: delta quantity (no absolute value)	decimal	7
MDEPRO.EGR:AUST	Scrap in tertiary quantity unit Note: delta quantity (no absolute value)	decimal	7
MDEPRO.EGR:NCHB	Rework quantity in base quantity unit Note: delta quantity (no absolute value)	decimal	7
MDEPRO.EGR:NCHP	Rework quantity in primary quantity unit Note: delta quantity (no absolute value)	decimal	7
MDEPRO.EGR:NCHS	Rework quantity in secondary quantity unit Note: delta quantity (no absolute value)	decimal	7
MDEPRO.EGR:NCHT	Rework quantity in tertiary quantity unit Note: delta quantity (no absolute value)	decimal	7
MDEPRO.EGR:PRBB	Problem quantity in base quantity unit Note: delta quantity (no absolute value)	decimal	7
MDEPRO.EGR:PRBP	Problem quantity in primary quantity unit Note: delta quantity (no absolute value)	decimal	7
MDEPRO.EGR:PRBS	Problem quantity in secondary quantity unit Note: delta quantity (no absolute value)	decimal	7
MDEPRO.EGR:PRBT	Problem quantity in tertiary quantity unit Note: delta quantity (no absolute value)	decimal	7
MDEPRO.EGE:GUTP	Base quantity unit	char	3
MDEPRO.EGE:GUTP	Primary quantity unit	char	3
MDEPRO.EGE:GUTS	Secondary quantity unit	char	3
MDEPRO.EGE:GUTT	Tertiary quantity unit	char	3
MDEPRO.EGR:HUB	Machine strokes Note: delta quantity (no absolute value)	decimal	7
MDEPRO.CTR:1	Counter 1: yield (parts) Note: absolute value (accumulated for the shift)	integer	7
MDEPRO.CTR:2	Counter 2: machine strokes Note: absolute value (accumulated for the shift)	integer	7
MDEPRO.CTR:3	Counter 3: scrap Note: absolute value (accumulated for the shift)	integer	7

Identification	Description	DB type	Length
MDEPRO.CTR:4	Counter 4 Note: absolute value (accumulated for the shift)	integer	7
MDEPRO.CTR:5	Counter 5 Note: absolute value (accumulated for the shift)	integer	7
MDEPRO.CTR:6	Counter 6 Note: absolute value (accumulated for the shift)	integer	7
MDEPRO.BEM	Comment The comment is created and edited in a separate table: event_dlg_data The table is identified via the ID MDEPRO.VERWEIS <u>Storage:</u> Comment = event_dlg_data.dlg_data Identified via the ID ereignis. dlg_data_verweis = event_dlg_data.verweis or event_dlg_data.ev_verweis= ereignis.verweis	Char	60

9 HYDRA Production Data Manager MPL - Data Collection

9.1 Please note for the posting dialogs described

The HYDRA-MPL dialogs are only useful in connection with the corresponding HYDRA-BDE dialogs (please also see the associated BDE documentation).

All mandatory fields are highlighted in gray. All other fields are optional and are only processed if they are filled out.

9.2 Batch Postings

9.2.1 Batch change (DLG=CA_WL)

<table>
<tr><th>ID</th><th>Type/max. field length</th><th colspan="2">Description</th></tr>
<tr><td>ANR=</td><td>C16</td><td colspan="2">Order number with operation number</td></tr>
<tr><td>MNR=</td><td>N8/C8</td><td colspan="2">Machine number (numeric/alphanumeric)</td></tr>
<tr><td>PNR=</td><td>C10</td><td rowspan="2">alter-
native</td><td>Personnel number</td></tr>
<tr><td>KNR=</td><td>C10</td><td>Badge number</td></tr>
<tr><td>CNR=</td><td>C20</td><td colspan="2">Batch number</td></tr>
<tr><td>EGR:GUT=</td><td>DEC</td><td colspan="2">Recorded yield</td></tr>
<tr><td>EGE:GUT=</td><td>C4</td><td colspan="2">Unit of yield</td></tr>
<tr><td>KLASSE=</td><td>C1</td><td colspan="2">“G” yield
“A” scrap
“O” on hold
“N” rework</td></tr>
<tr><td>STA=</td><td>C1</td><td colspan="2">Status of output batch
F= with residual quantity (default with KLASSE=G)
S= blocked (default with KLASSE=A)</td></tr>
<tr><td>ZLO=</td><td>C12</td><td colspan="2">Target location, output batch</td></tr>
<tr><td>EGG:AUS=</td><td>NUM</td><td colspan="2">Scrap reason KLASSE=A
KLASSE=O EGG:PRB
KLASSE=N EGG:NCH</td></tr>
</table>

ID	Type/max. field length	Description
TPE=	C10	Transport unit
LHW=	C20	Note, output batch
CALT1= to CALT20=	-	Alternative batch number 1-20 max. field length: CALT1-4 - C20 CALT5-14 - C40 CALT15-18 - C100 CALT19-20 - C512
ATTR:1 = to ATTR:10 =	--	The definition of which values are filed in additional attributes is configured for each material type in HYDRA
ATTR:101= to ATTR:140=	C40	Alphanumeric batch attributes
ATTR:201= to ATTR:220=	NUM	Numeric batch attributes
ATTR:301= to ATTR:320=	DEC	Decimal batch attributes
CNR.USRFLD	C8	User field key
CNR.FU:1 – CNR.FU:66	-	User fields

Please note:

There are other, similar dialogs regarding output batches (CA_AN & CA_AB) in HYDRA. They are only required to keep data within the event maintenance function at the console.

To change output batches, the CA_WL dialog always has to be sent! Except for the batch number (CNR) all information refers to the currently active HYDRA batch, which is to be logged off with this posting. The batch number defines which ID the HYDRA batch gets that is to be started, once the current HYDRA batch has been completed.

The same values are supported for log off operation (DLG=A_AB) and interrupt OP (DLG=A_UN).

1) Example: Change batch and indicate the quantity

```
DLG=CA_WL|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100451210200|CNR=X912536177|EGR:
GUT=1|
```

2) Example: Change batch and indicate the quantity and unit

```
DLG=CA_WL|USR=2106|DAT=02/17/2000|ZEI=47972|KNR=111111|ANR=AAA2100451210200|CNR=X912536177|EGR:
GUT=1|EGE:GUT=ST|
```

9.2.1.1 Log output batch on (DLG=CA_AN)

ID	Type/max. field length	Description	
ANR=	C16	Order number with operation number	
MNR=	N8/C8	Machine number (numeric/alphanumeric)	
PNR=	C10	alter-native	Personnel number
KNR=	C10		Badge number
CNR=	C20	Batch number	
CALT1= to CALT20=	-	Alternative batch number 1-20 max. field length: CALT1-4 - C20 CALT5-14 - C40 CALT15-18 - C100 CALT19-20 - C512	

Please note: Only possible if the operation runs on the machine.

9.2.1.2 Log output batch off (DLG=CA_AB)

ID	Type/max. field length	Description	
ANR=	C16	Order number with operation number	
MNR=	N8/C8	Machine number (numeric/alphanumeric)	
PNR=	C10	alter-native	Personnel number
KNR=	C10		Badge number
CNR=	C20	Batch number	
EGR:GUT=	DEC	Recorded yield	
EGE:GUT=	C4	Unit of yield	
KLASSE=	C1	“G” yield “A” scrap “O” on hold “N” rework	

ID	Type/max. field length	Description
STA=	C1	Status of output batch F= with residual quantity (default with KLASSE=G) S= blocked (default with KLASSE=A)
EGG:AUS=	NUM	Scrap reason KLASSE=A KLASSE=O EGG:PRB KLASSE=N EGG:NCH
ZLO=	C12	Destination, output batch
TPE=	C10	Transport unit
LHW=	C20	Note, output batch
CALT1= to CALT20=	-	Alternative batch number 1-20 max. field length: CALT1-4 - C20 CALT5-14 - C40 CALT15-18 - C100 CALT19-20 - C512
ATTR:1 = to ATTR:10 =	--	The definition of which values are filed in additional attributes is configured for each material type in HYDRA
ATTR:101= to ATTR:140=	C40	Alphanumeric batch attributes
ATTR:201= to ATTR:220=	NUM	Numeric batch attributes
ATTR:301= to ATTR:320=	DEC	Decimal batch attributes
CNR.USRFLD	C8	User field key
CNR.FU:1 – CNR.FU:66	-	User fields

Please note: Only possible if the operation runs on the machine.

9.2.2 Log input batch on (DLG=CE_AN)

ID	Type/max. field length	Description
ANR=	C16	Order number with operation number

ID	Type/max. field length	Description	
MNR=	N8/C8	Machine number (numeric/alphanumeric)	
PNR=	C10	alter-native	Personnel number
KNR=	C10		Badge number
CNR=	C20	Batch number of the input batch	
SLP	C10	Bill of material item of component	
ATK	C40	Article	

9.2.3 Log input batch off (DLG=CE_AB)

ID	Type/max. field length	Description	
ANR=	C16	Order number with operation number	
MNR=	N8/C8	Machine number (numeric/alphanumeric)	
PNR=	C10	alter-native	Personnel number
KNR=	C10		Badge number
CNR=	C20	Batch number of input batch	
LHW=	C20	Notes on the input batch to be logged off	
STA=	C1	Status of the input batch to be logged off F = with residual quantity S = blocked A = processed	
EGR:REST=	DEC	Residual quantity of the input batch	
EGE:REST=	C4	Unit	
SLP	C10	Bill of material item of the component	
EGR:VERB=	DEC	Consumption of input batch As an alternative to EGR:REST/EGE:REST	
EGE:VERB=	C4	Unit As alternative to EGR:REST/EGE:REST	

If a batch is registered several times in parallel as an input batch and the consumption is to be recorded, the fields EGR:VERB and EGE:VERB must be used for customer-specific implementations.

9.2.4 Repost batch (DLG=C_UMB)

ID	Type/max. field length	Description
ZLO=	C12	Destination, output batch
MNR=	N8/C8	Machine number (numeric/alphanumeric) Only if BEHÄLTERVERWALTUNG (container management) is active
CNR=	C20	Batch
DLL=		Batch number via entry
MOD=	C1	K mode: customer batch mode (objective: assignment of another customer batch) L mode: Delete mode V mode: “forward” mode R mode: “backward” mode
TECHINFO=	C20	Technical info: entry/barcode (mode K only) Will no longer be supported from MPL 7.2.1 on
ATTR:1=	--	G mode: consecutive number, incremented by the terminal
ATTR:2=	--	G mode: Priority R mode: Index from continuous counter
HSDAT=	mm/dd/yyyy	All batches of the same date/time (mode R only)
HSZEIT=	seconds	All batches of the same date/time (mode R only)
STA=	C1	Status (F)free or (S)blocked, default = F

9.2.5 Goods receipt batch (DLG=C_GEN)

<table>
<tr><th>ID</th><th>Type/max. field length</th><th colspan="2">Description</th></tr>
<tr><td>ANR=</td><td>C16</td><td colspan="2">Order number with operation number</td></tr>
<tr><td>MNR=</td><td>N8/C8</td><td colspan="2">Machine number (numeric/alphanumeric)</td></tr>
<tr><td>PNR=</td><td>C10</td><td rowspan="2">alter-native</td><td>Personnel number</td></tr>
<tr><td>KNR=</td><td>C10</td><td>Badge number</td></tr>
<tr><td>CNR=</td><td>C20</td><td colspan="2">Batch number (*)</td></tr>
<tr><td>EGR:GUT=</td><td>N8</td><td colspan="2">Batch quantity</td></tr>
<tr><td>EGE:GUT=</td><td>C3</td><td colspan="2">Unit of batch quantity</td></tr>
<tr><td>ATK=</td><td>C40</td><td colspan="2">Material/article</td></tr>
<tr><td>HZTYP=</td><td>C10</td><td colspan="2">Material type according to the configuration in HYDRA</td></tr>
<tr><td>KLASSE=</td><td>C1</td><td colspan="2">"G" yield or "A" scrap</td></tr>
<tr><td>EGG:AUS=</td><td>N3</td><td colspan="2">Scrap reason</td></tr>
<tr><td>ZLO=</td><td>C12</td><td colspan="2">Destination</td></tr>
<tr><td>TPE=</td><td>C10</td><td colspan="2">Transport unit</td></tr>
<tr><td>LHW=</td><td>C20</td><td colspan="2">Note, goods receipt batch</td></tr>
<tr><td>STA=</td><td>C1</td><td colspan="2">Batch status (F = free, S = blocked)</td></tr>
<tr><td>QST:2=</td><td>C1</td><td colspan="2">Quality status (G = blocked, F = free)</td></tr>
<tr><td>QST=</td><td>C1</td><td colspan="2">Manual quality status (G = blocked, F = free)</td></tr>
<tr><td>MATST=</td><td>C1</td><td colspan="2">Material status (e.g. V = packed)</td></tr>
<tr><td>CNR:ALT1 – 5 =</td><td>C20 -40</td><td colspan="2">Alternative batch number 1-5
maximum lenght:
CNR:ALT1-4 - C20
CNR:ALT5 - C40</td></tr>
<tr><td>MCNR=</td><td>C20</td><td colspan="2">Mother batch number (e.g. reference from collective batch)</td></tr>
<tr><td>ATTR:1 = to ATTR:11 =</td><td>--</td><td colspan="2">The definition of which values are filed in additional attributes is configured for each material type in HYDRA</td></tr>
</table>

ID	Type/max. field length	Description
ATTR:101= to ATTR:140=	C40	Alphanumeric batch attributes
ATTR:201= to ATTR:220=	NUM	Numeric batch attributes
ATTR:301= to ATTR:320=	DEC	Decimal batch attributes
RESART=	C4	Reservation type (OP = operation, AK = order)
RESVAL=	C40	Reservation value (e.g. operation number)
RESBEM=	C100	Reservation comment
CNR:BREITE=	DEC	Material width (conversion factor for MPLRF-BP)
CNR:RFAGVFA=	DEC	Mass per unit area (conversion factor for MPLRF-BP)
CNR:RFSTKF=	DEC	Surface per piece (conversion factor for MPLRF-BP)
CNR:SAPCNR=	C10	PPS batch number
CNR:TECHINFO =	C20	Technical info
CNR:VVDAT=	DATUM	Availability date
CNR:VVZEI=	ZEIT	Availability time
CNR:VFDAT=	DATUM	Expiry date
CNR:VFZEI=	ZEIT	Expiry time
CNR:WDAT=	DATUM	Warning date
CNR:WZEI=	ZEIT	Warning time
CNR:LAGORT=	C12	PPS storage location
CNR:EXTCNR=	C10	External batch number

(*) = The batch number does not have to be indicated if automatic batch number generation is enabled.

9.2.6 Consumption posting (DLG=A_VERB)

ID	Type/max. field length	Description

ID	Type/max. field length	Description	
MNR=	N8/C8	Machine number (numeric/alphanumeric) where the OP is running	
ANR=	C16	Order number with operation number (only running OP possible)	
SLP	C10	Bill of material item of the component	
ATK=	C40	Material/article of the component	
PNR=	C10	alter-native	PNR=
KNR=	C10		KNR=
EGR:VERB=	N8	Consumption quantity	
EGE:VERB=	C3	Unit of consumption quantity	
CNR=	C20	Batch number (*) for batch-related input material	

* The batch number is only optionally taken over to the consumption movement.

9.2.7 Create/change batches (DLG=CNR.MODIFY)

By way of the “CNR.MODIFY“ Bapi HYDRA batches can be created or existing batches may be changed.

ID	Type/max. field length	Description
CNR.CNR=	C20	Internal batch number
CNR.DLL=	C20	Run-through batch number
CNR.CNR:Z=	C20	Target batch number with CNR.COPY
CNR.ATK=	C40	Material number
CNR.ATKBEZ=	C40	Material designation
CNR.SGR:GUT=	DEC	Batch quantity
CNR.SGR:REST=	DEC	Residual quantity of the batch
CNR.SGE:GUT=	C3	Unit of the batch quantity
CNR.MNR=	N8/C8	Machine number
CNR:ATTR:11=	C40	Order number with operation number

ID	Type/max. field length	Description
CNR.STA=	C1	Batch status (F = free, S = blocked)
CNR.CKL=	C1	Batch class G = yield A = scrap O = on hold N = rework
CNR.TST=	C1	Transport status (L = delivered)
CNR.QST=	C1	Quality status (G = blocked, F = free)
CNR.QSTMANU=	C1	Manual quality status (G = blocked, F = free)
CNR.MATST=	C1	Material status (e.g. V = packed)
CNR.MATTYPART=	C10	Material type
CNR.OPT:REST=	C1	J – "batch has still got residual quantity" indicator N – batch has no residual quantity
CNR.FIR=	C4	Company
CNR.HSDAT=	DATUM	Manufacturing date
CNR.HSZEI=	ZEIT	Manufacturing time
CNR.VVDAT=	DATUM	Availability date
CNR.VVZEI=	ZEIT	Availability time
CNR.VFDAT=	DATUM	Expiry date
CNR.VFZEI=	ZEIT	Expiry time
CNR.WDAT=	DATUM	Warning date
CNR.WZEI=	ZEIT	Warning time
CNR.CSTWDAT=	DATUM	Date of the last status change
CNR.CSTWZEI=	ZEIT	Time of the last status change
CNR.MATPUF=	C12	Material buffer
CNR.MATTYP=	C10	Material type
CNR.TPE=	C10	Transport unit
CNR.BEM=	C20	Batch note

ID	Type/max. field length	Description
CNR.PNR=	C10	Personnel number
CNR.TECHINFO=	C20	Technical info
CNR.EGG:AUS=	N8	Scrap reason
CNR.GR=	N8	Scrap reason (synonymous with CNR.EGG:AUS)
CNR.GRTXT=	N8	Reference to scrap reason text
CNR.USR=	N8	User number
CNR.LAGORT=	C12	PPS storage location
CNR.LAGPZ=	C12	PPS storage bin
CNR.SAPCNR=	C10	PPS batch number
CNR.RESART=	C4	Reservation type (OP = operation, AK = order)
CNR.RESVAL=	C40	Reservation value (e.g. operation number)
CNR.RESBEM=	C100	Reservation comment
CNR.BREITE=	DEC	Material width (conversion factor for MPLRF-BP)
CNR.RFAGVFA=	DEC	Mass per unit area (conversion factor for MPLRF-BP)
CNR.RFSTKF=	DEC	Surface per piece (conversion factor for MPLRF-BP)
CNR.EGR:1 – 6=	DEC	Quantity, activity 1-6
CNR.RGR:1 – 6 =	DEC	Residual quantity, activity 1-6
CNR.EGE:1 – 6 =	C3	Unit, activity 1-6
CNR.CNR:ALT1 – 20 =	C20	Alternative batch number 1-20
CNR.EXTCNR=	C10	External batch number
CNR.MCNR=	C20	Mother batch number (e.g. reference from collective batch)
CNR.AUART=	C5	Order type
CNR.UMRFAKTP:Z=	DEC	Denominator – conversion factor relating to OP quantity units
CNR.UMRFAKTP:N=	DEC	Numerator – conversion factor relating to OP quantity units
CNR.UMRFAKTS:Z=	DEC	Denominator – conversion factor relating to OP quantity units
CNR.UMRFAKTS:N=	DEC	Numerator – conversion factor relating to OP quantity units
CNR.UMRFAKTT:Z=	DEC	Denominator – conversion factor relating to OP quantity units

ID	Type/max. field length	Description
CNR.UMRFAKTT:N=	DEC	Numerator – conversion factor relating to OP quantity units
CNR.ZUORD:1 – 6 =	C1	Indicator to assign “MPL activity account → ADE quantity unit (e.g. P = primary quantity)
ATTR:101 - ATTR:140	C40	Alphanumeric batch attributes
ATTR:201 - ATTR:220	NUM	Numeric batch attributes
ATTR:301 - ATTR:320	DEC	Decimal batch attributes
CNR.USRFLD	C8	User field key
CNR.FU:1 – CNR.FU:66	-	User fields

9.2.8 Goods movement (DLG=C_MBEW)

Using the command DLG=C_MBEW batches may be created or reposted. Moreover, the quantity and status of an existing batch may be changed as well.

ID	Type/max. field length	Description	
ANR=	C16	Order number with operation number	
MNR=	N8/C8	Machine number (numeric/alphanumeric)	
PNR=	C10	alternative	Personnel number
KNR=	C10		Badge number
CNR=	**C20**	**Batch number (*)**	
SGR:GUT= / EGR:GUT= / RGR:GUT=	DEC	Batch quantity RGR:GUT – change residual quantity only	
SGE:GUT= / EGE:GUT= / RGE:GUT=	C3	Unit of batch quantity RGE:GUT – change residual quantity only	
ATK=	C40	Material/article	
ATKBEZ=	C40	Material designation	

ID	Type/max. field length	Description
HZTYP=	C10	Material type of the batch
KLASSE=	C1	"G" yield or “A” scrap
EGG:AUS=	N3	Scrap reason with KLASSE=A
STA=	C1	Status (F) free or (S) blocked, by default = F
QST=	C1	Manual quality status (G – blocked, F – free)
MATST=	C1	Material status (e.g. V = packed)
ZLO=	C12	Destination
CNR:LAGORT=	C12	PPS storage location
TPE=	C10	Transport unit
LHW= / BEM=	C20	Note/comment
ATTR:1 = to ATTR:11 =	--	The definition of which values are filed in additional attributes is defined for each material type in HYDRA
CNR:ALT1 – 5 =	C20	Alternative batch number 1-5
EXTCNR=	C10	External batch number
MCNR=	C10	Mother batch number (e.g. reference from collective batch)
CNR:HSDAT=	DATUM	Manufacturing date
CNR:HSZEI=	ZEIT	Manufacturing time
CNR:VVDAT=	DATUM	Availability date
CNR:VVZEI=	ZEIT	Availability time
CNR:VFDAT=	DATUM	Expiry date
CNR:VFZEI=	ZEIT	Expiry time
CNR:WDAT=	DATUM	Warning date
CNR:WZEI=	ZEIT	Warning time
SAPCNR=	C10	PPS batch number
RESART=	C4	Reservation type (OP = operation, AU = order)
RESVAL=	C40	Reservation value (e.g. operation number)
RESBEM=	C100	Reservation comment

ID	Type/max. field length	Description
CNR:BREITE=	DEC	Material width (conversion factor with MPLRF-BP)
CNR:RFAGVFA=	DEC	Mass per unit area (conversion factor with MPLRF-BP)
CNR:RFSTKF=	DEC	Surface per piece (conversion factor with MPLRF-BP)
UMRFAKTP:Z=	DEC	Denominator – conversion factor relating to OP quantity units
UMRFAKTP:N=	DEC	Numerator – conversion factor relating to OP quantity units
UMRFAKTS:Z=	DEC	Denominator – conversion factor relating to OP quantity units
UMRFAKTS:N=	DEC	Numerator – conversion factor relating to OP quantity units
UMRFAKTT:Z=	DEC	Denominator – conversion factor relating to OP quantity units
UMRFAKTT:N=	DEC	Numerator – conversion factor relating to OP quantity units
ZUORD:1 – 6 =	C1	Indicator to assign “MPL activity account → ADE quantity unit (e.g. P = primary quantity)
MENGE1 – MENGE6 =	DEC	Activity 1-6
EINH1 – EINH6=	C3	Unit of activity 1-6
ATTR:101= to ATTR:140=	C40	Alphanumeric batch attributes
ATTR:201= to ATTR:220=	NUM	Numeric batch attributes
ATTR:301= to ATTR:320=	DEC	Decimal batch attributes

9.2.9 Change batch status (DLG=C_STA)

The status of an existing batch can be changed using the DLG=C_STA command.

<table>
<tr><th>ID</th><th>Type/max. field length</th><th colspan="2">Description</th></tr>
<tr><td>ANR=</td><td>C16</td><td colspan="2">Order number with operation number</td></tr>
<tr><td>MNR=</td><td>N8/C8</td><td colspan="2">Machine number (numeric/alphanumeric)</td></tr>
<tr><td>PNR=</td><td>C10</td><td rowspan="2">alternative</td><td>Personnel number</td></tr>
<tr><td>KNR=</td><td>C10</td><td>Badge number</td></tr>
</table>

ID	Type/max. field length	Description
CNR=	**C20**	**Batch number**
KLASSE=	C1	"G" yield or “A” scrap
STA=	C1	Status (F) free or (S) blocked, by default = F
QST=	C1	Manual quality status (G – blocked, F – free)
MATST=	C1	Material status (e.g. V = packed)
EGG:AUS= / GR=	N3	Scrap reason with KLASSE = A

9.2.10 Input batch change (DLG=CE_WL)

Input batches can be changed using the command DLG=CE_WL. Input batches are always changed with respect to the bill of material.

ID	Type/max. field length	Description	
ANR	C16	Order number with operation number	
MNR	N8/C8	Machine number (numeric/alphanumeric)	
PNR	C10	alter-native	Personnel number
KNR	C10		Badge number
CNRAB	C20	Batch number of the input batch to be logged **OFF**	
CNR	C20	Batch number of the input batch to be logged **ON**	
SLP	C10	Bill of material item of the component (if not transferred the BOM item (SLP) is set to 0)	
ATK	C40	Article	
LHW	C20	Note on the input batch to be logged off	
STA	C1	Status of the input batch to be logged off	

ID	Type/max. field length	Description
		F= with residual quantity S= blocked A= processed
EGR:REST	N8	Remaining quantity of the input batch to be logged off
EGE:REST	C4	Unit of remaining quantity
EGR:VERB	N8	Consumption of input batch Alternative to EGR:REST/EGE:REST
EGE:VERB	C4	Unit Alternative to EGR:REST/EGE:REST

9.3 Reading of MPL Data

9.3.1 Material list / batch information

The material/batch list is provided by the command DLG=LIST;13 and filed in the HYDRADIR\spool\ directory.

Structure of dialog data:

"DLG=LIST;13|DATEI={file name }|DAT=...|ZEI=...|USR=...|MOD=..."

a.) Reading of the material list for machine and order:

Parameter: MOD=M|MNR=Machine|ANR=Order number

→ Machine is required for still running batches
→ Order is required for planned materials (if necessary with batch assignment) of the order

b.) Reading of the batch info:

Parameter: MOD=L|CNR=batch number

c.) Reading of preceding batches (output batches) of an order **(including the current output batch):**

Parameter: MOD=A|ANR=order number

d.) Running batches of all machines of the terminal

Parameter: MOD=T

The list includes the following data:

ID	Field designation	Description
MNR	Machine	Machine number
ANR	Order	Order and OP [and type]
ATK	Article	Input material number (**)
ATKBEZ	Des. of final article	Designation of input material (**)
HZTYP	Semi-finished article type	Semi-finished article type (**)
SLP	BOM item	Bill of material item (*)
EMENGE	Required quantity	Quantity of required material for each unit of the output material (*)
EINH	Unit	Unit of required quantity (**)
TECHINFO	Techn.Info	Technical info relating to mat. (**)
DLL	Cs. ba. No.	Run-through batch number
DLLKZ	Cs.-batch ID	Run-through batch ID
CNR	Batch number	Batch number (***)
LHW	Batch note	Batch note (***)
TPE	TPUnit [spec.]	Transport unit (***)
HSDAT	Manuf. date	Manufacturing: date (***)
HSZEI	Manuf. time	Manufacturing: time (***)
VVDAT	Date availability	Availability: date (***)
VVZEI	Time availability	Availability: time (***)
VFDAT	Date expiry	Date of expiry: date (***)
VFZEI	Time expiry	Date of expiry: time (***)
WARNDAT	Warning date	Warning date (***)
WARNZEI	Warning time	Warning time (***)
SGR:GUT	Target	Batch quantity (***) = Target quantity for input batches = Recorded quantity for output batchespos. booking as output batch
SGR:REST	Residue	Remaining quantity of the batch (***) = Actual quantity of the batch: - when logging output batch off or transferring it SS == quantity of batch - neg. booking as input batch
RMENGEKZ	RQ	Remaining quantity indicator (***)

ALTMENGE	Alt.qty	Alternative quantity (***)
ALTEINH	Alt.unit	Alternative quantity unit (***)
ZLO	Destination	Destination (***)
CST	Batch status	Batch status (***)
CSTWDAT	Batch ch.date	Date of status change (***)
CSTWZEI	Batch ch.time	Time of status change (***)
PNR	PersNo.	Personnel number (***)
ABKZ	LogoffID	Logoff ID (****) “X“ for forced logoff
AGCST	OPBST	Batch status of operation: 0 Batch is currently not logged on to order (preceding batches) 1 Batch is currently logged on to order
ATTR:1 to ATTR:11		Customer-specific Batch attribute 1 to Batch attribute 11
HZBEZ	Material type des.	
MINLGZ	Min. storage time	
WARNGRENZ	Warning limit	
VFALLGRENZ	Expiry limit	
HZART	SF type	
LOSGR	Lot size	
LOSEINH	Unit	
LAY	Layout	
TICKETANZ	Number of tickets	
OPT:VFCNRA	Expiry date from input batch	
OPT:VFCNRE	Expiry date for output batch	
OPT:MULTICNR	Can be logged on several times	
OPT:INBESTVERB	Invent. char.	
PROZAGAB	Tolerance limit AGAB	
UTGAGAB	abs. tolerance limit AGAB	
OPT:VERE	Hand batch no. down	
OPT:AGZGVERB	REB-OP-Ref.	OP reference when batches are reposted
OPT:AGZGGEN	GR-OP-Ref.	OP reference when goods receipt batches are created
PLAUS:MATOK	Plaus. MatOK	

PLAUS:EMATOK	Plaus. EMatOK	
OPT:CNREA	Valid for 1 outp. batch	
LOCAL:1 to LOCAL:2	LOCAL	
CKL	Class	
EGG:AUS	Scrap reason	
TST	Transport status	
ART	Indic.	
SCHNEIDNR	Cut number	
SLOS	Collective batch	
LOSANZ	No. of batches	
PBANDNR	Prod. line no.	
CNR:RESVAL	Reservation	
TR:BREITE	Width of DR	
TR:ANZ_GES	Total no. outp. batches	
ISTATK	Material	
ISTATKBEZ	Material des.	
ATKDIFF	Art. diff.	Planned article and actual article are different (J/N/F) J: different N: not different F: free component
ISTANR	Order	
ATK:2	2nd material	
BEDARF	Requirement quantity	
VERB	Consumption	
TREST	Theo. remain. requirements	
REST:VZ	Sign	
STA_KOMBI	Combined status	
SAPCNR	SAP batch	
OPT:VGRCNR	Required qty. batch-related	
OPT:ERSBAR	Replaceable	
OPT:WZW	Bat.change req.	If the input batch of this component is changed the output batch needs to be changed.
STA:GEW	Entry of weight	
AGBEZ.|	Des. of final article	
LS1	Activity 1	
LS1:REST	Rem. quantity 1	
LS1:EINH	Unit 1	

.....	....	
LS6	Activity 6	
LS6:REST	Rem. quantity 6	
LS6:EINH	Unit 6	
EGR:LEN	Rework	
SLS	Level	
SLS:M	Mother level	
BEM	Comment	
ALT1 to ALT5	Altern. batch no. 1 to altern. batch no. 5	
SNR	Serial number	(*****)
HULEVEL	HU level	(*****)

(*) = Fields are not filled out for information relating to batches!!!

(**) = When it comes to information based on batches, the field is determined from the pool of batches, otherwise from material data.

(***) = Fields are not filled out if no batch is currently logged on for input material!!!

(****) = The batch number is determined subject to the command:

Command (a): Available batches and the materials planned for the order are read for the machine (without batch that is currently logged on).

Command (b): is transferred!

Command (c): output batch number

Example:

DLG=LIST;13|MOD=A|ANR=001122330020|DAT=10/11/2000|ZEI=40000|USR=101|DATEI= ./spool/ml_list.101|

(*****) = Only available as of HYDRA-MPL product version 7.2.5.

9.3.2 Material buffer

The list of material buffers is provided using the command DLG=LIST;49 and filed in the HYDRADIR\spool\ directory.

Structure of dialog data:

“DLG=LIST;49|DATEI={file name }|DAT=...|ZEI=...|USR=...|...“

Parameter: none

The list includes the following data:

ID	Field designation	Description
MATPUF	Material buffer	
BEZ	Designation	
OPT:PKORB	Wastebasket	
OPT:INBESTVERB	Calculate as inventory	
ART	Buffer model	
OPT:LAGVERB	Stock posting buffer	
OPT:VIRTLAG	Virtual stock buffer	
ABT	Department	
BER	Area	
KST	Cost center	
FIR	Company	
LAGORT	Storage location	
BEM	Comment	
DAUER	Duration	
TYP	Buffer type	
ZLO	Receiving storage location	
MPUFF	Material buffer	
HARCMATPUF	Hierarchical buffer	

Example:

DLG=LIST;49|DAT=02/11/2005|ZEI=40000|USR=101|DATEI= ./spool/mpuff_list.101|

9.3.3 Material types

Material types are provided using the command DLG=LIST;21 and filed in the HYDRADIR\spool\ directory.

Structure of dialog data:

"DLG= LIST;21|DATEI={file name}|DAT=...|ZEI=...|USR=..."

Parameter: AKRO=<dyn. user field columns e.g. AKRO=FU:23>

The list includes the following data:

ID	Field designation	Description
HZTYP	Material type	Material type
HZBEZ	Material type des.	Designation of the material type
MINLGZ	Min. storage time	Minimum storage time
WARNGRENZ	Warning limit	Warning limit

ID	Field designation	Description
VFALLGRENZ	Expiry limit	Expiry limit
HZART	SF type	Material type
LOSGR	Lot size	Standard lot size that is suggested as “default value” for a new generation
EINH	Unit	Unit
LAY	Layout	Layout that is used for ticket printing
TICKETANZ	Number of tickets	Number of tickets
OPT:VFCNRA	Expiry date from input batch	Flag whether the individual expiry date can be used to determine the expiry date of the output batch.
OPT:VFCNRE	Expiry date for output batch	Flag whether the least expiry date of appropriate batches is to be used to determine the individual expiry date.
OPT:MULTICNR	Can be logged on several times	Flag whether identical material types can be logged on several times.
OPT:INBESTVERB	Invent. char.	Flag for inventory management
PROZAGAB	Tolerance limit AGAB	Tolerance limit for OP off (%)
UTGAGAB	abs. tolerance limit AGAB	Absolute value for tolerance limit
OPT:VERE	Hand batch no. down	Flag for inheriting the batch number
OPT:AGZGVERB	REB-OP-Ref.	Flag whether an OP reference is required when reposting batches
OPT:AGZGGEN	GR-OP-Ref.	Flag whether an OP reference is required when generating batches
PLAUS:MATOK	Plaus. MatOK	Flag whether all input batches need to be logged on for booking a batch
PLAUS:EMATOK	Plaus. EMatOK	
OPT:CNREA	Valid for 1 outp. batch	Flag whether input batch exactly applies for 1 output batch
TPE	Transp. unit	Transport unit
OPT:VGRCNR	Required qty. batch-related	MPLRF: Material usage relating to output batch or operation
OPT:SNR	Serial numbers required	default=N (*)
OPT:GENHU	Generation of HU	default=N (*)

Example:

DLG=LIST;21|DATEI=./spool/hztyp_list.101|DAT=08/16/2007|ZEI=46918|USR=2101

(*) = Only available as of HYDRA-MPL product version 7.2.5

9.3.4 Transport unit

Transport units are provided using the command DLG=LIST;52 and filed in the HYDRADIR\spool\ directory.

Structure of dialog data:

“DLG=LIST;52|DATEI={file name}|DAT=...|ZEI=...|USR=...“

Parameter: none

The list includes the following data:

ID	Field designation	Description
HZTYP	Material type	Material type
TPE	Transp. unit	Transport unit
BEZ	Transp. unit des.	Designation of transport unit
VORG	Default transp. Unit	“Default” transport unit
ANZ	Lot size	Lot size
EINH	Unit	Unit
INBESTVERB	Invent. char.	Inventory flag
TPE.ANZ	No. of transport units	Number of transport units
BREITE	Width	Width dimension
BREITE:EINH	Width unit	Width dimension unit
HOEHE	Height	Height dimension
HOEHE:EINH	Height unit	Height dimension unit
LEN	Length	Length dimension
LEN:EINH	Length unit	Length dimension unit
GEW	Weight	Weight dimension
GEW:EINH	Weight unit	Weight dimension unit
COLOR:F	Foreground color	Foreground text color
COLOR:B	Background color	Background color

Example:

DLG=LIST;52|DATEI=./spool/tpe_list.101|DAT=08/16/2007|ZEI=46918|USR=2101

9.3.5 Component list

Components are provided by the command DLG=LIST;74 and filed in the HYDRADIR\spool\ directory.

Structure of dialog data:

“DLG=LIST;74|DATEI={file name}|DAT=...|ZEI=...|USR=...“

Parameter: ANR=order number

→ Order is required for planned materials (if necessary incl. batch assignment) of the order.

The list includes the following data:

ID	Field designation	Description
ART	Type	Component type
ATK	Article	Material number
BEZ	Designation	Customer-specific
BEZ:2	Designation	Customer-specific
SGR:GUT	Required quantity	Required quantity relating to the production of 1 article in primary quantity unit at the operation
MENGE:BED	Demand	Requirements quantity
SGE:GUT	Unit	Unit of required quantity
ATKBEZ	Article	Designation of the component (material, document, etc.)
SLP	BOM item	Position of components (bill of material item)
LAGORT	Storage location	Storage location SAP or storage type
LAGPZ	Rack compartment	Storage bin SAP
PATH	Path	For documents: reference to HY_PATH
DATEI	File name	For documents: file name
RESTYP	Type	**MAT = material**

Example:

DLG=LIST;74|ANR=TEST1|DATEI=./spool/komp_list.101|DAT=08/16/2007|ZEI=46918|USR=210
1

9.3.6 Batch attributes of a material type

Batch attributes are provided using the command DLG=LIST;67 and filed in the HYDRADIR\spool\ directory.

Table: hz_atgen

Structure of dialog data:

“DLG=LIST;67|DATEI={file name}|DAT=...|ZEI=...|USR=...|MOD=...| MOD2=...|ANR=…“

a.) Start via console

Parameter: MOD2=K

IDs are displayed without ":" → e.g. OPT_PRN

b.) Read batch attributes

Parameter: MOD=L|CNR=12345678

c.) Read operation attributes

Parameter: MOD=A|ANR=1234567890

The list includes the following data:

ID	Field designation	Description
KENNUNG	ID	ID of attribute
ATTR_VAL	Attribute value	Value of the attribute
MATTYP	Material type	Material type
IDX	Index	Completes the unique key (with HZ_TYP)
OPT:SIB	Display	Visible on console: J/N flag whether the attribute is to be displayed or not (applies for all console views)

ID	Field designation	Description
OPT:PRN	Print	Identifier whether the attribute is to be printed on the batch ticket or not (J/N)
POS:SIB	Display position	Position within display order
POS:PRN	Print position	Position within the ticket print order
TXT	Header	Description of the attribute
EINH	Trailer	Unit of the attribute
OPT:ERF	Enter	Identifier how the attribute is recorded (J/N)
POS:ERF	Entry position	Position within the order of entry
TYP:ERF	Type	Type of entry field - N → Numeric - C → Text field - F → Decimal value (This type might cause that a numeric value is saved in a text database field, if intended)
LEN:ERF	Entry length	Field length. The maximum length might be determined by the attribute database field in which the value is to be saved.
IDX:ERF	Entry index	Index of the material attribute to be recorded. Corresponds to the number of the batch attribute (relevant for entry = "J")
NKS:ERF	Decimal places	Number of decimal places to be recorded for decimal attributes

Example:

DLG=LIST;67|DATEI=spool/hyu2111.tmp|DAT=08/17/2007|ZEI=34897|USR=2111|MOD=A|ANR =005001720290

9.3.7 Batch logs (MPL-PRO)

Batch logs are provided via the DLG=CNRPROT.LIST command and filed in the HYDRADIR\spool\ directory.

Table: mpl_los_prot

License: MPL-PRO

Structure of dialog data:

"DLG=LIST;CNRPROT.LIST|DATEI={file name}|DAT=...|ZEI=...|USR=..."

Parameter: none

The list includes the following data:

ID	Field designation	Description
CNR	-	Batch number
VERWEIS	-	Reference to the data record
DLL	-	Run-through batch no.
GR	-	Reason
GRTEXT	-	Reason text
BEM	-	Comment/commentary
PNR	-	Personnel number
KNR	-	Badge number
ATTR_1	-	Attribute 1
ATTR_2	-	Attribute 2
ATTR_3	-	Attribute 3
ATTR_4	-	Attribute 4
ATTR_5	-	Attribute 5
ATTR_6	-	Attribute 6
ATTR_7	-	Attribute 7
ATTR_8	-	Attribute 8
ATTR_9	-	Attribute 9
ATTR_10	-	Attribute 10
ATTR_11	-	Attribute 11
ATTR_12	-	Attribute 12
ATTR_13	-	Attribute 13
ATTR_14	-	Attribute 14
ATTR_15	-	Attribute 15
MNR	-	Machine
ANR	-	Order + operation
AUNR	-	Order
AGNR	-	Operation
BEARB	-	Editor
BEARBDAT	-	Date of last change
BEARBZEI	-	Time of last change

9.4 Process of Changing Output Batches

9.4.1 Input batch data

To find out which input materials are logged on, the material list has to be polled cyclically:

DLG=LISTE;13|MOD=M|MNR=Machine|ANR=order number....

If no input batch is logged on for planned materials the batch number (CNR) fields, etc. are not filled out. These fields are only filled out if input batches are currently logged on.

9.4.2 Output batch data

To read data of the currently running output batch, the batch list has to be read as follows:

DLG=LISTE;13|MOD=A|ANR=order number

9.4.3 Output batch change

The following dialog has to be sent to HYDRA to execute a batch change:

DLG=CA_WL|ANR=order number |CNR=next batch number|....

The batch list has to be requested anew to be able to print tickets, once output batches have been changed ("DLG=LISTE13;MOD=L;CNR=.. ")

9.4.4 Job end

The following dialogs have to be sent to HYDRA to perform a job end:

DLG=CA_WL|.... - to log the last batch off

DLG=A_AB|… - to log the order off

DLG=A_UN|… - to interrupt the order

9.5 Packing and Palletizing (MPL-PAL)

Please note: All functions described in this section are only available as of MPL product version 7.2.5.

9.5.1 Assign batches (DLG=CE_AN_PA)

ID	Type/max. field length	Description	
ANR=	C16	Order number with operation number	
MNR=	N8/C8	Machine number (numeric/alphanumeric)	
PNR=	C10	alter-native	Personnel number
KNR=	C10		Badge number
CNR=	**C20**	**Batch number of the input batch**	

9.5.2 Delete batch assignment (DLG=CE_DEL_PA)

ID	Type/max. field length	Description	
ANR=	C16	Order number with operation number	
MNR=	N8/C8	Machine number (numeric/alphanumeric)	
PNR=	C10	alter-native	Personnel number
KNR=	C10		Badge number
CNR:PA=	C20	Batch number TPU batch	
CNR=	C20	Batch number of the assigned input batch	

9.5.3 Complete TPU (DLG=CA_WL_PA)

ID	Type/max. field length	Description	
ANR=	**C16**	**Order number with operation number**	
MNR=	**N8/C8**	**Machine number (numeric/alphanumeric)**	
PNR=	C10	alter-native	PNR=
KNR=	C10		KNR=
CNR=	**C20**	**New batch number for the next TPU**	
EGR:GUT=	DEC	Net weight	
EGE:GUT=	C4	Unit of net weight	
EGR:AUS=	DEC	Net weight for status scrap	

ID	Type/max. field length	Description
EGE:AUS=	C4	Unit of net weight
KLASSE=	C1	"G" yield or “A” scrap
STA=	C1	Status of output batch F= with remaining quantity (by default for KLASSE=G) S= blocked (by default for KLASSE=A)
ZLO=	C12	Destination of output batch
TPE=	C10	Transport unit
LHW=	C20	Optional: output batch note
CALT1= to CALT5=	C20	Alternative batch number 1 - 5
ATTR:1 = to ATTR:11 =	--	The definition of which values are filed in additional attributes is configured for each material type in HYDRA
ATTR:101= to ATTR:140=	C40	Alphanumeric batch attributes
ATTR:201= to ATTR:220=	NUM	Numeric batch attributes
ATTR:301= to ATTR:320=	DEC	Decimal batch attributes

9.5.4 List with assigned batches for active TPU

Data are provided using the command ***DLG=LIST;u_l_mpl_hu_elose*** and filed in the HYDRADIR\spool\ directory.

License: MPL-PAL

Structure of dialog data:

„DLG=LIST;u_l_mpl_hu_elose|DATEI={file name}|DAT=...|ZEI=...|USR=...“

Parameter: MNR=<machine>|ANR=<operation>

The list includes the following data:

ID	Field designation	Description
MNR	Machine	Machine number
ANR	Order	Order and OP
ATK	Article	Material number
ATKBEZ	Des. of final article	Material designation
CNR	-	Batch number
ACNR	Package	Batch number TPU
HSDAT	Date manuf.	Manufacturing: date
HSZEI	Time manuf.	Manufacturing: time
MENGE	Quantity	Batch quantity of individual batch
EINH	Unit	Unit of individual batch
LS1 – LS6	Activity 1 – 6	Activity 1 - 6
LS1:EINH – LS6:EINH	Unit activity 1 – 6	Unit activity 1 - 6

9.6 Annex

9.6.1 Summary of MPL field data

The following table provides an overview of field data including structure and brief example. Some points overlap with descriptions already explained in other PDM documents.

ID	Description	Structure	Example
MNR	Machine number	...|MNR={machine number }|...	...|MNR=100|...
MGRP	Machine group	...|MGRP={machine group }|...	...|MGRP=100|...
ANR	Order number	...|ANR={order}|...	...|ANR=47110010|...
AUNR	Order number without OP	...|AUNR={order number without OP }|...	...|AUNR=4711|...
AGNR	Operation number	...|AGNR={OP}|...	...|AGNR=0010|...
ATK	Article number	...|ATK={article number }|...	
EGR:*	Entry size Manually recorded size	...|EGR:{type}={value}|... Types of entry sizes: RPA01, RPA02, .., RPA12 DAUER, PDAUER HUB, GUT, AUS, LEN, GEW The recorded value is accumulated	..|EGR:GUT=10. |. ...|EGR:AUS=1 |...

		onto the respective account within the database	
EGE:*	Unit for entry sizes	...|EGE:{type}={unit}|... For types see EGR	EGE:GUT=ST
EGG:*	Reasons for entry sizes	...|EGG:{type}={reason }|... For types see EGR	EGG:AUS=1
SGR:*	Target size	...|SGR:{type}={value}|... SGR:GUT = target quantity order/OP For types see EGR	...|SGR:GUT=1126|
MST	Machine status	...|MST={machine status}|...	..|MST=1 |..
PNR	Personnel number	...|PNR={personnel number }|...	...|PNR=999999|...
KNR	Badge number	...|KNR={badge number}|...	...|KNR=9999|...
CNR	Batch number	...|CNR={batch number}	...|CNR=998877|...
BZW	Booking obligation	Optional parameter if set several plausibility checks are deactivated. If not sent: = N [...|BZW={ booking obligation}|...]	...|BZW=J|...
LHW	Info on batch	Info on batch (C20)	|LHW={info on batch}|...
ZLO	Destination	Destination	...|ZLO=121234|...
TPE	Transport unit	Transport unit (C10)	...|TPE=1234|...
SLP	Bill of material item	Bill of material item (C6)	...|SLP=AA34|...
ATTR:*	Additional batch attributes	...|ATTR:{nr}]={recorded attribute}|.. nr = 1 ..11 Field types; 1 - 4 : Integer 4 – 6 : Double 7 : Char 4 8, 9 : Char 10 10, 11: Char 20	
STN	Station number	...|STN ={station number }|...	...|STN=1|..

LOSANZ	Number of parallel output batches	...|LOSANZ ={number of batches}|...	...|LOSANZ=5|...
ATTR:101 - ATTR:140	Alphanumeric batch attributes	C40	...|ATTR:101=TXT| …
ATTR:201 - ATTR:220	Numeric batch attributes	NUM	...|ATTR:201=1| …
ATTR:301 - ATTR:320	Decimal batch attributes	DEC	...|ATTR:301=1.123| …

10 HYDRA Production Data Manager MPL - Master Data

10.1 Please note for the basic dialogs described

All mandatory fields are provided with the addition “PK” (primary key = key field). All other fields are optional and are processed if they are filled out.

10.2 Quantity Changes

10.2.1 Quantity change affecting several products

10.2.1.1 DLG=ACCMNT.CHANGE

Using the BAPI calls described in this section, it is possible to record and post a quantity change for different dependent products. The Bapi is implemented as script Bapi (b_accmnt.hsc).

Dependent products

Product	Bapi	Description
MDE	MDEPRO.CHANGEQTY	Edit MDE log records
ADE	ADEPRO.UPDATE	Edit ADE log records
MPL	CNR.UPDATE	Generate material movements/change batch status

BAPI call

ID	Content / {type}	Description
DLG	ACCMNT.CHANGE	Realize quantity change
ACCMNT.DLG	Dialog	Executing initial activity (e.g. CNR.UPDATE)
ACCMNT.MNR	Machine	Machine number
ACCMNT.ANR	Operation	OP number
ACCMNT.CNR	Batch	Batch number
ADEPRO.EGR:GUTP ADEPRO.EGR:AUSP CNR.SGR:GUT	Quantity	Depends on initial command

Return

ID	Content / {type}	Description

Validation checks

Error codes	Description
2027	Unknown mode
105	Parameters are missing

10.3 MPL – Master Data

10.3.1 MPL Setup

See “setup“

10.3.2 Material types

10.3.2.1 DLG=MATTYP.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT

The configuration of material types can be edited using the BAPI calls described in this section.

Tables

Table	Key field	Description
hz_typen	hz_typ MATTYP.MATTYP	Material type
hz_atgen		Delete corresponding attributes with DELETE

BAPI call

ID	Content / {type}	Description
DLG	MATTYP.INSERT	Create material type
	MATTYP.UPDATE	Change material type
	MATTYP.DELETE	Delete material type
	MATTYP.COPY	Copy material type
	MATTYP.LOCK	Block processing of the material type
	MATTYP.UNLOCK	Unblock material type after processing
	MATTYP.NEW	Read specification for new material type
	MATTYP.SELECT	Select material type
MATTYP.MATTYP	{C10}	Material type
MATTYP.MATTYP:Z	{C10}	New (target) material type for COPY
…	…	For further fields, please refer to the HYD-HDB documentation about above-mentioned tables.

Return

ID	Content / {type}	Description
MATTYP.MATTYP	{C10}	MATTYP.INSERT, MATTYP.COPY: Return of material type of the configuration created

Validation checks

Error codes	Description
2707	MATTYP.DELETE:Reference to the mat_mattyp table still exists

10.3.3 Material buffer

10.3.3.1 DLG=MATPUF.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT

The configuration of material buffers can be edited using the BAPI calls described in this section.

Tables

Table	Key field	Description
mat_puffer	mat_puf MATPUF.MATPUF	Material buffer

BAPI call

ID	Content / {type}	Description
DLG	MATPUF.INSERT	Create material buffer
	MATPUF.UPDATE	Change material buffer
	MATPUF.DELETE	Delete material buffer
	MATPUF.COPY	Copy material buffer
	MATPUF.LOCK	Block processing of the material buffer
	MATPUF.UNLOCK	Unblock material buffer after processing
	MATPUF.NEW	Read specification for new material buffers
	MATPUF.SELECT	Select material buffer
MATPUF.MATPUF	{C10}	Material buffer
MATPUF.MATPUF:Z	{C10}	New (target) material buffer for COPY

…	…	For further fields, please refer to the HYD-HDB documentation about the above-mentioned tables

Return

ID	Content / {type}	Description
MATPUF.MATPUF	{C10}	MATPUF.INSERT, MATPUF.COPY: Return of the material type of the configuration created

Validation checks

Error codes	Description
2700	DELETE Material buffer is still assigned to a machine
2701	DELETE Material buffer still refers to another hierarchy
2702	DELETE Material buffer still refers to a transport unit
2703	MATPUF.INSERT/UPDATE: Reference to hierarchy is wrong
2704	MATPUF.UPDATE: Reference to hierarchy wrong (too small)
2705	MATPUF.INSERT/UPDATE:Target location does not exist in the los_zielorte table
2730	DELETE Material buffer is still used within batch stock

10.3.4 Transport units

10.3.4.1 DLG=TPE.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT

The configuration of transport units can be updated using the BAPI calls described in this section.

Tables

Table	Key field	Description
hz_typen	tpe	Transport unit

	TPE.TPE	

BAPI call

ID	Content / {type}	Description
DLG	TPE.INSERT	Create transport unit
	TPE.UPDATE	Change transport unit
	TPE.DELETE	Delete transport unit
	TPE.COPY	Copy transport unit
	TPE.LOCK	Block processing of the transport unit
	TPE.UNLOCK	Unblock transport unit after processing
	TPE.NEW	Read specification for new transport unit
	TPE.SELECT	Select transport unit
TPE.TPE	{C10}	Transport unit
TPE.TPE:Z	{C10}	New (target) transport unit for COPY
…	…	For further fields, please see the HYD-HDB documentation about above-mentioned tables

Return

ID	Content / {type}	Description
TPE.TPE	{C10}	TPE.INSERT, TPE.COPY: Return of transport unit of the configuration created

Validation checks

Error codes	Description

10.4 HYDRA-MPL – Movement Data

10.4.1 Batch stock

10.4.1.1 DLG=CNR.INSERT, UPDATE, DELETE, MODIFY, COPY, LOCK, UNLOCK, NEW, SELECT

The batch stock can be edited using the BAPI calls described in this section.

Tables

Table	Key field	Description
los_bestand a_los_bestand	losnr CNR.CNR	Internal batch number a_los_bestand → Archive table
los_attribute		Batch attributes for a batch

BAPI call

ID	Content / {type}	Description
DLG	CNR.INSERT	Create batch
	CNR.UPDATE	Change batch
	CNR.DELETE	Delete batch
	CNR.COPY	Copy batch
	CNR.LOCK	Block processing of the batch
	CNR.UNLOCK	Unblock batch after processing
	CNR.NEW	Read specification for new batches
	CNR.SELECT	Select batch
	CNR.MODIFY	Create or change batch
	CNR.SPLITCREATE	Split batch
	CNR. SUMMARIZE	Batch merge
CNR.CNR	C20	Internal batch number ‚'/empty – A new batch number (prefix P) is assigned at the server via User=0, if configured like that in the setup
CNR.DLL	C20	Throughput batch number
CNR.CNR:Z	C20	New (target) batch for CNR.COPY

CNR.ATK	C40	Material number
CNR.ATKBEZ	C40	Material designation
CNR.SGR:GUT	DEC	Batch quantity
CNR.SGR:REST	DEC	Remaining quantity of the batch
CNR.SGE:GUT	C3	Unit of batch quantity
CNR.MNR	N8/C8	Machine number
CNR:ATTR:11	C40	Order number with operation number
CNR.STA	C1	Batch status (F = free, S = blocked)
CNR.CKL	C1	Batch class G = yield A = scrap O = on hold N = rework
CNR.TST	C1	Transport status (L = delivered)
CNR.QST	C1	Quality status (G – blocked, F – free)
CNR.QSTMANU	C1	Manual quality status (G – blocked, F – free)
CNR.MATST	C1	Material status (e.g. V = packed)
CNR.MATTYPART	C10	Material type
CNR.OPT:REST	C1	J – “Batch has still got residual quantity“ flag N – “Batch has not got residual quantity”
CNR.FIR	C4	Company
CNR.HSDAT	DATUM	Manufacturing date
CNR.HSZEI	ZEIT	Manufacturing time
CNR.VVDAT	DATUM	Availability date
CNR.VVZEI	ZEIT	Availability time
CNR.VFDAT	DATUM	Expiry date
CNR.VFZEI	ZEIT	Expiry time
CNR.WDAT	DATUM	Warning date
CNR.WZEI	ZEIT	Warning time
CNR.CSTWDAT	DATUM	Date of last status change
CNR.CSTWZEI	ZEIT	Time of last status change

CNR.MATPUF	C12	Material buffer
CNR.MATTYP	C10	Material type
CNR.TPE	C10	Transport unit
CNR.BEM	C20	Info on batch
CNR.PNR	C10	Personnel number
CNR.TECHINFO	C20	Technical info
CNR.EGG:AUS	N8	Scrap reason
CNR.GR	N8	Scrap reason (synonymous to CNR.EGG:AUS)
CNR.GRTXT	N8	Reference to scrap reason text
CNR.USR	N8	User number
CNR.LAGORT	C12	PPS storage location
CNR.LAGPZ	C12	PPS storage bin
CNR.SAPCNR	C10	PPS batch number
CNR.RESART	C4	Reservation type (AG = operation, AK = order)
CNR.RESVAL	C40	Value of reservation (e.g. operation number)
CNR.RESBEM	C100	Reservation comment
CNR.BREITE	DEC	Material width (conversion factor for MPLRF-BP)
CNR.RFAGVFA	DEC	Mass per unit area (conversion factor for MPLRF-BP)
CNR.RFSTKF	DEC	Surface per piece (conversion factor for MPLRF-BP)
CNR.OPT:SLOS	C1	J – batch is a merged batch
CNR.ANZ:SLOS	N8	For a merged batch, the number of the assigned individual batches
CNR.OPT:SLOSTYP	C1	J – Batch is a merged batch
CNR.OPT:SLOSCTRL	C1	J – Batch is a merged batch
CNR.OPT:MBEW	C1	J – Generate goods movement
CNR.OPT:MDC	C1	J – Generate additional goods movement to change quantities
CNR.MOD	C1	For CNR.SPITCREATE only: L – Batch quantity (otherwise split as default residual quantity)
CNR.MENGE	DEC	For CNR.SPITCREATE only:

		Quantity
CNR.EGR:1 – 6	DEC	Quantity activity 1 – 6
CNR.RGR:1 – 6	DEC	Residual quantity activity 1 – 6
CNR.EGE:1 – 6	C3	Unit activity 1 – 6
CNR.CNR:ALT1 – 20	-	Alternative batch number 1 – 20 max. field length: CNR.CNR:ALT1-4 - C20 CNR.CNR:ALT5-14 - C40 CNR.CNR:ALT15-18 - C100 CNR.CNR:ALT19-20 - C512
CNR.EXTCNR	C10	External batch number
CNR.MCNR	C10	Mother batch number (e.g. reference from merged batch)
CNR.AUART	C5	Order type
CNR.UMRFAKTP:Z	DEC	Denominator – conversion factor relating to OP quantity units
CNR.UMRFAKTP:N	DEC	Numerator – conversion factor relating to OP quantity units
CNR.UMRFAKTS:Z	DEC	Denominator – conversion factor relating to OP quantity units
CNR.UMRFAKTS:N	DEC	Numerator – conversion factor relating to OP quantity units
CNR.UMRFAKTT:Z	DEC	Denominator – conversion factor relating OP quantity units
CNR.UMRFAKTT:N	DEC	Numerator – conversion factor relating to OP quantity units
CNR.ZUORD:1 – 6	C1	Flag for assignment “MPL activity account → ADE quantity unit (e.g. P = primary quantity)
ATTR=101 - ATTR=140	C40	Alphanumeric batch attributes (in los_attribute table)
ATTR=201 - ATTR=220	NUM	Numeric batch attributes (in los_attribute table)
ATTR=301 - ATTR=320	DEC	Decimal batch attributes (in los_attribute table)
CNR.USRFLD	C8	User field key
CNR.FU:1 – CNR.FU:66	-	User fields

CNR.PROTEVENT	C10	Event used in the batch history. Using this key, the console can show a text defined for the event (from the reference list ELOS.EREIGNIS1).
CNR.SETEVDATA	C1	J – Transferred data are written in *event_dlg_data*
CNR.EVDATA:1 – CNR.EVDATA:n	C∞ (Restricted to a maximum of 1000 characters by *event_dlg_data.dlg_data*)	These data are written in the file *event_dlg_data.* CNR.SETEVDATA=J must be configured. These data are shown in the detail view of the batch history at the console.
CNR.EVDATABEZ:1 – CNR.EVDATABEZ:n	C∞ (Restricted to a maximum of 1000 characters by *event_dlg_data.dlg_data*)	Designations for CNR.EVDATA:1 – CNR.EVDATA:n. These are console IDs of the reference list for ELOS.EVDATABEZ. These data are **not** shown in the detail view of the batch history at the console.

Return

ID	Content / {type}	Description
CNR.CNR	{C20}	CNR.INSERT, CNR.COPY: Return of batches of the configuration created

Validation checks

Error codes	Description

10.4.1.2 DLG=CNR.SPLITCREATE

BAPI call

ID	Content / {type}	Description
DLG	CNR.SPLITCREATE	Split batch
CNR.CNR	C20	Batch number to be split
CNR.MENGE:**x**	DEC	Batch quantity of future splits e.g. "|CNR.MENGE:1=1.0| CNR.MENGE:2=2.0|"
CNR.GR:**x**	NUM	Scrap reason of future splits
CNR.CKL:**x**	C1	Batch class (G = yield, A = scrap) of future splits
CNR.MOD	C1	L – Batch transfer (a new batch is generated from the original batch using the "same" data) R – Post residual quantity onto a new batch. The original batch is "processed" and has the quantity 0. B – Reduce existing batch by the total of split quantities (remaining quantity and activities)

OPT.CNRGEN_VIA_BAPI	C1	J – New batches are generated by Bapi CNRGEN.CREATECNR N – New batches are generated by the internal function “erzeuge_losnr_sicher“.

BAPI call return values

ID	Content / {type}	Description
CNR:**x**	C20	All generated splits

10.4.1.3 DLG=CNR.SUMMARIZE

ID	Content / {type}	Description
DLG	CNR.SUMMARIZE	Merge batch
CNR.CNR	C20	Only with **CNR.MOD=B** Involved batch that is increased by the total quantity.
CNR.CNR:**x**		All batches that are merged. Please note: In **CNR.MOD=B** the batch to be increased is transferred as CNR.CNR.
CNR.MOD	C1	B – Post quantities to existing batch (CNR.CNR) N – Generate new batch and post quantities to this batch

BAPI call return values

ID	Content / {type}	Description
CNR.CNR:NEW	C20	Only with **CNR.MOD=N:** New batch number

10.4.1.4 DLG=CNR.NEWCNR

A new batch number can be requested using the BAPI calls described in this section (number range User=0).

BAPI call

ID	Content / {type}	Description
DLG	CNR.NEWCNR	Request new batch number (prefix P)

10.4.1.5 DLG=CNR.RESTORE

A batch from the batch archive may be saved back into the online table using the BAPI call described in this section.

BAPI call

ID	Content / {type}	Description
DLG	CNR.RESTORE	Retrieve batch from archive

10.4.1.6 DLG=CNRGEN.CREATENR

BAPI call

ID	Content / {type}	Description
DLG	CNRGEN.CREATENR	Generate new batch
CNRGEN.MATTYP	C10	Material type (fix “SYSTEM“) → FOR FUTURE USE
CNRGEN.TYP	C1	Type of the batch to be generated H → HU batch P → Production batch W → Goods receipt batch A prefix that might be transferred will be ignored if a type is transferred. The type takes priority.
CNRGEN.PRAEFIX	C10	Batch number prefix Overwrites setup configuration
CNRGEN.TNR	N8	Terminal user for number range (Default = 0) Overwrites setup configuration

CNRGEN.LEN	N8	Length of batch number Overwrites setup configuration
CNRGEN.TYP	C1	Batch type H → Packing station batch (prefix fix "HU") P → Produced batch (prefix from setup) W → Goods receipt batch (prefix from setup) A prefix that might be transferred will be ignored if a type is transferred.

Return

ID	Content / {type}	Description
CNR	C20	Generated batch number

10.4.1.7 DLG=CNRBAUM.INSERT

Batch assignments can be created by the BAPI calls described in this section.

Batch assignments describe the connection between input and output batches. Batch assignments have been designed to trace back material relating to batches.

Tables

Table	Key field	Description
los_zuordnung	CNRBAUM.CNR:E CNRBAUM.CNR:A	Batch assignments

BAPI call

ID	Content / {Type}	Description
DLG	CNRBAUM.INSERT	Create batch assignments
CNRBAUM.CNR:E	Batch number {C20}	Internal batch number: input batch
CNRBAUM.CNR:A	Batch number {C20}	Internal batch number: output batch
CNRBAUM.DLL:E	Batch number {C20}	Batch number: input batch
CNRBAUM.DLL:A	Batch number {C20}	Batch number: output batch
CNRBAUM.ATK:E	Article {C40}	Material number: input batch

CNRBAUM.ATK:A	Article {C40}	Material number: output batch
CNRBAUM.HZTYP:E	Material type {C10}	Material type: input batch
CNRBAUM.HZTYP:A	Material type {C10}	Material type: output batch
CNRBAUM.SAPCNR:E	ERP batch {C40}	ERP batch: input batch
CNRBAUM.SAPCNR:A	ERP batch {C40}	ERP batch: output batch
CNRBAUM.MNR	Machine / {C10}	Machine number
CNRBAUM.ANR	Operation / {C40}	Operation number
CNRBAUM.SLP	BOM item / {C10}	BOM item of input batch from component list
CNRBAUM.ART	Type of BOM item / {C1}	Type of BOM item of input batch from component list (e.g. M for material)
CNRBAUM.HSDAT	Date	Date of output batch logoff
CNRBAUM.HSZEI	Time	Time of output batch logoff
CNRBAUM.HSANDAT	Date	Date of output batch logon
CNRBAUM.HSANZEI	Time	Time of output batch logon
CNRBAUM.EANDAT	Date	Date of input batch logon
CNRBAUM.EANZEI	Time	Time of input batch logon
CNRBAUM.EABDAT	Date	Date of input batch logoff
CNRBAUM.EABZEI	Time	Time of input batch logoff

Return

ID	Content / {Type}	Description

Validation checks

Error codes	Description

10.4.2 Material movements

10.4.2.1 DLG=MBEW.INSERT

Material movements may be created by means of the BAPI calls described in this section. The Bapi has been implemented as script Bapi (b_mbew.hsc).

Material movements may only be created by INSERT, as they can be regarded as delta quantity for a stock or status (batch stock).

A negative algebraic sign must precede the quantity for goods issues (movement 261).

Uploads to PPS:

Goods receipts are uploaded to PPS using the ZWEI segment structure.

Goods issues and return transfers are uploaded to PPS using the ZWAU segment structure.

Tables

Table	Key field	Description
event_mlb	--	Material movements

BAP call

ID	Content / {type}	Description
DLG	MBEW.INSERT	Create material movement
MBEW.SAPBWART	Movement type / {C3}	**Goods receipt:** 101 – Goods receipt 102 – Cancellation for goods receipt 262 – Return transfer 525 – Goods receipt, blocking material 531 – Scrap material **Goods issue:** 261 – Goods issue 262 – Cancellation for goods issue
MBEW.EVENT	Event / {C10}	CMM_E - Goods receipt CMM_A – Goods issue CMM_R – Return transfer
MBEW.KLASSE	{C1}	fix “C“ for batch-related movement
MBEW.PRIO	{N4}	fix “9999“
MBEW.DAT	Date	May alternatively also be transferred with DAT=<> / ZEI=<>.

MBEW.ZEI	Time	
MBEW.DLG	Executing command / {C10}	e.g. MBEW
MBEW.MNR	Machine / {C10}	Machine number
MBEW.ANR	Operation / {C40}	Operation number → Determination of SAP order from backlog of orders
MBEW.ATK	Article / {C40}	Article number
MBEW.ATKBEZ	Designation / {C40}	Article designation
MBEW.SAPANR	{C12}	SAP order from backlog of orders ‚'/empty – definition via OP
MBEW.CNR	Batch / {C20}	Batch number for batch relation (internal batch number) Determination of PPS batch results from batch stock
MBEW.DLL	Run-through batch / {C20}	Run-through batch number for batch reference
MBEW.LAGORT	{C10}	Storage location (e.g. from material buffer of machine)
MBEW.ZLO	{C10}	Material buffer (e.g. material buffer of machine)
MBEW.SAPCNR	{C10}	PPS batch ‚'/empty – Determination via batch
MBEW.ATTR:6	Material type / {C10}	Material type to control uploads to PPS Reference check of hz_typen table
MBEW.OPT:RCK	Upload / {C1}	J – Upload movement to PPS N – Do not upload movement to PPS ‚'/empty – Determine upload flag via material type
MBEW.STA:RCK	Uploaded / {C1}	J – identify movement as being uploaded N – Movement has not yet been uploaded
MBEW.CST	{C1}	Batch status (e.g. "F" for free)
MBEW.CKL	{C1}	Batch class (e.g. "G" for yield)
MBEW.PNR	{C10}	Personnel number
MBEW.TNR	{N8}	Terminal user
MBEW.BEARB	{C10}	Editor
MBEW.EGR:1	{DEC}	Quantity in batch quantity unit
MBEW.TYP:1	{C10}	Quantity type e.g. "GUT" for goods receipt, yield (101) "VGR" for consumption quantity (261)

MBEW.EINH:1	{C3}	Quantity unit
MBEW.GR:1	{C10}	Quantity reason (e.g. blocking reason)
MBEW.EGR:2	{DEC}	Movement 531 only: Scrap in batch quantity unit
MBEW.TYP:2	{C10}	Movement 531 only: Quantity type = “AUS“
MBEW.EINH:2	{C3}	Movement 531 only: Quantity unit
MBEW.GR:2	{C10}	Movement 531 only: Scrap reason
MBEW.EGR:5	{DEC}	Quantity of activity field 3 (for batch relation)
MBEW.EGR:6	{DEC}	Quantity of activity field 4 (for batch relation)
MBEW.EGR:7	{DEC}	Quantity of activity field 5 (for batch relation)
MBEW.EGR:8	{DEC}	Quantity of activity field 8 (for batch relation)
MBEW.TYP:5 - 8	{C10}	Quantity type e.g. “GUT“ for goods receipt, yield (101) “VGR“ for consumption quantity (261)
MBEW.EINH:5 – 8	{C3}	Quantity unit (with batch relation from activity field)
MBEW.GR:5 - 8	{C10}	Quantity reason
MBEW.ERWSTA:1 – MBEW.ERWSTA:6	{C1}	Used customer-specifically
MBEW.VERARB:1 – MBEW.VERARB:4	{C1}	Used customer-specifically
MBEW.OPT:EAUS	{C1}	J – Flag for final issue effected When a component is withdrawn the last time in this order
MBEW.OPT:ENDLIEF	{C1}	J – Flag for final delivery for the last goods receipt of the order
…	…	For further fields, please refer to the HYD-HDB documentation about above-mentioned tables

Return

ID	Content / {type}	Description

Validation checks

Error codes	Description
2709	Material type does not exist

10.4.3 Cutting plan

10.4.3.1 DLG=BAHNVERT.CREATE

Using the BAPI calls described in this section, allows for cutting plans to be created for an order. Cutting plans are only used if the “cutting reels” machine type is used along with coil based manufacturing. They describe the web-shaped structure of the output material of an OP. If a cutting plan already exists for an order and machine it will be deleted beforehand.

Tables

Table	Key field	Description
mpl_bahnverteilung		Cutting plan (header record)
mpl_bahnlayout		Cut layout (detailed record)

BAPI call

ID	Content / {type}	Description
DLG	**BAHNVERT.CREATE**	Create cutting plan
BAHNVERT.MANR	Operation / {C40}	Operation number of mother OP
BAHNVERT.MNR	Machine / {C10}	Machine number
BAHNVERT.BAHNVERT	Cutting plan/ {C10}	Unique key of cutting plan
BAHNVERT.ART	Cut type / {C1}	‘V’ – prepared cutting plan ‘M’ – creation of measurement (without OP reference)
BAHNVERT.BEZ	Designation / {C20}	Designation
BAHNVERT.ANZBAHN:GES	Total of cuts / NUM	Total of cuts within the production order (parent and child OP)
ANR#1= to **ANR#n**	OP / {C40}	Operation number (overall key)
TYP#1= to **TYP#n**	Model / {C1}	H – Parent OP L – Child OP
BAHNART#1= to **BAHNART#n**	Type / {C1}	L – Left cut R – Right cut B – Cut /web
BAHNNR#1= to **BAHNNR#n**	Number / NUM	Cut number 1 – n
BREITE:S#1= to	Target width / DEC	Target width of a cut in mm

BREITE:S#n		
BREITE:I#1= to BREITE:I#n	Actual width / DEC	Actual width of a cut in mm

Return

ID	Content / {type}	Description

Validation checks

Error codes	Description

10.5 Transport management (MPL-TRA)

10.5.1 Create transport order

10.5.1.1 DLG=TRANR.CREATE

Using the BAPI calls described in this section, it is possible to create a transportation order for batches or resources.

Dependent products

Product	Bapi	Description
ADE	ANR.COPY	Create order
MPL	CNR.UPDATE	Batch status change

BAPI-Call

ID	Content / {type}	Description
DLG	TRANR.CREATE	Create transport order
TRANR.ATK	{C40}	Material number
TRANR.AUART	{C5}	Order type
TRANR.SMP	{C12}	Source material buffer
TRANR.TMP	{C12}	Target material buffer

TRANR.MNR	{C10}	Machine number
TRANR.SANR	{C40}	Trigger operation
TRANR.CNR	{C20}	Batch number for transport
TRANR.RES	{C20}	Resource number for transport
TRANR.RESTYP	{C10}	Resource type for transport
TRANR.SGR:P	{DEC}	Target quantity
TRANR.SGE:P	{C3}	Unit
TRANR.DATFB	{DAT}	Planned start date
TRANR.DATSE	{DAT}	Planned finish date
TRANR.CALLDLG	{C20}	Internal dialog action (e.g. CA_WL)
TRANR.ASYNC	{C1}	J = transport orders are created asynchronously

Return

ID	Content / {type}	Description
RET.AUNR	Order	Order number of created transport order

Validation checks

Error codes	Description
7043	Work plan not found
1690	Material buffer not found
2817	Oder type not found
3201	Assigned resource not found
7039	Resource is not in source material buffer
7038	Batch is not in source material buffer
1612	Batch not found
1594	batch status not valid
3636	Batch is locked
3632	material number of batch is not valid
7041	Batch is still in transport

10.5.2 Reserve transport order

10.5.2.1 DLG=TRANR.RESERVE

Using the BAPI calls described in this section, it is possible to reserve transportation orders for machines.

Dependent products

Product	Bapi	Description
ADE	ANR.SETSTATUS	Change order status
MPL	CNR.UPDATE	Change batch status

BAPI call

ID	Content / {type}	Description
DLG	TRANR.RESERVE	Reserve transport order
TRANR.MNR	Machine {C10}	Machine number
TRANR.ANR	Operation {C40}	Operation of the transport order

Return

ID	Content / {type}	Description

Validation checks

Error codes	Description
3702	Operation not found
7037	Operation is not in status initial
2613	Wrong operation status
107	Order type not valid

10.5.3 Start transportation order

10.5.3.1 DLG=TRANR.START

Using the BAPI calls described in this section, it is possible to start transportation order at machine.

Dependent products

Product	Bapi	Description
ADE	A_AN	Start operation
MPL	CNR.UPDATE	batch status change
WRM	RES_STATUS	resource status change

BAPI-Call

ID	Content / {type}	Description
DLG	TRANR.START	Start transportation order
TRANR.ATK	{C40}	Material number
TRANR.SMP	{C12}	Source material buffer
TRANR.TMP	{C12}	Target material buffer
TRANR.MNR	{C10}	Machine number
TRANR.ANR	{C40}	Operation
TRANR.CNR	{C20}	batch number for transport
TRANR.RES	{C20}	Resource number for transport
TRANR.RESTYP	{C10}	Resource type for transport

Return

ID	Content / {type}	Description

Validation checks

Error codes	Description
7039	Resource is not in source material buffer
7038	Batch is not in source material buffer

10.5.4 Finish transportation order

10.5.4.1 DLG=TRANR.END

Using the BAPI calls described in this section, it is possible to finish a running transportation order at machine.

Dependent products

Product	Bapi	Description
ADE	A_AB	Finish transportation order
MPL	CNR.UPDATE C_UMB	batch status change batch replace
WRM	RES_STATUS RES_UMB	Resource status change Move resource

BAPI-Call

ID	Content / {type}	Description
DLG	TRANR.END	Finish transportation order
TRANR.TMP	{C12}	Target material buffer
TRANR.MNR	{C10}	Machine number
TRANR.ANR	{C40}	Operation of transportation order
TRANR.EGR:GUT	{DEC}	Quantity

Return

ID	Content / {type}	Description

Validation checks

Error codes	Description

11 HYDRA Production Data Manager PDV - Master Data

11.1 Please note for the basic dialogs described

All mandatory fields are provided with the addition “PK” (primary key = key field). All other fields are optional and are processed if they are filled out.

11.2 Events

11.2.1 Edit event (DLG=PDVEVENTCFG.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, SELECT)

PDV events can be edited using the BAPI calls described in this section. HYDRA-PDV events may be recorded as general events at a machine and are saved in a corresponding log as soon as they occur.

Tables

Table	Key field	Description
pdv_event_cfg	event PDVEVENTCFG. EVENT	Event ID (PK) in the configuration table for PDV events

BAPI call

ID	Content/{type}	Description
DLG	PDVEVENTCFG. INSERT	Create PDV event
	PDVEVENTCFG. UPDATE	Change PDV event
	PDVEVENTCFG. DELETE	Delete PDV event
	PDVEVENTCFG. COPY	Copy PDV event
	PDVEVENTCFG. LOCK	Block PDV event to be processed
	PDVEVENTCFG. UNLOCK	Unblock PDV event after processing
	PDVEVENTCFG. SELECT	Select PDV event

PDVEVENTCFG.EVENT	{C50}	PK PDV event ID
PDVEVENTCFG.EVENT:Z	{C50}	PK new (target) PDV event ID for COPY
PDVEVENTCFG.CAPTION	{C100}	Description of the PDV event
PDVEVENTCFG.ALERT	{C1}	Flag whether the event is an alarm or not. Possible values: “Y” (yes, event is an alarm) or “N” (no, event is no alarm)
PDVEVENTCFG.ALERT:DURATION	{N}	Duration for the alarm signal when the PDV event occurs in seconds

Return

ID	Content/{type}	Description
PDVEVENTCFG.EVENT	{C50}	PDVEVENTCFG.INSERT, PDVEVENTCFG.COPY: Return of PDV event ID of the configuration created

Plausibility checks

Error codes	Description

11.2.2 List of events (DLG=PDVEVENTCFG.LIST)

This BAPI call lists the PDV events configured in HYDRA including their configuration.

Tables

Table	Key field	Description
pdv_event_cfg	event PDVEVENTCFG.EVENT	Event ID (PK) in the configuration table for PDV events

BAPI call

ID	Content	Description
DLG	PDVEVENTCFG.LIST	List of PDV events
PDVEVENTCFG.EVENT	{C50}	Optional restriction to a certain PDV event
DATEI	{C256}	Specification of the file name for the list

Return

ID	Content	Description
—	—	—

Plausibility checks

Error codes	Description
1656	The file assigned to the name DATEI cannot be written

11.3 Logical Channels

11.3.1 Edit logical channels (DLG=LOGCHAN.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, SELECT)

Logical channels required for HYDRA-PDV collection may be edited using the BAPI calls described in this section. Logical channels represent the central configuration for assigning physical interfaces of a machine (measuring channels) to logistic HYDRA values (characteristics, PDV events …).

By assigning them to a machine and a terminal, assigned characteristics or PDV events can be accessed at the corresponding place of entry. Master data of logical channels are put to the second under version control.

Tables

Table	Key field	Description
pdv_logic_chan	masch_nr LOGCHAN.MNR	Machine number (PK) of logical channels
pdv_logic_chan	capture_id LOGCHAN.TNR	Terminal number (PK) of logical channels
pdv_logic_chan	chan_nr LOGCHAN.LOGK	Number of the physical channel (PK) of logical channels
pdv_logic_chan	Fkey LOGCHAN.ID	ID of the logistic value (process parameter ID or PDV event) (PK) of logical channels
pdv_logic_chan	type LOGCHAN.TYP	Channel type, which logistic type it is about (process parameter or PDV event) (PK) of logical channels
pdv_logic_chan	valid_from LOGCHAN.VALID FROM	Valid from for master data put under version control (PK)

BAPI call

ID	Content/{type}	Description
DLG	LOGCHAN.INSERT	Create logical channel
	LOGCHAN.UPDATE	Change logical channel
	LOGCHAN.DELETE	Delete logical channel
	LOGCHAN.COPY	Copy logical channel
	LOGCHAN.LOCK	Block logical channel to be processed
	LOGCHAN.UNLOCK	Unblock logical channel after processing
	LOGCHAN.SELECT	Select logical channel
LOGCHAN.MNR	{C20}	PK machine number
LOGCHAN.TNR	{SHORT}	PK terminal number
LOGCHAN.LOGK	{NUM}	PK channel number
LOGCHAN.ID	{C50}	PK reference ID of recorded channel data
LOGCHAN.TYP	{C2}	PK type of reference ID of the recorded channel (process parameter or PDV event)
LOGCHAN.VALIDFROM	{DATETIME}	PK valid from point in time for master data put under version control
LOGCHAN.MNR:Z	{C20}	PK new (target) machine number for COPY
LOGCHAN.TNR:Z	{SHORT}	PK new (target) terminal number for COPY
LOGCHAN.LOGK:Z	{NUM}	PK new (target) channel number for COPY
LOGCHAN.ID:Z	{C50}	PK new (target) reference ID of recorded channel data for COPY
LOGCHAN.VALIDFROM:Z	{DATETIME}	PK new (target) valid from point in time for master data put under version control for COPY
…	…	For further fields, please refer to the HYD-HDB documentation

		about above-mentioned tables

Return

ID	Content/{type}	Description
LOGCHAN.MNR	{C20}	LOGCHAN.INSERT, LOGCHAN.COPY: Return of the keys of the configured PK channel number
LOGCHAN.TNR	{SHORT}	LOGCHAN.INSERT, LOGCHAN.COPY: Return of the keys of the configured PK channel number
LOGCHAN.LOGK	{NUM}	LOGCHAN.INSERT, LOGCHAN.COPY: Return of the keys of the configured PK channel number
LOGCHAN.ID	{C50}	LOGCHAN.INSERT, LOGCHAN.COPY: Return of the keys of the configured PK channel number
LOGCHAN.TYP	{C2}	LOGCHAN.INSERT, LOGCHAN.COPY: Return of the keys of the configured PK channel number
LOGCHAN.VALIDFROM	{DATETIME}	LOGCHAN.INSERT, LOGCHAN.COPY: Return of the keys of the configured PK channel number

Plausibility checks

Error codes	Description
550	Unknown channel reference type: Only process parameters (PP) and PDV events are supported as references.
551	The PDV event transferred is not available
552	Anonymous characteristics from TNT configuration only support measured values (MW) as channel type
553	Channel numbers between 0 and 9999 are valid only
554	Cycle times greater than 0 are allowed only
555	Wrong channel direction. Input and output channels are supported only.
556	Alarm channels are not supported for the channel data type indicated. This is only possible for OTG, UTG, OPEG or UPEG (UTL, LTL, UPAL, LPAL).
2039	Machine is not available
1668	Terminal is not available

11.3.2 List of logical channels (DLG=LOGCHAN.LIST)

This BAPI call lists the logical channels configured in HYDRA including their configuration.

Tables

Table	Key field	Description
pdv_logic_chan	masch_nr LOGCHAN.MNR	Machine number (PK) of logical channels
pdv_logic_chan	capture_id LOGCHAN.TNR	Terminal number (PK) of logical channels
pdv_logic_chan	chan_nr LOGCHAN.LOGK	Number of the physical channel (PK) of the logical channels
pdv_logic_chan	fkey LOGCHAN.ID	ID of the logistic value (process parameter ID or PDV event) (PK) of logic channels
pdv_logic_chan	type LOGCHAN.TYP	Channel type, which logistic type it is about (process parameter or PDV event) (PK) of logic channels
pdv_logic_chan	valid_from LOGCHAN.VALID FROM	Valid from for master data put under version control (PK)

BAPI call

ID	Content	Description
DLG	LOGCHAN.LIST	List of logical channels
DATEI	{C256}	Specification of the file name for the list

Return

ID	Content	Description
—	—	—

Plausibility checks

Error codes	Description
1656	The file assigned to the name DATEI cannot be written

11.4 Characteristic Attribute

11.4.1 Edit characteristic attributes (DLG=PAUMMAUSP.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

Characteristic attributes can be edited by way of the BAPI calls described in this section. They are required to record the necessary configurations of an inspection order characteristic for the generation of samples. In this context, the different configuration versions of an inspection order characteristic are put under version control for a certain period of time. Thus, it is possible to trace back, which configuration applied for a characteristic at a certain point in time.

Tables

Table	Key field	Description
caq_paumm_ausp	rec_type RECTYP	Rectype in characteristic attribute (PK)
caq_paumm_ausp	bereich BER	Area in characteristic attribute (PK)
caq_paumm_ausp	pruefanf_nr PANNR	Inspection requirement number in characteristic attribute (PK)
caq_paumm_ausp	pruefauft_nr PAUNR	Inspection order number in characteristic attribute (PK)
caq_paumm_ausp	afo AFO	Order sequence in characteristic attribute (PK)
caq_paumm_ausp	maschine_nr MNR	Machine number in characteristic attribute (PK)
caq_paumm_ausp	valid_from_ts VALIDFROM	Valid from point in time in characteristic attribute (PK)

BAPI call

ID	Content/{type)	Description
DLG	PAUMMAUSP.INSERT	Create characteristic attribute
	PAUMMAUSP.UPDATE	Change characteristic attribute
	PAUMMAUSP.D	Delete characteristic attribute

	ELETE	
	PAUMMAUSP.COPY	Copy characteristic attribute
	PAUMMAUSP.LOCK	Block characteristic attribute to be processed
	PAUMMAUSP.UNLOCK	Unblock characteristic attribute after processing
	PAUMMAUSP.SELECT	Select characteristic attribute
PAUMMAUSP.RECTYP	{C10}	Rectyp (PK)
PAUMMAUSP.BER	{C10}	CAQ area (PK)
PAUMMAUSP.PANNR	{NUM}	Inspection request number in characteristic attribute (PK)
PAUMMAUSP.PAUNR	{NUM}	Inspection order number in characteristic attribute (PK)
PAUMMAUSP.AFO	{NUM}	Order sequence in characteristic attribute (PK)
PAUMMAUSP.MNR	{C20}	Machine number in characteristic attribute (PK)
PAUMMAUSP.VALIDFROM	{DATETIME}	Valid from point in time in characteristic attribute (PK)
PAUMMAUSP.RECTYP	{C10}	PK new (target) Rectyp for COPY
PAUMMAUSP.BER	{C10}	PK new (target) CAQ area for COPY
PAUMMAUSP.PANNR	{NUM}	PK new (target) inspection request number for COPY
PAUMMAUSP.PAUNR	{NUM}	PK new (target) inspection order number for COPY
PAUMMAUSP.AFO	{NUM}	PK new (target) order sequence for COPY
PAUMMAUSP.MNR	{C20}	PK new (target) machine number for COPY
PAUMMAUSP.VALIDFROM	{DATETIME}	PK new (target) “colname designation” for COPY (column name)

…	…	For further fields, please refer to the HYD-HDB documentation about above-mentioned tables

Return

ID	Content/{type}	Description
caq_paumm_ausp	{C10}	PAUMMAUSP.INSERT, PAUMMAUSP.COPY: Rectype in characteristic attribute (PK)
caq_paumm_ausp	{C10}	PAUMMAUSP.INSERT, PAUMMAUSP.COPY: Area in characteristic attribute (PK)
caq_paumm_ausp	{NUM}	PAUMMAUSP.INSERT, PAUMMAUSP.COPY: Inspection requirement number in characteristic attribute (PK)
caq_paumm_ausp	{NUM}	PAUMMAUSP.INSERT, PAUMMAUSP.COPY: Inspection order number in characteristic attribute (PK)
caq_paumm_ausp	{NUM}	PAUMMAUSP.INSERT, PAUMMAUSP.COPY: Order sequence in characteristic attribute (PK)
caq_paumm_ausp	{C20}	PAUMMAUSP.INSERT, PAUMMAUSP.COPY: Machine number in characteristic attribute (PK)
caq_paumm_ausp	{DATETIME}	PAUMMAUSP.INSERT, PAUMMAUSP.COPY: Valid from point in time in characteristic attribute (PK)

Plausibility checks

Error codes	Description
—	—

11.4.2 List of characteristic attributes (DLG=PAUMMAUSP.LIST)

This BAPI call lists the characteristic attributes configured in HYDRA including their configuration.

Tables

Table	Key field	Description
caq_paumm_ausp	rec_type RECTYP	Rectype in characteristic attribute (PK)
caq_paumm_ausp	bereich BER	Area in characteristic attribute (PK)
caq_paumm_ausp	pruefanf_nr PANNR	Inspection requirement number in characteristic attribute (PK)
caq_paumm_ausp	pruefauft_nr PAUNR	Inspection order number in characteristic attribute (PK)

caq_paumm_ausp	afo AFO	Order sequence in characteristic attribute (PK)
caq_paumm_ausp	maschine_nr MNR	Machine in characteristic attribute (PK)
caq_paumm_ausp	valid_from_ts VALIDFROM	Valid from period of the characteristic attribute (PK)

BAPI call

ID	Content	Description
DLG	PAUMMAUSP.LIST	List of characteristic attributes
DATEI	{C256}	Specification of the file name for the list

Return

ID	Content	Description
—	—	—

Plausibility checks

Error codes	Description
1656	The file assigned to the name DATEI cannot be written

12 HYDRA Production Data Manager WRM - Data Collection

12.1 Please note for the posting dialogs described

All mandatory fields are displayed in gray; all other fields are optional and are processed, provided that they are filled out.

12.2 WRM Editing Dialogs

This section describes the editing dialogs to map the terminal function with respect to the resource status.

12.2.1 Log resource on (DLG=RES_AN)

Using this dialog a resource can be logged on to an order. Depending on the configuration of the corresponding resource type, quantities and times are then also posted onto the resource that is logged on.

ID	Type/max. field length	Description	
DLG	RES_AN	Log resource on	
DAT	{mm/dd/yyyy}	The date needs to be indicated to log a resource on.	
ZEI	{seconds}	Time, please see DAT.	
RESID	N10	Resource ID, unique number to identify the resource in database tables . Alternative specification for RESTYP and RES	
RESTYP	C4	Key 1 of the double key to uniquely identify a resource. Resource type. Alternative specification for RESVERWEIS	
RES	C40	Key 2 of the double key to uniquely identify a resource. Name of the resource. Alternative specification for RESVERWEIS	
MNR	C20	Machine number	
ANR	C40	Order number	
PNR	C10	alter-native	Personnel number
KNR	C10		Badge number
RESVER	C20	Version ID	
KOMMENTAR	C500	Comment on the logon process, is saved for the event (event_res) in the event_dlg_data table .	

12.2.2 Log resource off (DLG=RES_AB)

By way of this dialog a resource can be logged off from an order. Depending on the configuration of the corresponding resource type, quantities and times are also posted onto the resource that is logged off.

ID	Type/max. field length	Description	
DLG	RES_AB	Log resource off	
DAT	{mm/dd/yyyy}	The date needs to be entered to log a resource off.	
ZEI	{seconds}	Time, please see DAT.	
RESID	N10	Resource ID, unique number to identify the resource in database tables. Alternative specification for RESTYP and RES	
RESTYP	C4	Key 1 of the double key to uniquely identify a resource. Resource type. Alternative specification for RESVERWEIS	
RES	C40	Key 2 of the double key to uniquely identify a resource. Resource name. Alternative specification for RESVERWEIS	
MNR	C20	Machine number	
ANR	C40	Order number	
PNR	C10	alter-native	Personnel number
KNR	C10		Badge number
RESVER	C20	Version ID	
KOMMENTAR	C500	Comment on the logoff process is saved for the event (event_res) in the event_dlg_data table.	

12.2.3 Set resource status (DLG=RES_STATUS)

The status of a resource can be reset by way of this dialog. The old and the new status are compared with each other and corresponding plausibility checks are performed. Once the plausibility check has been realized with success, affected tables are updated/specified.

Resource allocation in res_ress_belegung is updated respectively. If a resource is blocked, i.e. it gets a status with the ID "verarb_planung" != "K" an entry is made in res_ress_belegung. In this context, the date (DATB/ZEIB and DATE/ZEIE) is taken into account. If the current point in time is within the specified period of time an entry is made. In case the end date is prior to this period of time, it is ignored. Provided that the start date lies in the future, it is entered with the date lines specified.

In case the “bill of material processing” license (WRM-STL or DNC-STL) is available, superordinate resources are blocked collectively according to the bill of material by increasing the collective block counter by 1 for superordinate resources. When the resource is released, the collective block counter of the superior resource is reduced by 1.

Block: All superordinate resources. Collective block + 1.

Unblock: All superior resources. Collective block - 1..

The PROD=x ID has been configured to switch to a defined status. If this value is entered a transferred status is ignored. The system then searches for the status, which is assigned to exactly this ID in res_status_zuord.prod and takes this status as new specification. The prod ID is always unique for each “type – family” combination. If this is not the case, it is a configuration error. In this case the last status found is used. This technology actually renders the RES_FREI and RES_ABSTA dialogs redundant. They may be replaced by DLG=RES_STATUS|PROD=F (=RES_FREI) or DLG=RES_STATUS|PROD=B (=RES_ABSTA).

<table>
<tr><th>ID</th><th>Type/max. field length</th><th colspan="2">Description</th></tr>
<tr><td>DLG</td><td>RES_STATUS</td><td colspan="2">Status change</td></tr>
<tr><td>DAT</td><td>{mm/dd/yyyy}</td><td colspan="2">The date specification is very important for setting statuses. As due to this date and the other date specifications it can automatically be recognized whether the status is set immediately or in future.</td></tr>
<tr><td>ZEI</td><td>{seconds}</td><td colspan="2">Time; please see DAT.</td></tr>
<tr><td>RESID</td><td>N10</td><td colspan="2">Resource ID, unique number to identify the resource in database tables.
Alternative specification for RESTYP and RES</td></tr>
<tr><td>RESTYP</td><td>C4</td><td colspan="2">Key 1 of the double key to uniquely identify a resource. Resource type.
Alternative specification for RESVERWEIS</td></tr>
<tr><td>RES</td><td>C40</td><td colspan="2">Key 2 of the double key to uniquely identify a resource. Resource name.
Alternative specification for RESVERWEIS</td></tr>
<tr><td>RESSTA</td><td>N10</td><td colspan="2">Target status of the resource</td></tr>
<tr><td>PNR</td><td>C10</td><td rowspan="2">alter-
native</td><td>Personnel number</td></tr>
<tr><td>KNR</td><td>C10</td><td>Badge number</td></tr>
<tr><td>RESVER</td><td>C20</td><td colspan="2">Version ID</td></tr>
<tr><td>PROD</td><td>C1
{F|B|U|…}</td><td colspan="2">ID whether the status is to be determined in the database or not. If nothing is entered the transferred status applies. If entered the appropriate status is searched in . res_status_zuord.prod.
At the moment the following is configured:
F = Release status (status when a resource is released)
B = Logoff status (status when a resource is logged off - A_AB)
U = Upload status (status when a resource is uploaded - RES_UPLOAD)</td></tr>
<tr><td>DATB</td><td>{mm/dd/yyyy}</td><td colspan="2">Start date for realizing a status in the future. Format mm/dd/yyyy in local time zone
If nothing is entered or if the ID is left out or the date lies in the past the status is reset immediately.</td></tr>
<tr><td>ZEIB</td><td>{seconds}</td><td colspan="2">Start time for resetting a status in the future. Normally 0.</td></tr>
<tr><td>DATE</td><td>{mm/dd/yyyy}</td><td colspan="2">End date for resetting a status with limited validity. If this date is finally exceeded a monitoring process recognizes that the status has expired and sets it to the only possible status that has the status “free”.</td></tr>
</table>

ID	Type/max. field length	Description
		Format mm/dd/yyyy in local time zone If nothing is entered or the ID is left out an unlimited validity period is set (max. date).
ZEIE	{seconds}	End time for a status change with limited validity. If empty or = 0. Then automatic change over to 86400.
ZLO	C12	Receiving storage location. If empty the default value from status configuration is used as storage location .
KOMMENTAR	C500	Comment on the status change is saved for the event (event_res) in the event_dlg_data table.

12.2.4 Release resource (DLG=RES_FREI)

With this dialog a resource is changed over to a released status. There is only one resource status with the ID “released” for each type or family .

The process exactly corresponds to starting a RES_STATUS dialog with preselected release status. Consequently, the start process and behavior is the same as for RES_STATUS. By using the ID "PROD=**F**" for the dialog RES_STATUS, it is even to be preferred to the RES_FREI dialog. Example: DLG=RES_STATUS|PROD=**F**|…

ID	Type/max. field length	Description
DLG	RES_FREI	Status change
RESSTA	N10	NO ENTRY!!!
Rest like RES_STATUS		

12.2.5 Change resource status to “status after logging OP off” (DLG=RES_ABSTA)

This dialog changes a resource status to a “status after logging OP off”, i.e. a resource switches to this status if it is logged off/interrupted with an order. There is only one resource status with the "Abmelde_Status" ID for each type or family .

The procedure exactly corresponds to starting a RES_STATUS dialog with preselected logoff status. Thus, the start procedure and behavior is just the same as for RES_STATUS. By using the "PROD=**B**" ID for the RES_STATUS dialog it is even to be preferred to the RES_ABSTA dialog. Example: DLG=RES_STATUS|PROD=**B**|…

ID	Type/max. field length	Description
DLG	RES_ABSTA	Status change
RESSTA	N10	No Entry!!!
Remainder like RES_STATUS		

12.2.6 Repost resource (DLG=RES_UMB)

This dialog reposts a resource onto another storage location.

ID	Type/max. field length	Description
DLG	RES_UMB	Change of storage location
ZLO	C12	New receiving storage location (target). If nothing is entered the default value from status configuration is used as storage location.
KOMMENTAR	C500	Comment on the changed storage location; is saved for the event (event_res) in the table event_dlg_data.

12.2.7 Mount resource (DLG=RES_EIN)

A resource may be added to another resource using this dialog. From a technical point of view, a BOM relationship is established between mother and daughter resource.

Quantities or times are not posted onto the resource by this dialog. But the event is recorded and shown in many different evaluations/reports.

ID	Type/max. field length	Description
DLG	RES_EIN	Log resource off
DAT	{mm/dd/yyyy}	The date needs to be specified for logging a resource off.
ZEI	{seconds}	Time; see DAT.
RESTYP:M	C4	Key 1 of the double key to uniquely identify a resource. Type of the mother resource.
RES:M	C40	Key 2 of the double key to uniquely identify a resource. Name of the mother resource.
RESTYP:T	C4	Key 1 of the double key to uniquely identify a resource. Type of the daughter resource.
RES:T	C40	Key 2 of the double key to uniquley identify a resource. Name

ID	Type/max. field length	Description	
		of the daughter resource.	
PNR	C10	alter-native	Personnel number
KNR	C10		Badge number
KOMMENTAR	C500	Logoff comment is saved for the event (event_res) in the table event_dlg_data.	

12.2.8 Demount resource (DLG=RES_AUS)

A resource can be demounted from another resource using this dialog. From a technical point of view, the BOM relationship between the transferred mother and daughter resource is removed/deleted.
Quantities or times are not posted onto the resource by this dialog. But the actual event is recorded and shown in different reports/evaluations.

ID	Type/max. field length	Description	
DLG	RES_AUS	Log resource off	
DAT	{mm/dd/yyyy}	The date needs to be specified for logging a resource off.	
ZEI	{seconds}	Time; see DAT.	
RESTYP:M	C4	Key 1 of the double key to uniquely identify a resource. Type of the mother resource.	
RES:M	C40	Key 2 of the double key to uniquely identify a resource. Name of the mother resource.	
RESTYP:T	C4	Key 1 of the double key to uniquely identify a resource. Type of the daughter resource.	
RES:T	C40	Key 2 of the double key to uniquely identify a resource. Name of the mother resource.	
PNR	C10	alter-native	Personnel number
KNR	C10		Badge number
KOMMENTAR	C500	Logoff comment is saved for the event (event_res) in the table event_dlg_data.	

12.3 DNC Dialogs

This section describes special dialogs to manage resources of the DNC type. These are functions to upload and download resources. As DNC resources also include program files in addition to resource data records, these program files are also transferred along with these operations. This establishes the connection between the local data room (local hard disk directory) and the HYDRA server data room. Depending on its configuration, the HYDRA server data room can also be connected with an external programming system.

12.3.1 Load DNC resource to the machine (DLG=RES_DOWNL)

Using this dialog a DNC resource (res_typ.dnc_verarbeitung != "K") is logically “downloaded” to a terminal, i.e. the dialog performs different plausibility checks and the logging function in event_res.

Files are not copied with RES_DOWNL. An external copy process is required for this. RES_DOWNL exclusively provides for database processing.

Consequently, a logical, complete download process looks as follows:

1. Validity check (through RES_DOWNL|VERB=N| …)
2. Files are copied to the terminal, which is performed by external functions (not HYDDI).
3. Posting of the download (through RES_DOWNL, possibly with BZW=J)

Using different “definitions” of the IDs leads to different behaviors of the dialog. The “VERB“ ID differentiates between “J“ = complete download incl. posting (by default) and “N” = only plausibility check without posting. Using the "BZW=J" (booking obligation) ID avoids all plausibility checks. This is reasonable if started several times, after starting with VERB=N.

Several resources can be transferred in one start process. That is why a number variable is planned for RESID:n or RES:n and RESTYP:n. If only one resource is transferred, the number variable can be omitted. If several resources are transferred they have to be indicated completely including their respective RESID or RESTYP-RES combination. At the moment a maximum of 30 resources can be transferred. Further resources are ignored. All plausibility checks are made for the resources specified and all resources are recorded separately in event_res. However, VERB and BZW have to be configured respectively.

The DNC-BP license is required to use this dialog.

ID	Type/max. field length	Description
DLG	RES_DOWNL	Download
VERB	C1	N = only validity check, without posting J = validity check and posting (by default)

ID	Type/max. field length	Description
DAT	{mm/dd/yyyy}	Date specification for the download.
ZEI	{seconds}	Time for the download
RESVERWEIS:n or RESID:n	N10	Resource ID, unique number to identify the resource in database tables. Alternative specification for RESTYP and RES. n = 1..30 (complete entry required). RESID takes priority over RESTYP+RES.
RESTYP:n	C4	Key 1 of the double key to uniquely identify a resource. Resource type. Has always to be indicated together with RES. Alternative specification for RESVERWEIS. n = 1..30 (complete entry required).
RES:n	C40	Key 2 of the double key to uniquely identify a resource. Resource name. Has always to be indicated together with RESTYP. Alternative specification for RESVERWEIS. n = 1..30 (complete entry required).
BEARB	C10	Editor
PNR	C10	alter-native: PNR
KNR	C10	alter-native: KNR
RESVER:n	C20	Version ID (not used as key) n = 1..30 (complete entry required).
BZW	C1	J = Booking obligation, i.e. no plausibility checks N = No booking obligation (by default)
OPT_DNCVERARB	C1	ID whether DNC resources are concerned ("L", "E", "O") or not ("K"). "K"= no DNC resource "L"=local "E"=external "O"=optimized Currently not in use. If not indicated the dialog determines this value on the basis of the resource type of resource 1.
MNR	N8/C8	Machine
KOMMENTAR	C255	Download comment. Is saved for the event (event_res) in the event_dlg_data table. A reference is saved in event_res. If the resource is indicated several times the comment is saved only once. All new event_res entries (for each resource specified) then refer to it.

12.3.2 Upload DNC resource from the machine (DLG=RES_UPLOAD)

This dialog logically uploads a DNC resource (res_typ.dnc_verarbeitung != "K") from a terminal, i.e. the dialog takes over different plausibility checks, creates a new resource incl. attributes, if required, and records it in event_res.

Files are not copied with RES_UPLOAD. This has to be realized by way of an external copy process. RES_ UPLOAD only provides for database processing.

Consequently, a logical, complete upload process looks as follows:

1) Validity checking (through RES_UPLOAD|VERB=N| …)

2. Files are copied from the terminal into the HYDRA namespace, which is performed by functions provided externally (not HYDDI).

3. Posting of the upload (through RES_UPLOAD, possibly with BZW=J). If required, a new resource incl. user fields is created in this context. In case of optimization (res_typen.dnc_verarbeitung=”O”), the resource is additionally set to “optimized” and the resource status is switched to the PROD=”U” status (upload).

Using different “definitions” of the IDs leads to different behaviors of the dialog. The “VERB“ ID differentiates between “J“ = complete upload incl. posting and creation of resources (by default) and “N” = only validity checking without posting. Using the "BZW=J" (posting required) ID avoids all validity checks. This is useful if started several times, after starting with VERB=N.

Two different behaviors can be defined by the MOD mode. MOD=N (= new) and MOD=U (= update). When “N” is selected, it is checked, among other things, whether the resource already exists and can be created anew. In case it has not yet been created, the new resource is saved incl. the user fields transferred (RES.INSERT and several RESATTR.INSERT). A new resource does not have to be created for MOD=U, the optimization flag is set in the resource instead (RES.UPDATE) and if defined the status is switched to the status assigned to the PROD=U ID (DLG=RES_STATUS|PROD=U|…).

The upload is recorded in event_res. However, VERB and BZW have to be configured respectively.

This dialog requires the additional function "DNC-VGN".

ID	Type/max. field length	Description
DLG	RES_UPLOAD	Upload
VERB	C1	N = plausibility check only, without posting J = plausibility check and posting (by default)
DAT	{mm/dd/yyyy}	Date specification for the upload.

ID	Type/max. field length	Description
ZEI	{seconds}	Time for the upload
MOD	C1	ID to differentiate whether a DNC resource is to be "U" = changed or "N" = newly created
RESVERWEIS or RESID	N10	Resource ID, unique number to identify the resource in database tables. Alternative specification for RESTYP and RES. Valid for MOD="U" only (=Update), if "N" this ID is only determined when the resource is created.
RESTYP	C4	Key 1 of the double key to uniquely identify a resource. Resource type. Always to be indicated together with RES. Alternative specification for RESVERWEIS if MOD="U".
RES	C40	Key 2 of the double key to uniquely identify a resource. Resource name. Always to be indicated along with RESTYP. Alternative specification for RESVERWEIS if MOD="U".
BEARB	C10	Editor
PNR	C10	alter-native: PNR
KNR	C10	alter-native: KNR
RESVER	C20	Version ID (not used as key)
RESFAMID	N10	Resource family identifier (serial) for creating a new resource. If both values (RESFAMID and RESFAM) remain empty no family is defined for the resource.
RESFAM	C20	Resource family ID, provided that the resource family identifier is not entered.
BZW	C1	J = Booking obligation, i.e. no plausibility checks N = No booking obligation (by default)
OPT_DNCVERARB	C1	Identifier whether it is a DNC resource ("L", "E", "O") or not ("K"). "K"=no DNC resource "L"=local "E"=external "O"=optimized If not indicated the dialog determines this value on the basis of the resource type of the resource. "O" triggers additional activities when it comes to posting.
MNR	N8/C8	Machine
SPEICHORT:DATA	C128	Name of the DNC file without path and file extension
IDX:1..4	N10	A maximum of 4 user fied data. User field index. This field has to be configured correctly by the assigned user field key of the resource. The value entered here corresponds to the field index of the database.
FIL:1..4	C20	A maximum of 4 user field data. Data content of the user field. Please also see info on "IDX1…4". The value entered here corresponds to the data field that matches the specified field index of the database.

ID	Type/max. field length	Description
KOMMENTAR	C500	Upload comment. Is saved for the event (event_res) in the event_dlg_data table. A reference is saved in event_res. If the resource is indicated several times the comment is saved only once. All new event_res entries (for each resource specified) then refer to it.

12.4 Lists for DNC-Data

To operate DNC resources there are special lists to read DNC information at the workplace (terminal) of the machine. These lists have also been configured to ensure conformity and to perform required inspections.

12.4.1 Enhancement of BDE lists

12.4.2 Machines – DNC family (DLG=LIST;82)

The machine defines which DNC family is assigned to the machine. Corresponding attribute configurations and, as a result, validity checks and functions are connected by these admissible families.

Moreover, the list provides a complete description of attributes! Consequently, user attributes do not have to be read separately.

Structure of dialog data:

“DLG=LIST;82|DATEI={file name}|DAT=...|ZEI=...|USR=...|MOD=...“

Parameter:

MOD=M|MNR=... Finding the family of a machine

MOD=T|TNR=... Finding the family of a terminal

Result list:

ID	Type/max. field length	Description
MNR	C20	Machine
RESFAM	C20	Resource family
RESFAM:BEZL	C60	Designation
RESTYPID	C4	Resource type identification
VORG	C1	J/N specification (= default family)
IDX:x	Integer	ID of the user field (free user attributes) within this user field key Numeric 1…n → Value corresponds to value FELD IDX of resource attributes
BEZK:x	C20	Designation, e.g. “DIAM”
BEZL:x	C80	Long designation, e.g. “diameter”
POS:x	Short	Display position/sequence
FKT_PLAUS:x	C10	ID for algo. plausibility check „=“ = IST_GLEICH check [level 1, other checks follow]
CFG_1:x … CFG_10:x	C10	1-4 for "DNC" 1: Filter (for filter dialog at the terminal and for the entry when uploading a new program) 2: Mandatory field for the DNC upload 3: Specification for the DNC upload

ID	Type/max. field length	Description
		4: Read-only for the DNC upload
KENN:x	C20	Acronym by which the field is addressed. This acronym is used in formulas to reference it as parameter, for example. Customer IDs are assigned within the individual namespace, i.e. the system automatically prefixes a “U:”. e.g. MNR
EINH:x	C3	Unit
TYP:x	C1	NUM, TEXT; DEC
FMT:x	C20	Format for console display (optional)
NKS:x	Short	If DEC, decimal places
LEN:x	Short	Field length
VON:x	DEC 18,6	For NUM/DEC data type: min. value range
BIS:x	DEC 18,6	For NUM/DEC data type: min. value range
ZEICH:x	C100	Which characters are allowed for the TEXT data type?
VORG_ EXPR:x	C50	Default value that is, for example, used for calculating formulas if the initial field is not available or empty.
VORG_EDIT:x	C80	Default value for new input

12.4.3 Loadable DNC programs (DLG=LIST;83)

By defining attributes of the DNC family and assigning them to the machine, it is determined which DNC programs currently saved in the system may be transferred to the terminal. The list provides all programs that are admissible or currently planned for the terminal. This restriction is improved through the resource attributes defined and the corresponding plausibility checks. In this context, restrictions are also predetermined by the operation that is logged on. The bills of material defined for DNC packages are broken down to their elements and posted in the list.

Structure of dialog data:

„DLG=LIST;83|DATEI={file name}|DAT=...|ZEI=...|USR=...|MOD=...“

Parameter:

MOD=A|... DNC programs filtered by order relation

MOD=P|... DNC programs filtered via the program name

MOD=D|... DNC programs without order relation. Direct reading on the resource stock. Selection via DNC family

The selection parameters are:

- TNR

- MNR
- ANR
- RES

Valid combinations:

	TNR	MNR	ANR	RES
MOD:A	No	No	Yes	No
MOD:P	No	Yes	No	Yes
MOD:D	Yes	Yes	No	No

ID	Type/max. field length	Description
MNR	C20	**MOD:A:** if started with parameter MNR or TNR then machine on which the OP is planned. **MOD:P:** Transferred machine **MOD:D:** Transferred machine
ANR	C40	**MOD:A:** OP which this resource is required for. **MOD:P:** empty **MOD:D:** empty
RESTYPID	C4	Type ID
RESTYP	C20	Resource type
RESTYP_BEZL	C60	Resource type designation
RES	C40	Resource
VERWEIS	N10	Resource ID
RESFAM	C20	DNC family
RESFAM_BEZL	C60	DNC family
OPT_DATEIVORH	C1	Optimized program available
BEARBDAT	{mm/dd/yyyy}	Date
BEARBZEI	{seconds}	Time

ID	Type/max. field length	Description
OPT_DNCVERARB	C3	DNC processing with optimization procedure
OPT_DATEIBASIERT	C1	Optimization, file-based
OPT_EXTGUELT	C5	Valid file extension
OPT_EXTOPTIM	C5	Optimized file extension
OPT_PATH	C128	Path of opt. file
DATEI	C128	File name
DATEI_SIZE	N	File size
DATEI_LOKAL	C1	Locally available
AKTIV	C1	Active
OPT_VERSION	C1	Optimized
RESSTA	N10	Status
RESSTABEZ	C60	Status designation
OPT_ERF	C1	Blocked
SSPERR	C1	Collective block
SSTA	N10	Collective status
SSTABEZ	C60	Designation of collective status
COLOR		Color
META_RES	C1	Package J/N
RESTYP:M	C4	ResTyp parent
VERWEIS:M	N10	ResID parent
RES:M	C20	ResNr parent

Comments:

- The configuration “anzeige_teminal J/N“ of the resource must be set to “J“. Otherwise, the element or package is not displayed. However, if the element is part of a package, it is displayed anyway.

- For package elements data are taken over 1:1 from resource types, i.e. the package values are not “handed down” to the elements

➔ With respect to the configuration, it has to be ensured that resource types of the main programs and subprograms are configured correctly. The “package” setting at the resource type is not relevant in this context.

12.5 WRM Maintenance Dialog

This section deals with special dialogs for managing resource maintenances. These are functions to set the maintenance status, activate and deactivate maintenances as well as to restart interval maintenances.

12.5.1 Maintenance status and activation (DLG=RES_WART)

By way of this dialog

- the status of a resource maintenance is set (MOD=Z)
- a maintenance is activated or deactivated (MOD=A)
- a resource maintenance is reset and prepared for the next maintenance (MOD=R)

In all 3 modes current maintenance data are first saved as event. These events are used for logging and reports.

Depending on the mode, different functions are performed:

- Set the resource maintenance status (MOD=Z). The WARTSTA parameter transfers the new status that is to be set. The status is set for the maintenance stated.
- Activate or deactivate (MOD=A) the maintenance. The AKTIV (J/N) parameter indicates whether the maintenance is to be activated or deactivated.
- Reset resource maintenance (MOD=R). All actual values are set to 0 and target values are calculated for the next maintenance. The AUFSATZ (S/I) parameter indicates whether the new calculation is to be based on the old target value “S” or on the current actual value “I”. The LWART:DAT, LWART:ZEI and LWART:USR parameters save the date, time and person of the maintenance made.

Finally, a comment is saved for all modes.

ID	Type/max. field length	Description
DLG	RES_WART	Set maintenance data
MOD	C1	Z = set status A = activate/deactivate R = reset
DAT	{mm/dd/yyyy}	Date specification for the download.
ZEI	{seconds}	Time for the download
RESVERWEIS or RESID	N10	Resource ID, unique number to identify the resource in database tables. Alternative specification for RESTYP and RES. RESID takes priority over the RESTYP+RES specification.

ID	Type/max. field length	Description	
RESTYP	C4	Key 1 of the double key to uniquely identify a resource. Resource type. Always to be stated along with RES. Alternative specification for RESVERWEIS.	
RES	C40	Key 2 of the double key to uniquely identify a resource. Resource name. Always to be stated along with RESTYP. Alternative specification for RESVERWEIS.	
WARTVERWEIS	N10	Reference to the defined maintenance → event_res.info_03	
WARTVERWEIS	N10	Reference to the defined maintenance → event_res.info_03	
PNR	C10	alter-native	Personnel number
KNR	C10		Badge number
RESVER	C20	Version ID (not used as key)	
ZUSTAND	N10	New status [value range 0 to 3]	
AKTIV	C1	New active status: J=enable, N=disable	
KOMMENTAR	C255	Comment	
LWART:DAT	DATE	Date of last maintenance	
LWART:ZEI	TIME	Time of last maintenance	
LWART:USR	C10	Person of last maintenance	

12.6 WRM Measures Dialog

This section describes special dialogs that have been configured to manage resource measures. At the moment this is a function to define a new measure for a resource. At the same time this activity is recorded for reports in HYDRA.

12.6.1 Activate measure (DLG=RES_MASS)

Using this dialog, a measure is activated, i.e. the entered measure is defined within the resource status.

The provided comment is saved.

<table>
<tr><th>ID</th><th>Type/max. field length</th><th colspan="2">Description</th></tr>
<tr><td>DLG</td><td>RES_MASS</td><td colspan="2">Set measure</td></tr>
<tr><td>DAT</td><td>{mm/dd/yyyy}</td><td colspan="2">Date specification for the download.</td></tr>
<tr><td>ZEI</td><td>{seconds}</td><td colspan="2">Time for the download</td></tr>
<tr><td>RESVERWEIS
or
RESID</td><td>N10</td><td colspan="2">Resource ID, unique number to identify the resource in database tables. Alternative specification for RESTYP and RES.
RESID takes priority over the RESTYP+RES specification.</td></tr>
<tr><td>RESTYP</td><td>C4</td><td colspan="2">Key 1 of the double key to uniquely identify a resource. Resource type. Always to be entered along with RES. Alternative specification for RESVERWEIS.</td></tr>
<tr><td>RES</td><td>C40</td><td colspan="2">Key 2 of the double key to uniquely identify a resource. Resource name. Always to be entered along with RESTYP. Alternative specification for RESVERWEIS.</td></tr>
<tr><td>PNR</td><td>C10</td><td rowspan="2">alter-native</td><td>Personnel number</td></tr>
<tr><td>KNR</td><td>C10</td><td>Badge number</td></tr>
<tr><td>RESVER</td><td>C20</td><td colspan="2">Version ID (not used as key)</td></tr>
<tr><td>MASSNR</td><td>N10</td><td colspan="2">Measure number</td></tr>
</table>

12.7 Lists for Resource Data

12.7.1 Resource list (DLG=LIST;115)

The list is used to select a resource.

Structure of dialog data:

“DLG=LIST;115|DATEI={file_name}|DAT=...|ZEI=...|USR=...|RESTYP=...|RESFAM=...|RES=…|ANR=…|AKRO=…|MOD=…|“

Parameter:

Parameters are optional and support wildcard characters (%).

RESTYP =... restrict the list by a resource type

RESFAM =... restrict the list by a resource family. The designation is to be indicated here, not the ID.

RES =... Restrict the list by a resource.

ANR =... Show only resources that are planned for the order.

AKRO=… Fast user fields are available as dynamic fields. When starting the list, they can be requested, separated by semicolon using the AKRO=FU:1 to FU:66 ID.

MOD=M : As set union the list includes all resources that are either logged on/active to/on the transferred order (ANR=) or that are planned for this order. In this case, only resources are selected, which may be logged on explicitly (configuration of the resource stock) or that are to be displayed in the resource list (configuration of the resource type).

OPT:BEDRES=N: The list does **not** include required resources. This option allows for the resource list to be requested without required resources. All other values, except for “N” mean that the list also includes required resources. This function cannot be combined with MOD=M.

Example:

hymw -u2020

-c"DLG=LIST;115|DATEI=l.lst|DAT=today|ZEI=now|USR=2020|RESTYP=WNR| RESFAM= Soft%|RES=066% |AKRO=FU:12;FU:20|"

Result list:

ID	**Type/max. field length**	**Description**
RESTYP	C4	Resource type
RESFAM	C20	Resource family
RES	C40	Resource number
BEZ	C40	Designation
RESSTA	Integer	Current status
RESVERWEIS	Integer	Internal ID
RESVER	C20	Resource version
RESSTATXT	C20	Resource status text
RESTYPBEZ	C20	Resource type designation
RESFAMBEZ	C20	Resource family designation
COLOR	Integer	Color
MATPUF:I	12	Current storage location
MNR	C20	Active machine
ANR	C40	Active operation
BZG:RESTYP	C4	Reference resource type
BZG:RES	C40	Reference resource
SLP	Integer	Bill of material item

12.7.2 Resource Status List (DLG=LIST;116)

List of all possible statuses for a resource.

Structure of dialog data:

“DLG=LIST;116|DATEI={file_name}|DAT=...|ZEI=...|USR=...|RESTYP=...|RESFAM=...|RES=…“

Parameter:

Parameters are optional.

RESTYP =... Restrict the list by a resource type

RESFAM =... Restrict the list by a resource family. The designation is to be indicated here, not the ID.

RES =... Restrict the list by a resource.

Example:

hymw -u2020

-c"DLG=LIST;116|DATEI=l.lst|DAT=today|ZEI=now|USR=2020|RES=0668258-WKS|RESTYP=WNR|RESFAM= Soft|"

Result list:

ID	Type/max. field length	Description
OPT:SIBTNR	C1	Display of the resource at the terminal
STUFE	Integer	Authorization level
COLOR	Integer	Status color
MATPUF	C12	Storage location when setting the status
PRIO	Integer	Priority
PKENN	C1	Posting indicator
RESFAMID	Integer	Resource family ID
RESTYP	C4	Resource type
RESSTA	Integer	Status
BEZ	C20	Status designation
OPT:ERF	C1	Collection/processing
OPT:PLAN	C1	HLS assignment

12.7.3 List of measures (DLG=LIST;117)

List of all possible measures for a resource.

Structure of dialog data:

“DLG=LIST;117|DATEI={file_name}|DAT=...|ZEI=...|USR=...|RESFAM=...|RESFAMID=...|RES=…“

Parameter:

Parameters are optional.

RESFAM =... Restrict the list by a resource family (designation).

RESFAMID =... Restrict the list by a resource family (ID).

RES =... Restrict the list by a resource.

Example:

hymw -u2020

-c"DLG=LIST;117|DATEI=l.lst|DAT=today|ZEI=now|USR=2020|RES=0668257-WKS|RESFAMID=10001|RESFAM=Handih.|"

Result list:

ID	Type/max. field length	Description
BEZK	C20	Short designation
BEZL	C60	Long designation
DATE	Datum	Date valid till
DATB	Datum	Date valid from
ZEIE	Integer	Time valid till
ZEIB	Integer	Time valid from
BEM	C200	Comment
MASSNR	Integer	ID of measure
RESFAM	C20	Resource family
RESFAMID	Integer	Reference to resource family
RESTYP	C4	Resource type
TYP	C1	Type of measure
VAB	C15	Responsibility area

12.7.4 Resource types (DLG=LIST;118)

List of all possible resource types.

Structure of dialog data:

“DLG=LIST;118|DATEI={file name}|DAT=...|ZEI=...|USR=...|“

Parameter:

none

Example:

hymw -u2020 -c "DLG=LIST;118|DATEI=l.lst|DAT=today|ZEI=now|USR=2020|"

Result list:

ID	Type/max. field length	Description
OPT:AZSWSIB	C10	Show in reports
OPT:VERB	C1	Posting of the resource
PLAUS:BEL	C1	Assignment within HYDRA-HLS
RESTYPEXT	C10	Designation
BEZL	C60	Long designation
VERB:BMK1..11	Integer	Posting for operating hours
OPT:DATEI	C1	File-based resource
OPT:DNC	C1	Indicator for DNC processing
EXT:GUELITG	C5	File extension
EXT:OPTIM	C5	File extension, optimized files

ID	Type/max. field length	Description
OPT:AUTOANMELD	C1	Log resource on/off with order
CFG:1..10	C1	Configuration settings
PATH	C8	Reference to the hy_path table
AUTOANLAG	C1	Create resource automatically
RESTYP	C4	Resource type
OPT:VBR	C1	Control
USRFLD	C20	Assigned user field ID

12.7.5 Resource family (DLG=LIST;119)

List of all possible resource families.

Structure of dialog data:

“DLG=LIST;119|DATEI={file name}|DAT=...|ZEI=...|USR=...|RESTYP=...|“

Parameter:

Parameters are optional.

RESTYP =... Restrict the list by a resource type

Example:

hymw -u2020

-c"DLG=LIST;119|DATEI=cb.lst|DAT=today|ZEI=now|USR=2020|RESTYP=WNR|"

Result list:

ID	Type/max. field length	Description
RESFAM	C20	Resource family
BEZL	C60	Long designation
VORG:RES	C50	Default specification of resource name
OPT:KOPIE	C1	Performance when copying the resource
OPT:EINST	C1	Parameters for Viewer
FKT:SPEICHORT	Integer	Autom. generation of names
CFG:1..10	C1	Configuration parameters
RESFAMID	Integer	ID of the resource family
RESTYP	C4	Resource type
SPEICHART	C1	Storage type of the family
USRFLD	C20	Assigned user field ID
VAB	C15	Responsibility area
OPT:RESVER	C10	Version procedure
VIEWER:DATA	C10	Viewer to display the resource file
VIEWER:EINST	C10	Parameter fields for the viewer

12.7.6 Resource maintenance (DLG=LIST;120)

List of all possible resource maintenances.

Structure of dialog data:

“DLG=LIST;120|DATEI={file name}|DAT=...|ZEI=...|USR=...|ANR=…|AKTIV=…|BEZ=...|“

Parameter:

Parameters are optional and support wildcard characters (%).

ANR =... Restrict the list by an order

AKTIV =... Restrict the list by active or inactive

BEZ =... Restrict the list using the maintenance designation

Example:

hymw -u2020

-c"DLG=LIST;119|DATEI=cb.lst|DAT=today|ZEI=now|USR=2020|RESTYP=WNR|"

Result list:

ID	Type/max. field length	Description
AKTIV	C1	Active maintenance
APUNR	C40	Work plan
ANR	C40	Order number
BEZ	C60	Maintenance designation
OPT:BZG	C1	Reference of values
DATE	Datum	For maintenance monitoring
DATB	Datum	For maintenance monitoring
INFO:1..7	C80	Info value
ANZ:I	Integer	Actual number
BSTD:I	Integer	Actual operating hours
TAKT:I	Integer	Actual cycle
BDEJMOD	Integer	Year model
WARTKL	C10	Classification of maintenance
LWART:DAT	Datum	Date of last maintenance
LWART:ZEI	Integer	Time of last maintenance
LWART:USR	C10	Person of last maintenance
ANZ:N	Integer	Quantity till next maintenance
BSTD:N	Integer	Operating hours, next maintenance
TAKT:N	Integer	Cycles, next maintenance
TG:N	Datum	Date of next maintenance
RESVERWEIS	Integer	Reference to resource
WARTG:1	Integer	Threshold value for level 1
WARTG:2	Integer	Threshold value for level 2
WARTG:3	Integer	Threshold value for level 3
ANZ:S	Integer	Target number

ID	Type/max. field length	Description
BSTD:S	Integer	Target operating hours
TAKT:S	Integer	Target cycles
TG:S	Integer	Target days
ART	C2	Maintenance type
VERWEIS	Integer	Consecutive no. of maintenance activity
WARTSTA	Integer	Maintenance status

12.7.7 List of maintenance activities (DLG=LIST;91)

List of all possible maintenance activities.

Structure of dialog data:

"DLG=LIST;91|DATEI={file name}|DAT=...|ZEI=...|USR=...|MNR=…|RES=…|RESTYP=...|MOD=…|"

Parameter:

<u>In the machine mode (MOD=M)</u> :

MNR =... Generate list for the transferred machine

<u>In the resource mode (MOD=R) :</u>

RES =... Generate list for the transferred resource + resource type

RESTYP =... Generate list for the transferred resource + resource type

Example:

hymw -u2020

-c"DLG=LIST;91|DATEI=wtk.lst|DAT=today|ZEI=now|USR=2020|MOD=M| MNR=MASCH100|"

Result list:

ID	Type/max. field length	Description
MNR	C40	Machine
RESVERWEIS	Integer	Resource ID
VERWEIS	Integer	Maintenance number
BEZ	C20	Designation of the maintenance
ART	C2	Type of maintenance
WARTSTA	Integer	Status
WARTKL	C10	Class
AKTWERT	Integer	Current value
NAEWERT	Integer	Next value
WARTG:1 … 3	Integer	Threshold value 1 – 3
INFO:1 … 7	C80	Info text 1 – 7
LWART:DAT	Datum	Date of last maintenance
LWART:ZEI	Integer	Time of last maintenance
LWART:USR		User of last maintenance
RES	C40	Resource no.
RESTYP	C4	Resource type

ID	Type/max. field length	Description
KST	10	Cost center
OPT:ONCE	C1	Non-recurring maintenance (J/N)

12.7.8 List of resource comments(DLG=LIST;133)

List of all comments for a resource.

Structure of dialog data:

“DLG=LIST;133|DATEI={FileName}|DAT=...|ZEI=...|USR=...|RESTYP =...|RES=…|DATB=…|ANZ=…|“

Parameter:

RES =... Generate list for resource.

RESTYP =... Restrict list by resource type.

DATB =... Restrict list by date range
(only show comments that are “younger” than DATB).

ANZ =... Restrict the number of columns of the list (by default 100).

Example:

hymw -u2020

-c"DLG=LIST;133|DATEI=kommentare.lst|DAT=today|ZEI=now|USR=2020| RES=0668257-WKS|RESTYP=WNR|DATB=12/12/2007|ANZ=10"

Result list:

ID	Type/max. field length	Description
RES	C40	Resource number
RESTYP	C4	Resource type
BEZ	C20	Resource name
DAT	Datum	Date of the comment
ZEI	Integer	Time of the comment
PNR	C10	The editor’s personnel number
PNAME	C40	The editor’s name
BEM	C500	Comment

12.7.9 List of registered resources (DLG=LIST;129)

The list can be used to determine the resources that are logged on.

Structure of dialog data:

“DLG=LIST;129|DATEI={FileName}|DAT=...|ZEI=...|USR=...|MOD=…|“

Parameter:

MOD=M : In the M mode only the topmost machine designations of resources are selected for the machine.

Without MOD: If the list is requested without Mod=M all resources are displayed that are logged on to the machine. Duplicates are removed (e.g. if the resource refers to a machine and is logged on to an order).

Example:

hymw -u2020

-c"DLG=LIST;129|DATEI=l.lst|DAT=today|ZEI=now|USR=2020|MOD=M|MNR=100|"

Result list:

ID	Type/max. field length	Description
MNR	C20	Machine
ANR	C40	Order
RES	C40	Resource number
RESBEZ	C40	Resource name
RESTYP	C4	Resource type
RESTYPBEZ	C20	Resource type name
RESFAMBEZ	C20	Resource family name
RESSTA	Integer	Current status

12.7.10 Combined list of production resources and tools: batch and resources (DLG=LIST;132)

Materials/batches and resources can also be requested in a combined list:

- Resources from the resource list (DLG=LIST;115)
- Materials/batches from the list “material list/batch information“ (DLG=LIST;13)

For further details on these two lists, please refer to the section entitled “resource list” in this documentation or the section “material list/batch information” in the documentation entitled MPL-PDM-DC.

Structure of dialog data:

“DLG=LIST;132|DATEI={FileName}|MNR=…|ANR=…|FHM.ID=…|DAT=...|ZEI=...|USR=...|MOD=…|“

Parameter:

MOD=
K(OMBI)
List of all production resources and tools – material/batches and resources – planned for the OP and logged on to the OP/machine

R(ES)
The list only includes the production resources and tools resources planned for the OP and logged on to the OP/machine

M(AT)
The list only includes the production resources and tools material/batches planned for the OP and logged on to the OP/machine

I(NFO)
List including a single-row information (if the key is not unique, several rows, if necessary) for a specific production resource and tool – batches and/or resources

FHM.ID=
With single-row information it is the key for the selection (resource or batch number)

MNR=
Requesting machine

ANR=
Requesting order/OP

Example:

hymw -u2020

-c"DLG=LIST;132|DATEI=l.lst|DAT=today|ZEI=now|USR=2020|MOD=K|MNR=100|ANR=47110100"

Provides all materials/batches and resources logged on to machine 100 and OP 47110100 or that are planned for OP 47110100.

The result list includes the fields of both lists:

Prefix	**Description**
MAT.*	The prefix MAT provides all data fields of the batch list DLG=LIST;13. For further information on the individual fields refer to the section “material list/batch information” in the documentation entitled MPL-PDM-DC. These fields are initially assigned (0 or empty) for resource entries.
RES.*	The prefix RES provides all data fields of the resource list DLG=LIST;115 For further information on the individual fields refer to the section entitled “resource list “ in this documentation. These fields are initially assigned (0 or empty) for material/batch entries.
FHM.IDTYP	Key field resource type MAT for materials/batches != MAT for resource (can be WNR, DOC, etc.)
FHM.ID	Key field ID For batches → batch number CNR from list 12 For resources → resource number RES from list 115
FHM.SLP	Key field BOM item BOM item from list 13 or 115
FHM.ATK	Key field For batches → Material ATK from 13 For resources → reference resource BZG:RES from list 115
FHM.BEZ	Key field designation For batches → ATKBEZ from list 13 For resources → BEZ from list 115
RES.AKTIV	J → Resource has already be logged on N → Resource not yet logged on
FHM.VIS	Processing flag for the HYDRA terminal

Prefix	Description

13 HYDRA Production Data Manager WRM - Master Data

13.1 Note on the descriptions of the input dialogs

All fields that are mandatory and must be specified are highlighted using a gray background color or marked using (PK). All other fields are optional and are processed when passed.

13.2 Resources

13.2.1 Edit resources (DLG=RES.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

You use these dialogs to edit the master data of resources and free resource attributes. You can create and change resources.

Tables

Table	Key field	Description
res_bestand	res_typ RES.RESTYP	Resource type (PK) in resource stock
res_bestand	res_nr RES.RES	Resource number (PK) in resource stock
res_bestand	verweis RES.VERWEIS	Internal serial number
res_status	res_typ RES.RESTYP	Resource type (PK) in resource status
res_status	res_nr RES.RES	Resource number (PK) in resource status
res_status	verweis RES.VERWEIS	Internal serial number, reference to res_bestand.verweis

BAPI call

Identification	Content / {type}	Description
DLG	RES.INSERT	Create resource
	RES.UPDATE	Change resource
	RES.DELETE	Delete resource
	RES.COPY	Copy resource

	RES.LOCK	Lock resource for editing
	RES.UNLOCK	Unlock resource after editing
	RES.NEW	Read specification for new resource
	RES.SELECT	Select resource
RES.RESTYP	{C4}	PK resource type
RES.RES	{C40}	PK resource number
RES.VERWEIS	{N10}	Instead of using RES.RESTYP and RES.RES, you can also call the resource using the serial key RES.VERWEIS.
RES.RESTYP:Z	{C4}	PK new (target) resource type for COPY
RES.RES:Z	{C40}	PK new (target) resource number for COPY
RES.BEZ:Z	{C40}	New (target) resource name for COPY
…	…	For information on further fields, refer to the documentation of the database schema of the above-listed tables. For further information, see the section above.

Validation checks

Error codes	Description
3204	The resource specified in the dialog is already stored in the resource stock.
3220	Resource is logged on or locked
3253	The resource cannot be deleted because it is still used in a resource list relation.
4110	For this resource, a requirement is still available.
3254	The resource cannot be deleted because it is still used as production resource/tool.
7000	Resource records of type "resource logon" or "resource logoff" are available for the resource. For this reason, the resource cannot be deleted.
3261	The resource cannot be deleted because a maintenance calendar is stored for the resource.
3214	The resource type specified is not available in the system.
3200	No resource or resource ID has been transferred to the dialog.
1661	A value relevant for processing is missing.
101	General error message that is displayed when the selected data (tables or files) is not available.
1669	Data with the same key fields already exist. It is possible that you cannot see the

Error codes	Description
	data because you are not authorized.
3231	The check confirming that the resource is not active fails; i.e. the resource is currently used.
1803	You are not authorized for this responsibility area.
3263	The resource family status is now invalid because the resource family has been changed.
4101	The specified resource is included in a resource list.
3228	The check confirming that the resource is logged on fails; i.e. the resource is currently logged off.
3260	You cannot create or change the resource because the DNC file name is already assigned.

13.2.2 Resource list (DLG=RES.LIST)

This BAPI call creates a resource list. You can narrow down the selection and call the list for one resource, a family, a resource type or combinations of these fields. The free attributes must be read afterwards.

Tables

Table	Key field	Description
res_bestand	res_typ RES.RESTYP	Resource type (PK) in resource stock
res_bestand	res_nr RES.RES	Resource number (PK) in resource stock
res_bestand	verweis RES.VERWEIS	Internal serial number
res_status	res_typ RES.RESTYP	Resource type (PK) in resource status
res_status	res_nr RES.RES	Resource number (PK) in resource status
res_status	verweis RES.VERWEIS	Internal serial number, reference to res_bestand.verweis

BAPI call

Identification	Contents	Description
DLG	RES.LIST	Resource list

RES.RESTYP	{C4}	Search criterion resource type
RES.RES	{C40}	Search criterion resource number
RES.RESFAMID	{N10}	Search criterion resource family
RES.VERWEIS	{N10}	Search criterion serial key
…	…	…
DATEI	{C256}	Specification of the file name for the list

13.3 Free attributes

13.3.1 Edit free attributes (DLG=RESATTR.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK, NEW, SELECT)

Use the resource type or the resource family to define the resource attributes that you want to edit. The field contents of these fields are entered and managed separately.

When a resource is created (RES.INSERT), a field definition is assigned to this resource, which depends of the relevant resource family or resource type. The key to this field definition is stored in the data field RES.USRFLD,

The field definition of the resource attributes are read via list USRFLDELEM.LIST.

The values entered for the resource are listed in RESATTR.LIST. Both lists have USRFLD and IDX as key. The field definition and the entered attributes are parallel entries; i.e. in one list the column names are listed, in the other the entered values.

Use the dialogs RESATTR.INSERT and RESATTR.UPDATE to enter attribute values, or RESATTR.DELETE to delete values.

If free attributes are used, these attributes must be created after the resource or deleted before the resource.

Tables

Table	Key field	Description
res_attribute	res_id RESATTR.RESVERWEIS	Serial ID (PK) of resource in the resource attributes FK (foreign key) of resource stock (res_bestand.verweis)
res_attribute	feld_id	Field ID (PK) in the resource attributes

Table	Key field	Description
	RESATTR.IDX	Foreign key (FK) of user fields (hyd_userfieldelem.feldid)

BAPI call

Identification	Content / {type}	Description
DLG	RESATTR.INSERT	Create resource attribute
	RESATTR.UPDATE	Change resource attribute
	RESATTR.DELETE	Delete resource attribute
	RESATTR.COPY	Copy resource attribute
	RESATTR.LOCK	Lock resource attribute for editing
	RESATTR.UNLOCK	Unlock resource attribute after editing
	RESATTR.NEW	Read specification for new resource attribute
	RESATTR.SELECT	Select resource attribute
RESATTR.RESVERWEIS	{N10}	Serial ID (PK) of the resource of the resource attribute
RESATTR.IDX	{N2}	Field ID (PK) in the resource attributes
RESATTR.RESTYP	{C4}	Instead of using RESATTR.RESVERWEIS (serial key), you can also call the resource using the resource type RESATTR.RESTYP and the resource number RESATTR.RESNR.
RESATTR.RESNR	{C40}	See above
RESATTR.RESVERWEIS:Z	{N10}	New serial ID (PK) of the resource of the resource attribute for COPY
RESATTR.IDX:Z	{N2}	New field ID (PK) of the resource attribute
RESATTR.ATTR	{C80}	Attribute value
…	…	For information on further fields, refer to the documentation of the database schema of the above-listed tables. For further information, see the section above.

Validation checks

Error codes	Description
3201	You must transfer an existing resource number with resource type or a valid resource ID to the dialog.
1661	A value relevant for processing is missing: • You must either specify the serial ID (RESATTR.RESVERWEIS) or the resource type (RES.RESTYP) and the resource number (RES.RES). • You must specify the field ID (RESATTR.IDX) • You must specify the serial ID (RESATTR.RESVERWEIS:Z) with COPY
101	An attribute for the specified key does not exist.
1669	An attribute for the specified key does already exist.
1666	The data record is currently locked by another user.

Note

Make sure that the below configurations are made before you create attributes (RESATTR.INSERT):

1. Create a user field key for the resource type (e.g. DNC) of the resource.
2. To this user field key, assign the user fields for which you want to create attributes. Make sure that the field ID of the user field matches the field ID of the attribute (RESATTR.IDX).
3. You can assign the user field key via the resource type (e.g. DNC) or the resource family(ies).

13.3.2 List of resource attributes (DLG=RESATTR.LIST)

This BAPI call generates a list with the user-defined attributes of the resource. The defined attributes are numbered. The description of the attributes can be read using the field definitions.

Tables

Table	Key field	Description
res_attribute	res_id RESATTR.RESVERWEIS	Serial ID (PK) of resource in the resource attributes Foreign key (FK) of resource stock (res_bestand.verweis)
res_attribute	feld_id RESATTR.IDX	Field ID (PK) in the resource attributes Foreign key (FK) of user fields (hyd_userfieldelem.feldid)

BAPI call

Identification	Contents	Description

Identification	Contents	Description
DLG	RESATTR.LIST	List of resource attributes
RESATTR.RESVERWEIS	{N10}	Serial ID (PK) of the resource of the resource attribute
RES.RESTYP	{C4}	Instead of using RESATTR.RESVERWEIS (serial key), you can also call the resource using the resource type RESATTR.RESTYP and the resource number RESATTR.RES.
RES.RES	{C40}	See above
DATEI	{C256}	Specification of the file name for the list

13.3.3 List of field definitions (DLG= USRFLDELEM.LIST)

This BAPI call creates a list of field definitions. You can optionally narrow down the list to USRFLD.

Tables

Table	Key field	Description
hyd_userfieldelem	objekt_typ USRFLDELEM.OBJTYP	WRM: resource type of the resource
hyd_userfieldelem	usrfld_key USRFLDELEM.USRFLD	WRM: user field key

BAPI call

Identification	Contents	Description
DLG	USRFELDELEM.LIST	User field list
OBJTYP	{C20}	WRM: The object type is identical to the resource type.
USRFLD	{C6}	WRM: user field key (optional)
DATEI	{C256}	Specification of the file name for the list

13.4 Resource list

13.4.1 Edit resource list (DLG=RESLIST.INSERT, DELETE)

You can create a multi-level structure for the resources in HYDRA. To this end, you can specify a resource list for a resource. This list, also called "package", specifies the components included in the resource. The resource list also integrates the existing relations between the components. A complete tree structure with all components is integrated. A faulty use of this function can cause structural errors in the resource function, which can lead to a hang in HYDRA. We highly recommend to consult MPDV.

To read the resource list entries, a list command is available. This list command can be used at one level of the list or recursively for the complete structure of the resource list.

Note: The above is only true for the master data of the resource list. Once you have created an order and you are interested in the bill of materials, you must call the relevant lists of the BDE.

BAPI call

Identification	Content / {type}	Description
DLG	RESLIST.INSERT	Create resource list entry
	RESLIST.DELETE	Delete resource list entry
RESLIST.RESTYP:M	{C4}	PK: resource type of higher-level resource (M is for "mother")
RESLIST.RESNR:M	{C40}	PK: resource number of higher-level resource (M is for "mother") - alternative entry instead of RESVERWEIS:M
RESLIST.RESVERWEIS:M	{N10}	PK: ID of higher-level resource - alternative entry instead of RESNR:M
RESLIST.RESTYP:T	{C4}	PK: resource type of lower-level resource

Identification	Content / {type}	Description
RESLIST.RESNR:T	{C40}	PK: resource number of lower-level resource (T is for "*Tochter* = daughter") - alternative entry instead of RESVERWEIS:T
RESLIST.RESVERWEIS:T	{N10}	PK: ID of lower-level resource - alternative entry instead of RESNR:T
RESLIST.POS	{N10}	Display position / sequence (with RESLIST.INSERT)
RESLIST.MENGE	{N18.6}	Quantity ratio to mother (with RESLIST.INSERT) May only be set with anonymous resources to a value > 1. Otherwise the value must be 1.
MOD	{C1}	Delete mode (with RESLIST.DELETE) E: Single resource list relation is deleted. G: The complete resource list below a specified mother resource is deleted.
DELFIRSTLEVEL	{C1}	Can only be used with MOD=G. The lower-level resource lists are not deleted. J: Only the first level of the resource list is deleted.

Validation checks

Error codes	Description
3202	The resource specified in the dialog (entry without resource type) is available multiple times in the resource stock with different resource types each.
1666	The data record is currently locked by another user.
1661	A value relevant for processing is missing. • Invalid mode (MOD) is specified. • The lower-level resource has not been specified (RESLIST.RESVERWEIS:T or RESLIST.RESTYP:T and RESLIST.RESNR:T) • The higher-level resource has not been specified (RESLIST.RESVERWEIS:M or RESLIST.RESTYP:M and RESLIST.RESNR:M) • The resource type has not been specified (RESLIST.RESTYP:T or RESLIST.RESTYP:M)
	RESSOURCE_NICHT_VORHANDEN
4101	The specified resource is included in a resource list.
101	General error message that is displayed when the selected data (tables or files) is not available.
1669	Data with the same key fields already exist. It is possible that you cannot see the

Error codes	Description
	data because you are not authorized.

13.4.2 Resource list (DLG=RESLIST.LIST)

This BAPI call creates a resource list.

BAPI call

Identification	Contents	Description
DLG	RESLIST.LIST	Resource list
RESLIST.RESTYP:M	{C4}	PK: resource type of higher-level resource (M is for "mother")
RESLIST.RESNR:M	{C40}	PK: resource number of higher-level resource (M is for "mother") - alternative entry instead of RESVERWEIS:M
RESLIST.RESVERWEIS:M	{N10}	PK: ID of higher-level resource - alternative entry instead of RESNR:M
MOD	{C1}	MOD=E: Resource list with one level only from the mother on. Here you can find the resources of the direct level below. MOD=B: Complete structure of the resource list starting from mother. All relations below the mother are recursively read. Here, you can find all related resources in the resource list.
…	…	…
DATEI	{C256}	Specification of the file name for the list

Result list

The result list provides the data fields listed in section 13.4.1 Edit resource list (DLG=RESLIST UPDATE,).

13.5 Assignment to required resources

13.5.1 Edit assignments to required resources (DLG=RESBEDRES.INSERT, DELETE, COPY,)

You can manage required resources in HYDRA that are used as replacements. When the resources are logged on, concrete resources are logged on for these required resources, which are assigned to the required resource. You can assign one or several (replacement) resources to one required resource.

BAPI call

Identification	Content / {type}	Description
DLG	RESBEDRES.INSERT	Create an assignment to a required resource
	RESBEDRES.DELETE	Delete an assignment to a required resource
	RESBEDREST.COPY	Copy an assignment to a required resource
RESBEDRES.RESTYP:M	{C4}	PK: resource type of the required resource
RESBEDRES.RES:M	{C40}	PK: resource number of required resource
RESBEDRES.RESTYP:T	{C4}	PK: resource type of the assigned resource
RESBEDRES.RES:T	{C40}	PK: resource number of the assigned resource
RESBEDRES.BEM	{C100}	Comment
RESBEDRES.RESTYP:MZ	{C4}	With COPY: resource type of the target required resource
RESBEDRES.RES:MZ	{C40}	With COPY: resource number of the target required resource
RESBEDRES.RESTYP:TZ	{C4}	With COPY: resource type of the assigned target resource
RESBEDRES.RES:TZ	{C40}	With COPY: resource number of the assigned target resource
MOD	{C1}	Delete mode (with RESBEDRES.DELETE) E: Single assignment to a required resource is deleted. G: For a specified required resource, all assigned (replacement) resources are deleted

		Copy mode (with RESBEDRES.COPY) E: Single assignment to a required resource is copied. F: All missing assignments of a specified required resource are copied to the target required resource. A: All assignments of a specified required resource are copied to the target required resource.

Validation checks

Error codes	Description
4103	The required resource specified in the dialog is not configured as required resource in the resource stock.
3201	The required resource specified in the dialog or the assigned resource is not available in the resource stock.
4101	The required resource specified in the dialog is already included in a resource list.
4102	The assigned resource specified in the dialog has the type anonymous resource or required resource.
1666	The data record is currently locked by another user.
1661	A value relevant for processing is missing. • Invalid mode (MOD) is specified. • The lower-level resource has not been specified (RESBEDRES.RESTYP:T and RESBEDRES.RESNR:T) • The higher-level resource has not been specified (RESBEDRES.RESTYP:M and RESBEDRES.RESNR:M). • The resource type has not been specified (RESBEDRES.RESTYP:T or RESBEDRES.RESTYP:M)
101	General error message that is displayed when the selected data (tables or files) is not available.
1669	Data with the same key fields already exist. It is possible that you cannot see the data because you are not authorized.

13.6 Resource families

13.6.1 Edit resource families (DLG=RESFAM.INSERT, UPDATE, DELETE, LOCK, UNLOCK, NEW, SELECT)

You use these dialogs to edit the master data of resource families. You can create and change resource families. You can only delete a resource family, if no resource is assigned to this family.

Tables

Table	Key field	Description
res_familien	bezeichnung RESFAM.RESFAM	Resource family (PK). The ID must be unique in all resource types.
res_familien	res_typ RESFAM.RESTYP	If you create a resource family, you must define the resource type.

BAPI call

Identification	Content / {type}	Description
DLG	RESFAM.INSERT	Create resource family
	RESFAM.UPDATE	Change resource family
	RESFAM.DELETE	Delete resource family
	RESFAM.LOCK	Lock resource family and prevent editing
	RESFAM.UNLOCK	Remove lock of resource family after editing
	RESFAM.NEW	Read specification for new resource family
	RESFAM.SELECT	Select resource family
RESFAM.RESFAM	{C8}	PK "resource family"
RESFAM.RESFAMID	{N8}	Internal ID of resource family.
RESFAM.RESTYP	{C4}	Resource type of the resource family. If you create a new resource family, you must define the resource type.
RESFAM.BEZL	{C30}	Name of the resource family
RESFAM.VAB	{C15}	Responsibility area
RESFAMB.USRFLD	{C8}	Reference to a user field key

Validation checks

Error codes	Description
1661	A value is missing that is required for processing. Missing parameter if no short description has been transferred (RESFAM.BEZK).
1661	A value is missing that is required for processing. Missing parameter if no resource type (RESFAM.RESTYP) has been passed on creating a resource family (RESFAM.INSERT).
101	General error message that is displayed when the selected data (tables or files) is not available.
1669	Another resource family with the same short description already exists (RESFAM.BEZK). Data with the same key fields already exist. It is possible that you cannot see the data because you are not authorized.
3243	The resource family is assigned to at least one resource. To delete a resource family, you must first remove all assignments in the resource configuration.
3241	The user field key transferred (RESFAM.USRFLD) does not exist.
1803	You are not authorized for this responsibility area (RESFAM.VAB).

13.7 Resource maintenances

13.7.1 Edit resource maintenances (DLG=RESWART.INSERT, UPDATE, DELETE, COPY, LOCK, UNLOCK)

You use these dialogs to edit the master data of activities in the *Activity calendar.* You can create, change, copy and delete activities.

Note

To reset a maintenance, use the dialog RES_WART.

Tables

Table	Key field	Description
res_wartungen		The table contains the master data of the activities.

BAPI call

Identification	Content / {type}	Description
DLG	RESWART.INSERT	Create activity
	RESWART.UPDATE	Change activity
	RESWART.DELETE	Delete activity
	RESWART.COPY	Copy activity
	RESWART.LOCK	Lock activity for editing
	RESWART.UNLOCK	Unlock activity after editing
RESWART.AKTTYP	{C1}	Activity type Possible values: empty = maintenance, K = calibration
RESWART.WARTKL	{C10}	Maintenance category
RESWART.KTR	{C40}	Specification of the cost object
RESWART.TAKT:I	{N18.6}	Cycles recorded so far (for the specified resource). Must be set if RESWART-ART = T
RESWART.TAKT:S	{N18.6}	Interval target cycles. Must be set if RESWART-ART = T
RESWART.TAKT:N	{N18.6}	Target cycles (absolute); the next maintenance activity is due after the specified number of target cycles. Must be set if RESWART-ART = T
RESWART.BSTD:S	{N10}	Hours of operation recorded so far (for the specified resource). Must be set if RESWART-ART = B
RESWART.BSTD:N	{N10}	Interval hours of operation. Must be set if RESWART-ART = B
RESWART.BSTD:I	{N10}	Hours of operation (absolute); the next maintenance activity is due after the specified number of operating hours. Must be set if RESWART-ART = B
RESWART.TG:S		Date of next activity. Must be set if RESWART-ART = Z
RESWART.TG:N	{N4}	Interval days. Must be set if RESWART-ART = Z
RESWART.BEZ	{C60}	Name of activity

RESWART.PLANAUNR	{C40}	Specification of planned order to which the activity is assigned
RESWART.PRJNR	{C25}	Specification of project number to which the activity is assigned.
RESWART.OPT:BZG	{C1}	Specification of the reference. Possible values: G = total or A = refers to order
RESWART.RESVERWEIS	{N10}	Internal resource identification number the activity refers to
RESWART.WARTG:1	{N10}	Threshold value for level 1
RESWART.WARTG:2	{N10}	Threshold value for level 2
RESWART.WARTG:3	{N10}	Threshold value for level 3
RESWART.ART	{C2}	Maintenance type T = based on cycles B = based on hours of operation Z = based on time (days)
RESWART.DATB	mm/dd/yyyy	Specification of date for "Valid from"
RESWART.DATE	mm/dd/yyyy	Specification of date for "Valid to"
RESWART.INFO:1 bis 5	{C80}	Information fields 1 to 5
…	…	For information on further fields, refer to the documentation of the database schema of the above-listed tables. For further information, see the section above.

Validation checks

Error codes	Description
3200	No resource or resource ID has been transferred to the dialog.
1661	A value relevant for processing is missing.
101	General error message that is displayed when the selected data (tables or files) is not available.
1669	Data with the same key fields already exist. It is possible that you cannot see the

Error codes	Description
	data because you are not authorized.
3231	The check confirming that the resource is not active fails; i.e. the resource is currently used.
1803	You are not authorized for this responsibility area.
3263	The resource family status is now invalid because the resource family has been changed.
4101	The specified resource is included in a resource list.